# Extreme Series

# EXAM BOOK

Natural Science

University Entrance Examination

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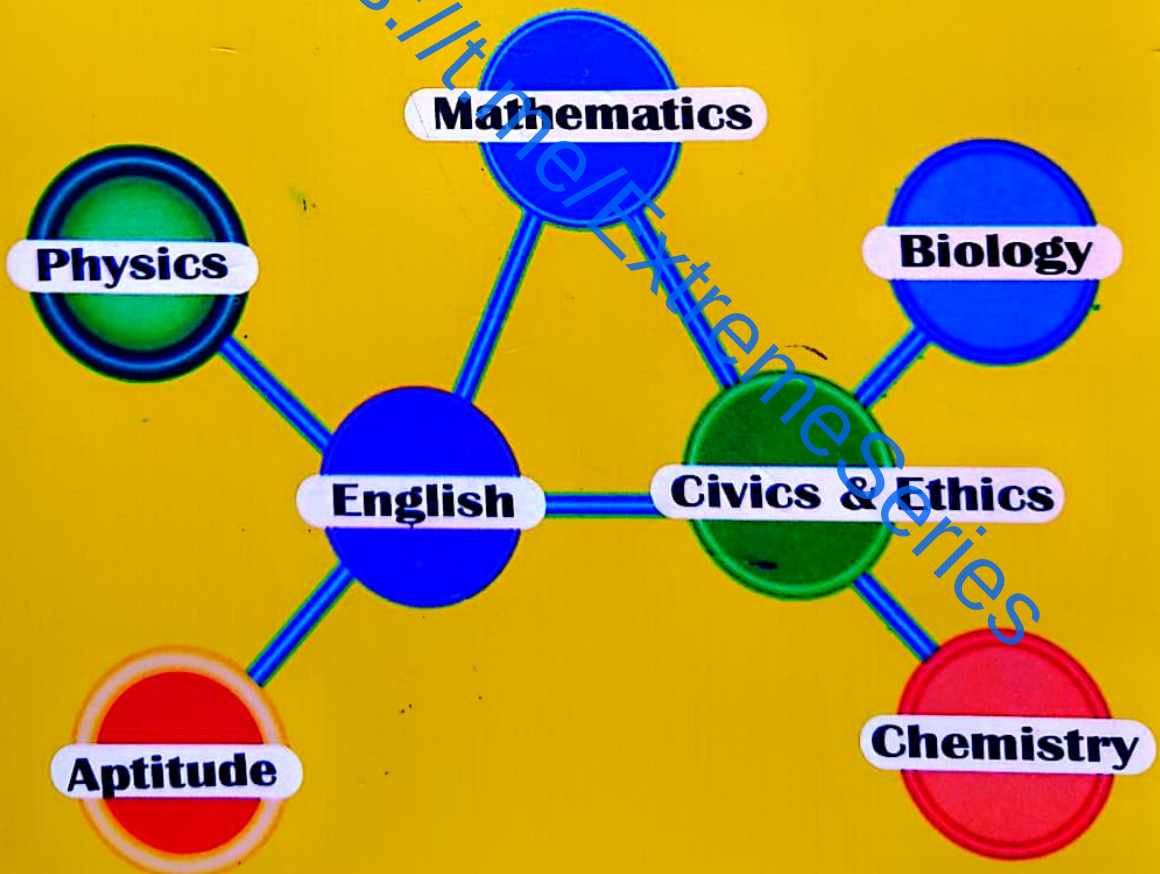

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## SECTION 1 – MATHEMATICS

1. Which of the following is rational function:

A. $f(x)=\dfrac{\sqrt{2x}-x+1092}{x^4+x^2+\pi}$ C. $h(x)=\dfrac{x^2-2^x+x}{x^4+x+1}$

B. $g(x)=3x^4-\sqrt{x^2+1}+2x$ D. $r(x)=\dfrac{x^2+\sin x+2}{x+1}$

***Solution:*** Rational function $=\dfrac{\text{polynomial}}{\text{polynomial}}$ **Answer: A**

2. If $P(x)=x-4$, $q(x)=x^2+8$ and $r(x)=x^2-2x-8$ then which of the following is the most simplified expression of $f(x)=\dfrac{x}{P(x)}-\dfrac{q(x)}{r(x)}$ :

A. $\dfrac{x}{x-4}-\dfrac{q(x)}{r(x)}$ C. $\dfrac{2}{x+2}$

B. $\dfrac{8x}{x-4}$ D. $\dfrac{2x+8}{x^2-2x-8}$

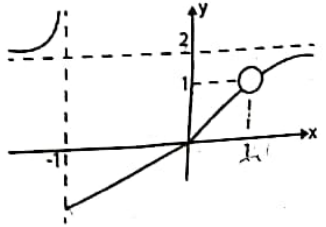
***Solution:*** $f(x)=\dfrac{x}{x-4}-\dfrac{x^2+8}{x^2-2x-8}\Rightarrow f(x)=\dfrac{x}{x-4}-\dfrac{x^2+8}{(x-4)(x+2)}$

$\Rightarrow f(x)=\dfrac{x(x+2)-(x^2+8)}{(x-4)(x+2)}$ ← multiply $\dfrac{x}{x-4}$ by $\dfrac{x+2}{x+2}$

$\therefore f(x)=\dfrac{x^2+2x-x^2-8}{(x-4)(x+2)}=\dfrac{2(x-4)}{(x-4)(x+2)}=\dfrac{2}{x+2}$,

**Answer: C**

3. Which of the following functions called most likely be sketched as in the figure?




A. $f(x)=\dfrac{2x}{x+1}$

B. $f(x)=\dfrac{2x(x-1)}{x^2-1}$

C. $f(x)=\dfrac{2}{x+1}$

**Solution:** The graph of $f(x)=\frac{2x(x-1)}{x^2-1}=\frac{2x}{x+1}$ has:

- Vertical asymptote $x = -2$
- Horizontal asymptote, $y = 2$
- Hole at $x = 1$ and $x \to 1, y \to 1$
- $x$ and $y$ intercept at (0, 0)

**Answer: B**

4. Let $\ell$ be the line containing (1, 5) and its angle of inclination be $\theta = 135°$. Which of the following is the distance form $\ell$ to (-1, 1)?

A. $2\sqrt{2}$ B. $3\sqrt{2}$ C. $4\sqrt{2}$ D. $6\sqrt{2}$

**Solution:**

Suppose slope of $\ell = \tan\theta = \tan(135°) = -1$.

Equation of $\ell$: $y-5=(-1)(x-1) \Leftrightarrow \ell: x+y-6=0$.

The distance from $\ell$ to $(-1, 1)$ is given by:

$$\frac{|Ax+By+C|}{\sqrt{A^2+B^2}}=\frac{|-1+1-6|}{\sqrt{(-1)^2+1^2}}=\frac{6}{\sqrt{2}}=3\sqrt{2}$$

**Answer: B**

5. The equation of circle with center at (2, −1) and tangent to the $y$−axis.

A. $(x+2)^2+(y-1)^2=1$ C. $(x-2)^2+(y+1)^2=4$
B. $(x-2)^2+(y+1)^2=1$ D. $(x+2)^2+(y-1)=4$

**Solution:**

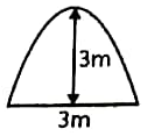
If the circle tangent to $y$-axis with center (h, k) then the radius will be $r = h$. ∴ The equn. of circle: $(x-h)^2+(y-k)^2=h^2$
$\Rightarrow (x-2)^2+(y+1)^2=4$

**Answer: C**

**Note:** If tangent to the x-axis use $(x-h)^2+(y-k)^2=k^2$.

6. Consider a gate whose shape is a parabolic arch of height 3m and base width 3m. If the origin of the coordinate system is at the vertex, the equation that represents the edge of the gate is:

A. $3x^2+3y=0$
B. $4x^2+3y=0$
C. $y=-3x^2+1$
D. $3y^2-4x=0$






**Solution:**

Since the vertex (0, 0) and the point $\left(\frac{3}{2},-3\right)$ on parabola

$\Rightarrow x^2=4py \Rightarrow \left(\frac{3}{2}\right)^2=4P(-3) \Rightarrow P=-\frac{3}{16}$

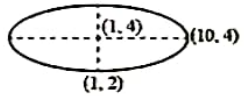
$x^2=4\left(-\frac{3}{16}\right)y \Rightarrow x^2=-\frac{3}{4}y \Leftrightarrow 4x^2+3y=0$

**Answer: B**

7. Let the center of an ellipse be at (1, 4) and two of its vertices be at (10, 4) and (1, 2). What is the equation of the ellipse:

A. $4(x-1)^2+81(y-4)^2=324$
B. $9(x-1)^2+4(y-4)^2=4$
C. $(x-1)^2+9(y-4)^2=4$
D. $2(x-1)^2+9(y-4)^2=4$

**Solution**: Distance between centers:
and major vertices, $a = 10 - 1 = 9$
and minor vertices, $b = 4 - 2 = 2$
Since it is $x$ ellipse




$$\therefore \frac{(x-1)^2}{81}+\frac{(y-4)^2}{4}=1 \Leftrightarrow 4(x-1)^2+81(y-4)=324$$

**Answer: A**

8. What is the length of the minor axis of $x^2+9y^2-2x=0$.

A. $\frac{1}{9}$ B. $\frac{2}{3}$ C. 2 D. 6

**Solution**: $x^2-2x+9y^2=0 \Rightarrow x^2-2x+1+9y^2=1$

$\Rightarrow (x-1)^2+9y^2=1 \Leftrightarrow \frac{(x-1)^2}{1}+\frac{(y-0)^2}{\frac{1}{9}}=1$, so that $a^2 = 1$ and

$b^2=\frac{1}{9} \Rightarrow b=\frac{1}{3} \Rightarrow 2b=\frac{2}{3} \leftarrow$ lengths of minor axis.

**Answer: B**

9. The equation $x^2-4y^2=-1$ represents:

A. A hyperbola with one of its foci, at the origin.
B. A parabola with vertex at the origin.

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C. A hyperbola with vertices at $\left(-\frac{1}{2},0\right)$ and $\left(\frac{1}{2},0\right)$

D. A hyperbola with foci at $\left(0,\frac{-\sqrt{5}}{2}\right)$ and $\left(0,\frac{\sqrt{5}}{2}\right)$

***Solution:*** $x^2-4y^2=-1 \Rightarrow 4y^2-x^2=1$, dividing by $(-1)$.

$\Rightarrow \frac{y^2}{\frac{1}{4}}-\frac{x^2}{1}=1 \Rightarrow a^2=\frac{1}{4} \Rightarrow a=\pm\frac{1}{2}$ and $b^2=1 \Rightarrow b=\pm 1$ and $c^2$

$=a^2+b^2 \Rightarrow c^2=\frac{1}{4}+1=\frac{5}{4} \Rightarrow c=\pm\frac{\sqrt{5}}{2}$

Since $y^2$ term comes first, thus it is $y$ – hyperbola, with vertices $\left(0,\frac{-1}{2}\right)$ and $\left(0,\frac{1}{2}\right)$, foci $\left(0,-\frac{\sqrt{5}}{2}\right)$ and $\left(0,\frac{\sqrt{5}}{2}\right)$.

**Answer: D**

10. Let P $(x)$: $x^2+x$ is positive, then $\neg(\exists x)$ P $(x)$ equal to:

A. $(\exists x)\neg(x^2+x>0)$ C. $(\exists x)(x^2+x\le 0)$

B. $(\forall x)(x^2+x<0)$ D. $(\forall x)(x^2+x\le 0)$

***Solution:*** $\neg(\exists x)(Px) \equiv (\forall x)\neg(\forall x)\neg P(x)$

$\equiv (\forall x)(x^2+x\le 0)$

**Answer: D**

11. If $(P \wedge Q) \Rightarrow (\neg S \vee R)$ is false, which of the following is true?

A. $S \Rightarrow R$ C. $(P \wedge \neg Q) \Leftrightarrow R$

B. $\neg P \vee \neg S$ D. $P \wedge R$

***Solution:***

$a \Rightarrow b \equiv F$ if and only if $b = F$, and $a \equiv T$

- $P \wedge Q \Rightarrow (\neg S \vee R) \equiv F$ if and only $\neg S \vee R \equiv F$ and $P \wedge Q \equiv T$
- $\neg S \vee R = F$ if and only if both $\neg S \equiv F$ and $R \equiv F$
- $P \wedge Q \equiv T$ when both $P \equiv T$ and $Q \equiv T$

A. $T \Rightarrow F \equiv F$, B. $F \vee F \equiv F$ C. $F \Leftrightarrow F \equiv T$ D. $T \wedge F \equiv F$

**Answer: C**



12. Let $A=\begin{pmatrix}2 & 3 & 5\\ 4 & 5 & 7\\ 8 & -3 & 3\end{pmatrix}$ and $B=\begin{pmatrix}-4 & 3 & -5\\ 2 & 1 & 7\\ 6 & 5 & 0\end{pmatrix}$.

If $C = AB + A^T$, then the entry of $C_{23}$ is:

A. 12 B. 18 C. 22 D. 18

***Solution:*** $C = AB + A^T = \begin{pmatrix}2 & 3 & 5\\ 4 & 5 & 7\\ 8 & -3 & 3\end{pmatrix}\begin{pmatrix}-4 & 3 & -5\\ 2 & 1 & 7\\ 6 & 5 & 0\end{pmatrix} + (-3)$

$\Rightarrow C_{23} = (457)\begin{pmatrix}-5\\ 7\\ 0\end{pmatrix} + (-3) = -20+35+(-3) = 12$

**Answer: A**

13. Given matrix $B=\begin{pmatrix}2 & -1\\ 3 & 0\end{pmatrix}$, what is a matrix A if $AB = A + B$:

A. $\frac{1}{2}\begin{pmatrix}1 & 1\\ 3 & -3\end{pmatrix}$ C. $\frac{1}{2}\begin{pmatrix}1 & 1\\ -3 & 3\end{pmatrix}$

B. $\frac{1}{2}\begin{pmatrix}1 & 2\\ -3 & 3\end{pmatrix}$ D. $\frac{1}{2}\begin{pmatrix}1 & -1\\ 3 & -3\end{pmatrix}$

***Solution:*** Let $A=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$ so that $AB = A + B$

$\Rightarrow \begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}2 & -1\\ 3 & 0\end{pmatrix} = \begin{pmatrix}a & b\\ c & d\end{pmatrix} + \begin{pmatrix}2 & -1\\ 3 & 0\end{pmatrix}$

$\Rightarrow \begin{pmatrix}2a+3b & -a+0\\ 2c+3d & -c+0\end{pmatrix} = \begin{pmatrix}a+2 & b-1\\ c+3 & d\end{pmatrix}$

$\Rightarrow \begin{cases}2a+3b=a+2 \Rightarrow a+3b-2=0 \ldots\ldots\ldots\ldots(*)\\ 2c+3d=c+3 \Rightarrow c+3d-3=0 \ldots\ldots\ldots\ldots(**)\end{cases}$

and $-a = b - 1 \Rightarrow a = -b + 1$

$-c = d$

Thus, put $a = -b + 1$ in $(*) \Rightarrow -b + 1 + 3b - 2 = 0$

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$\Rightarrow 2b-1=0 \Rightarrow b=\frac{1}{2}$ and put $d=-c$ in (**)

$\Rightarrow c+3(-c)-3=0 \Rightarrow c=-\frac{3}{2}$

Now, $a=-b+1=-\frac{1}{2}+1=\frac{1}{2}$ and $d=\frac{3}{2}$

$$\therefore A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}=\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{3}{2} & \frac{3}{2} \end{pmatrix}=\frac{1}{2}\begin{pmatrix} 1 & 1 \\ -3 & 3 \end{pmatrix}$$

**Answer: C**

14. For what value of x is $\begin{vmatrix} 1 & -2 \\ x & x-2 \end{vmatrix}=x+1$:

A. 1 B. $\frac{3}{2}$ C. $\frac{1}{2}$ D. $-1$

***Solution***: $(1)(x-2)-(-2)(x)=x+1$

$\Rightarrow 3x-2=x+1 \Rightarrow 2x=3 \Rightarrow x=\frac{3}{2}$

**Answer: B**

15. Let $A=\begin{pmatrix} a & a & ac \\ b & -b & bc \\ c & c & -c^2 \end{pmatrix}$ then det A is equal to:

A. $2abc^2$ B. $4abc^2$ C. $-4abc^3$ D. 0

***Solution***:

$$\det(A=(-1)^2 a\begin{vmatrix} -b & bc \\ c & -c^2 \end{vmatrix}+(-1)^3 a\begin{vmatrix} b & bc \\ c & -c^2 \end{vmatrix}+(-1)^4 ac\begin{vmatrix} b & -b \\ c & c \end{vmatrix}$$

$\Rightarrow |A|=a(bc^2-bc^2)-a(-bc^2-bc^2)+ac(bc+bc)$

$|A|=0+2abc^2+2abc^2=4abc^2$

**Answer: B**



16. Suppose A is $3\times 3$ matrix and $B=\begin{pmatrix} 0 & 4 & 1 \\ 5 & 0 & 6 \\ 0 & 4 & 2 \end{pmatrix}$ If det (A) = 2, the det $(A^{-1}B^t)$ is equal to:

A. $-10$ B. 40 C. $-40$ D. 10

***Solution***: $|A^{-1}B^t|=|A^{-1}||B^t|$ and $|A^{-1}|=\frac{1}{|A|}$

$$\Rightarrow |B^t|=|B|=0\begin{vmatrix} 0 & 6 \\ 4 & 2 \end{vmatrix}-4\begin{vmatrix} 5 & 6 \\ 0 & 2 \end{vmatrix}+1\begin{vmatrix} 5 & 0 \\ 0 & 4 \end{vmatrix}=-20 \text{ and}$$

$$|A^{-1}|=\frac{1}{|A|}=\frac{1}{2}$$

$$\therefore |A^{-1}B^t|=|A^{-1}||B^t|=\left(\frac{1}{2}\right)(-20)=-10$$

**Answer: A**

17. Let $A=\begin{pmatrix} k & 2 \\ 5 & -1 \end{pmatrix}$ which of the following is necessarily true?

A. For any number $k$, A is invertible.

B. If $k\neq 10$, then $A^{-1}=\frac{1}{-k-10}\begin{pmatrix} -1 & -2 \\ -5 & k \end{pmatrix}$

C. If $k\neq -10$, then $A^{-1}=\frac{1}{-k+10}\begin{pmatrix} 1 & 2 \\ 5 & -k \end{pmatrix}$

D. If $k\neq -10$, then A is invertible.

***Solution***: $|A|=-k-10$

Matrix A is invertible if $|A|\neq 0 \Rightarrow k\neq -10$ and

$$A^{-1}=\frac{1}{|A|}\text{adj.}A=\frac{1}{-k-10}\begin{pmatrix} -1 & -5 \\ -2 & k \end{pmatrix}$$

**Answer: D**

18. For what value k does the system $\begin{cases} kx+y=k \\ x+ky=k \end{cases}$ hav nique solution?

A. $k=1$ C. $k=0$

B. $k\in(-\infty,\infty)\setminus\{-1,1\}$ D. $k\in(-\infty,\infty)\setminus\{1\}$

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**Solution**: First $D = \begin{vmatrix} k & 1 \\ 1 & k \end{vmatrix} = k^2 - 1$, so that the system have unique solution if $k^2 - 1 \neq 0 \Rightarrow k \neq 1$, and $k \neq -1 \Rightarrow k \in (-\infty, \infty) \setminus \{-1, 1\}$.

**Answer: B**

19. The system of equations $\begin{cases} x + y - z = 6 \\ x - 2y = -4 \\ x + 4y - az = b \end{cases}$ has no solution if:

A. $a = 2, b = 16$ C. $a = -2, b = 16$

B. $a = -2, b \neq 16$ D. $a = 2, b \neq 16$

**Solution**: First, $D = \begin{vmatrix} 1 & 1 & -1 \\ 1 & -2 & 0 \\ 1 & 4 & a \end{vmatrix}, D_Z = \begin{vmatrix} 1 & 1 & 6 \\ 1 & -2 & -4 \\ 1 & 4 & b \end{vmatrix}$

- $D = (1)\begin{vmatrix} -2 & 0 \\ 4 & a \end{vmatrix} - 1\begin{vmatrix} 1 & 0 \\ 1 & a \end{vmatrix} + (-1)^4(-1)\begin{vmatrix} 1 & -2 \\ 1 & 4 \end{vmatrix}$

$= -2a - a - 6 = -3a - 6$

- $D_Z = (-1)^2(1)\begin{vmatrix} -2 & -4 \\ 4 & b \end{vmatrix} + (-1)^3(1)\begin{vmatrix} 1 & -4 \\ 1 & b \end{vmatrix} + (-1)^4 6 \begin{vmatrix} 1 & -2 \\ 1 & 4 \end{vmatrix}$

$\Rightarrow D_Z = -2b + 16 - b - 4 + 24 + 12 = -3b + 48$

They system has no solution if $D = 0, D_Z \neq 0, D_x \neq 0$.

Therefore $D = 0 \Rightarrow -3a - 6 = 0 \Rightarrow a = -2$

$D_Z \neq o \Rightarrow -3b + 48 \neq 0 \neq b \neq 16$

**Answer: B**

**Note**: unique solution if $D \neq 0, D_X \neq 0, D_Y \neq 0$ and $D_Z \neq 0$ i.e., $a \neq -2$ and $b \neq 16$.

- Infinite solution if $D = 0, D_X = D_Y = D_Z = 0$ i.e., $a = -2$ and $b = 16$.

20. Let ABCD be a parallelogram and $\overline{AC}$ and $\overline{BD}$ be its diagonals. Which of the following is not true?

A. $\overline{AB} = \overline{AC} - \overline{BC}$ C. $\overline{AC} + \overline{BD} = 2\overline{BC}$

B. $\overline{AB} = \overline{BC} - \overline{BD}$ D. $\overline{CD} + \overline{BD} = \overline{AD}$



**Solution**:

A. $\overline{AB} = \overline{AC} - \overline{BC} \Rightarrow \overline{AC} = \overline{AB} + \overline{BC}$ from triangle law.

B. $\overline{AB} = \overline{BC} - \overline{BD} \Rightarrow \overline{BC} = \overline{AB} + \overline{BD} = \overline{AD}$

$\Rightarrow \overline{BC} = \overline{AD} \leftarrow$ opposite side of // gram.

$\therefore \overline{AB} = \overline{BC} - \overline{BD}$

C. $\overline{AC} + \overline{BD} = 2\overline{BC} \Rightarrow \overline{BC} = \frac{\overline{AC}}{2} + \frac{\overline{BD}}{2} = \overline{OC} + \overline{BD}$

$\Rightarrow \overline{BO} + \overline{OC} = \overline{BC} \therefore \overline{AC} + \overline{BD} = 2\overline{BC}$

D. $\overline{CD} + \overline{BD} = \overline{AD} \Rightarrow \overline{CD} - \overline{AD} = -\overline{BD}$

$\Rightarrow \overline{CA} = -\overline{BD} \longleftarrow$ False

**Answer: D**

21. Let A = (0, 1), B = (2, 7) and C = (a, b) be points in the coordinate plane. What is (a, b) so that $\overline{AC} = \overline{AB} + 2\overline{CA}$

Explanation
- $\overline{AB} - \overline{CB} = \overline{AC}$
- $\overline{DB} = -\overline{BD}$

A. $\left(\frac{1}{3}, 2\right)$ B. (1, 3)

C. (2, 3) D. $\left(\frac{2}{3}; 3\right)$

**Solution**: $\overline{AC} = (a - 0, b - 1), \overline{AB} = (2 - 0, 7 - 1) = 2, 6$

$\Rightarrow (a - 0, b - 1) = (2, 6) + 2(-a, 1 - b)$

$\Rightarrow (a, b - 1) = (2 - 2a, 6 + 2 - 2b)$

- $a = 2 - 2a$ and $b - 1 = 8 - 2b$
- $3a = 2$ and $3b = 9$

$\Rightarrow a = \frac{2}{3}$ and $b = 3$

$\therefore (a, b) = \left(\frac{2}{3}, 3\right)$ **Answer: D**

22. If $\bar{u} = 2i - j, \|\bar{v}\| = 5$ and $\bar{u}$ and $\bar{v}$ are perpendicular vectors then $\|\bar{u} + 3\bar{v}\|$ is equal to:

A. $\sqrt{55}$ B. $9\sqrt{5}$ C. $\sqrt{230}$ D. 20

**Solution**: Recall $\|\bar{a} + \bar{b}\| = \sqrt{(\bar{a} + \bar{b}).(\bar{a} + \bar{b})}$

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$$\therefore \|\vec{u}+3\vec{v}\| = \sqrt{(\vec{u}+3\vec{v}).(\vec{u}+3\vec{v})}$$
$$= \sqrt{|\vec{u}|^2 + 6|\vec{u}||v|\cos(90^o) + 9|v|^2}$$
$$= \sqrt{\sqrt{5}^2 + 0 + 9(5)^2} = \sqrt{230}$$

**Answer: C**

23. Let $\ell$ be a line given by $x - 2y = 0$. A parametric vector equation of the line that passes through (5, 0) and perpendicular to $\ell$ is:

A. $(x, y) = (5, 0) + t(1, 2), t \in (-\infty, \infty)$
B. $(x, y) = (5, 0) + t(1, -2), t \geq 0$
C. $(x, y) = (5 + t, -2t), t \in (-\infty, \infty)$
D. $(x, y) = (5 - t, 2t), t \geq 0$

***Solution***: $x - 2y = 0 \Rightarrow x = 2y \Rightarrow y = \frac{1}{2}x \Rightarrow \left(x, \frac{1}{2}x\right) = t(2, 1)$. Let

$\vec{v} = (a, b) \leftarrow$ direction vector perpendicular to $\ell$

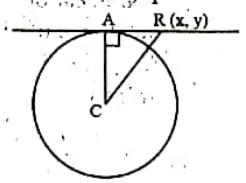
$\Rightarrow (a, b).(2, 1) = 0 \Rightarrow 2a + b = 0 \Rightarrow b = -2a$
$\Rightarrow (a, -2a) = a(1, -2)$
$\Rightarrow \vec{v} = t(1, -2)$,
$\therefore (x, y) = (5, 0) + t(1, -2) \leftarrow$ vector equation.
$\therefore (x, y) = (5 + t, -2t) \leftarrow$ parametric vector equn.

**Answer: C**

24. Consider circle that passes through A = (2, 3) and centered at C = (1, 1). The equation of the line tangent to the circle at A is:

A. $3x + y = 9$ C. $3x - 2y = 0$
B. $x + 2y = 8$ D. $x + y = 5$

***Solution***: Let R = (x, y) point on the line A = (2, 3) point of tangency on circle C = (1, 1) center of circle.




$\Rightarrow \overline{RA}.\overline{AC} = 0 \Rightarrow (2 - x, 3 - y).(-1, -2) = 0$

$\Rightarrow x - 2 + 2y - 6 = 0 \Rightarrow x + 2y = 8$

**Answer: B**



25. If the equation of a line $\ell$ is given by the dot product: $(x - 1, y + 3).(2, -1) = 0$, which of the following is the equation of the circle with center at (−2, 1) and $\ell$ is its tangent line?

A. $(x + 2)^2 + (y - 1)^2 = 2\sqrt{5}$
B. $(x + 2)^2 + (y - 1)^2 = 10$
C. $(x + 2)^2 + (y - 1)^2 = 5\sqrt{2}$
D. $(x + 2)^2 + (y - 1)^2 = 20$

***Solution***: $(x - 1, y + 3).(2, 1) = 2x - 2 - y - 3 = 0 \Rightarrow 2x - y = 5$

The perpendicular distance (radius) from line $\ell$: $2x - y - 5 = 0 + 0$ center of circle (−2, 1) is given by

$$r = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} = \frac{|-4 + (-1)(1) - 5|}{\sqrt{4+1}} = \frac{10}{\sqrt{5}} = 2\sqrt{5}$$

$\therefore$ Equation of circle: $(x + 2)^2 + (y - 1)^2 = 20$, **Answer: D**

26. Suppose anti-clock wise rotation about the origin through 90° is followed by reflection in the x-axis. What is the image of the ellipse given by $\frac{1}{4}(x-1)^2 + (y+2)^2 = 1$ under the two transformations?

A. $4(x - 2)^2 + (y + 1)^2 = 4$ C. $(x - 1)^2 + 4(y + 2)^2 = 4$
B. $4(x - 2)^2 + (y - 1)^2 = 4$ D. $(x + 1)^2 + 4(y - 2)^2 = 4$

***Solution***: Center $(1, -2) \xrightarrow{R_{90^o}} (2, 1) \leftarrow$ rotation with 90°
$M(2, 1) = (2, -1) \leftarrow$ reflection on x-axis.
$x -$ ellipse become y-ellipse.
$\Rightarrow$ Image will be $4(x - 2)^2 + (y + 1)^2 = 4$.

**Answer: A**

27. A translation takes $x^2 + (y + 1)^2 = 5$ to $(x - 2)^2 + y^2 = 5$. What is the image of (1, 3) under this transformation?

A. (3, 4) B. (3, 2) C. (−1, 2) D. (−2, 3)

***Solution***:
Center of $x^2 + (y + 1)^2 = 5$ is (0, −1)
Center of $(x - 2)^2 + y^2 = 5$ is (2, 0)
$T(0, -1) = (2, 0)$, $\vec{v} = (2 - 0, 0 - (-1)) = (2, 1) \leftarrow$ vector
$\therefore T(1, 3) \xrightarrow{\vec{v}=(2,1)} (2 + 1, 3 + 1) = (3, 4)$

**Answer: A**

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28. Which of the following is a point of intersection of the sphere $(x-1)^2+(y-2)^2+(z+1)^2=6$ and the $y$-axis?

A. $(0,-2,0)$ B. $(0,2,0)$ C. $(0,-4,0)$ D. $(0,4,0)$

***Solution:*** $(0, y, 0)\leftarrow$ the set of all point in $y$-axis.

$\Rightarrow (0-1)^2+(y-2)^2+(0+1)^2=6 \Rightarrow (y-2)^2=4$

$\Rightarrow y-2=2$ or $y-2=-2$

$\Rightarrow y=4$ or $y=0$

$\Rightarrow (0,4,0)$ or $(0,0,0)$ pt. of inter section. **Answer: D**

29. If $\theta = \arccos\left(\frac{1}{2}\right)$, what is the value of $\tan\theta$?

A. $\frac{1}{\sqrt{3}}$ B. $\sqrt{3}$ C. $\frac{\sqrt{3}}{2}$ D. 1

***Solution:*** $\theta=\cos^{-1}\left(\frac{1}{2}\right)\Rightarrow \cos\theta=\frac{1}{2}$, $\sin\theta=\sqrt{1-\frac{1}{4}}=\frac{\sqrt{3}}{2}$

$\tan\theta=\frac{\sin\theta}{\cos\theta}=\frac{\sqrt{3}}{2}.(2)=\sqrt{3}$

**Answer: B**

30. If $x=\arcsin\left(\frac{3}{4}\right)$, the value of $\cos 2x$ is:

A. $\frac{3}{4}$ B. $\frac{-3}{4}$ C. $\frac{-1}{8}$ D. $\frac{1}{8}$

***Solution:*** $x=\sin^{-1}\left(\frac{3}{4}\right)\Rightarrow \sin x=\frac{3}{4}$, $\cos x=\sqrt{1-\left(\frac{3}{4}\right)^2}=\frac{\sqrt{7}}{4}$

$\therefore \cos 2x=\cos(x+x)=\cos^2 x-\sin^2 x=\frac{7}{16}-\frac{9}{16}=\frac{-2}{16}=\frac{-1}{8}$

**Answer: C**

31. Let $f(x)=-3\sin\left(\frac{1}{2}x\right)$ which of the following is true?

A. The amplitude o the graph of $f$ is $-3$.

B. The range of $f$ is $[-1, 1]$.

C. $f$ is decreasing on $[-2\pi, 0]$.

D. $f$ is increasing on $[\pi, 3\pi]$.



***Solution:***

- Amplitude, $|a|=|-3|=3$
- Range is $[-3, 3]$
- f is increasing on: $\frac{1}{2}x=\left[\frac{\pi}{2},\frac{3\pi}{2}\right]\Rightarrow[\pi, 3\pi]$

**Answer: D**

32. Which of the following is equal to $\tan\left(x-\frac{\pi}{4}\right)$ if $\cos x\neq 0$.

A. $\frac{\sin x-\cos x}{\sin x+\cos x}$ C. $\frac{\sin x+\cos x}{\sin x-\cos x}$

B. $\frac{\cos x-\sin x}{\cos x+\sin x}$ D. $\frac{\cos x+\sin x}{\cos x-\sin x}$

***Solution:***

$$\tan\left(x-\frac{\pi}{4}\right)=\frac{\tan x-\tan\frac{\pi}{4}}{1+\tan x\tan\frac{\pi}{4}}=\frac{\tan x-1}{1+\tan x}$$

$$\Rightarrow \frac{\tan x-1}{1+\tan x}=\frac{\frac{\sin x}{\cos}-1}{1+\frac{\sin x}{\cos x}}=\frac{\sin x-\cos x}{\sin x+\cos x}$$

**Answer: A**

33. In the interval $[0, 2\pi]$, the solution set of $\cos^2 x+\sin x\cos x=1$.

A. $\left\{0,\frac{\pi}{4},\pi\frac{5\pi}{4}\right\}$ C. $\left\{0,\frac{\pi}{4},\frac{3\pi}{4},\pi\right\}$

B. $\{0, \pi\}$ D. $\left\{0,\frac{\pi}{4},\pi\right\}$

***Solution:*** $\Rightarrow \cos^2 x+\sin x\cos x-1=0$

$\Rightarrow -\sin^2 x+\sin x\cos x=0\Rightarrow \sin x(-\sin x+\cos x)=0$

$\Rightarrow \sin x=0$ or $\cos x-\sin x=0\Rightarrow \cos x=\sin x$

$\Rightarrow x=\pi k$ or $\tan x=1\Rightarrow x=\frac{\pi}{4},\frac{\pi}{4}+\pi=\frac{5\pi}{4}$

$\therefore S.S=\left\{0,\frac{\pi}{4},\pi,\frac{5\pi}{4}\right\}$ **Answer: A**

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34. The expression $\frac{\cos A}{1+\sin A}+\frac{\cos A}{1-\sin A}$ is equal to:

A. 2 csc A C. 2 cos A
B. 2 sec A D. 2 sin A

***Solution:*** $\cos A\left[\frac{1}{1+\sin A}+\frac{1}{1-\sin A}\right]=\cos A\left[\frac{1-\sin A+\sin A}{1-\sin^2 A}\right]$

$\Rightarrow \cos A\left[\frac{2}{\cos^2 A}\right]=2\left[\frac{1}{\cos A}\right]=2\sec A$

**Answer: B**

35. In the set of complex numbers, the solution set of $x^2 - 4x + 5 = 0$ is:

***Solution:*** use $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{4\pm\sqrt{16-20}}{2}=2\pm 2i$

$\therefore$ S.S = {2 + 2i, 2 – 2i}

36. Let z be complex number, such that $\left|\frac{3z}{-4+3i}\right|=(2-i)\left(\overline{2-i}\right)$, then the modulus of z, is equal to:

A. $\frac{25}{3}$ B. $\frac{10}{3}$ C. $\frac{14}{3}$ D. $\frac{25}{7}$

***Solution:*** $\left|\frac{3z}{-4+3i}\right|=(2-i)(2+i)\Rightarrow\frac{3|z|}{\sqrt{4^2+3^2}}=4+1=5$

$\Rightarrow\frac{3|z|}{5}=5\Rightarrow 3|z|=25\Rightarrow|z|=\frac{25}{3}$ **Answer: A**

37. In the set of complex numbers, find the solution set of $z^3 - 8=0$.

***Solution:*** $z^3-8=0\Rightarrow(z-2)(z^2+2z+4)=0$

$\Rightarrow z-2=0$ or $z^2+2z+4=0$

$\Rightarrow z=2$ or $z=\frac{-2\pm\sqrt{4-16}}{2}=-1\pm\sqrt{3}\,i$

$\therefore$ S.S = {2, $-1+\sqrt{3}\,i$, $-1-\sqrt{3}\,i$} ← in rectangular form

In polar form, $z = x + yi = r(\cos\theta + i\sin\theta)$

$\Rightarrow z_1=-1+\sqrt{3}\,i\Rightarrow|z_1|=r=\sqrt{1^2+\sqrt{3}^2}=2$,

$\theta=\tan^{-1}\left(-\sqrt{3}\right)=\frac{2\pi}{3}\Rightarrow z_2=-1-\sqrt{3}\,i$



$\Rightarrow|z_2|=r=\sqrt{1^2+\sqrt{3}^2}=2,\theta=\tan^{-1}\left(\frac{-\sqrt{3}}{-1}\right)=\frac{4\pi}{3}$

Therefore the polar form:

$S.S=\left\{2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right),2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\right\}$

38. Solve, $z^3 + 8z = 0$, where z = x + yi

***Solution:*** $z(z^2+8)=0\Rightarrow z(z-\sqrt{8}i)\left(z+\sqrt{8}i\right)=0$

$\Rightarrow z=0$, $z=\sqrt{8}\,i$ or $z=-\sqrt{8}i$

$\therefore$ S.S = $\left\{0,2\sqrt{2}i,-2\sqrt{2}i\right\}$

39. If k- the term of a sequence is $a_k=\frac{(-1)^k(2-k)}{2k-1}$ what is its next term $a_{k+1}$ when k is even:

A. $\frac{k-1}{2k+1}$ C. $\frac{k-3}{2k+1}$
B. $\frac{3-k}{2k+3}$ D. $\frac{1-k}{2k+1}$

***Solution:*** $a_{k+1}=\frac{(-1)^{k+1}(2-(k+1))}{2(k+1)-1}=\frac{(-1)^k(-1)(1-k)}{2k+1}$

$\Rightarrow a_{k+1}=\frac{-(1-k)}{2k+1}=\frac{k-1}{2k+1}$, if $(-1)^k=1$, for $k=2, 4, 6, \ldots$

**Answer: A**

40. Consider the sequence $\{a_n\}$ where the first term is $a_1 = 2$ and $a_{n+1} = a_n + 4$ for all $n \geq 1$. Then, the sum $a_3 + a_4 + a_5 + \ldots a_{35}$ is equal to:

A. 2460 B. 2458 C. 2442 D. 2450

***Solution:*** $d=\frac{a_{n+1}-a_n}{n+1-n}=\frac{(a_n+4)-a_n}{1}=4$

$\Rightarrow$ Total term from $a_3$ to $a_{35}$ is $35 - 3 + 1 = 33$ and

$a_3 = A_1 = a_1 + (n-1)d = 2 + 2(4) = 10$

$\therefore S_{33}=\frac{33}{2}(2A_1+32d)=\frac{33}{2}(2(10)+32(4))=2442$

**Answer: C**

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41. Suppose a geometric sequence $\{G_n\}_n^{\infty}$ has terms $G_{10} = 192$ and $G_6 = 12$. Which of the following is equal to $G_2$?

A. $\frac{3}{2}$ B. $\frac{3}{4}$ C. $\frac{3}{8}$ D. $\frac{3}{16}$

***Solution:*** $\frac{G_{10}}{G_6} = \frac{G_1 r^9}{G_1 r^5} = \frac{192}{12} \Rightarrow r^4 = 16 \Rightarrow r = 2$

From $G_6$ we have, $G_6 = G_1 r^5 \Rightarrow 12 = G_1 (2)^5 \Rightarrow G_1 = \frac{12}{32} = \frac{3}{8}$

$G_2 = G_1 r \Rightarrow G_2 = \left(\frac{3}{8}\right)(2) = \frac{6}{8} = \frac{3}{4}$

**Answer: B**

42. The sum of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3^n)^2}$ is equal to:

A. $\frac{9}{8}$ B. $\frac{9}{10}$ C. $\frac{-9}{10}$ D. $\frac{-1}{10}$

***Solution:*** $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3^n)^2} = \frac{-1}{3^2} + \frac{1}{3^4} - \frac{1}{3^6} + \ldots, G_1 = \frac{-1}{9}, r = -\frac{1}{9}$

$\Rightarrow S_{\infty} = \frac{G_1}{1-r} = \frac{\frac{-1}{9}}{1+\frac{1}{9}} = \left(-\frac{1}{9}\right)\left(\frac{9}{10}\right) = \frac{-1}{10}$

**Answer: D**

43. Which of the following is true about $\left\{2 + \frac{(-1)^n}{n+1}\right\}_{n=1}^{\infty}$ ?

A. The least upper bounds of the sequence is $\frac{7}{3}$.

B. The sequence is monotonic.

C. 1.6 is its lower bound.

D. $2 - \frac{1}{15}$ is term of sequence.

***Solution:*** If $n = 1, 3, 5, 7, \ldots$



$\Rightarrow \left\{2 + \frac{(-1)^n}{n+1}\right\} = \left\{2 + \frac{(-1)}{n+1}\right\} = \left\{2 - \frac{1}{2}, 2 - \frac{1}{4}, \ldots 2\right\}$

If $n = 2, 4, 6, 8, \ldots$ then $\left\{2 + \frac{1}{n+1}\right\} = \left\{2 + \frac{1}{3}, 2 + \frac{1}{7}, \ldots 2\right\}$

$\therefore \frac{3}{2} \le \left\{2 + \frac{(-1)^n}{n+1}\right\} \le \frac{7}{3}$, so that

glb $= \frac{3}{2}$, and lub $= \frac{7}{3}$

**Answer: A**

44. Which of the following is a divergent sequence?

A. $\left\{\left(\sqrt{2}\right)^{-n}\right\}$ B. $\left\{5^{\frac{1}{n}}\right\}$ C. $\left\{\left(\frac{\pi}{3}\right)^n\right\}$ D. $\left\{1 + \frac{(-1)^n}{7^n}\right\}$

***Solution:*** $\lim_{n\to\infty}\left(\frac{1}{\sqrt{2}^n}\right) = 0 \leftarrow$ convergent sequence

- $\lim_{n\to\infty} 5^{\frac{1}{n}} = 1 \leftarrow$ convergent
- $\lim_{n\to\infty}\left(1 + \frac{(-1)^n}{7^n}\right) = 1 \leftarrow$ convergent

But $\lim_{n\to\infty}\left(\frac{\pi}{3}\right)^n$ doesn't exist because $\frac{\pi}{3} > 1$

$\therefore \left\{\left(\frac{\pi}{2}\right)^n\right\}$ diverges

**Answer: C**

45. $\lim_{n\to\infty} \frac{2n^2 - n + 1}{6n^2 + 1}$ is equal to:

A. 2 B. $\frac{1}{3}$ C. $\frac{2}{3}$ D. $\infty$

***Solution:*** Dividing numerator and denominator by $n^2$.

$\Rightarrow \lim_{n\to\infty} \frac{2 - \frac{1}{n} + \frac{1}{n^2}}{6 + \frac{1}{n^2}} = \frac{2 - 0 + 0}{6 + 0} = \frac{2}{6} = \frac{1}{3}$

**Answer: B**

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46. $\lim_{x\to 1}\frac{x^3+x-2}{x^2-x}$ is equal to:

A. −2 B. 1 C. 4 D. doesn't exist

**Solution:** $\lim_{x\to 1}\frac{x^3+x-2}{x^2-x}=\lim_{x\to 1}\frac{(x-1)(x^2+x+2)}{(x-1)(x)}=1+1+2=4$

**Answer: C**

47. What is the left hand limit, $\lim_{x\to 0}\frac{|x|-xe^x}{x}$:

A. 0 B. −2 C. −1 D. Doesn't exist

**Solution:** Note, $|x|=-x$ if $x<0$

$\Rightarrow \lim_{x\to 0^-}\frac{|x|-xe^x}{x}=\lim_{x\to 0^-}\frac{-x-xe^x}{x}=\lim_{x\to 0^-}\frac{x(-1-e^x)}{x}$

$=-1-1=-2$

**Answer: B**

48. Let $f(x)=\begin{cases}a-\cos x, x\le 0\\ 3e^x-a, x>0\end{cases}$ for what value of a $f$ is continuous function.

A. 1 B. $\frac{3}{2}$ C. 2 D. 3

**Solution:** $\lim_{x\to 0} f(x)=f(0)\Rightarrow \lim_{x\to 0^+} f(x)=f(0)$

$\Rightarrow 3-a=a-1\Rightarrow 4=2a\Rightarrow a=2,$

**Answer: C**

49. If $f(x)=x^3$, then $\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$ is equal to:

A. 4 B. 1 C. 12 D. ∞

**Solution:** Note, $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$f(x)=x^3\Rightarrow f'(x)=3x^2\Rightarrow f'(2)=3(2^2)=12$

$\therefore f'(2)=\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}=12$

**Answer: C**

50. Which of the following is the equation of the line tangent to the graph of $f(x)=\begin{cases}x-\pi, x\le 0\\ 1-2\sin x, x>0\end{cases}$ at $(\pi, f(\pi))$

A. $y=2x-2\pi+1$ C. $y=-2x+2\pi+1$

B. $y=2x-2\pi-1$ D. $y=-2x+2\pi-1$



**Solution:** $y-f(\pi)=f'(\pi)(x-\pi)\Rightarrow f'(x)=\begin{cases}1, x<0\\ -2\cos x, x>0\end{cases}$

$\Rightarrow f'(\pi)=-2\cos(\pi)=(-2)(-1)=2$

$\therefore y-1=2(x-\pi)\Rightarrow y=2x-2\pi+1$

**Answer: A**

51. Suppose $f(x)$ is differentiable at all $x\in(-\infty,\infty)$ and the graph of $y=f'(x)$ is as shown in the figure below. Which of the following is true:

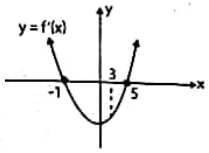
A. $f(x)$ increasing on $[3,\infty\}$

B. $f(x)$ attains its minimum value at $x=3$

C. $f(x)$ decrease on $[-1, 5]$

D. $f(x)$ attains its maximum value at $x=5$

**Solution:** $f'(x)\ge 0$, on $(-\infty,-1]$ and $[5,\infty)$




$\therefore f(x)$ is increasing on $(-\infty,-1]$ and $[5,\infty)$

- $f'(x)\le 0$, on $[-1, 5]$

$\therefore f(x)$ is decreasing on $[-1, 5]$

- $f'(-1)=f'(5)=0$
- $f'$ changes the sign from (+) to (−) at $x=-1$

  $\therefore f(x)$ has local maximum at $x=-1$.
- $f'$ changes the sign from (−) to (+) at $x=5$

  $\therefore f$ has local minimum at $x=5$.

**Answer: C**

52. Let $f(x)=x^3-12x+1$ be defined on $[0, 3]$. What is the range of f:

A. $(-\infty,\infty)$ C. $[-15, 1]$

B. $(-\infty, 1]$ D. $[-15, 17]$

**Solution:** To find critical number set $f'(x)=0$

$\Rightarrow 3x^2-12=0\Rightarrow x^2=\frac{12}{3}=4\Rightarrow x=2\in[0,3]$

$\Rightarrow f(0)=0-0+1=1\leftarrow$ maximum of f

$\Rightarrow f(2)=8-24+1=-15\leftarrow$ minimum of f

$\Rightarrow f(3)=27-36+1=-8$

$\therefore$ The range of f is $[-15, 1]$

**Answer: C**

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53. If $f(x) = \ln(2 + \sec x)$, then $f'(x)$ is equal to:

A. $\dfrac{\tan x}{2+\sec x}$ C. $\dfrac{\cos x}{1+2\cos x}$

B. $\dfrac{\tan x}{2+\cos x}$ D. $\dfrac{\tan x}{1+2\cos x}$

**Solution:** $f'(x) = \dfrac{d}{dx}f(x) = \dfrac{d}{dx}\ln u = \dfrac{1}{u} = \dfrac{du}{dx}$, let $u = 2 + \sec x \Rightarrow \dfrac{du}{dx} = \sec x.\tan x$ so that:

$$\frac{d}{dx}\ln(u) = \frac{1}{u}\frac{du}{dx} = \frac{\sec x \tan x}{2+\sec x}$$

$$f'(x) = \frac{(\sec x \times \tan x).\cos x}{2\cos x + 1} = \frac{\tan x}{2\cos x + 1}$$

**Answer: D**

54. Which of the following is the set of all critical numbers of $f(x) = \begin{cases} x-1, x<1 \\ x^2 - x, x \geq 1 \end{cases}$:

A. $\{1\}$ B. $\left\{\frac{1}{2}\right\}$ C. $\left\{\frac{1}{2}, 1\right\}$ D. $\{\}$

**Solution:** $f'(x) = \begin{cases} 1, x<1 \\ 2x-1, x \geq 1 \end{cases}$

Set $f'(x) = 0 \Rightarrow f'(x) \neq 0$, for all $x \in [1, \infty) \Rightarrow x = \dfrac{1}{2} \notin [1, \infty)$

$\therefore f$ has no critical number

**Answer: D**

55. Let $h(x) = x + 3^x$ and $F(x) = (f \circ h)(x)$ such that $f'(4) = -1$. Then $F'(1) =$ _____

A. $1 + 3\ln 3$ C. $4 + 12\ln 3$

B. $-1 - 3\ln 3$ D. $-4 - 12\ln 3$

**Solution:** $F'(x) = f'(h(x)).h'(x)$ and $h(1) = 1 + 3 = 4$

$\Rightarrow h'(x) = 1 + 3^x \ln 3 \Rightarrow h'(1) = 1 + 3\ln 3$

$\therefore F'(1) = f'(h(1)).h'(1) = f'(4).(1 + 3\ln 3) = -1 - 3\ln 3$,

**Answer: B**



56. The rate of certain chemical reaction is given by $r(x) = kx(a - x)$ for $0 \leq x$, $a > 0$ and $k > 0$. For what value of $x$ does the rate of reaction increase?

A. $0 \leq x \leq \dfrac{a}{2}$ C. $-\infty \leq x \leq \dfrac{a}{2}$

B. $\dfrac{a}{2} \leq x \leq a$ D. $\dfrac{a}{2} \leq x \leq \dfrac{3a}{4}$

**Solution:** $r(x) = akx - kx^2 \Rightarrow r'(x) = ak - 2kx$

$\Rightarrow r'(x) \geq 0 \Rightarrow ak - 2kx \geq 0 \Rightarrow 2kx \leq ak$

$\Rightarrow x \leq \dfrac{a}{2} \Rightarrow 0 \leq x \leq \dfrac{a}{2}$

**Answer: A**

57. Which of the following is equal to $\int_0^{\pi/3} \dfrac{\tan x}{\cos x} dx$:

A. $\dfrac{1}{2}$ B. 1 C. $-1$ D. $-1 + \dfrac{2\sqrt{3}}{3}$

**Solution:** $\int_0^{\pi/3} \dfrac{\tan x}{\cos x} dx = \int_0^{\pi/3} \dfrac{\sin x}{\cos^2 x} dx$,

Let $u = \cos x$, $du = -\sin x\, dx$

$$\therefore \int_1^{\frac{1}{2}} \frac{-du}{u^2} = \frac{1}{u}\Big|_1^{\frac{1}{2}} = 2 - 1 = 1$$

58. The area of the region bounded between the graph of $f(x) = x^3$ and the $x$-axis form $x = -1$ to $x = 2$ is:

A. $\dfrac{15}{4}$ B. $\dfrac{17}{4}$ C. 4 D. 16

**Solution:** $\int_{-1}^{0} f(x)dx + \int_0^2 f(x)dx = -\int_{-1}^{0} x^3 dx + \int_0^2 x^3 dx$

$$\Rightarrow -\left(\frac{x^4}{4}\right)\Big|_{-1}^{0} + \left(\frac{x^4}{4}\right)\Big|_0^2 = 0 - \left(0 - \frac{1}{4}\right) + \left(\frac{2^4}{4} - 0\right) = \frac{1}{4} + 4 = \frac{17}{4}$$

**Answer: B**

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59. $\int \frac{(e^x+1)^2}{e^x}dx$ equal to:

A. $e^x+e^{-x}+2x+c$ C. $\frac{1}{3}(e^x+1)^3+c$

B. $e^x-e^{-x}+2x+c$ D. $\frac{1}{3}(1-e^{-x})^3+c$

**Solution:** $\Leftrightarrow \int \frac{e^{2x}+e^x+1}{e^x}dx$

$\Rightarrow \int (e^x+e^{-x}+2)dx = e^x-e^{-x}+2x+c$

**Answer: B**



1. What are the respective values of A, B and C so that $\frac{x-1}{x^3+x}=\frac{A}{x}+\frac{Bx+C}{x^2+1}$?

A. $-1, 1, 1$ B. $-1, 1, 2$ C. $1, -2, 0$ D. $3, 1, 2$

**Solution:**

Multiplying both side by $x(x^2+1)$

$\Rightarrow x-1=A(x^2+1)+Bx^2+Cx,$

$Ox^2+x-1=Ax^2+Bx^2+Cx+A$ ← comparing the coefficient of variable with left side.

$\Rightarrow A+B=0$ and $Cx=x \Rightarrow C=1$, and $A=-1, B=1$

**Answer: A**

2. The solution set of $1-\frac{5}{x^2+4}=\frac{x-1}{x^2-x}-\frac{1}{x}$.

A. $\{1,-1\}$ B. $\{2\}$ C. $\{-1\}$ D. $\{1, 2, -\}$

**Solution:** Multiply by LCM both side by $(x^2+4)(x^2-x)x$, gives.

$\Rightarrow (x^2+4)(x^2-x)(x)-5(x^2-x)x$

$=(x^{2+}+4)(x-1)x-(x^2+4)(x^2-x)$

$\Rightarrow (x^2-x)x[x^2+4-5]=(x^2+4)[x^2-x-(x^2-x)]$

$\Rightarrow (x^2-x)(x^2-1)=0 \Leftrightarrow x(x-1)(x-1)(x+1)=0$

$\Rightarrow x=0, x=1, x=-1$

$\therefore$ The solution set $=\{-1\}$ since Domain $=\{x: x\in\mathbb{R}\mid 0, 1\}$

**Answer: C**

3. The simplified form of $\frac{t-\frac{1}{t}}{1+\frac{t}{1-t}}\div\frac{t^2-2t+1}{t}, t\neq 0,1$.

A. $t+1$ B. $t-1$ C. $1-t$ D. $-t-1$

**Solution:** The expression the same as

$$\Rightarrow \left(\frac{\frac{t^2-1}{t}}{\frac{1-t+t}{1-t}}\right)\cdot\frac{t}{(t-1)^2} \Leftrightarrow \frac{(t-1)(1-t)(t+1)}{t}\cdot\frac{t}{(t-1)(t-1)}$$

$\Rightarrow -(t+1)=-t-1,$

**Answer: D**

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4. Which of the following is not true about the graph of $g(x)=\frac{x^2-1}{x^2+1}$?

A. The range of $g$ is $(-\infty, 1)$.
B. g is an even function.
C. $y=1$ is H. A.
D. As $x \to \infty, g(x) \to 1$

***Solution***: To find the range, put $y = g(x)$ and solve for $x$.

$$\Rightarrow y=\frac{x^2-1}{x^2+1} \Leftrightarrow x^2y+y=x^2-1 \Rightarrow y+1=x^2-x^2y$$

$$\Leftrightarrow x^2(1-y)=y+1 \Rightarrow x^2=\frac{y+1}{1-y} \Rightarrow x=\sqrt{\frac{1+y}{1-y}}$$

A. The range, $=\left\{y:\frac{1+y}{1-y}\geq 0\right\}=\{y: -1 \leq y<1\}$

B. $g(-x)=g(x) \leftarrow$ it is even.

C. $\frac{x^2-1}{x^2+1}=\frac{x^2+1}{x^2+1}-\frac{2}{x^2+1}=1-\frac{2}{x^2+1}$, $\therefore y=1 \leftarrow$ H.A

D. As $x \to \infty, g(x) \to 1$, from c

**Answer: A**

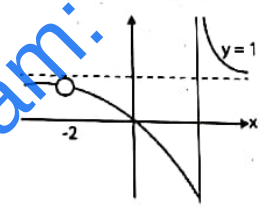
5. Which of the following functions could most likely be drawn in the figure below?

A. $f(x)=\frac{x}{x-2}$

B. $f(x)=\frac{x^2+2x}{x^2-4}$

C. $\frac{x+1}{x-2}$

D. $\frac{x^3+2x}{x^2-4}$




***Solution***: $\frac{x}{x-2}$ and $\frac{x+1}{x-2}$ has no hole at $x = -2$ and $\frac{x^3+2x}{x^2-4}$ has no horizontal asymptote thus, $f(x)=\frac{x^2+2x}{x^2-4}=\frac{x(x+2)}{(x-2)(x+2)}$ satisfy the graph.

**Answer: B**



6. The equation of the circle tangent to the x-axis at (6, 0) whose center is on the line $y=\frac{1}{2}x$ is:

A. $(x-6)^2+y^2=9$
B. $(x-6)^2+(y-3)^2=36$
C. $(x-6)^2+y^2=36$
D. $(x-6)^2+(y-3)^2=9$

***Solution***: Here,

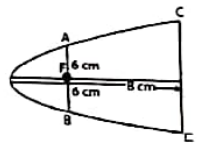
$\Rightarrow$center(h, k)$=\left(h, \frac{1}{2}h\right)=(2k, k)$, since lies on $y = 2x$

$\Rightarrow (x-h)^2+(y-k)^2=r^2 \leftarrow$ eqn of circle
$\Rightarrow (x-h)^2+(y-k)^2=k^2 \leftarrow$ tangent to x-axis
$\Rightarrow (x-2k)^2+(y-k)^2=k^2 \leftarrow$ put $h = 2k$
$\Rightarrow (6-2k)^2+(0-k)^2=k^2 \leftarrow$ (6, 0) on circle
$\Rightarrow (6-2k)^2+k^2=k^2 \Leftrightarrow (6-2k)^2=0$
$\Rightarrow k = 3$, thus (6, 3) is center
$\therefore (x-6)^2+(y-3)^2=9$

**Answer: D**

7. Across section of a parabolic reflector is shown below. The opening at the focus, AB is 12 cm. What is the diameter of the opening $\overline{CD}$, 8 cm from the vertex.

A. $8\sqrt{6}$ cm
B. 12 cm
C. 9 cm
D. $6\sqrt{8}$ cm




***Solution***: $\overline{AB} = 4P = 12 \Rightarrow P = 3$; $4Px = y^2 \leftarrow$ the eqn of parabola
$\Rightarrow 4(3)x=y^2 \Leftrightarrow y^2=12x \leftarrow$ eqn of parabola.
$y^2 = 12(8) = 96 \Rightarrow y=\pm 4\sqrt{6}$
$\Rightarrow \overline{CD}=2y=2\left(4\sqrt{6}\right)=8\sqrt{6}$

**Answer: A**

8. The equation $3x^2-6x=y^2$ represents:

A. an ellipse with center at (1, 0)
B. a hyperbola with center (1, 0)
C. an ellipse with center at (–1, 0)
D. a hyperbola with one of its foci at (1, 0)

***Solution***: $3x^2-6x=y^2 \Leftrightarrow 3x^2-y^2-6x=0$
$\Leftrightarrow 3x^2-6x-y^2=0 \Rightarrow 3(x^2-2x)-y^2=0$
$\Rightarrow 3(x^2-2x+1)-y^2=0+3 \leftarrow$ completing square

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$\Rightarrow 3(x-1)^2 - y^2 = 3 \Leftrightarrow (x-1)^2 - \frac{y^2}{3} = 1$

$\Rightarrow$ hyperbola with center (1, 0) **Answer: B**

9. A mirror has an elliptic edge. If the shortest and longest distance between any two points on the edge of the mirror through the geometric center of the mirror are respectively 6 cm and 10 cm, which of the following is a possible equation of the edge of the mirror?

A. $\frac{x^2}{9} + \frac{(y-1)}{5} = 1$ C. $\frac{(x+1)^2}{36} + \frac{y^2}{100} = 1$

B. $\frac{(x-1)^2}{25} + \frac{y^2}{9} = 1$ D. $\frac{(x+1)^2}{36} + \frac{(y-1)^2}{25} = 1$

***Solution***: $2a = 10$ cm and $2b = 6$ cm

$\Rightarrow a = 5$ cm and $b = 3$cm $\Rightarrow$ so that $a^2 = 25$ and $b^2 = 9$

$\Rightarrow$ The possible equation will be $\frac{(x-1)^2}{25} + \frac{y^2}{9} = 1$

**Answer: B**

10. If $(\neg p \vee Q) \Rightarrow (R \wedge \neg R)$ is true then which of the following is necessarily true?

A. $R \Rightarrow Q$ C. $P \wedge \neg R$

B. $(Q \Rightarrow R) \wedge R$ D. $(Q \wedge R) \Rightarrow P$

***Solution***: since $R \wedge \neg R \equiv F$. Hence $\neg p \vee Q = F$,

$\therefore \neg P \equiv F,\ Q \equiv F, P \equiv T$

$\therefore (Q \wedge R) \Rightarrow P \equiv T,$ **Answer: D**

11. Suppose $A = \begin{pmatrix} 4 & 0 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 2 \\ 6 & -4 \end{pmatrix}$. If $x$ is $2 \times 2$ matrix such that $x + \frac{1}{2}B^t = AB$, then $x$ is equal to:

A. $\begin{pmatrix} 7 & 5 \\ 5 & -4 \end{pmatrix}$ B. $\begin{pmatrix} 5 & -3 \\ 5 & 0 \end{pmatrix}$ C. $\begin{pmatrix} 5 & -3 \\ 5 & -2 \end{pmatrix}$ D. $\begin{pmatrix} 7 & 5 \\ 7 & 0 \end{pmatrix}$

***Solution***: $x + \frac{1}{2}B' = AB \Rightarrow x = AB - \frac{1}{2}(B')$

$\Rightarrow x = \begin{pmatrix} 4 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 2 \\ 6 & -4 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 2 & 6 \\ 2 & -4 \end{pmatrix}$



$\Rightarrow x = \begin{pmatrix} 8 & 8 \\ 8 & -2 \end{pmatrix} - \begin{pmatrix} 1 & 3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 7 & 5 \\ 7 & 0 \end{pmatrix}$ **Answer: D**

12. $\begin{vmatrix} 1 & -a & b-a \\ 1 & b & a+b \\ 0 & c & c+c \end{vmatrix}$ is equal to:

A. 0 B. $-2ac$ C. $2bc$. D. $2c(b-a)$

***Solution***: Adding or multiplying of row (column) of matrix from another row (or column) the value of the determinant remains unchanged. Therefore the given matrix determinant is the same as:

$$\begin{vmatrix} 1 & -a & b \\ 1 & b & a \\ 0 & c & c \end{vmatrix} = 0\begin{vmatrix} -a & b \\ b & a \end{vmatrix} + c(-1)\begin{vmatrix} 1 & b \\ 1 & a \end{vmatrix} + c\begin{vmatrix} 1 & -a \\ 1 & b \end{vmatrix}$$

$\Rightarrow 0 - c(a-b) + c(b+a) = 2bc$ **Answer: C**

13. If $\theta$ is the angle between the vectors $\vec{A} = 2\vec{i} - \vec{j}$ and $\vec{B} = \vec{i} + \vec{j}$, then $\cos\theta$ is _____

A. $\frac{1}{\sqrt{10}}$ B. $\frac{1}{2}$ C. $\frac{\sqrt{3}}{2}$ D. $\frac{2}{\sqrt{5}}$

***Solution***: $\cos\theta = \frac{\vec{A}.\vec{B}}{|\vec{A}.\vec{B}|} = \frac{2-1}{\sqrt{5}\sqrt{2}} = \frac{1}{\sqrt{10}}$

**Answer: A**

14. If $\vec{A} = (3, -3)$ and $\vec{B} = (1, -3)$, what is the unit vector in the direction of $\vec{C} = 3\vec{A} - \vec{B}$.

A. $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ C. $\left(\frac{4}{5}, \frac{-3}{5}\right)$

B. $\left(\frac{1}{\sqrt{10}}, \frac{-3}{\sqrt{10}}\right)$ D. $\left(\frac{-6}{10}, \frac{8}{10}\right)$

***Solution***:

$$\frac{\vec{C}}{|\vec{C}|} = \frac{3(3,-3) - (1,-3)}{|3\vec{A} - \vec{B}|} = \frac{(8,-6)}{\sqrt{8^2 + 6^2}} = \left(\frac{4}{5}, \frac{-3}{5}\right)$$

**Answer: C**

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15. If $\vec{u}$ is vector in the direction as vector $\vec{v} = (-3, 4)$ having length 3 times the length of $\vec{v}$, what is the length of the vector $\vec{u} - \frac{3}{2}\vec{v} - 2\vec{j}$, where $\vec{j}$ is unit vector in the direction of the positive y-axis.

A. $\frac{1}{2}\sqrt{185}$ B. $\frac{1}{2}\sqrt{145}$ C. $\frac{21}{2}$ D. $\frac{17}{2}$

***Solution***: Let $\vec{w} = 3(-3,4) - \frac{3}{2}(-3,4) - 2(0,1) = \left(\frac{-9}{2}, 4\right)$

$\therefore |\vec{w}| = \sqrt{\left(\frac{-9}{2}\right)^2 + (4)^2} = \sqrt{\frac{81}{4} + 16} = \frac{1}{2}\sqrt{145}$ **Answer: B**

16. Consider a circle given by $x^2 + y^2 = 2$ and the line $\ell$ given by parametric vector equation $(x, y) = (2, 0) + t(-1, 1)$. Which of the following is true?

A. $\ell$ is tangent to the circle at $\left(\frac{1}{2}, \frac{\sqrt{7}}{2}\right)$.

B. $\ell$ intersects with the circle at two distinct point

C. $\ell$ and the circle have no common point

D. The distance from the center of the circle to $\ell$ is $\sqrt{2}$.

***Solution***: $(x, y) = (2, 0) + t(-1, 1)$ ← parametric eqn of $\ell$

$\Rightarrow x = 2 - t$ and $y = 0 + t$

$\Rightarrow x = 2 - y$ ← solving for $t$

$\Rightarrow x + y = 2$ ← eqn of $\ell$

Eqn of circle and the line $\ell$ intersect at $x^2 + y^2 = 2$

$\Leftrightarrow x^2 + (2 - x)^2 = 2$ ← put $y = 2 - x \Leftrightarrow x^2 + x^2 - 4x + 4 = 2$

$\Leftrightarrow 2x^2 - 4x + 2 = 0 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1, y = 1$

$\therefore (1, 1)$ ← point of tangency of $\ell$ to circle

The distance from center (0, 0) of circle to $\ell$ to: $x + y - 2 = 0$ is

$\frac{|0 + 0 - 2|}{\sqrt{1^2 + 1^2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

**Answer: D**



17. The reflection of the parabola $y = x^2 + 1$ in the line $y = -x$.

A. $y = -x^2 - 1$ C. $x = y^2 + 1$

B. $y = x^2 - 1$ D. $x = -y^2 - 1$

***Solution***: Recall, reflection in the line $y = -x$

• M $((x, y)) = (-y, -x)$, so that $y \to -x$, $x \to -y \Rightarrow -x = (-y)^2 + 1$

$\Rightarrow -x = y^2 + 1 \Leftrightarrow x = -y^2 - 1$ **Answer: D**

18. If $\theta = \arcsin\left(\frac{1}{2}\right)$ then $\tan(2\theta)$ is equal to:

A. $\frac{1}{\sqrt{3}}$ B. 1 C. $\sqrt{3}$ D. $\frac{\sqrt{3}}{2}$

***Solution***: $\theta = \sin^{-1}\left(\frac{1}{2}\right) \Leftrightarrow \sin\theta = \frac{1}{2} \Rightarrow \cos^2\theta = 1 - \sin^2\theta =$

$1 - \frac{1}{4} = \frac{3}{4} \Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$

$\therefore \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1}{\sqrt{3}} \Rightarrow \tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta} = \frac{\frac{2}{\sqrt{3}}}{1 - \frac{1}{3}} = \sqrt{3}$

**Answer: C**

19. What is the amplitude and period, respectively of the graph of $f(x) = 2 - 3\sin\left(\frac{x + \pi}{4}\right)$?

A. $2, \frac{\pi}{4}$ B. $-3, 8\pi$ C. $3, \frac{\pi}{2}$ D. $3, 8\pi$

***Solution***: Amplitude, $|a| = |-3| = 3$

Period, $P = \frac{2\pi}{k} = \frac{2\pi}{\frac{1}{4}} = 8\pi$ **Answer: D**

20. The simplified form $(1 + \tan^2 x)\sin\left(x - \frac{3}{2}\pi\right)$.

A. $\sec x$ B. $\cos x$ C. $-\sec x$ D. $-\cos x$

***Solution***: $1 + \tan^2 x = \sec^2 x$ and

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$$\sin\left(x-\frac{3\pi}{2}\right)=\sin x\cos\left(\frac{3\pi}{2}\right)-\sin\frac{3\pi}{2}\cos x=\cos x.$$

$$\therefore (\sec^2 x)\cos x=\frac{\cos x}{\cos^2 x}=\frac{1}{\cos x}=\sec x$$ **Answer: A**

21. The solution set of $\sqrt{3}\sin 2x=\cos 2x$ on $[0,\pi]$.

A. $\left\{\frac{\pi}{6},\frac{\pi}{12}\right\}$ B. $\left\{\frac{\pi}{6},\frac{\pi}{3}\right\}$ C. $\left\{\frac{\pi}{12},\frac{7\pi}{12}\right\}$ D. $\left\{\frac{\pi}{6},\frac{2\pi}{3}\right\}$

***Solution***: $\frac{\sqrt{3}\sin 2x}{\cos 2x}=\frac{\cos 2x}{\cos 2x}\Rightarrow\tan 2x=\frac{1}{\sqrt{3}}$

$\Rightarrow 2x=\frac{\pi}{6},\frac{\pi}{6}+\pi k\Rightarrow x=\frac{\pi}{12},\frac{\pi}{12}+\frac{\pi}{2}k$, $\Rightarrow$ when $k=1$, the solution set $=\left\{\frac{\pi}{12},\frac{7\pi}{12}\right\}$ **Answer: C**

22. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_{n+1}=-na_n$, $a_1=1$; $n=1,2,3,\ldots$ which of the following is not true about the sequence.

A. $a_3=2$ C. $a_n>0$ if n is odd

B. $a_4=-3$ D. $a_n<0$ if n is even

***Solution***: $a_{1+1}=a_2=-a_1=-1\,(1)=-1$

$a_{2+1}=a_3=-2a_2=-2\,(-1)=2$

$a_{3+1}=a_4=-3a_3=-3\,(2)=-6$ **Answer: B**

23. Let $S_n=4n^2$, $n=1,2,3,\ldots$ be the sum of first n-terms of a sequence $\{a_n\}_{n=1}^{\infty}$. Then $a_3$:

A. 8 B. 12 C. 16 D. 20

***Solution***: $a_n=S_n-S_{n-1}\Rightarrow a_3=S_3-S_2=4\,(3)^2-4(2)^2=36-16=20$ **Answer: D**

24. Let $\{a_n\}_{n\geq 1}$ be sequence $a_1=2$, and $a_n=3a_{n-1}$ for all $n\geq 2$. Then $\sum_{i=1}^{20}a_i$ is equal to:

A. $3^{20}-1$ B. $1-\frac{1}{3^{20}}$ C. $1-\frac{1}{2^{21}}$ D. $3^{-20}-1$

***Solution***: $S_{20}=\frac{2[3^{20}-1]}{3-1}=3^{20}-1$ **Answer: A**



25. $\lim_{n\to\infty}\frac{n+\cos 3n}{2-n}$ equal:

A. $-1$ B. 0 C. 1 D. 3

***Solution***: $\lim_{n\to\infty}\frac{\frac{n+\cos 3n}{n}}{\frac{2-n}{n}}=\lim_{n\to\infty}\frac{1+\frac{\cos 3n}{n}}{\frac{2}{n}-1}=\frac{1+0}{0-1}=-1$

**Answer: A**

26. The left hand limit, $\lim_{x\to 0^-}\frac{|x|-2xe^x}{x}$ is equal to:

A. 0 B. $-2$ C. $-3$ D. doesn't exist

***Solution***: $x\to 0^-$ means $x<0\Rightarrow |x|=-x$

$\Rightarrow\lim_{x\to 0^-}\frac{-x-2xe^x}{x}=\lim_{x\to 0^-}(-1-2e^x)=-1-2=-3$

**Answer: C**

27. $\lim_{x\to\pi}\frac{x\sin 2x}{x-\pi}$ equal to _____

A. $2\pi$ B. $\pi$ C. 0 D. 1

***Solution***: Let $y=x-\pi\Rightarrow x=y+\pi, x\to\pi, y\to 0$

$$\therefore\lim_{y\to 0}(y+\pi)\frac{\sin(2y+2\pi)}{y}=\lim_{y\to 0}\left[\frac{(\sin 2y\cos 2\pi)}{y}+\frac{\cos 2y\sin 2\pi}{y}\right].$$

**Answer: A**

28. Let $f(x)=\frac{\sqrt{x}-2}{\frac{-4}{x}+1}$. Then $\lim_{x\to 4^+}f(x)$ is equal to

A. 0 B. $\frac{1}{4}$ C. 1 D. $-\infty$

***Solution***: $\lim_{x\to 4^+}\left(\frac{\sqrt{x}-2}{x-4}\right)x=\lim_{x\to 4^+}\frac{(\sqrt{x}-2)(\sqrt{x}+2)}{(x-4)(\sqrt{x}+2)}(x)$

$\Rightarrow\lim_{x\to 4+}\frac{(x-4)x}{(x-4)\sqrt{x}+2}=\lim_{x\to 4+}\frac{x}{\sqrt{x}+2}=\frac{4}{\sqrt{4}+2}=\frac{4}{2+2}=\frac{4}{4}=1$

**Answer: C**

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29. $\lim_{x\to 2}\frac{6x-12}{x^2-x-2}$ is equal to:

A. 0 B. 2 C. 1 D. $\infty$

***Solution***: $\lim_{x\to 2}\frac{6x-12}{x^2-x-2}=\lim_{x\to 2}\frac{6(x-2)}{(x-2)(x+1)}=\lim_{x\to 2}\frac{6}{x+1}=\frac{6}{3}=2$

**Answer: B**

30. Let $f(x)=\begin{cases}3a-x \text{ if } x<1\\ b+1 \text{ if } x=1\\ ax \text{ if } x>1\end{cases}$ then the value of b:

A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. $-1$ D. 3

***Solution***: since it is continuous at x = 1.

$\Rightarrow \lim_{x\to 1^+} f(x)=\lim_{x\to 1^-} f(x)\Leftrightarrow \lim_{x\to 1^+} ax=\lim_{x\to 1^-} 3a-x$

$\Leftrightarrow a=3a-1\Rightarrow -2a=-1\Rightarrow a=\frac{1}{2}$ and

- $\lim_{x\to 1^+} f(x)=f(1)\Leftrightarrow \lim_{x\to 1^+} ax=b+1\Rightarrow a=b+1$

$\Leftrightarrow \frac{1}{2}=b+1\Rightarrow b=-\frac{1}{2}$

**Answer: A**

31. On which of the following interval is f continuous where

$f(x)=\begin{cases}\frac{x-x^2}{x-1}, x<0\\ 3-x,\ x\geq 0\end{cases}$:

A. $(-\infty, 0]$ B. $[-2, 1]$ C. $[-1, 2]$ D. $[0, \infty)$

***Solution***: **To show continuity in $(-\infty, 0]$**

i) $\lim_{x\to 0^-} f(x)=f(0)\Rightarrow \lim_{x\to 0^-}\frac{x-x^2}{x-1}=3-0\Leftrightarrow \lim_{x\to 0^-}\frac{x(1-x)}{x-1}\Rightarrow 0\neq 3,$

Therefore it is not continuous from the left to 0.

**To show continuity on $(-\infty, 0)$ let $C\in(-\infty, 0)$**

$\lim_{x\to C} f(x)=f(C)\Leftrightarrow \frac{C-C^2}{C-1}=f(C)$

$\therefore f$ is cont. on $(-\infty, 0)$ but not cont. $(-\infty, 0]$.

**Answer: A**



ii) $f$ is continuous on $[0, \infty)$ because

$\lim_{x\to 0^+} f(x)=f(0)\Leftrightarrow \lim_{x\to 0^+} 3-x=f(0)\Rightarrow 3=3$ and let

$C\in[0, \infty)$, so that $\lim_{x\to C} f(x)=f(C)\Rightarrow 3-C=3-C$

$\therefore$ f is cont. on $[0, \infty)$

**Note:** f is continuous on $(-\infty, 0)$ and $[0, \infty)$.

**Answer: D**

32. If $f(x)=x^5+2x$, then $\lim_{h\to 0}\frac{f(1+h)-f(1)}{h}$ is equal to

A. 1 B. 7 C. 5 D. $\infty$

***Solution***: Recall $f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$ it is concept of derivative. So that $f'(1)=\lim_{h\to 0}\frac{f(1+h)-f(1)}{h}$

$\Rightarrow f'(x)=5x^4+2\Rightarrow f'(1)=5+2=7$

**Answer: B**

33. Let $f(x)=e^{3x-1}$ and $g(x)=\tan x$. If $F(x)=xf(x)+f(g(x))$. Then F '(0):

A. $3e^{-1}$ B. $4e^{-1}$ C. $\frac{1}{e}\left(1+\frac{1}{e}\right)$ D. $\frac{1}{e}\left(1+\frac{3}{e}\right)$

***Solution***: $f(x)=e^{3x-1}\Rightarrow f'(x)=e^{3x-1}(3)=3e^{3x-1}$, $g(x)=\tan x$

$\Rightarrow g'(x)=\sec^2 x$. Then: $F(x)=xf(x)+f(g(x))$

$\Rightarrow F'(x)=x'f(x)+xf'(x)+f'(g(x))g'(x)$

$\Rightarrow F'(x)=1.f(x)+xf'(x)+f'(g(0))g'(0)$

$\Rightarrow F'(0)=f(0)+0f'(0)+f'(g(0))g'(0)=e^{-1}+0+f'(0)(1)$

$=e^{-1}+e^{-1}(3)=4e^{-1}$

**Answer: B**

34. Let $f(x)=g\left(\sqrt{4x+1}\right)$. If $g'(3)=\frac{1}{2}$, then f ' (2) is equal to:

A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{1}{2}\sqrt{5}$ D. $\frac{1}{2}\sqrt{13}$

***Solution***: $f'(x)=g'\left(\sqrt{4x+1}\right).\left(\sqrt{4x+1}\right)'(4x+1)'$

$\Rightarrow f'(x)=g'\left(\sqrt{4x+1}\right)\frac{1}{2}(4x+1)^{-\frac{1}{2}}(4)$

$\Rightarrow f'(2)=g'(3)(2)\left(\frac{1}{9}\right)=\left(\frac{1}{2}\right)(2)\frac{1}{3}=\frac{1}{3}$

**Answer: B**

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35. The second derivative of $f(x) = \ln(x^2 + 1)$.

A. $\dfrac{1-2x^2}{x^2+1}$ C. $\dfrac{-2(x^2-1)}{(x^2+1)^2}$

B. $\dfrac{-2(x^2-1)}{x^2+1}$ D. $\dfrac{2-4x^2}{(x^2+1)^2}$

***Solution***: $f(x) = \ln(x^2+1) \Rightarrow$

$$f'(x) = \frac{2x}{x^2+1} \Rightarrow f''(x) = \frac{2(x^2+1)-2x(2x)}{(x^2+1)^2} = \frac{-2(x^2-1)}{(x^2+1)^2}$$

**Answer: C**

36. Let $f(x) = x\,|x|$, for all $x \in (-\infty, \infty)$. Which of the following is true?

A. $f$ is not differentiable at x = 0
B. $f'(x) = 2|x|$ at each $x \in (-\infty, \infty)$
C. $f$ is differentiable only on $(0, \infty)$
D. $f'(x) = 2x$ at each $x \in (-\infty, \infty)$

***Solution***: $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \Rightarrow f(x) = x|x| = \begin{cases} x^2, x \geq 0 \\ -x^2, x < 0 \end{cases}$

$$\Rightarrow f'(x) = \begin{cases} 2x, x \geq 0 \\ -2x, x < 0 \end{cases} \Rightarrow f'(x) = 2|x|$$

$\therefore f$ is differentiable on $(-\infty, \infty)$

**Answer: B**

37. Let $f(x) = x^4 - 6x^2 + 1$ on $[-2, 3]$. The local maximum value of $f$.

A. $f(0)$ B. $f(\sqrt{3})$ C. $f(3)$ D. $f(-\sqrt{3})$

***Solution***: $f'(x) = 4x^3 - 12x \Rightarrow 4x(x^2 - 3) = 0$, $x = 0$, $x = \sqrt{3}$, $-\sqrt{3} \leftarrow$ critical number

- $f(-2) = 16 - 6(4) + 1 = -6$
- $f(3) = 3^4 - 6(3^2) + 1 = 81 - 54 + 1 = 28 \leftarrow$ local maximum
- $f(0) = 1$
- $f(-\sqrt{3}) = 9 - 6(3) + 1 = -8 \leftarrow$ local minimum
- $f(\sqrt{3}) = -8$

$\therefore f(3) = 28$ is local max

**Answer: C**



38. Let $f(x) = x^4 - \dfrac{16}{3}x^3 + 6x^2$, for all $x \in (-\infty, \infty)$ what is the range of $f$.

A. $(-\infty, \infty)$ B. $[0, \infty)$ C. $[-9, \infty)$ D. $\left[\dfrac{5}{3}, -9\right]$

***Solution***: $f'(x) = 4x^3 - 16x^2 + 12x \Leftrightarrow 4x(x^2 - 4x + 3) = 0$

$\Leftrightarrow 4x(x-3)(x-1) = 0 \Rightarrow x = 0, x = 1, x = 3$

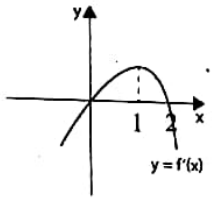
$f(0) = 0, f(1) = 1 - \dfrac{16}{3} + 6 = \dfrac{5}{3}$

$f(3) = 81 - 144 + 54 = -9 \leftarrow$ the minimum value

$\therefore$ The range of f $[-9, \infty)$

**Answer: C**

39. Suppose $f(x)$ is differentiable at all $x \in (-\infty, \infty)$ and the graph of $y = f'(x)$ is as shown in the figure below. Which of the following is true?




A. $f(x)$ is increasing on $(-\infty, 1)$
B. $f(x)$ attains its maximum value at $x = 2$
C. $f(x)$ attains its maximum value at $x = 1$
D. $f(x)$ is decreasing on $(0, 2)$

***Solution***: From the sign of derivative of $y = f'(x)$

| | $-\infty$ | 0 | 2 | $\infty$ |
|---|---|---|---|---|
| $f'(x)$ | − | + | − | |
| conclusion about $f(x)$ | decrease | increase | decrease | |

Hence:

- $f(x)$ is increasing on $(0, 2)$
- $f(x)$ has local minimum at $x = 0$
- $f(x)$ has maximum value at $x = 2$
- $f(x)$ is decreasing $(-\infty, 0)$ and $(2, \infty)$

**Answer: B**

40. The radius of a right circular cylinder of height 10 cm increases at a rate of 3 cm/min. What is the rate of change of the volume of cylinder when the radius is 8 cm?

A. $80\pi cm^3/min$ C. $480\ \pi cm^3/min$

B. $320\pi cm^3/min$ D. $640\ \pi\ cm^3/min$

***Solution***: $\dfrac{dr}{dt} = 3$ cm/min, h = 10 cm, $v = \pi r^2 h$

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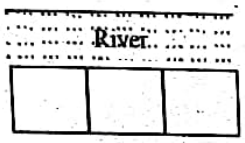
$\Rightarrow v = \pi r^2 h \Rightarrow \frac{dv}{dt} = 20\pi r \frac{dr}{dt}$, when $r = 8$

$\Rightarrow \frac{dv}{dt} = 20\,\pi\,(8)(3)\ cm^3/min = 480\ \pi cm^3/min$ **Answer: C**

41. A farmer wants to make three identical rectangular enclosures (same width and same length) along a straight river, as in the diagram shown below. If he has a fencing material of 1,200 meters and the sides along the river need no fence, what should be the dimensions (length $\ell$ and width w) of each enclosure if the total area of the enclosures is to be maximized?




A. $\ell = 100$ m, w = 225 m

B. $\ell = 300$ m, w = 75 m

C. $\ell = 200$m, w 200 m

D. $\ell = 200$m, w = 150 m

***Solution***: Let $x$ be each segment of length $\ell$ let $y$ be the width

Perimeter: $3x + 4y = 1200 \Rightarrow 4y = 1200 - 3x = y = 300 - \frac{3}{4}x$

Area $= (3x)(y) \Rightarrow A(x) = (3x)\left(300 - \frac{3}{4}x\right) = 900x - \frac{9x^2}{4}$

$A'(x) = 900 - \frac{9}{2}x \Rightarrow A'(x) = 0$ if $x = 200$ m and

$y = w = 300 - \frac{3}{4}(200) = 150$ m **Answer: D**

42. Which of the following is an anti-derivative of $f(x) = \frac{3}{x\sqrt{x}}$?

A. $\frac{6}{\sqrt{x}} + C$ B. $-6\sqrt{x} + C$ C. $\frac{-6}{\sqrt{x}} + c$ D. $\frac{-3}{2\sqrt{x}} + 3$

***Solution***:

$f(x) = \frac{3}{x\sqrt{x}} \Rightarrow f(x) = 3x^{-3/2}$.



An anti derivative of $f(x)$ is $\frac{3x^{-1/2}}{\frac{-3}{2}+1} = (3)(-2)x^{-1/2} = \frac{-6}{\sqrt{x}} + C$

**Answer: C**

43. Which of the following is equal to $\int \frac{x^2-1}{x^3+x}dx$?

A. $\ln|x^2+1| + \ln|x| + C$

B. $\frac{1}{2}\ln(x^2+1) - \ln|x| + C$

C. $\ln(x^2+1) - \ln|x| + C$

D. $2\ln(x^2+1) - \ln|x| + C$

***Solution***: $\frac{x^2-1}{x^3+x} = \frac{x^2-1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$

$\Rightarrow x^2 - 1 = A(x^2+1) + Bx^2 + Cx \leftarrow$ Multiply both side by $x$

$x(x^2+1) = x^2 - 1 = Ax^2 + Bx^2 + A + Cx$

$\Rightarrow x^2 - 1 = (A+B)x^2 + Cx + A \leftarrow$ factoring $x^2$

$\Rightarrow A + B = 1$, and $A = -1$, $C = 0$ and $B = 2$

Now, $\int \frac{x^2-1}{x^3+x}dx = \int \frac{-dx}{x} + \int \frac{2xdx}{x^2+1}$. Note $\int \frac{-dx}{x} = -\ln x$

To find $\int \frac{2xdx}{x^2+1}$, $u = x^2 + 1 \Rightarrow du = 2x\,dx$

$\Rightarrow \int \frac{du}{u} = \ln|u| = \ln(x^2+1)$

$\therefore \int \frac{x^2-1}{x^3+x}dx = \ln|x| = \ln(x^2+1) + C \Rightarrow \ln(x^2+1) - \ln|x| + C$

**Answer: C**

44. $\int (2 + \sin^2 x)\cot x\,dx$ is equal to:

A. $2\csc x - \sec^2 x + C$ C. $2\ln|\sin x| - \cos^2 x + C$

B. $2\ln|\sin x| + \frac{1}{2}\sin^2 x + C$ D. $2\csc x + \sin x\cos x + C$

***Solution***: $\int (2 + \sin^2 x)\cot x\,dx = \int (2\cot x + \sin x\cos x)dx$

$\Rightarrow \int \frac{2\cos x}{\sin x} + \int \sin x\cos x\,dx$, let $u = \sin x \Rightarrow du = \cos x\,dx$

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$$\Rightarrow \int \frac{2\,du}{u} + \int u\,du = 2\ln|u| + \frac{u^2}{2} + C = 2\ln|\sin x| + \frac{\sin^2 x}{2} + C$$

**Answer: B**

45. $\int_1^2 x\left(\frac{1}{x^2} + 2\ln x\right)dx$ is equal to:

A. $2\ln 2 - 1$ C. $2\ln 2 - \frac{5}{2}$

B. $5\ln 2 + \frac{5}{2}$ D. $5\ln 2 - \frac{3}{2}$

***Solution:*** $\int_1^2\left(\frac{1}{x} + 2x\ln x\right)dx = \int_1^2 \frac{dx}{x} + \int_1^2 2x\ln x\,dx$

To find $\int 2x\ln x\,dx$, let $u = \ln x$, $dv = 2x\,dx \Rightarrow du = \frac{1}{x}, v = x^2$

$$\Rightarrow \int_1^2 2x\ln x\,dx = x^2\ln x - \int x\,dx = x^2\ln x - \frac{x^2}{2}\Big|_1^2 = 4\ln 2 - \frac{3}{2}$$

$$\therefore \int_1^2 \frac{dx}{x} + \int_1^2 2x\ln x\,dx = \ln x\Big|_1^2 + 4\ln 2 - \frac{3}{2} = 5\ln 2 - \frac{3}{2}$$

**Answer: D**

46. $\int_0^1 \frac{dx}{e^{-x}(1+e^x)}$ is equal to:

A. $\ln(e-1)$ B. $\ln(1+e) - \ln 2$ C. $\ln(2+e)$ D. $\ln 2$

***Solution:***

$\int_0^1 \frac{e^x dx}{(1+e^x)}$, since $\frac{1}{e^{-x}} = e^x$ let $u = 1 + e^x \Rightarrow du = e^x dx$

$$\Rightarrow \int \frac{du}{u} = \ln u = \ln(1+e^x)\Big|_0^1 = \ln(1+e) - \ln 2$$

**Answer: B**

47. The area of the region bounded between the graph of $y = x^2$ and $y = \sqrt{x}$ is:

A. 35 q unit B. $\frac{1}{3}$ sq. unit C. $\frac{2}{3}$ sq. unit D. 1 sq. unit



***Solution:***

The intersection $\begin{cases} y = x^2 \\ y = \sqrt{x} \end{cases}$ is (0, 0) and (1, 1).

$$\therefore A = \int_0^1 \left(\sqrt{x} - x^2\right)dx = \frac{2}{3}x^{3/2} - \frac{x^3}{3}\Big|_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3} \text{ sq. unit}$$

48. Which one of the following points is outside the sphere $x^2 + y^2 + z^2 - 4z = 0$?

**Answer: B**

A. (0, −1, 1) B. (−2, 0, 2) C. (1, −1, 3) D. (−1, 2, 0)

***Solution:*** $x^2 + y^2 + z^2 - 4z + 4 = 4 \Rightarrow (x-0)^2 + (y-0)^2 + (z-2)^2 = 4$

A. (0, −1, 1): $0^2 + 1^2 + 1^2 = 2 < 4$
∴ (0, −1, 1) lies inside the sphere.

B. (−2, 0, 2): $(-2)^2 + 0^2 + 0^2 = 4 = 4$
∴ (−2, 0, 2) lies on the sphere.

C. (1, −1, 3): $1^2 + (-1)^2 + (3-2) = 3 < 4$
∴ (1, −1, 3) lies inside sphere.

D. (−1, 2, 0): $(-1)^2 + (2)^2 + 0 = 5 > 4$
∴ (−1, 2, 0) lies outside sphere

49. Which of the following is true about the points of intersection of the sphere of radius 3 units with center at (−1, 3, 2) and z axis.

**Answer: D**

A. (0, 0, 3) is a point of intersection
B. There are two points of intersection.
C. There is no point of intersection.
D. There are many points of intersection.

***Solution:*** $(x+1)^2 + (y-3)^2 + (z-2)^2 = 3^2 \leftarrow$ eqn of sphere.
Point on z axis is (0, 0, z). So that intersection of sphere with z axis:

$\Rightarrow (0+1)^2 + (0-3)^2 + (z-2)^2 = 9 \Rightarrow (z-2)^2 = 9 - 10 = -1$
$\Rightarrow (z-2)^2 = -1 \leftarrow$ no point of intersection

**Answer: C**

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1. If $x<0$, then the simplest form of $f(x)=\dfrac{x-|x|}{x}$ is equal to:

A. $2x$ B. $2$ C. $-2$ D. $0$

***Solution***: $|x|=-x$ if $x<0 \Rightarrow f(x)=\dfrac{x-(-x)}{x}=\dfrac{2x}{x}=2$

**Answer: B**

2. If $f(x)=\dfrac{\sqrt{x+2}}{x+2}$, and $g(x)=\dfrac{1}{x}-2$, then $f(g(x))$ is equal to:

A. $\sqrt{x}-2$ B. $\sqrt{x}+2$ C. $\sqrt{x}$ D. $\dfrac{\sqrt{x}}{x}$

***Solution***:

$$f(g(x))=\frac{\sqrt{g(x)+2}}{g(x)+2}=\frac{\sqrt{\frac{1}{x}-2+2}}{\frac{1}{x}-2+2}=\frac{x}{\sqrt{x}}=\sqrt{x}$$

**Answer: C**

3. If $f(x)=\ln\left(\dfrac{x}{x-1}+2\right)$, for $x>1$, then the inverse of the function $f$

A. $g(x)=\dfrac{e^x-2}{e^x-3}$ C. $g(x)=\dfrac{e^x}{e^x+1}$

B. $g(x)=\dfrac{e^x-2}{e^x+1}$ D. $e^{\frac{x}{x-1}}-2$

***Solution***:

To find $f^{-1}$, set: $x\to y, y\to x$

$\Rightarrow y=\ln\left(\dfrac{x+2(x-1)}{x-1}\right)=\ln\left(\dfrac{3x-2}{x-1}\right)$, so $x\to y, y\to x$

$\Rightarrow x=\ln\left(\dfrac{3y-2}{y-1}\right)\Rightarrow e^x=\dfrac{3y-2}{y-1}\Rightarrow e^xy-e^x=3y-2$

$\Rightarrow e^xy-3y=e^x-2\Leftrightarrow y(e^x-3)=e^x-2\Rightarrow y=\dfrac{e^x-2}{e^x-3}$

**Answer: A**



4. The simplified form of $\dfrac{4-(a-2)^2}{a^4+2a^2+1}\div\dfrac{1-\dfrac{1}{a^2+1}}{a}$.

A. $\dfrac{4-a}{a}$ B. $\dfrac{-a}{a^2+1}$ C. $\dfrac{-1}{a^2(a^2+1)}$ D. $\dfrac{4-a}{a^2+1}$

***Solution***: The expression the same as:

$$\Leftrightarrow\frac{4-(a^2-4a+4)}{(a^2+1)^2}\div\frac{a^2+1-1}{(a^2+1)a}=\frac{-a^2+4a}{(a^2+1)^2}\cdot\frac{a(a^2+1)}{a^2}$$

$$\Rightarrow\frac{a(4-a)}{a^2}\cdot\frac{a(a^2+1)}{(a^2+1)^2}=\frac{4-a}{a^2+1}$$

**Answer: D**

5. If $f(x)=\dfrac{x-1}{(x-2)^2(x+1)}$ which of the following is true about f.

A. Its graph has an oblique asymptote.
B. The graph of f does not meet its asymptote.
C. As, $x\to -1^+, f(x)\to -\infty$
D. As $x\to 2^-, f(x)\to -\infty$

***Solution***: Here

A. Has no oblique asymptote since the degree of numerator less than denominator

B. $y=0\leftarrow$ H.A, so that the graph meet horizontal asymptote at $x=1$

C. $x\to -1^+, \dfrac{-1^+-1}{(-1-2)^2(-1^++1)}=\dfrac{-2}{(9)(0^+)}=-\infty\Rightarrow f(x)\to -\infty$

D. $x\to 2^-, f(x)=\dfrac{2-1}{(2^--2)^2(2+1)}=\dfrac{1}{(0^+)(3)}=+\infty$

$\therefore x\to 2^-, f(x)\to\infty$,

**Answer: C**

6. Line $\ell$ passes through (0, 5) and (–5, 0). What is the angle between the y-axis and $\ell$ in radian measure?

A. $\dfrac{\pi}{4}$ B. $\dfrac{\pi}{3}$ C. $\dfrac{\pi}{2}$ D. $\dfrac{3}{4}\pi$

***Solution***: Slope of $\ell=\dfrac{5-0}{0-(-5)}=\dfrac{5}{5}=1$

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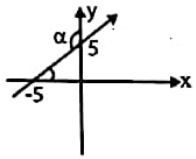



Angle of inclination, $\tan\theta = 1$

$\Rightarrow \theta = \frac{\pi}{4}$ ← the angle measured from positive x-axis to line $\ell$.

$\Rightarrow \alpha = \pi - \frac{\pi}{4} = \frac{3}{4}\pi$ ← the angle between y-axis and $\ell$ measured C.C.W from y to $\ell$. **Answer: D**

7. Consider a circle whose center is on the $x$-axis. If a line given by $y = x$ is tangent to the circle at point (2, 2), what is the equation of the circle?

A. $x^2 + y^2 = 8$ C. $(x-4)^2 + y^2 = 8$
B. $(x-2)^2 + y^2 = 4$ D. $(x-1)^2 + y^2 = 5$

***Solution***: Let the center (a, 0)

- Slope of the line $y = x$ is 1.
- Slope of the line passing through (2, 2) and (a, 0) is −1.

$$\Rightarrow \frac{2-0}{2-a} = -1 \Rightarrow 2 = -2 + a \Rightarrow a = 4$$

$\therefore$ The center of circle is (4, 0).

- The radius $r = \sqrt{(2-0)^2 + (2-4)^2} = \sqrt{8} \Rightarrow r^2 = 8$

$\Rightarrow$ Equation of circle: $(x-4)^2 + (y-0)^2 = 8$ **Answer: C**

8. What is the vertex and the equation of the direction, respectively of the parabola $x + y^2 + 2y + 1 = 0$?

A. $(0, -1), x = -\frac{1}{4}$ C. $(0, -1), x = \frac{1}{4}$
B. $(-1, 0), x = -\frac{1}{4}$ D. $(-1, 0), y = \frac{1}{4}$

***Solution***: Here $x + y^2 + 2y + 1 = 0 \Rightarrow y^2 + 2y + 1 = -x$

$\Rightarrow (y+1)^2 = -x \Rightarrow 4p = |-1| \Rightarrow p = \frac{1}{4}$ ← focal length

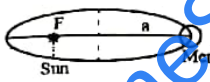
Vertex (0, −1) and directix, $x = \frac{1}{4}$ **Answer: C**

9. The orbit of Mercury around the sun forms an ellipse with eccentricity 0.206, length of the major axis $1.16 \times 10^8$ km and the sun at one focus. The best approximation of the maximum distance from Mercury to the sun.



A. $7.596 \times 10^7$ km C. $8.695 \times 10^7$ km
B. $5.695 \times 10^7$ km D. $6.995 \times 10^7$ km

***Solution***: Let 2a = distance between major axis.
2c = distance between focus
e = eccentricity




$\Rightarrow 2a = 1.16 \times 10^8$ km $\Rightarrow a = 5.8 \times 10^7$ km

$e = \frac{c}{a} \Rightarrow c = ea = (0.206)(5.8 \times 10^7 \text{ km}) = 1.195 \times 10^7$ km

The maximum distance, $a + c = 5.8 \times 10^7 + 1.95 \times 10^7 \approx 6.995 \times 10^7$ km

Note: the minimum distance $a - c = (5.8 - 1.195) \times 10^7 = 4.61 \times 10^7$ km **Answer: D**

10. Suppose proposition $p \Rightarrow \neg q$ is false (F), which of the following is true?

A. $\neg q \wedge (p \Rightarrow q)$ C. $(\neg q \vee p) \Leftrightarrow q$
B. $\neg p \vee (q \Rightarrow \neg p)$ D. $(p \wedge q) \Leftrightarrow \neg p$

***Solution***: $p \Rightarrow \neg q \equiv F$, if $p \equiv T$, $\neg q \equiv F$, and $q \equiv T$ $(\neg q \vee p) \Leftrightarrow q \equiv (F \vee T) \Leftrightarrow T \equiv T \Leftrightarrow T \equiv T$

**Answer: C**

11. Suppose $e(x) \equiv x$ is even, $p(x) \equiv x$ is prime, $d(x) \equiv x$ is divisible by 2. Which one of the following has the truth value F on the set of natural number?

A. $(\exists x)[e(x) \wedge \neg d(x)]$ C. $(\exists x)[e(x) \wedge \neg d(x)]$
B. $(\forall x)[e(x) \Rightarrow d(x)]$ D. $(\forall x)[e(x) \vee \neg d(x)]$

***Solution***: A. True, let $x = 2$
B. True D. True

**Answer: C**

12. Consider the following argument form production is high if rain continues. Rain does not continue. Therefore, either production is low or rain continues. Let p ≡ production is low, q ≡ rain continues. The following table is also given about p and q.

| Row | p | q | ¬p | ¬q | ¬p⇒q | q⇒¬p | p ∨q |
|---|---|---|---|---|---|---|---|
| 1 | T | T | F | F | T | F | T |
| 2 | T | F | F | T | T | T | T |
| 3 | F | T | T | F | T | T | T |
| 4 | F | F | T | T | F | T | F |

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Which of the following is necessary true:

A. argument form is valid due to row 2
B. argument form is valid due to row 2 and 3
C. argument form is invalid due to row 4
D. argument form is invalid due to rows 1 and 3

***Solution***: Argument form: $q \Rightarrow \neg p, \neg q \vdash p \vee q$. The premise $q \Rightarrow \neg p, \neg q$ are true in row 4 but the conclusion $p \vee q$ is false in row 4.

**Answer: C**

13. If $x$ and $y$ are non-negative integers which of the following is not true?

A. $(\forall x)(\exists x)(y > x^2 - 1)$ C. $(\exists y)(\forall x)(y \leq x^2 - 1)$
B. $(\exists x)(\forall x)\ (y > x^2 - 1)$ D. $(\exists y)(\exists x)(y \leq x^2 - 1)$

***Solution***:

A. Let $x = 2 \Rightarrow y > 3$, True
B. False, let fixed $x = 1 \Rightarrow y > 0$ is false, because when $y = 0$
C. True, let fixed number be $y_o \Rightarrow y_o \geq x^2 - 1 \Rightarrow y_o + 1$. Since $x$ and $y$ are non-negative so that let $y_o = 0 \Rightarrow x^2 \geq 1$ is true for all $x$.

**Answer: B**

14. A school has three class rooms for grade for grade 11 namely 11A, 11B and 11C. The number of students in these class rooms is 28, 20 and 22 respectively. All the students took an examination and the average score of the students of 11A, 11B and 11C is 60, 70 and 70 respectively. What is the average score in this examination for all grade 11 students?

A. 66 B. 66.67 C. 65 D. 65.67

***Solution***: The combined mean $\bar{x}$ is calculated by

$$\bar{x} = \frac{60 \times 28 + 20 \times 70 + 22 \times 70}{28 + 20 + 22} = \frac{4620}{70} = 66$$

**Answer: A**

15. You are given a data on the age of students, in a primary school.

| Age | 8 | 10 | 11 | 12 | 13 |
|---|---|---|---|---|---|
| Number of students | 5 | 5 | 8 | 10 | 12 |

Which of the following is NOT true about the data?

A. the median is 10.5 C. the mean is 10.5
B. the mode is 10 D. the range is 5

***Solution***: Median of raw data is given by



$$M_d = \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{ item if } n \text{ is odd} \\ \dfrac{\left[\left(\frac{n}{2}\right)^{th} \text{ item} + \left(\frac{n}{2}+1\right)^{th} \text{ item}\right]}{2} \text{ item if } n \text{ is even} \end{cases}$$

A. median $= \dfrac{\left[\left(\frac{40}{2}\right) + \left(\frac{40}{2}+1\right)\right]}{2}$ item $= \dfrac{10+11}{2} = 10.5$

B. mode $= 10 \leftarrow$ the most frequenct value

C. mean $= \dfrac{8\times5 + 10\times15 + 11\times8 + 12\times10 + 13\times2}{5+15+8+10+2} = 10.6$

D. range = max − min = 13 − 8 = 5

**Answer: C**

16. The following is a table of simple frequency distribution of a data with variable x.

| $x$ | 1 | 3 | 4 | 5 | 7 |
|---|---|---|---|---|---|
| Frequency | 2 | 5 | 6 | 5 | 2 |

The standard deviation of the data is equal to:

A. $\sqrt{3}$ B. 3 C. $\sqrt{2.3}$ D. $\sqrt{3.6}$

***Solution***: Standard deviation = $\sqrt{\text{variance}}$ and mean $(\bar{x}) = 4$

$$S.D = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i}}$$

$$= \sqrt{\frac{2(1-4)^2 + 5(3-4)^2 + 6(4-4)^2 + 5(5-4)^2 + 2(7-4)^2}{2+5+6+5+2}}$$

$$= \sqrt{\frac{18+5+0+5+18}{20}} = \sqrt{\frac{23}{10}} = \sqrt{2.3}$$

**Answer: C**

17. A three digit library identification card is to be printed from numbers 0, 1, 2, 3, 4, 5 in a such away that the first is non-zero and no number is to be repeated. How many such cards can be printed?

A. 100 B. 120 C. 150 D. 180

***Solution***: $5 \times 5 \times 4 = 100$

**Answer: A**

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18. A student needs to select 3 books form 3 different mathematics, 3 different physic and 1 history books. What is the probability that one of them is mathematics and the other two are either physics or history books?

A. $\frac{3}{35}$ B. $\frac{9}{25}$ C. $\frac{15}{35}$ D. $\frac{18}{35}$

***Solution***: Use combination rule

$$\Rightarrow \frac{C(3,1).C(3,1).C(1,1)}{C(7,3)} + \frac{C(3,1)C(3,2).C(1,0)}{C(7,3)} = \frac{9}{35} + \frac{9}{35} = \frac{18}{35}$$

**Answer: D**

19. Items produced by a certain company are subjected to two kinds of defect $D_1$ and $D_2$. Out of the total production if 5% have defect $D_1$, 10% have defect $D_2$ and 2% have both defects then what is the probability for an item to have defect $D_2$ given that it has defect $D_1$?

A. 0.2 B. 0.05 C. 0.1 D. 0.4

***Solution***: P ($D_1$ and $D_2$) = 0.02, P ($D_1$) = 0.05 $\Rightarrow$ P ($D_1$ and $D_2$) = P ($D_1$). P ($D_2/D_1$) $\Rightarrow$ 0.02 = (0.05) P ($D_2.D_1$)

$$\Rightarrow P(D_2/D_1) = \frac{0.02}{0.05} = \frac{2}{5} = 0.4$$

**Answer: D**

20. Let $A = \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -y \\ 0 & 3 \\ 1-x & 2 \end{pmatrix}$ such that $A + 2B^T = 0$. Then which of the following is the value of y?

A. 0 B. $\frac{-13}{2}$ C. $-8$ D. any real number

***Solution***: $B^T = \begin{pmatrix} 1 & 0 & 1-x \\ -y & 3 & 2 \end{pmatrix} \Rightarrow 2B^T = \begin{pmatrix} 2 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix}$

$$\Rightarrow A + 2B^T = \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix} + \begin{pmatrix} 2 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow x + (2 - 2x) = 0$ and $x + y + 6 = 0 \Rightarrow -x = -2 \Rightarrow x = 2$, so that to find y put $x = 2$ in $x + y + 6 = 0 \Rightarrow 2 + y + 6 = 0 \Rightarrow y + 8 = 0$

$\Rightarrow y = -8$

**Answer: C**



21. Let A and B be 3 × 3 matrices such that $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & \frac{1}{2} \end{pmatrix}$ and $|B| = \frac{1}{10}$ then $|2AB^T|$ equal to _

A. 1 B. 4 C. 100 D. 400

***Solution***: $|2AB^T| = 2^3 |A||B^T| = 8\left(2 \times 5 \times \frac{1}{2}\right)\left(\frac{1}{10}\right) = 4$

***Explanation***:

3 is the order to matrix $|A| = |A| = 2 \times 5 \times \frac{1}{2}$ ← determinant of triangular matrix.

22. What is the solution set of the following system of equations $\begin{cases} x + y + 2z = 1 \\ x + 2y + z = -2 \\ -2x - 2y - 4z = -2 \end{cases}$ ?

A. {(0, 1, 0)} C. {(−3km k+1,k)},k∈(−∞, ∞)

B. (−∞, ∞) D. {(3k, k − 1, k)}, k ∈(−∞, ∞)

***Solution***: $D = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ -2 & -2 & -4 \end{vmatrix} = -2\begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 0$ since row 1 and 3 are the same.

$D_x = \begin{vmatrix} 1 & 1 & 2 \\ 2 & 2 & 1 \\ -2 & -2 & -4 \end{vmatrix} = -2\begin{vmatrix} 1 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 0$, and $D_y = 0$, $D_z = 0$.

$\Rightarrow$ The system has infinite many solution.

$$-\begin{cases} x + y + 2z = 1 \\ x + 2y + z = 2 \end{cases} \quad \text{and} \quad +\begin{cases} x + 2y + z = 2 \\ -2x - 2y - 4z = -2 \end{cases}$$

$0 + -y + z = -1 \Rightarrow y = z + 1$ $\quad -x + 0 - 3z = 0 \Rightarrow x = -3z$

$\therefore$S.S = $\{(x, y, z)\} = \{(-3z, z + 1, z)\} = \{(-3k, k + 1, k)\}$ $k \in \mathbb{R}$

**Answer: C**

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23. If $\begin{pmatrix} \alpha & 2 & \beta \\ 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix}$ and the determinant of the coefficient matrix $|A| = -5$ then the value of $x$ is equal to:

A. 3 B. $\alpha + \beta$ C. $-5\alpha$ D. 5

***Solution***: Let $x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $C = \begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix}$, $|A| = -5 \Rightarrow Ax = C$

$\Rightarrow x = A^{-1}C$, But $A^{-1} = \frac{adj.A}{|A|} \Rightarrow x = \frac{adj.A}{|A|}(C)$ and to find adj. A first find the cofactor.

- $A_{11} = (-1)^{1+1}\begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1$ $\quad A_{21} = 2$
- $A_{12} = (-1)^{1+2}\begin{vmatrix} 2 & 3 \\ -1 & 1 \end{vmatrix} = -5$ $\quad A_{22} = (-2)^{2+2}\begin{vmatrix} \alpha & \beta \\ -1 & 1 \end{vmatrix} = \alpha + \beta$
- $A_{13} = (-1)^{1+3}\begin{vmatrix} 2 & 1 \\ -1 & 0 \end{vmatrix} = 1$ $\quad A_{23} = (-1)^{2+3}\begin{vmatrix} \alpha & 2 \\ -1 & 0 \end{vmatrix} = 2$
- $A_{31} = (-1)^{3+1}\begin{vmatrix} 2 & \beta \\ 1 & 3 \end{vmatrix} = 6 - \beta$ $\quad A_{32} = (-1)^{3+2}\begin{vmatrix} \alpha & \beta \\ 2 & 3 \end{vmatrix} = 2\beta - 3\alpha$
- $A_{33} = (-1)^{3+3}\begin{vmatrix} \alpha & 2 \\ 2 & 1 \end{vmatrix} = \alpha - 4$,
- Adjoins of A is transpose of cofactor.

$\Rightarrow$ cofactor of $A = \begin{pmatrix} 1 & -5 & 1 \\ 2 & \alpha+\beta & 2 \\ 6-\beta & 2\beta-3\alpha & \alpha-4 \end{pmatrix}$,

adj. $A = \begin{pmatrix} 1 & 2 & 6-\beta \\ -5 & \alpha+\beta & 2\beta-3\alpha \\ 1 & 2 & \alpha-4 \end{pmatrix}$



$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{adj.A}{|A|}(C) = \frac{-1}{5}\begin{pmatrix} 1 & 2 & 6-\beta \\ -5 & \alpha+\beta & 2\beta-3\alpha \\ 1 & 2 & \alpha-4 \end{pmatrix}$

$\Rightarrow x = \begin{pmatrix} \frac{-1}{5} & \frac{2}{5} & \frac{6-\beta}{5} \end{pmatrix}\begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix} = 1 + 2 + 0 = 3$

$\Rightarrow y = \begin{pmatrix} -1 & \frac{\alpha+\beta}{-5} & \frac{2\beta-3\alpha}{-5} \end{pmatrix}\begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix} = 5 - \alpha - \beta + 0 = 5 - (\alpha + \beta$

$\Rightarrow z = \begin{pmatrix} \frac{-1}{5} & \frac{-2}{5} & \frac{-\alpha+4}{5} \end{pmatrix}\begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix} = 1 - 2 + 0 = -1$

$\Rightarrow$S.S = $\{(x, y, z)\} = \{(3, 5 - (\alpha + \beta)$ or S.S = $\{(3, 5) - (\alpha + \beta), -1)\}$

**Answer: A**

24. In the set of complex numbers, the solution set of $x^2 - 2x + 5 = 0$ is:

A. $\phi$ C. $\{1 + 2i, 1 - 2i\}$

B. $\{2 + i, 2 - i\}$ D. $\{2 + 4i, 2 - 4i\}$

***Solution***: $x^2 - 2x + 5 = 0 \Leftrightarrow x^2 - 2x = -5 \Leftrightarrow x^2 - 2x + 1 = -5 + 1 \Rightarrow (x-1)^2 = -4 \Rightarrow x - 1 = \pm\sqrt{-4} = \pm 2i \Rightarrow$ and $x = 1 \pm 2i$

**Answer: C**

25. The simplified form of $\frac{4-3i}{3+4i} + \overline{1-2i}$ is equal _____

A. 1 + i B. 1 + 3i C. 2 – i D. 1 – 3i

***Solution***: $\frac{4-3i}{3+4i}\cdot\frac{3-4i}{3-4i} + 1 + 2i = \frac{(12-16i-9i-12)}{9+16} + 1 + 2i$

$\Rightarrow -i + 1 + 2i = 1 + i$,

**Answer: A**

26. Let z be complex number. Then the solution set of $z^3 - iz = 0$.

A. $\left\{\pm\frac{1}{\sqrt{2}}(1+i)\right\}$ C. $\left\{\pm\sqrt{2}(1+i)\right\}$

B. $\left\{0, \pm\frac{1}{\sqrt{2}}(1-i)\right\}$ D. $\left\{0, \pm\frac{1}{\sqrt{2}}(1-i)\right\}$

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**Solution**: $z^3 - iz = 0 \Leftrightarrow z(z^2 - i) = 0 \Rightarrow z = 0$ or $z^2 - i = 0$ but $z = x + yi$ so that $z^2 - i = 0 \Leftrightarrow z^2 = i \Rightarrow (x + yi)^2 = i$

$\Rightarrow x^2 + 2xyi - y^2 = i \leftarrow$ comparing the real part and i

$\Rightarrow x = \pm y$ and $2xy = 1$ put in $2xy = 1$

$\Rightarrow 2x^2 = 1 \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm\frac{1}{\sqrt{2}}$ and $y = \pm\frac{1}{\sqrt{2}}$

$\therefore z = x + yi = \pm\frac{1}{\sqrt{2}}(1 + i)$ and $z = x + yi = 0$

$\therefore S.S = \left\{0, \pm\frac{1}{\sqrt{2}}(1 + i)\right\}$ **Answer: D**

27. Which of the following is a vector that lies on the line through (0, 0) and (2, 4)?

A. $\bar{u} = (2, 1)$ C. $\bar{u} = \left(\frac{1}{2}, 2\right)$

B. $\bar{u} = (-1, -2)$ D. $\bar{u} = (-2, -6)$

**Solution**:

Let A = (0, 0) and B = (2, 4) $\Rightarrow \overrightarrow{AB} = (2 - 0, 4 - 0) = (2, 4)$ vector equation of $\ell$: $(x, y) = (0,0) + t(2, 4) = (2t, 4t)$, $t \in \mathbb{R}$

$\Rightarrow$ The set of all $(x, y) = (2t, 4t)$, if $t = -\frac{1}{2} \Rightarrow u = (-1, -2)$

**Answer: B**

28. Let $i$ and $j$ be the standard unit vector in the directions of positive x-axis and positive y-axis respectively and $\overrightarrow{AB}$ be a vector from the point A (−1, 0) to the point B (2, 2). If $\bar{v} = 3\overrightarrow{AB} - 3i + 2j$, then the unit vector in the direction of $\bar{v}$ is equal to:

A. $\left(\frac{3}{5}, \frac{4}{5}\right)$ B. $\left(\frac{-3}{5}, \frac{-4}{5}\right)$ C. $\left(\frac{-3}{5}, \frac{4}{5}\right)$ D. $\left(\frac{3}{5}, \frac{-4}{5}\right)$

**Solution**: $\overrightarrow{AB} = (2, 2) - (-1, 0) = (3, 2) = 3i + 2j$

$\bar{v} = 3(3i + 2j) - 3i + 2j = 6i + 8j = (6, 8)$

Unit vector in the direction of $\bar{v}$ is $\frac{\bar{v}}{|\bar{v}|} = \frac{(6,8)}{\sqrt{6^2 + 8^2}} = \left(\frac{3}{5}, \frac{4}{5}\right)$.

**Answer: A**

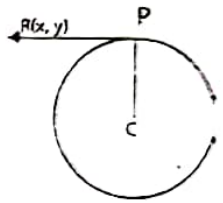


29. Which of the following is a vector equation of the line tangent to the circle $x^2 + y^2 + 2x - 7 = 0$?

A. $(x, y) = (0, 3) + \lambda(-1, 2)$ C. $(x, y) = (0, 3) + \lambda(1, -1)$

B. $(x, y) = (1, 2) + \lambda(2, -1)$ D. $(x, y) = (1, 2) + \lambda(-1, 2)$

**Solution**: $x^2 + y^2 + 2x - 7 = 0 \Rightarrow (x + 1)^2 + (y - 0)^2 = 8 \Rightarrow$ center C (−1, 0). Let R $(x, y)$ and P (1, 2) point on tangent line,




$\overrightarrow{CP} = (1 - (-1), (2 - 0) = (2, 2)$

$\overrightarrow{CP} \cdot \overrightarrow{PR} = 0 \Rightarrow (2, 2) \cdot (x - 1, y - 2) = 0$

$\Rightarrow 2x - 2 + 2y - 4 = 0 \Rightarrow 2x + 2y = 6$

$\Rightarrow x + y = 3$. The direction vector $\bar{v} = (a, b)$ parallel to the line $x + y = 3$ is $x + y = 0$

$\Rightarrow y = -x$, $\bar{v} = (a, b) = (a, -a) = \lambda(1, -1)$

Take any arbitrary point on $x + y = 3$ say (0, 3).

$\therefore$ Vector equation of $\ell$: $(x, y) = (0, 3) + \lambda(1, -1)$.

**Answer: C**

30. Let $\ell$ be the line whose equation $2x - y = 10$ whichy one of the following is the equation of the image of $\ell$ after reflection in the line $y = 2x - 5$ followed by a rotation through the angle of 90 about the origin?

A. $x + 2y = 0$ C. $x + 2y = 5$

B. $2x + y = 0$ D. $x - 2y = 5$

**Solution**: $\ell$: $y = 2x - 10$ is parallel to the reflecting (line): $y = 2x - 5$. Hence, the image $\ell'$: $y = 2x + k$, now we need to find $k$. Let $(a, b)$ be any point on $\ell$, say $(0, -10)$.

Let the image of $(0, -10) = (a', b')$ which lies on $\ell'$: $y = 2x + k$.

$\Rightarrow M((0, -10)) = (a', b') \Rightarrow \frac{b' + 10}{a' - 0} = -\frac{1}{2} \leftarrow$ slope of perpendicular bisector of the two line $y = 2x - 10$ and $y = 2x + k$.

$\Rightarrow a' = -2b' - 20$. Also, the mid-point of (0, −10) and $(a', b')$

$\Rightarrow \left(\frac{0 + a'}{2}, \frac{b' - 10}{2}\right)$ which lies on reflecting axis $y = 2x - 5$

$\Rightarrow \left(\frac{a'}{2}, \frac{b' - 10}{2}\right) \Rightarrow \frac{b' - 10}{2} = 2\left(\frac{a'}{2}\right) - 5$

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$\Rightarrow b' = 2a'$ but $a' = -2b' - 20 \Rightarrow a' = -2(2a') - 20$ but $a' = -2b' \Rightarrow a' = -2(2a') - 20 \Rightarrow 5a' = -20 \Rightarrow a' = -4$ and $b' = 2a' = 2$

$\therefore$ The image of M (0, −10) = (4, 8). Since (4, 8) lies on $y = 2x + k$, so that $\Rightarrow 8 = 2(4) + k \Rightarrow k = 0$.

$\therefore$ The image of $y = 2x - 10$ on reflecting line $y = 2x - 5$ is $y = 2x$. Now rotate $y = 2x$ on $\theta = 90°$ about (0, 0)

$\Rightarrow$ R $(x, y) = (-y, x) \Rightarrow x \to -y$, $y \to x$. Thus, the image after rotastion $\Rightarrow x = -2y \Leftrightarrow x + 2y = 0$ **Answer: A**

31. Let $\bar{a} = -i + 3k$ and $\bar{b} = -i + j$ be vector in the space. Which one of the following is the cosine of the angle between $\bar{a}$ and $\bar{a} - \bar{b}$.

A. $\frac{9}{10}$ B. $\frac{3}{5}$ C. $\frac{3}{\sqrt{10}}$ D. $\frac{-9}{10}$

***Solution***: $\bar{a} = -i + 0j + 3k$, and $\bar{b} = -i + j + 0k$. Let θ be angle between $\bar{a}$ and $\bar{a} - \bar{b} = \bar{a} - \bar{b} = 0i - j + 3k$, thus

$$\cos\theta = \frac{\bar{a}.(\bar{a}.\bar{b})}{|\bar{a}||\bar{a}-\bar{b}|} = \frac{(-i+0j+3k)}{\sqrt{1+9}} \cdot \frac{(0i-j+3k)}{\sqrt{1+9}} = \frac{9}{\sqrt{10}}$$

i.e. $\cos\theta = \frac{9}{\sqrt{10}}$

**Answer: A**

32. Let $\bar{v} = 3i - 4k$, when $\bar{i}$ and $\bar{k}$ are the standared unit vectors in the directions of positive x-axis and positive z-axis respectively, and $\overline{AB}$ be a vector from the point A (0, 1, 2) to a point B in space. If $\overline{AB}$ is parallel to $\bar{v}$ and $\|\overline{AB}\| = 10$, then point B is at:

A. (−6, −1, 10) C. (6, 1, −6)
B. (6, −1, −10) D. (−6, −1, 6)

***Solution***: Let B = (x, y, z). Since

$\overline{AB} \| \bar{V} \Rightarrow \overline{AB} = t\bar{v} \Rightarrow |\overline{AB}| = |t\bar{v}|$

$\Rightarrow t\bar{v} = t(3, 0, -4) \Rightarrow (3t, 0, -4t)$

$\Rightarrow |t\bar{v}| = 10 \Rightarrow \sqrt{9t^2 + 0 + 16t^2} = \sqrt{25t^2} = 5t$

$\Rightarrow 5t = 10 \Rightarrow t = 2$



$\Rightarrow \overline{AB} = \bar{B} - \bar{A} = (x, y, z) - (0, 1, 2) = (3, 0, -4)$

$\Rightarrow x = 6, y - 1 = 0 \Rightarrow y = 1$ and $z - 2 = -8 \Rightarrow z = -6$

$\Rightarrow$ B = (x, y, z) = (6, 1, −6) **Answer: C**

33. Let $\ell_1$ and $\ell_2$ be two lines in space intersecting at the origin, (0, 0, 0). If $l_1$ and $l_2$ pass through point A (1, 1, 0) and B (0, 1, 1) respectively, then the angle between $\ell_1$ and $\ell_2$ is equal to:

A. 30° B. 45° C. 60° D. 90°

***Solution***: Let origin be O = (0, 0, 0) $\Rightarrow \overline{OA} = (1, 1, 0)$ and $\overline{OB} = (0, 1, 1)$. Let θ be angle between $\overline{OA} = \ell_1$ and $\ell_2 = \overline{OB}$ is

$$\cos\theta = \frac{\overline{OA}.\overline{OB}}{|\overline{OA}||\overline{OB}|} = \frac{(1,1,0).(0,1,1)}{\sqrt{1+1+0}\sqrt{0+1+1}} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

**Answer: C**

34. The simplified form of $\csc\left(\frac{\pi}{2} - x\right)\cos x - \sin^2 x$.

A. $2\cos x$ B. $\cos^2 x$ C. $2\sin x$ D. $\sin 2x$

***Solution***: $\csc\left(\frac{\pi}{2} - x\right)\cos x - \sin^2 x = (\sec x)\cos x - \sin^2 x$

$\Rightarrow 1 - \sin^2 x = \cos^2 x$ **Answer: B**

35. $\cos(\tan^{-1}(0.5))$ is equal to:

A. $\frac{1}{\sqrt{5}}$ B. $\frac{1}{\sqrt{3}}$ C. $\frac{2}{\sqrt{5}}$ D. $\frac{2}{\sqrt{3}}$

***Solution***: Let $\theta = \tan^{-1}(0.5) \Rightarrow \tan\theta = \frac{1}{2} \Rightarrow \cos\theta = \frac{3}{\sqrt{5}}$ and $\sin\theta = \frac{1}{\sqrt{5}}$ **Answer: C**

36. A patrol that on a sea sailed from its station 10 km to south and then changed is course and sailed $5 + \sqrt{6}$ km in the direction N60°E. Then, what is the distance the boat should travel in order to return to its station by the short route?

A. $4\sqrt{6}$ km B. 9 km C. $3\sqrt{6}$ km D. 7 km

***Solution***: $AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos 60°$

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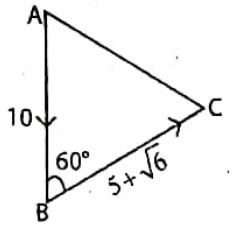


$$= 100 + \left(5+\sqrt{6}\right)^2 - 2(10)\left(5+\sqrt{6}\right)\left(\frac{1}{2}\right)$$

$$= 100 + 25 + 10\sqrt{6} + 6 - 50 - 10\sqrt{6} = 81$$

$$AC = \sqrt{81}\text{ km} = 9\text{ km}$$

**Answer: B**

37. Let $a_n = n + \cos n\pi$, $n = 1, 2, 3, \ldots$ be the $n^{th}$ term of a sequence. Then which of the following is true?

A. $a_5 = 6$ C. $a_7 + a_8 = 15$

B. $a_{10} = 9$ D. $a_7 - a_6 = 1$

***Solution***:

A. $a_5 = 5 + \cos(5\pi) = 5 + (-1) = 4$

B. $a_{10} = 9 + \cos(9\pi) = 9 + (-1) = 8$

C. $a_7 + a_8 = [7 + \cos(7\pi)] + [8 + \cos 8\pi] = 7 + (-1) + (8+1) = 15$

D. $a_7 + a_6 = [7 + \cos 7\pi] - [6 + \cos 6\pi] = 7 + (-1) - [6+1] = 1$

**Answer: C**

38. If $\{A_n\}$ is an arithmetic sequence with first term $A_1 = 5$ and $A_5 = 21$, then $\sum_{n=1}^{20} A_n$.

A. 760 B. 780 C. 860 D. 870

***Solution***: Use, $d = \frac{A_k - A_m}{k-m} \Rightarrow d = \frac{A_5 - A_1}{5-1} = \frac{21-5}{4} = 4$

Use, $S_n = \frac{n}{2}[2A_1 + (n-1)d] \Rightarrow S_{20} = \frac{20}{2}[2(5) + (20-1)4] = 860$

**Answer: C**

39. What is the sum of the series $\sum_{n=0}^{\infty} 2^n 3^{2-n}$ ?

A. 6 B. 18 C. 27 D. $\infty$

***Solution***: $\sum_{n=0}^{\infty} 2^n 3^{2-n} = \sum_{n=0}^{\infty} 2^n . 3^{-n} . 3^2 = \sum_{n=0}^{\infty} 9\left(\frac{2}{3}\right)^n$

$\Rightarrow G_1 = 9, r = \frac{2}{3} \Rightarrow S = \frac{G_1}{1-r} = \frac{9}{1-\frac{2}{3}} = 9 \times 3 = 27$

**Answer: C**



40. A business association gets a net profit of Birr 3,000 at the end of each month. Just after the fifth year, its amount was Birr 230,000. What was the starting amount of the business?

A. Birr 47,000 C. Birr 50,000

B. Birr 53,000 D. Birr 56,000

***Solution***: Monthly profit = 3000 Birr

Yearly profit = $12 \times 300 = 36{,}000$ Birr

$\Rightarrow$ Total amount in 5 year = 230,000 (given)

Let $A_0$ = initial investment

Let $A_1$ = 1st year profit = 36,000

Let $A_2$ = 2nd year profit = $A_1 + 36{,}000$

$A_3$ = 3rd year profit = $A_2 + 36{,}000 \Rightarrow A_1 + 36{,}000 + 36{,}000$

Let $A_5$ = 5th year profit = $(36000)5 = 180{,}000$

$\Rightarrow$ Total profit in 5 year = 180,000 $\Rightarrow$ Total amount = initial + total profit (at 5 year) $230{,}000 = A_0 + 180{,}000$

$\Rightarrow A_0 = 230{,}000 - 180{,}000 = 50{,}000$

**Answer: C**

41. $\lim_{n\to\infty} \frac{3\sqrt{n} - 6n + 5}{4n+1}$ equal to:

A. $-\frac{3}{2}$ B. 0 C. $\frac{3}{4}$ D. $\frac{5}{4}$

***Solution***: Divide the numerator and denominator by n.

$$\lim_{n\to\infty} \frac{\frac{3\sqrt{n}}{n} - \frac{6n}{n} + \frac{5}{n}}{\frac{4n}{n} + \frac{1}{n}} = \lim_{n\to\infty} \frac{\frac{3}{n} - 6 + \frac{5}{n}}{4 + \frac{1}{n}} = \frac{0 - 6 + 0}{4 + 0} = -\frac{6}{4} = \frac{-3}{2}$$

**Answer: A**

42. $\lim_{x\to 1^-} \frac{x^4 - 1}{|x-1|}$ equal to:

A. 4 B. 0 C. −1 D. −4

***Solution***: $|x-1| = \begin{cases} x-1 \text{ if } x \geq 1 \\ -(x-1) \text{ if } x < 1 \end{cases}$ so that $\lim_{x\to 1^-} \frac{x^4-1}{|x-1|}$

$$= \lim_{x\to 5} \frac{(x-1)(x^2+1)(x+1)}{-(x-1)} = -\left(\lim_{x\to 1^-}(x^2+1)(x+1 = (-4)\right)$$

**Answer: D**

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43. $\lim_{x\to 0}\frac{x\ \csc\ 3x}{x+1}$ is equal to:

A. $\frac{1}{3}$ B. 0 C. 1 D. 3

**Solution:** $\lim_{x\to 0}\frac{x\ \csc\ 3x}{x+1}=\lim_{x\to 0}\frac{x}{\sin\ 3x}.\lim_{x\to 0}\frac{1}{x\ +\ 1}=\frac{1}{3}$

**Answer: A**

44. $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{2x+4}$ is equal to:

A. $e$ B. $e^2$ C. $e^4$ D. $\infty$

**Solution:** $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{2x+4}=\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{2x}.\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{4}=e^2.1=e^2$

**Answer: B**

45. Let $f(x)=\begin{cases}\frac{a}{x^2+3} & \text{if } x\le 1\\ \frac{x-1}{\sqrt{x}-1} & \text{if } x>1\end{cases}$ what is the value of a if f is continuous at $x=1$.

A. 0 B. 2 C. 4 D. 8

**Solution:** Since f is continuous at $x=1$.

$\Rightarrow \lim_{x\to 1^+} f(x)=\lim_{x\to 1^-} f(x)\Rightarrow \lim_{x\to 1^+}\frac{x-1}{\sqrt{x}-1}=\lim_{x\to 1^-}\frac{a}{x^2+3}$

$\lim_{x\to 1^+}\frac{(x-1)(\sqrt{x}+1)}{x-1}=\frac{a}{4}\Rightarrow 2=\frac{a}{4}\Rightarrow a=8$ **Answer: D**

46. If $f(x)=\pi^2$ then $f'(1)$ is equal to:

A. $2\pi$ B. $e^2$ C. 1 D. 0

**Solution:** $f(x)=\pi^2$ is constant function $\Rightarrow f'(x)=0\Rightarrow f'(1)=0$

**Answer: D**

47. Let $f(x)=\frac{6x}{x+a}$, for what value of a is $f'(a)=1$.

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. 3



**Solution:**

$f'(x)=\frac{6(x+a)-6x}{(x+a)^2}\Rightarrow f'(x)=\frac{6a}{(x+a)^2}\Rightarrow f'(a)=\frac{6a}{4a^2}=1$

$\Leftrightarrow 4a=6\Rightarrow a=\frac{6}{4}=\frac{3}{2}$ **Answer: C**

48. Which one of the following is necessarily true about a function $f(x)$?

A. If $f$ is continuous at $x=a$, then f is differentiable at $x=a$.

B. If $f$ is not differentiable at $x=a$ then $\lim_{x\to a^-} f(x)\neq\lim_{x\to a^+} f(x)$.

C. If f is differentiable at x = a then $\lim_{x\to a^-} f(x)=f(a)=\lim_{x\to a^+} f(x)$.

D. If the derivative f '(a) = 0, then f attains its maximum value at $x=a$.

**Solution:**

A. False, example, $f(x)=|x-1|$, cont. at $x=1$ but not differentiable at $x=1$.

B. False

C. True, if function $f$ is differentiable at $x=a$ then f is continuous at $x=a$.

D. False, example $f(x)=x^3+5\Rightarrow f'(x)=3x^2$. $f'$ has the same sing for all $x$.

**Answer: C**

49. If $g(x)=\frac{f(x)}{x+1}+(f(x))^2, f(1)=8$ and $f'(1)=2$ then $g'(1)$ is equal to:

A. 36 B. 31 C. 25 D. 16

**Solution:** We have $f(1)=8$ and $f'(1)=2$.

$g'(x)=\frac{f'(x)(x+1)-f(x)(x+1)'}{(x+1)^2}+2f(x)f'(x)$

$\Rightarrow g'(1)=\frac{f'(1)(1+1)-f(1)(1)}{(1+1)^2}+2f(1)f'(1)$

$=\frac{(2)(2)-8}{4}+2(8)(2)=-1+32=31$

**Answer: B**

50. If $f(x)=xe^{3x}-\cos(2x)$, then $f''(0)$ is equal to:

A. 0 B. 2 C. 6 D. 10

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**Solution:**

$f'(x) = (x')\,e^{3x} + x\,(e^{3x})' + \sin(2x)\,.\,2 = e^{3x} + 3 \times e^{3x} + 2\sin(2x)$

$f''(x) = 3e^{3x} + 3e^{3x} + (3)(3x)e^{3x} + 4\cos(2x)$

$f''(0) = 3 + 3 + 0 + 4 = 10$ **Answer: D**

51. At what value (s) of x does $f(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x + 1$ have a local maximum?

A. $x = 2$ B. $x = -1$ C. $x = 5$ D. $x = 2, x = -1$

**Solution:** $f'(x) = x^2 - x - 2 = (x - 2)(x + 1)$ critical number,

$f'(x) = 0 \Rightarrow x = 2$ and $x = -1$.

$f''(x) = 2x - 1 \Rightarrow f''(2) = 4 - 1 = 3 > 0 \Rightarrow x = 2$ is local minimum value and $f''(-1) = -2 - 1 = -3 < 0 \Rightarrow$ at $x = -1$ is local maximum

**Answer: B**

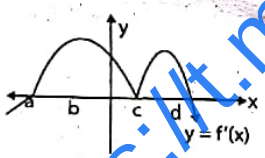
52. Look at the following graph of $f'(x)$. Which of the following is true about $f$?

A. $f$ is increasing on $[c, d]$

B. $f$ is decreasing on $[b, c]$

C. $f$ has local minimum at $d$

D. $f$ has a local extreme value at $c$




**Solution:** Remember that:

- If $f'(x) > 0$ for all $x$ in (m, n), then f is increasing on $[m, n]$.
- If $f'(x) < 0$ for all $x$ in (m, n) then f is decreasing on $[m, n]$.
- If $f'(x_o) = 0$ and $f'$ changes the sign:
  - i) from (+) to (–) then f has local max at $x_o$
  - ii) from (–) to (+) then f has local min at $x_o$

Now, from the above graph we have:

| Interval | $-\infty < x < a$ | $a < x < b$ | $b < x < c$ | $c < x < d$ | $d < x < \infty$ |
|---|---|---|---|---|---|
| sign of $f'(x)$ | – | + | + | + | – |
| Conclusion of $y = f(x)$ | decreasing | increasing | decreasing | increasing | Decreasing |

- $f$ is increasing on $[c, d]$, $[a, b]$, $f$ is decreasing on $[b, c]$
- $f$ has local minimum at $x = a$
- $f$ has local maximum at $x = d$
- no local extreme value at b and c because no sign change

**Answer: A**



53. On which of the following interval is the graph of $f(x) = \ln(x^2 + 1)$ concave upward?

A. $(-\infty, -1][1, \infty)$ B. $[0, \infty)$ C. $(-\infty, 0]$ D. $(-1, 1)$

**Solution:** If $f''(x) > 0$ for all $x$ in I, then the graph of $f$ concave upward in I. If $f''(x) < 0$ for all $x$ in I, then graph of $f$ is concave down ward in I.

Now that,

$f'(x) = \frac{2x}{x^2 + 1}$ and $f''(x) = \frac{2(x^2 + 1) - 4x^2}{(x^2 + 1)^2} = \frac{2 - 2x^2}{(x^2 + 1)^2}$

Because $f''(x) = 0$ when $x = \pm 1$, so that test $f''$ in $(-\infty, -1)$, $(-1, 1)$ and $(1, \infty)$.

| | $-\infty$ | $-1$ | 1 | $\infty$ |
|---|---|---|---|---|
| $2 - 2x^2$ | – | + | – | |
| $(x^2 + 1)^2$ | + | + | + | |
| $f''(x)$ | – | + | – | |
| conclusion of $f$ | concave down | concave upward | concave down ward | |

$\therefore f$ is concave down ward on $(-\infty, -1)$ and $(1, \infty)$.

$\therefore f$ is concave upward on $(-1, 1)$ **Answer: D**

54. Water is poured into a cylindrical tanker of radius 5 meters at a rate of 10 meter cube/min. What is the rate of change of the height of the level of water when it rises to 3 meters?

A. $\frac{3}{25\pi}$ m/min C. $\frac{1}{2\pi}$ m/min

B. $\frac{2}{5\pi}$ m/min D. $\frac{3}{5\pi}$ m/min$^{-1}$

**Solution:** $v = \pi r^2 h \Rightarrow v = \pi(5^2)h = 25\pi h$

$\Rightarrow \frac{dv}{dt} = 25\pi \frac{dh}{dt}$ ← derivate w.r.t time

$\Rightarrow 10m^3 / \min = 25\pi \frac{dh}{dt} \Rightarrow \frac{10}{25\pi} = \frac{2}{5\pi} m / \min$ **Answer: B**

55. A rectangular field of length $\ell$ and width (w) meters for $w < \ell$ has perimeter 400 meters. If a circular region of area $w^2$ is to be reserved for office purpose what should be the length of eth field so that the area of the remaining region is maximum.

A. 50m B. 100m C. 120m D. 150m

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**Solution:**

i) A (rectangle) = lw ii) A (reserve) = $w^2$

iii) P (perimeter of rectangle) = $2w + 2\ell = 400$

$\Rightarrow w + \ell = 200 \Rightarrow \ell = 200 - w$

iv) The remaining area A = A (rectangle) – A (reserve)

$\Rightarrow A(\ell, w) = \ell w - w^2 \Rightarrow A(w) = (200 - w)w - w^2$

$= 200w - w^2 - w^2 \Rightarrow A(w) = 200w - 2w^2$

$\Rightarrow A'(w) = 200 - 4w \Rightarrow A'(w) = 0 \Rightarrow 200 - 4w = 0 \Rightarrow 4w = 200$

$\Rightarrow w = 50$ and the length ($\ell$) of the field, $\ell = 200 - w$

$\Rightarrow \ell = 200 - 50 = 150$

**Answer: D**

56. The anti derivative of f (x) = tan x.

A. $\frac{1}{2}\tan^2 x$ B. $\sec^2 x$ C. $-\ln(\cos x)$ D. $\ln|\sec x|$

**Solution:** $\int \tan x dx = \int \frac{\sin x}{\cos x} dx$, Let $u = \cos x \Rightarrow du = -\sin x\, dx$

$\Rightarrow \int \frac{-du}{u} = -\ln|u| + c \Rightarrow -\ln|\cos x| + c$

$\Rightarrow \ln(\cos x)^{-1} = \ln\left(\frac{1}{\cos x}\right) = \ln|\sec x|$

**Answer: D**

57. $\int \frac{x}{x^2 + 2x + 1} dx$ equal to:

A. $\ln(x^2 + 2x + 1) + c$ C. $\ln(x^2 + 2x + 1) - \frac{1}{x+1} + c$

B. $\ln|x+1| + \frac{1}{x+1} + c$ D. $\ln(x^2 + 2x + 1) + \frac{1}{x+1} + c$

**Solution:** $\int \frac{x}{x^2 + 2x + 1} dx = \int \frac{x}{(x+1)^2} dx$, Let $u = x + 1$

$\Rightarrow x = u - 1 \Rightarrow du = dx$

$\Rightarrow \int \frac{u-1}{u^2} du = \int \frac{1}{u} du - \int \frac{du}{u^2} = \ln|u| + \frac{1}{u} + c$

$\Rightarrow \ln|x+1| + \frac{1}{x+1} + c$

**Answer: B**



58. $\int \frac{e^{2x} - 4x}{xe^{2x}} dx$ is equal to _____

A. $\ln|x| + 2e^{-2x} + c$ C. $\ln|x| + 4e^{-2x} + c$

B. $\ln|x| - 4e^{2x} + c$ D. $-\frac{1}{x^2} + 2e^{-2x} + c$

**Solution:** $\int \frac{e^{2x} - 4x}{xe^{2x}} dx = \int \left(\frac{e^{2x}}{xe^{2x}} - \frac{4x}{xe^{2x}}\right) dx$

$= \int \frac{dx}{x} - \int \frac{4}{e^{2x}} dx = \ln|x| + 2e^{-2x} + c$

Explanation: $\boxed{\int e^{mx} dx = \frac{e^{mx}}{m}}$

**Answer: A**

59. $\int_0^1 \frac{(x-1)^2}{x+1}$ equal to:

A. $-\frac{3}{2} + \ln 6$ C. $-\frac{5}{2} + 4\ln 2$

B. $-\frac{5}{2} + \ln 15$ D. $\frac{5}{2} + \ln 16$

**Solution:** $\int_0^1 \frac{(x-1)^2}{x+1} dx = \int_0^1 \frac{x^2 - 2x + 1}{x+1} dx = \int_0^1 \left(x - 3 + \frac{4}{x+1}\right) dx$

$\Rightarrow \frac{x^2}{2} - 3x + 4\ln|x+1| \Big|_0^1 = \frac{1}{2} - 3 + 4\ln 2 = -\frac{5}{2} + 4\ln 2$

**Answer: C**

60. If $f(x) = \begin{cases} \sin x, & x \le 0 \\ 3x\sqrt{x^2+1} & x > 0 \end{cases}$ then $\int_{-\frac{\pi}{2}}^{\sqrt{8}} f(x) dx$ equal:

A. 17 B. 25 C. 27 D. $3\sqrt{8}$

**Solution:** $\int_{-\frac{\pi}{2}}^{\sqrt{8}} f(x) dx = \int_{-\frac{\pi}{2}}^{0} f(x) dx + \int_0^{\sqrt{8}} f(x) dx$

$= \int_{-\frac{\pi}{2}}^{0} \sin x\, dx + \int_0^{\sqrt{8}} 3x\sqrt{x^2+1}\, dx$

Let $u = \sqrt{x^2+1} \Rightarrow u^2 = x^2 + 1 \Rightarrow 2udu = 2x\, dx \Rightarrow udu = x\, dx$ when $x = \sqrt{8}$ then $u = \sqrt{8+1} = 3$, $x = 0$, then $u = \sqrt{0+1} = 1$

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$$\int_{-\frac{\pi}{2}}^{0}\sin x\,dx+\int_{1}^{3}3u^2du=-\cos x\Big|_{-\frac{\pi}{2}}^{0}+3\frac{u^3}{3}\Big|_{1}^{3}$$

$=-1-0+3^3-1^3=25$ **Answer: B**

61. The volume of the solid which is generated when the region bounded by $y=\sqrt{x+1}$ and x-axis from $x=0$ to $x=2$ is rotated about x-axis is equal to:

A. $4\pi$ B. $3\pi$ C. $\frac{4}{3}\pi$ D. $\frac{3}{4}\pi$

***Solution:*** Volume $=\int_a^b\pi[f(x)]^2dx=\int_0^2\pi\left(\sqrt{x+1}\right)^2dx$

$$\Rightarrow\pi\int_0^2(x+1)dx=\pi\left[\frac{x^2}{2}+x\right]_0^2=\pi\left[\frac{4}{2}+2\right]=4\pi$$

**Answer: A**

62. If a sphere with center C (0, 1, 1) intersects the z-axis at P (0, 0, 3) then the radius of the sphere is equal to:

A. 5 B. 3 C. $\sqrt{3}$ D. $\sqrt{5}$

***Solution:*** Equation of sphere:

$\Rightarrow(x-0)^2+(y-1)^2+(z-1)^2=r^2\Rightarrow(0-0)^2+(0-1)^2+(3-1)^2=r^2$

$\Rightarrow r^2=5\Rightarrow r=\sqrt{5}$ **Answer: D**

63. Suppose A and B are the end points of a diameter of the sphere with equation $x^2+y^2+(z+2)^2=1$. If A $(1, 0, -2)$ then B is equal to:

A. $(0, 1, -2)$ B. $(0, 0, -1)$ C. $(-1, 0, -2)$ D. $(0, -1, -2)$

***Solution:*** Let center of sphere P $(0, 0, -2)$. Let B $= (x, y, z)$ and P $(0, 0, -2)$ is mid pt. of AB.

$$\Rightarrow\left(\frac{x+1}{2},\frac{y+0}{2},\frac{z-0}{2}\right)=(0,0,-2)\Rightarrow x+1=\Rightarrow x=-1,y=0,$$

$$\frac{z-2}{2}=-2\Rightarrow z-2=-4\Rightarrow z=-4+2=-2$$

$\therefore B(x, y, z)=(-1, 0, -2)$

**Answer: C**



1. If $f(x)=\frac{1}{e^x+1}$, then which of the following is equal to $f^{-1}(x)$

A. $\ln(1-x)-\ln(x)$ C. $\ln\left(\frac{1}{x+1}\right)$

B. $e^{-x}$ D. $\frac{1}{x^{-x}+1}$

***Solution:***

Remember that, to find $f^{-1}(x)$, first interchange $x\to y$ and $y\to x$.

Then $y=\frac{1}{e^x+1}\Rightarrow x=\frac{1}{e^y+1}\Rightarrow xe^y+x=1$

$$\Rightarrow xe^y=1-x\Rightarrow e^y=\frac{1-x}{x}$$

$$\Rightarrow y=\ln\left(\frac{1-x}{x}\right)\Rightarrow f^{-1}(x)=\ln(1-x)-\ln x.$$ **Answer: A**

2. Which one of the following function has no vertical asymptote?

A. $f(x)=\ln(x+1)$ C. $f(x=\frac{x^2-9}{x-3}$

B. $f(x)=\frac{x^2+1}{x^3+8}$ D. $f(x)=\frac{x-1}{x^3-x}$

***Solution:***

A. $x+1=0\Rightarrow x=-1\leftarrow$ vertical asymptote of $\ln(x+1)$

B. $x^3+8=0\Rightarrow x=-2\leftarrow$ Vertical asymptote.

C. $\frac{x^2-9}{x-3}=\frac{(x-3)(x+3)}{x-3}=x+3,x\neq3$ No vertical asymptote

D. $\frac{x-1}{x(x-1)}=\frac{1}{x},x\neq0,1$ thus $x=0\leftarrow$ Vertical asymptote.

**Answer: C**

3. The value of $|x|+2x$ if $x<0$

A. $-3x$ B. $3x$ C. $-x$ D. $x$

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**Solution:** $x<0 \Rightarrow |x| = -x$, thus

$|x|+2x = -x+2x = -x+2x = x$ **Answer: D**

4. Given $f(x) = \ln(x-1)$ and $g(x) = \sqrt{1-2x}$, then the domain of fog is equal to _____

A. $\{x \in \mathbb{R} : x > 1\}$ C. $\{x \in \mathbb{R} : x < 0\}$

B. $\left\{x \in \mathbb{R} : x \le \frac{1}{2}\right\}$ D. $\left\{x \in \mathbb{R} : x > \frac{1}{2}\right\}$.

**Solution:** Remember that, to find the domain of (fog)(x). A number x is in the domain of (fog)(x) is and only if both g(x) and $f(g(x))$ are defined. Hence, $g(x) = \sqrt{1-2x}$ is defined on

$\{x : 1-2x \ge 0\} = \left\{x : x \le \frac{1}{2}\right\}$

$f(g(x)) = \ln(g(x)-1) = \ln\left(\sqrt{1-2x}-1\right)$.

is defined $\sqrt{1-2x}-1 > 0 = \sqrt{1-2x} > 1 \Rightarrow 1-2x > 1 \Rightarrow x < 0$

Therefore both $g(x)$ and $f(g(x))$ is defined on $x < 0$,

$\Rightarrow$ Domain f(g(x)) $\{x : x < 0\}$. **Answer: C**

- **Exercise left for you**: what is the domain (gof)?

5. The solution set of $\frac{2}{x} - \frac{x-2}{x^2-2x} = 1 - \frac{2x-2}{3x-2}$?

A. $\{1,-2\}$ B. $\{1,2\}$ C. $\{-1\}$ D. $\{1\}$

**Solution:** The expression is the same as

$\left(\frac{2}{x}\right)\left(\frac{x-2}{x-2}\right) - \frac{x-2}{x^2-2x} = \frac{3x-2-2x+2}{3x-2}; x \ne 0, 2, \frac{2}{3}$

$\Leftrightarrow \frac{x-2}{x(x-2)} = \frac{x}{3x-2}$

$\Rightarrow \frac{1}{x} = \frac{x}{3x-2} \Leftrightarrow 3x-2 = x^2 \Rightarrow x^2-3x+2 = 0$

$\Rightarrow (x-2)(x-1) = 0 \Rightarrow x = 2$, x = 1, but $x \ne 2$

Therefore: S.S = {1}

**Answer: D**



6. If a line with angle of inclination of $\frac{3\pi}{4}$ passes through (0,1), which one of the following is the equation of the line

A. $y = -x+1$ B. $y = x+1$ C. $y = -x-1$ D. $y = x-1$

**Solution:** We know that Slope $= \tan\left(\frac{3\pi}{4}\right) = -1$

$\Rightarrow$ equation of line $l$: $y-1 = -1(x-0) \Rightarrow y = y = -x+1$

7. If $x^2 - 6x + y^2 + k = 0$ is equation of circle with radius 2, then what is the value of $k$?

A. 13 B. 5 C. 4 D. -4

**Solution:** Use completing square method

$\Rightarrow x^2 - 6x + 9 + y^2 = -k + 9$ ← Adding 9 to both sides

$\Rightarrow (x-3)^2 + (y-0)^2 = r^2 = -k+9$, but $r = 2 \Rightarrow r^2 = -k+9$

$\Leftrightarrow 4 = -k+9 \Rightarrow k = 5$. **Answer: B**

8. A parabola with focus at (3,-1) has directrix, $y = 3$, then the equation of parabola

A. $(x-3)^2 = -4(y+1)$ C. $(x-3)^2 = 4(y+1)$

B. $(x-3)^2 = -8(y-1)$ D. $(x-3)^2 = 8(y-1)$

**Solution:**

Since the focus below the directrix, $y=3$ then the parabola open down ward. Equation of parabola is given by $(x-h)^2 = -4p(y-k)$

- Distance between focus and directrix is $|2p| \Rightarrow 2p = 3-(-1) = 4 \Rightarrow p = 2$
- Vertex $(h, k) = (3,-1+2) = (3,1) \Rightarrow (x-3)^2 = -4(2)(y-1)$

$\Rightarrow (x-3)^2 = -8(y-1)$

**Answer: B**

9. **Exercise left for you**: Find the directrix, focus, of question number 8 for choice A, C and D.

10. A satellite moves along a hyperbolic curve whose horizontal transverse axis is 24km and an asymptote, $y = \frac{5}{12}x + 2$ then, what is the eccentricity of the hyperparabola?

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A. $\frac{5}{13}$ B. $\frac{12}{13}$ C. $\frac{13}{12}$ D. $\frac{15}{3}$

***Solution:***

The distance between vertices of hyperbola (transverse axis) is $2a=24 \Rightarrow a=12$.

The equation with center (h, k) is given by: $\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$

- Asymptote given by $y-k=\pm\frac{b}{a}(x-h)$ with a=12. But we have: $y=\frac{5}{12}x+2$, so that $b=5, a=12$
- Eccentricity (e)

$$y(e)=\frac{c}{a}=\frac{\sqrt{a^2+b^2}}{a}=\frac{\sqrt{144+25}}{12}=\frac{13}{12}$$

$$y(e)=\frac{13}{12}$$

**Answer: C**

11. ***Exercise left for you (Tip).*** From question number 10, what is the distance between foci?

12. Let p, q and r be proposition such that $p \Rightarrow (r \vee \neg q)$ is false. Then, which one of the following proposition is true?

A. $p \Rightarrow r$ B. $\neg p \Leftrightarrow q$

C. $\neg r \Rightarrow q$ D. $q \Leftrightarrow r$

***Solution:*** $p \Rightarrow (r \vee \neg q)$ is false if and only if $r \vee \neg q \equiv F$ and $P \equiv T$, so that $(r \vee \neg q)$ is false if $r \equiv F$ and $\neg q \equiv F$, imply, $q \equiv T$, thus,

A. $p \Rightarrow r \equiv r \equiv T \Rightarrow F \equiv F$

B. $\neg p \Leftrightarrow q \equiv F \Leftrightarrow T \equiv F$

C. $\neg r \Rightarrow q \equiv T \Rightarrow T \equiv T$

D. $q \Leftrightarrow r \equiv T \Leftrightarrow F \equiv F$

**Answer: B**



13. Consider the following argument:
If he does not love her, she will not marry him. He loves her. Therefore she will marry him."
If '$P \equiv$ he love her' and '$q \equiv$ she will marry him' Which one of the following is the correct representation of the argument and its validity?

A. $p \Rightarrow \neg q, p \square q$; Valid argument

B. $p \Rightarrow q, p \square q$; Valid argument

C. $\neg p \Rightarrow \neg q, p \square q$; invalid argument

D. $p \Rightarrow q, p \square q$; Invalid argument

***Solution:*** $\neg p \Rightarrow \neg q, p \vdash q$

1) $p \equiv True \leftarrow$ premise

2) $\neg p \Rightarrow \neg q \equiv T \leftarrow$ premise

3) $\neg p \equiv F$ and $\neg q \equiv F$ then $q \equiv T$ and if $\neg p \equiv T$, then $q \equiv F$ thus, $\neg p \Rightarrow \neg p, p \vdash q$ is invalid,

**Answer: C**

14. For real number $x$ and $y$, which one of the following statements is true

A. $(\forall x)(\exists y)(x^2+y+1=0)$ C. $(\exists y)(\forall x)(x^2+y+1=0)$

B. $(\exists x)(\forall y)(x^2+y+1=0)$ D. $(\forall y)(\exists x)(x^2+y+1=0)$

***Solution:***

A. For every $x$, we can find $y$, that satisfies $x^2+y+1=0$ is true, $\Rightarrow y=-x^2-1$

B. There is fixed x which stands for all y that satisfies $x^2+y+1=0$ is false.

C. False

D. For every $y$, we can find x that satisfies $x^2+y+1=0$ is false.

Let $y=0 \Rightarrow -x^2-1=0 \Rightarrow -x^2=1 \Rightarrow x^2=-1$ is false.

**Answer: A**

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15. Among students who took a quiz, 15 students scored 6, 20 student scored 7, 10 students scored 8, 5 student scored 10. What is the average score of the student?

A. 7.8 B. 7.5 C. 7.2 D. 7.0

***Solution:***

$$\text{Average score} = \frac{15\times6+20\times7+10\times8+5\times10}{15+20+10+5}$$

$$= \frac{90+140+80+50}{50} = \frac{360}{50} = 7.2,$$ **Answer: C**

16. The market that students scored in an examination is grouped in class intervals shown in the following table.

| **lass interval (mark)** | 55 - 64 | 65 - 74 | 75 - 84 | 85 – 94 | 95 – 100 |
|---|---|---|---|---|---|
| **mber of student** | 8 | 12 | 20 | 6 | 4 |

What is the median of the mark?

A. 25.0 B. 75.5 C. 77.0 D. 79.5

***Solution:***

*Remember that the median of data is given by*

$$M_d = B_L + \left(\frac{\frac{n}{2} - Cf_b}{f_c}\right)i$$

- Median class is $\left(\frac{n}{2}\right)^{th} = \left(\frac{50}{2}\right)^{th} = (25)^{th}$ item, it lies on $3^{rd}$ class (75 - 84).
- Lower boundary of median class denoted by $B_L = 74.5$
- Cumulative frequency just before the median class denoted by $Cf_b = 8+12=20$
- Frequency of median class, $f_c = 20$
- Class interval (width), $i = 65\text{-}55 = 10$
- Total frequency, $n=50$

Therefore $Md = 74.5 + \left(\frac{\frac{50}{2} - 20}{20}\right)10 = 77.0$



**Answer: C**

- **Exercise left for you (Tip).** What is the mean, the mode, of the mark?

17. How many four digit even numbers can be formed from 1,2,3,4 and 5 if the number start with 3?

A. 40 B. 50 C. 100 D. 120

***Solution:***

Let the digit be ABCD but A = 3, is 1 way
B = we can fill either of student = 5 way
C = we can fill either of 5 digit = 5 way
D = we can fill either 2 or 4 = 2 way.
Thus, $1\times5\times5\times2 = 50$ way. **Answer: B**

- **Exercise left for you (Tip).** How many 5 digit even numbers can be formed from 1,2,3,4 and5 if the number starts with 3 and 1?

18. A committee consisting of 3 students is to be selected from 10 candidates among which 4 are girls. What is the probability that at least one girl is selected.

A. $\frac{5}{6}$ B. $\frac{2}{3}$ C. $\frac{1}{3}$ D. $\frac{1}{6}$

***Solution:*** Remember that P(At least one girl = 1 – P (All boys)

$$\text{P (All boys)} = \frac{c(6,3)c(4,0)}{c(10,3)} = \frac{20}{120} = \frac{1}{6}$$

$$\Rightarrow \text{P(At least one girl)} = 1 - \frac{1}{6} = \frac{5}{6}$$ **Answer: A**

19. A box contains 5 white, 6 red and 4 black balls of identical size. If 3 balls are randomly taken out of the box one after the other, what is the probability that the first ball is white and both the second and third balls are red?

A. $\frac{2}{15}$ B. $\frac{3}{15}$ C. $\frac{4}{75}$ D. $\frac{5}{91}$

***Solution:***

Let us assume that drawing is without replacement.
Let W = White, let Red = $R_2$, $R_3$ ($2^{nd}$, $3^{rd}$ draw) $\Rightarrow$ P(W $R_2$ $R_3$)

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$$= P(W).P(R_2/W).P(R_3/WR_2) = \frac{5}{15}.\frac{6}{14}.\frac{5}{13} = \frac{5}{91}$$

**Answer: D**

- Find the probability of getting the ball:
  A. All 3 are White
  B. The ball drawn in the order (Red, White, Black)

20. Suppose $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$. If X is a $2\times2$ matrix such that $AX - A^t = 2A$. then what is the value of X

A. $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ B. $\begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$ C. $\begin{pmatrix} 3 & 6 \\ 6 & 9 \end{pmatrix}$ D. $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

***Solution:*** Let $X=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $A^t=\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ then, $AX - A^t = 2A$

$$\Rightarrow \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 4 & 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a+2c & b+2d \\ 2a+3c & 2b+3d \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 6 & 9 \end{pmatrix}$$

$$\Rightarrow \begin{cases} a+2c=3 \\ 2a+3c=6 \end{cases} \text{ and } \begin{cases} b+2d=6 \\ 2b+3d=9 \end{cases}$$

$$\Rightarrow \underset{-c=0\Rightarrow a=3}{+\begin{cases} -2a-4c=-6 \\ 2a+3c=6 \end{cases}} \qquad \underset{-d=-3\Rightarrow d=3\Rightarrow b=0}{+\begin{cases} -2b-4d=-12 \\ 2b+3d=9 \end{cases}}$$

Thus, $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

**Answer: D**

21. Suppose that A and B are $3\times3$ matrices I is the identity matrix of order 3 such that AB=2I. If det B = |B|=6. What is the det $(A^t)$?

A. $\frac{1}{3}$ B. $\frac{4}{3}$ C. 12 D. 48



***Solution:*** Recall that $|AB|=|2I| \Rightarrow |A||B| = 8(1)$ since $|2I| = 2^3(1) = 8$, and 3 is order

$\Rightarrow |A|(6) = 8 \Rightarrow |A| = \frac{8}{6} = \frac{4}{3} \leftarrow |A^t| = |A|$

**Answer: B**

22. For what value of b does the parabola $P(x) = ax^2+x+b$, pass through the points (-1, 5) and (2, -1)

A. 9 B. 3. C. -3 D. -15

***Solution:***

$$\left.\begin{aligned} P(-1) = a-1+1+b = 5 \Rightarrow a+b=6 \\ P(2) = 4a+2+b = -1 \Rightarrow 4a+b=-3 \end{aligned}\right\}$$

$-3a = 9 \Rightarrow a = -3$ Thus, -3+b=6 $\Rightarrow$ b=9

**Answer: A**

23. Consider the system $\begin{cases} ax+y+z=1 \\ x+2y+4z=0 \\ 5x-y+z=0 \end{cases}$

If the determinant of the coefficient matrix is 2 then what is the solution of the system of equation:

A. $\left(3\alpha, \frac{19\alpha}{2}, \frac{-11\alpha}{2}\right)$ C. $\frac{3}{\alpha}, \frac{-19}{2}, \frac{11}{2}$

B. $\left(3, \frac{19}{2}, \frac{-11}{2}\right)$ D. $\left(\frac{3}{2}, \frac{19}{2}, \frac{-9}{2}\right)$

***Solution:*** D=2, remember, $|A^t| = |A|$

$$x = \frac{D_x}{D} = \frac{\begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & -1 & 1 \end{vmatrix}}{2} = \frac{\begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 4 & 1 \end{vmatrix}}{2} = \frac{\begin{vmatrix} 2 & -1 \\ 4 & 1 \end{vmatrix}}{2} = \frac{2+4}{2} = 3$$

$$Y = \frac{D_y}{D} = \frac{\begin{vmatrix} \alpha & 1 & 1 \\ 1 & 0 & 4 \\ 5 & 0 & 1 \end{vmatrix}}{2} = -\frac{\begin{vmatrix} 1 & \alpha & 1 \\ 0 & 1 & 4 \\ 0 & 5 & 1 \end{vmatrix}}{2} \leftarrow \text{Interchanging row } (R_1 \leftrightarrow R_2)$$

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$$y=\frac{D_y}{D}=-\frac{\begin{vmatrix}1&0&0\\ \alpha&1&5\\ 1&4&1\end{vmatrix}}{2}=\frac{\begin{vmatrix}1&5\\4&1\end{vmatrix}}{2}=-\frac{(1-20)}{2}=\frac{19}{2}$$

$$P_z=\begin{vmatrix}\alpha&1&1\\1&2&0\\5&-1&1\end{vmatrix}=1(-1-10)=-11$$

$$z=\frac{-11}{2}$$

$$\therefore S.S=\left\{\left(3,\frac{19}{2},\frac{-11}{2}\right)\right\}$$

**Answer: B**

24. Which one of the following is the simplest form of $|3+4i|-\frac{25i}{3+4i}$?

A. $5-5i$ B. $5+5i$ C. $1+3i$ D. $1-3i$

***Solution:*** We know that

$$|3+4i|=\sqrt{3^2+4^2}=5\Leftrightarrow 5-\frac{25i}{3+4i}\cdot\frac{3-4i}{3-4i}=5-\frac{75i-100}{25}$$

$$=5-4-3i=1-3i$$

**Answer: D**

25. In the set of complex numbers, which one of the following is the solution set of $z^3-iz^2+2z=0$?

A. {0} B. {0, -i} C) {0, -i, 2i} (D) {0, i, -2i}

***Solution:***

$z^3-iz^2+2z=0\Rightarrow z(z^2-iz+2)=0$

$\Rightarrow z\ (z^2-iz+2)=0\Rightarrow z=0$ or $z^2-iz+2=0$

$\Rightarrow z=i\pm\sqrt{(i)^2-4(2)}$

$\Rightarrow z=\frac{i+3i}{2}\Rightarrow z=4i$ or $z=2i^2$

$\therefore$ S.S = {0, -i, 2i}

**Answer: D**



26. If $z=\cos\left(\frac{\pi}{10}\right)+i\sin\left(\frac{\pi}{10}\right)$, then what is the value of $z^5$?

A. $\frac{\pi}{2}+\frac{\pi}{2}i$ B. $\frac{1}{2}+\frac{1}{2}i$ C. i D. 1 + i

***Solution:***

$$z^5=\left[\cos\left(\frac{\pi}{10}\right)+i\sin\left(\frac{5\pi}{10}\right)\right]=\cos\frac{\pi}{2}+i\sin\left(\frac{\pi}{2}\right)=0+i=i$$

**Answer: C**

27. If $\overline{U}=(-3,\ x)$ and $\overline{V}=(x,\ y-2)$ are vectors what is the value of y so that $\overline{U}+\overline{V}=3\overline{U}-\frac{1}{2}\overline{V}$

A. $\frac{2}{3}$ B. $\frac{-10}{3}$ C. $-4$ D. $\frac{-22}{3}$

***Solution:*** $\overline{U}+\overline{V}=(-3+x,x+y-2)=(-9,3x)-\left(\frac{1}{2}x,\frac{y-2}{2}\right)$

$$=\left(-3+x,x+y-2\right)=\left(-\frac{1}{2}x-9,3x-\frac{y+2}{2}\right)$$

$$\Rightarrow -3+x=-\frac{1}{2}x-9 \text{ and } x+y-2=3x-\frac{y+2}{2}$$

$\Rightarrow -12=3x\Rightarrow x=-4$ and $x+y-2=3x-\frac{y}{2}+1$, since

$x=-4\Rightarrow -4+y-2=-12-\frac{y}{2}+1$

$$\Rightarrow -6+y=-11-\frac{y}{2}\Rightarrow -6+11=-\frac{y}{2}-y$$

$$\Rightarrow -\frac{3y}{2}=5\Rightarrow y=\frac{-10}{3}$$

**Answer: B**

28. Suppose $\overline{A}=2j-k$ and $\overline{B}=5\overline{i}+15\overline{k}$, where $\overline{i},\overline{j}$ and $\overline{k}$ are the standard unit vectors in the directions of positive x, y and z axis respectively. Which one of the following is the unit vector in the direction of $\overline{A}+\frac{1}{5}\overline{B}$?

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A. $\frac{3}{5}\vec{i}+\frac{4}{5}\vec{k}$ C. $\frac{4}{5}\vec{j}-\frac{3}{5}\vec{k}$

B. $\frac{1}{3}\vec{i}+\frac{2}{3}\vec{j}+\frac{2}{3}\vec{k}$ D. $\frac{2}{3}\vec{i}-\frac{1}{3}\vec{j}+\frac{2}{3}\vec{k}$

***Solution:*** Unit vector in the direction of $\vec{A}+\frac{1}{5}\vec{B}$ is given by

$$\frac{\vec{A}+\frac{1}{5}\vec{B}}{\left|\vec{A}+\frac{1}{5}\vec{B}\right|}=\frac{2\vec{j}-\vec{k}+\frac{1}{5}\left(5\vec{i}+15\vec{k}\right)}{\left|2\vec{j}+\vec{i}+2\vec{k}\right|}$$

$$=\frac{i+2j+2k}{\sqrt{1+4+4}}=\frac{\vec{i}}{3}+\frac{2}{3}j+\frac{2}{3}\vec{k}$$ **Answer: B**

29. A line given by a vector equation $r(t)= (0,3) + t(1,1)$ is tangent to circle at point (0,3). If the radius of the circle is $\sqrt{2}$, which one of the following is the centre of circle?

A. (1,4) B. (1,-4) C. (-1,2) D. (1,2)

***Solution:*** The position vector of (0,3) and centre $(x, y)$ is $(x,y-3)$, since it is perpendicular to the direction vector t(1,1)

$\Rightarrow (x,y-3).(1,1)=0 \Rightarrow x+y-3=0 \Rightarrow x+y=3$, hence

A. $(1+4)=5$ B. $1+(-4)=-3$ D. $1+2=3$

**Answer: D**

30. Let $\vec{a}=2\vec{i}+(x-1)\vec{j}+\vec{k}$ and $\vec{c}=\vec{i}-\vec{j}+y\vec{k}$ be vectors.

If $\vec{a}.\vec{c}=0$ and $\|\vec{a}\|=3$ then the possible value of $y$?

A. -4 B. -1 C. 3 D. 4

***Solution:***

$\vec{a}.\vec{c}=0 \Rightarrow (2i+(x-1)j+k)(i-j+yk)=0 \Rightarrow 2-1(x-1)+y=0 \Rightarrow 2-x+1+y=0 \Rightarrow y-x=-3$,

However

$\|a\|=3 \Rightarrow \|a\|=\sqrt{4+(x-1)^2+1}=3 \Rightarrow 4+(x-1)^2+1=9 \Rightarrow (x-1)^2=9-5=4 \Rightarrow x-1=\pm\sqrt{4}=\pm 2 \Rightarrow x=3$ or $x=-1$ thus from $y-x=-3$ let $x=3 \Rightarrow y-3=-3 \Rightarrow y=0$ and Let $x=-1$ then $y+1=-3 \Rightarrow y=-3-1=-4$

**Answer: A**



31. If $\vec{A}$ is perpendicular to $\vec{B}$, what is the cosine of the angle between $\vec{A}$ and $\vec{A}-\vec{B}$

A. $\frac{\|\vec{A}-\vec{B}\|}{\|\vec{A}\|}$ C. $\frac{\|\vec{A}-\vec{B}\|}{\|\vec{B}\|}$

B. $\frac{\|\vec{A}\|}{\|\vec{A}-\vec{B}\|}$ D. $\frac{\|\vec{B}\|}{\|\vec{A}-\vec{B}\|}$

***Solution:***

$\vec{A}.(\vec{A}-\vec{B})=|\vec{A}||\vec{A}-\vec{B}|\cos\theta \Rightarrow \vec{A}.\vec{A}-\vec{A}.\vec{B}=|\vec{A}||\vec{A}-\vec{B}|\cos\theta$

$$\Rightarrow \cos\theta=\frac{\vec{A}.\vec{A}}{|\vec{A}||\vec{A}-\vec{B}|}=\frac{|\vec{A}|^2}{|\vec{A}||\vec{A}-\vec{B}|}=\frac{|A|}{|\vec{A}-\vec{B}|}$$

Note, If $\vec{A}\perp\vec{B}$ then $\vec{A}.\vec{B}=0$ **Answer: B**

32. Which one of the following is necessarily true?

A. If $|\vec{A}|=|\vec{B}|$, then $\vec{A}=\vec{B}$

B. If $|\vec{KA}|=K|\vec{A}|$, for any real number K

C. If $\vec{A}$ is parallel to $\vec{B}$, then $\vec{A}.\vec{B}=0$

D. If $\vec{U}$ is a unit vector in the direction of $\vec{A}$, then $\vec{A}.\vec{U}=|\vec{A}|$

***Solution:***

A. False, example, let $\vec{A}=3i+4j$ and $\vec{B}=0i+5j$

B. False, example: $K=-3<0$

C. False, $\vec{A}.\vec{B}=\pm|A||B|$

D. True **Answer: D**

33. Suppose P(1,2,1) and Q(1,0,2) are points in space and $\vec{A}=\vec{PQ}$. If $\vec{B}$ is parallel to $\vec{PQ}$ and $\vec{A}.\vec{B}=-10$, then which

A. $\vec{A}$ and $\vec{B}$ has the same direction. C. $|\vec{B}|=\frac{1}{10}|\vec{A}|$

B. $|\vec{B}|=10|\vec{A}|$ D. $|\vec{B}|=2|\vec{A}|$

***Solution:***

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$\vec{A} = \overrightarrow{PQ} = (1, 0, 2) - (1, 2, 1) = (0, -2, 1)$

$\vec{A} = \overrightarrow{PQ} \doteq \sqrt{0^2 + (-2)^2 + 1^2} = \sqrt{5}$

- If $\vec{B} \parallel \overrightarrow{PQ}$, then $\vec{B} = k\ \overrightarrow{PQ}$, $k \in \mathbb{R}$

  $\Rightarrow \vec{B} = k\ \vec{A} \Leftrightarrow |\vec{B}| = |k|\,|\vec{A}| = \sqrt{5}\,|k|$

- If $\vec{A}.\vec{B} = |\vec{A}|\ |\vec{B}| \cos\theta = -10$

  $\Leftrightarrow (\sqrt{5})(\sqrt{5})|k|\cos\theta = -10$

  $\Rightarrow 5|k|\cos\theta = -10 \Leftrightarrow |k| = 2$ and $\cos\theta = -1$

- Because $\vec{A}.\vec{B} = -10 < 0$ and $\vec{A} \parallel \vec{B}$, so that $\theta = \pi$

  $\Rightarrow \cos\theta = -1$, therefore $\vec{B} = 2\,|A|$. **Answer: D**

34. What is the image of the ellipse whose equation is $2(x+2)^2 + (y-1)^2 = 2$ under a translation that takes (2,1) to (4,0) followed by a rotation $90^0$.

A. $x^2 + 2y^2 = 2$ C. $2(x-4)^2 + y^2 = 2$

B. $2x^2 + y^2 = 2$ D. $(x-4)^2 + 2y^2 = 2$

***Solution:***

$2(x+2)^2 + (y-1)^2 = 2 \Rightarrow (x+2)^2 + \frac{(y-1)^2}{2} = 1$ ($y$ – ellipse)

- Translation of vector: $\vec{V} = (4-2, 0-1) = (2, -1)^2$
- Centre of ellipse (-2,1), (0,0), $\xrightarrow{\vec{V}=(2,-1)}$

The new ellipse becomes: $x^2 + \frac{y^2}{2} = 1 \Leftrightarrow 2x^2 + y^2 = 2$

- When the ellipse $2x^2 + y^2 = 1$ is rotated by $\theta = 90^0$

  $\Rightarrow$ $y$-ellipse $\xrightarrow{\theta = 90^0}$ x-ellipse

  $2x^2 + y^2 = 2 \xrightarrow{\theta = 90^0} x^2 + 2y^2 = 2$ **Answer: A**

35. If $f(x) = 2 - \frac{1}{2}\sin\left(\frac{\pi}{2} - x\right)$, then which one of the following is the amplitude and period of $f$, respectively?

A. $\frac{1}{2}$ and 4 B. $-\frac{1}{2}$ and 4 C. 2and $\Pi$ D. $\frac{1}{2}$ and $\Pi$



***Solution:*** The amplitude $\left|-\frac{1}{2}\right| = \frac{1}{2}$,

- Period is $\frac{2\pi}{K} = \frac{2\pi}{\frac{\pi}{2}} = 2\pi.\frac{2}{\pi} = 4$ **Answer: A**

36. **Exercise left for you (Tip).** Find the range, the maximum and minimum value.

37. Which one of the following is equal to $\sec\left(\frac{\pi}{2} - x\right)\sin^3 x + \cos 2x$?

A. 2cosx B. 2sinx C. $\cos^2 x$ D. $\sin^2 x$

***Solution:*** Recall that $\sec\left(\frac{\pi}{2} - x\right) = \csc x = \frac{1}{\sin x}$

$\Rightarrow \frac{\sin^3 x}{\sin x} + \cos^2 x - \sin^2 x = \cos^2 x$

**Answer: C**

38. The solution set of $\cos^2 x + \frac{1}{2}\sin 2x = 1$ in the interval $[0, 2\pi]$

A. $\left\{0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}\right\}$ C. $\{0, \pi, \}$

B. $\left\{0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi\right\}$ D. $\left\{0, \frac{\pi}{4}, \pi\right\}$

***Solution:*** We know that

$\sin 2x = \sin(x + x) = \sin x \cos x$ and $\cos^2 x = 1 - \sin^2 x$. Now

$\Leftrightarrow 1 - \sin^2 x + \frac{1}{2}(2)\sin x \cos x = 1 \Rightarrow \sin x \cos x - \sin^2 x = 0$

$\Rightarrow \sin x(\cos x - \sin x) = 0 \Leftrightarrow \sin x = 0$ or

$\cos x - \sin x = 0 \Rightarrow \sin x = 0 \Rightarrow x = 0, \pi$ or

$\cos x = \sin x \Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$

$s.s = \left\{0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}\right\}$

**Answer: A**

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39. What is $\cot(\arcsin x)$ if $0 < x < 1$

A. $\frac{x}{\sqrt{1-x^2}}$ C. $\sqrt{1-x^2}$

B. $\frac{\sqrt{1-x^2}}{x}$ D. $\frac{1}{\sqrt{1-x^2}}$

***Solution:*** $\cot(\sin^{-1}x)$ let $\sin^{-1}x=\theta \Rightarrow \sin\theta = x$ we know that:

$$\cos^2\theta+\sin^2\theta=1 \Rightarrow \frac{\cos^2}{\sin^2\theta}+1=\frac{1}{\sin^2\theta}$$

$$\cot^2\theta+1=\frac{1}{x^2} \Rightarrow \cot^2\theta=\frac{1}{x^2}-1=\frac{1-x^2}{x^2}$$

$$\Rightarrow \cot\theta=\sqrt{\frac{1-x^2}{x^2}}=\frac{\sqrt{1-x^2}}{x}$$

**Answer: B**

40. If $\{a_n\}$ is a sequence such that $a_1=2$ and $a_n+1=a_n+4$ for all $n\geq 1$ then $\sum_{n=1}^{35}a_n$ is equal to:

A. 2460 B. 2458 C. 2450 D. 2442

***Solution:***

$\Rightarrow a_2=a_{1+1}=a_1+4=2+4=6 \Rightarrow a_3=a_{2+1}=a_2+4$

$=6+4=10 \Rightarrow d=10-6=4$

Hence, $\sum_{n=1}^{35}a_n=a_1+a_2+a_3+....+a_{35} \Rightarrow S_n=\frac{n}{2}[2A_1+(n-1)d]$

hence $s_{35}=\frac{35}{2}[2(2)+(35-1)4]=2450$

**Answer: C**

41. The population of certain country is currently 80 million with growth rate of 25 per year.

**Given** $(0.02)^9=5.12\times10^{-16}$, $(0.02)^{10}=1.024\times10^{-17}$

$(1.02)^9=1.19$ and $(1.02)^{10}=1.22$ which one of the following is the best approximation of the population after 10 year (in million)?

A. 81.3 million C. 95.2 million

B. 86.8 million D. 97.6 million

***Solution:***

$G_1=80$ million, rate $=2\%=0.02$, after, t-year, the proportion



$G_t=G_1(1+\text{rate})^t$

$\Rightarrow G_{10}=80(1+0.02)^{10}=80(1.02)^{10}$ million $=97.6$ million

**Answer: D**

42. What is the $50^{th}$ term of the sequence 3,10,17,24,31,...?

A. 310 B. 346 C. 510 D. 531

***Solution:*** $A_n=A_1+(n-1)d \Rightarrow A_{50}=3+(50-1)7=346$

**Answer: B**

43. What is the sum of the series $\sum_{n=1}^{\infty}\frac{2^{2n+1}}{5^{n-1}}$

A. 40 B. 20 C. 10 D. 0

***Solution:***

$$\sum_{n=1}^{\infty}\frac{2^{2n}.2}{5^n.5^{-1}}=10\sum_{n=1}^{\infty}\left(\frac{4}{5}\right)^n, G_1=\frac{4}{5}, r=\frac{4}{5}<1$$

$$\Rightarrow 10\sum_{n=1}^{\infty}\left(\frac{4}{5}\right)=10\left(\frac{G_1}{1-r}\right)=10\left(\frac{\frac{4}{5}}{1-\frac{4}{5}}\right)=40$$

**Answer: A**

44. Which one of the following is equal to $\lim_{n\to\infty}\left(\frac{1-n-3n^2}{6n^2+1}\right)$?

A. $\frac{1}{6}$ B. $-\frac{1}{2}$ C. $-\frac{1}{6}$ D. $-\infty$

***Solution:*** Dividing the numerator and denominator by $n^2$,

$$\Rightarrow \lim_{n\to\infty}\left(\frac{\frac{1}{n^2}-\frac{1}{n}-3}{6+\frac{1}{n^2}}\right)=\frac{0-0-3}{6+0}=-\frac{1}{2}$$

**Answer: B**

45. $\lim_{x\to1}\frac{\sqrt{x}-1}{x^2-1}$ is equal to

A. $\infty$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. 0

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**Solution:** $\lim_{x\to 1}\frac{\sqrt{x}-1}{(x-1)(x+1)}=\lim_{x\to 1}\frac{\sqrt{x}-1}{(x-1)(x+1)}\cdot\frac{\sqrt{x}+1}{\sqrt{x}+1}$

$=\lim_{x\to 1}\frac{x-1}{(x-1)(x+1)(\sqrt{x}+1)}=\frac{1}{(2)(2)}=\frac{1}{4}$

**Answer: C**

46. $\lim_{x\to 1}\frac{1-x}{1-\frac{1}{x^2}}=$_____

A. 1 B. 0 C. $-\frac{1}{2}$ D. Does not exist

**Solution:** $\lim_{x\to 1}\frac{(1-x)x^2}{x^2-1}=\lim_{x\to 1}\frac{(1-x)(x^2)}{(x-1)(x+1)}=-\frac{1}{2}$

**Answer: C**

47. $\lim_{x\to 1}\left(1+\frac{1}{x}\right)^{-\frac{x}{2}}$ is equal to

A. $\frac{1}{\sqrt{e}}$ B. $\sqrt{e}$ C. $e^{-2}$ D. $\infty$

**Solution:** $\lim_{x\to 1}\left(\left(1+\frac{1}{x}\right)^{x}\right)^{-\frac{1}{2}}=e^{-\frac{1}{2}}=\frac{1}{\sqrt{e}}$

**Answer: A**

48. Let $f(x)=\begin{cases}3^x+k \text{ if } x\le 0\\ \frac{3\sin(2x)}{x} \text{ if } x>0\end{cases}$

if $f$ is continuous at $x=0$, then what is the value of $k$?

A. 6 B. 5 C. 2 D. 0

**Solution:** Since it is continuous at $x=0$ implies $\lim_{x\to 0} f(x)$ exist

$\Rightarrow \lim_{x\to 0^+} f(x)=\lim_{x\to 0^-} f(x)$

$\Leftrightarrow \lim_{x\to 0^+}\frac{3\sin(2x)}{x}=\lim_{x\to 0^-}3^x+k\Rightarrow 3(2)=1+k\Rightarrow k=6-1=5$

**Answer: B**



49. If $f(x)=x^2+2\ln x$, then what is $\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$

A. 5 B. 4 C. 2 D. 0

**Solution:** Recall that $f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$

$\Rightarrow f'(x)=2x+\frac{2}{x}\Rightarrow f'(2)=4+\frac{2}{2}=5$

**Answer: A**

50. The equation of the line tangent to the graph of $f(x)=\frac{1}{x+1}+\cos x$ at (0, f(0))?

A. $x+y=1$ B. $x-y=-2$ C. $x+y=2$ D. $x+4y=2$

**Solution:** $f'(x)=-\frac{1}{(x+1)^2}-\sin x\Rightarrow f'(0)=-1-0=-1$

- Equation of tangent line is given by $y-f(0)=f'(0)(x-0)\Rightarrow y-2=-1(x)\Leftrightarrow y+x=2$

**Answer: C**

51. If $g(x)=xf(x)-\sqrt{f(x)}$ and $f(2)=f'(2)=4$ then which of the following is equal to, $g'(2)$

A. 11 B. 8 C. 2 D. 0

**Solution:**

$g(x)=xf(x)-(f(x))^{\frac{1}{2}}$ use the product and quotient rule.

$g'(x)=x'f(x)+xf'(x)-\frac{f'(x)}{2\sqrt{f(x)}}$

$\Rightarrow g'(2)=1.f(2)+2f'(2)-\frac{f'(2)}{2\sqrt{f(2)}}=4+2(4)-\frac{4}{4}=12-1=11$

**Answer: A**

52. Which one of the following is true about the derivative of $f(x)=x|x|$

A. $f$ is not differentiable at $x=0$

B. $f'(x)=2|x|$, for every $x\in(-\infty,\infty)$

C. $f'(x)=2x$, for every $x\in(-\infty,\infty)$

D. $f'(x)=|x|+x$, for every $x\in(-\infty,\infty)$

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**Solution:** Let $x \geq 0$, then, $|x| = x$. Then

$\Rightarrow f(x) = x|x| \Rightarrow f(x) = x^2 \Rightarrow f'(x) = 2x,\ x > 0,$

Let $x < 0$, then, $|x| = -x$.

Then $f(x) = x(-x) = -x^2 \Rightarrow f'(x) = -2x, x < 0,$

$$f'(x) = \begin{cases} 2x, x > 0 \\ -2x, x < o \end{cases} \Rightarrow f'(x) = 2|x| \text{ for all } x \in (-\infty, \infty)$$

**Answer: B**

53. If $f(x) = e^{2x} + x - 3\cos x$, then $f''(x)$?

A. $e^{2x} + 1 - 3\sin x$ C. $4e^{2x} - 3\cos x$

B. $e^{2x} + 1 + 3\sin x$ D. $4e^{2x} + 3\cos x$

**Solution:** $f'(x) = 2e^{2x} + 1 + 3\sin x \leftarrow (e^{2x})' = 2e^{2x}$

$f''(x) = 4e^{2x} + 3\cos x$

**Answer: D**

54. On which one of the following intervals does $f(x) = x^4 + 4x$ increasing on:

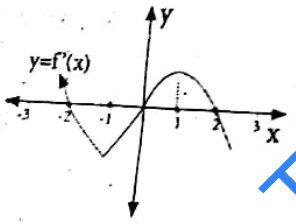
A. $(-\infty, -1]$ B. $(-\infty, 0]$ C. $[-1, \infty)$ D. $(-\infty, \infty)$

**Solution:** $f(x)$ is increasing on $[a,b]$ if $f'(x) \geq 0$ for all $x \in (a,b) \Rightarrow f'(x) = 4x^3 + 4 \geq 0 \Rightarrow 4(x^3 + 1) \geq 0$

For all $x \geq -1$, therefore $f$ is increasing on $[-1, \infty)$.

**Answer: C**

55. Suppose $f$ is differentiable on $(-\infty, \infty)$ and the graph of its derivative is as shown below




- Which one f the following is true about $f$

A. $f$ is decreasing on $(-\infty, \infty] \cup [1, \infty)$

B. $f$ has a local minimum at $x = -2$

C. $f$ is concave down $[0, \infty)$

D. $f$ is concave up on $(-1,1)$

**Solution:** $f'(x) \leq 0$ on $[-2,0] \cup [2, \infty) \Rightarrow f$ is decreasing



- $f'(x) \geq 0$. on $(-\infty, -2] \cup [0,2]$ so that $f$ is increasing.
- $f$ has local maximum at $x = -2$ because $f$ is increasing on $x \leq -2$ then decreasing on $x \geq -2$.
- The graph of $y = f'(x)$ is increasing on $(-1,1)$. so that $f''(x) > 0$ on $(-1,1)$ thus $f$ is concave up on $(-1,1)$.

**Answer: D**

56. A box seen below is to have a square base an open top and volume of 32 cubic units. If $x$ is the length of each side of its base and $y$ is its height, how many units should $x$ and $y$ be in order to make the box with the smallest amount of material.

A. $x = 4, y = 2$ C. $x = \sqrt{8}, y = 4$

B. $x = 2, y = 8$ D. $x = \sqrt{2}, y = 16$

**Solution:** Volume of box $= (B)(h) = x^2 y = 32 \Rightarrow y = \frac{32}{x^2}$.

surface area of box $A = 4xy + x^2$

$\Rightarrow A(x) = (4x)\left(\frac{32}{x^2}\right) + x^2 \Rightarrow A(x) = \frac{128}{x} + x^2$

$\Rightarrow A'(x) = -\frac{128}{x^2} + 2x$, set, $A'(x) = 0$

$\Rightarrow \frac{2x^3 - 128}{x^2} = 0 \Rightarrow 2x^3 - 128 = 0 \Rightarrow x^3 = 64 \Rightarrow x = 4$, then

$y = \frac{32}{16} = 2$

**Answer: A**

57. The volume V of a melting ice cube after t seconds is $V = 2000 - 40t + 0.2t^2 \left(\text{in cm}^3\right)$. How fast is the volume changing when t = 40 second?

A. $24 cm^3/\text{sec}$ B. $15 cm^3/\text{sec}$ C. $-15 cm^3/\text{sec}$ D. $-24 cm^3/\text{sec}$

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**Solution:** $\frac{dV}{dt} = -40 + 0.4t$ when t = 40se

$\Rightarrow -40 + 0.4(40) = -40 + 16 = -24\, cm^3/sec$ **Answer: D**

58. At which value(s) of $x$ does $f(x) = \frac{1}{4}x^4 - 2x^2$ have a local maximum.

A. $x = 4$ C. $x = -2$ and $x = 2$

B. $x = 0$ D. $x = 0$ and $x = 2$

**Solution:**

$f'(x) = x^3 - 4x \Rightarrow x(x^2 - 4) = 0 \Rightarrow x = 0, x = -2, x = 2$

- $f(0) = 0 - 0 = 0 \leftarrow$ Local maximum.
- $f(-2) = \frac{16}{4} - 8 = -4 \leftarrow$ Local minimum.
- $f(2) = -4$

**Answer: B**

59. What is an anti-derivative of $f(x) = \frac{2}{4x^2 + 4x + 1}$

A. $\frac{-1}{2x+1}$ C. $-\frac{2}{4x^2 + 4x + 1}$

B. $-\frac{2}{2x+1}$ D. $\ln(4x^2 + 4x + 1)$

**Solution:** Use $\int k(ax+b)^n dx = \frac{k}{a(n+1)}(ax+b)^{n+1} + C$

$\Rightarrow \int \frac{2}{4x^2 + 4x + 1} dx = \int 2(2x+1)^{-2} dx$

$= \frac{2}{2(-2+1)}(2x+1)^{-2+1} + C = -\frac{1}{2x+1}$ **Answer: A**

60. $\int (1+x)3^x dx$ is equal to

A. $(1+x)3^x - 3^x + C$

B. $(1+x)3^x + (\log 3^e)3^x + C$

C. $(1+x)3^x \log 3^e - (\log 3^e)^2 3^x + C$

D. $(1+x)3^x \log 3^e - 3x(\log e^3) + C$



**Solution:** $\int (1+x)3^x dx$, using table method

| Derivative | | Anti-derivative |
|---|---|---|
| 1+x | + | $3^x$ |
| 1 | - | $3^x \log 3^e$ |
| 0 | | $3^x (\log 3^e)^2$ |

$\Rightarrow \int (1+x)3^x dx = (1+x)3^x \log 3^e - 3^x (\log 3^e)^2 + C$ **Answer: C**

61. Which one of the following is equal to $\int \frac{\ln(xe^x)}{x} dx$

A. $\ln|x| + \frac{1}{2}e^x + C$ C. $\ln|x| + e^{2x} + C$

B. $\frac{1}{2}(\ln x)^2 + x + C$ D. $-\frac{1}{x^2} + (\ln x)^2 + x + C$

**Solution:** Recall that $\ln(xe^x) = \ln x + \ln e^x = \ln x + x$

$\Rightarrow \int \frac{\ln(xe^x)}{x} dx = \int \left(\frac{\ln x}{x} + \frac{x}{x}\right) dx = \int \left(\frac{\ln x}{x} + 1\right) dx$

$= \frac{(\ln x)^2}{2} + x + C$ **Answer: B**

62. $\int \frac{1}{x^2 + x} dx$ is equal to _____

A. $\ln|x^2 + x| + C$ C. $\ln|x| - \ln|x+1| + C$

B. $2\ln|x+1| + \ln|x| + C$ D. $\ln|x| + \ln|x+1| + C$

**Solution:** Using partial decomposition:

$\frac{1}{x^2 + x} = \frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1} \Leftrightarrow A(x+1) + Bx = 1$

- Let $x = 0 \Rightarrow A(0+1) + 0 = 1 \Rightarrow A = 1$
- Let $x = -1 \Rightarrow A(-1+1) + B(-1) = 1 \Rightarrow B = -1$

$\Rightarrow \int \frac{1}{x^2 + x} dx = \int \left(\frac{1}{x} - \frac{1}{x+1}\right) dx = \ln|x| - \ln|x+1| + C$

**Answer: C**

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63. The value of $\int_0^{\pi/2} 2x\cos x\,dx$ is equal to

A. $\Pi-2$ B. $\frac{\Pi}{2}+1$ C. $\Pi+2$ D. $\frac{\Pi}{2}-1$

***Solution:*** Use integration by part $\int udv = uv - \int vdu$

Let $u = 2x$, $du = 2dx$, $dv = \cos x\,dx$, $v = \sin x$

$$\Rightarrow \int_0^{2\Pi} 2x\cos x dx = 2x\sin x - \int 2\sin x dx = 2x\sin x + 2\cos x$$

$$\int_0^{\pi/2} = (\Pi\sin\frac{\Pi}{2} + 0) - (0 + 2\cos 0) = \Pi - 2$$ **Answer:** A

64. What is the area of the region between the graphs of $y = x^2$ and $y = -x+2$ for $0 \le x \le 2$

A. 3 B. 2 C. $\frac{3}{2}$ D. $\frac{2}{3}$

***Solution:*** Area, $A = \int_0^2 (-x+2)dx = -\frac{x^2}{2} + 2x\Big|_0^2 = -2+4 = 2$

**Answer:** B

65. Which one of the following points is closer to the sphere $x^2 + y^2 + Z^2 - 2x + 6Z + 9 = 0$

A. (1,0,0) B. (0,0,0) C. (0,-1,0) D. (0,0,-1)

***Solution***

$x^2 - 2x + y^2 + Z^2 + 6Z = -9 \Rightarrow x^2 - 2x + 1 + y^2 + Z^2 + 6Z + 9$

$= -9 + 9 + 1 \Rightarrow (x-1)^2 + (y-0)^2 + (Z+3)^2 = 1$,

distance between point and center (1,0,-3)

A. Distance between (1,0,0) and (1,0,-3) is $d = \sqrt{0+0+9} = \sqrt{9}$

B. Distance between (0,0,0) and (1,0,-3) is $d = \sqrt{1+0+9} = \sqrt{10}$

C. Distance between (0,0,-1) and (1,0,-3) is $d = \sqrt{1+0+4} = \sqrt{5}$, It is closest.

**Answer:** D



1. Which of the following expression is a polynomial expression?

A. $x^2 - 3x + \sin x$ C. $\frac{2+\pi}{1+\pi^2}$

B. $\frac{4x^3 + 12x^2 - x}{\pi x^2}$ D. $2 - 3x^{2/3} + 7x^{5/2} + 3x^{-1}$

***Hint:***

$\frac{2+\pi}{1+\pi^2}$ is constant – polynomial function. **Answer: C**

2. If $f(x) = \sqrt[3]{1+e^{-x}}$, then $f^{-1}(x)$ equals _____

A. $\ln\left(\frac{1}{x^3-1}\right)$ B. $\frac{1}{\ln(x^3-1)}$ C. $\ln(1-x^3)$ D. $(1+e^{-x})^3$

***Hint:*** Let $y = f(x)$, then $y = \sqrt[3]{1+e^{-x}}$

To find the inverse interchange $y \to x$ and $x \to y$

$\Rightarrow x = \sqrt[3]{1+e^{-y}} \Leftrightarrow x^3 = 1 + e^{-y}$

$\Rightarrow e^{-y} = x^3 - 1 \Leftrightarrow -y = \log_e^{(x^3-1)} = \ln(x^3-1)$

$\Rightarrow y = -\ln(x^3-1) = \ln(x^3-1)^{-1} = \ln\left(\frac{1}{x^3-1}\right)$

$\therefore f^{-1}(x) = \ln\left(\frac{1}{x^3-1}\right)$

**Answer: B**

3. If $f(x) = \frac{x+1}{x-1}$ and $f(a) = 5$, then $f(2a)$ is equal to:

A. 2 B. 4 C. 6 D. 8

***Hint:*** Here at:

$f(x) = \frac{x+1}{x-1}$ then $f(a) = \frac{a+1}{a-1} = 5$

$\Rightarrow a + 1 = 5a - 5 \Leftrightarrow -4a = -6;\ a = \frac{6}{4} = \frac{3}{2}$

$f(2a) = f\left(2\left(\frac{3}{2}\right)\right) = f(3)$

$\Rightarrow f(3) = \frac{3+1}{3-1} = \frac{4}{2} = 2$

**Answer: A**

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4. Let $f(x)=\dfrac{x^2-1}{x-x^2}$ which of the following is true about $f(x)$.

A. $x=0$ and $x=1$ are its vertical asymptotes.
B. $y=1$ is its horizontal asymptote.
C. $y=x-1$ is its oblique asymptote.
D. It is almost the same as the horizontal line $y=-1$ as $x\to\pm\infty$.

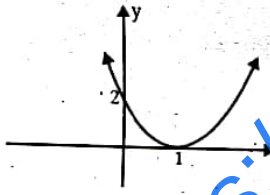
***Hint***:

$$y=\frac{x^2-1}{x-x^2}=\frac{(x-1)(x+1)}{x(1-x)}=-\left(\frac{x+1}{x}\right)=-1-\frac{1}{x}$$

i) $x=0$ is the vertical asymptote.
ii) $y=-1$ is the horizontal asymptote.
iii) as $x\to\pm\infty, y\to-1$

**Answer: D**

5. Which of the following equations below is represented by the following parabola?




A. $y=x^2+2$
B. $y=(2x-1)^2$
C. $y=2(x-1)^2$
D. $y=(2x+1)^2$

***Hint***:

$y=a(x-b)^2+c$

- The vertex of the parabola is (1, 0).
- The y – intercept is (0, 2).
- The value of c is 0.

$\therefore y=2(x-1)^2+0$

**Answer: C**

6. If the equation $(x-2)^2-(y-2)^2=1$ represents a hyperbola which one of the following represents equation of an asymptote to the hyperbola?

A. $y=4-x$ B. $x+y=1$ C. $x=2-y$ D. $x+2y=3$

***Hint***:

$$(x-2)^2-(y-2)^2=1\Leftrightarrow\frac{(x-2)^2}{1}-\frac{(y-2)^2}{1}=1$$

Asymptote: $y-2=\pm\dfrac{1}{1}(x-2)$

$\Leftrightarrow y-2=x-2$ and $y-2=-x+2$

$y-x=0$ and $y+x=4\Rightarrow y=4-x$

**Answer: A**



7. What is the distance from the origin to the line that passes through (1, 0) and (0, 1)?

A. $\frac{1}{2}\sqrt{2}$ B. 1 C. $\frac{1}{2}$ D. $\sqrt{2}$

***Hint***:

The equation of the line through (1, 0) and (0, 1) is $y-1=\left(\dfrac{0-1}{1-0}\right)(x-0)\Leftrightarrow x+y+1=0.$

The distance from a point $(x_o, y_o)$ to the line $Ax + By + C = 0$ is obtained by the formula:

$$d=\left|\frac{Ax_o+By_o+C}{\sqrt{A^2+B^2}}\right|=\left|\frac{1(0)+1(0)+1}{\sqrt{1^2+1^2}}\right|=\frac{1}{\sqrt{2}}=\frac{\sqrt{2}}{2}$$

**Answer: A**

8. If two lines $y = x$ and $y = x - 4$ are tangent to the circle at (2, 2) and (4, 0), respectively, then what is the equation of the circle?

A. $(x-2)^2+y^2=4$ C. $(x-3)^2+(y-1)^2=2$
B. $(x-4)^2+(y-2)^2=4$ D. $(x-1)^2+(y+1)^2=10$

***Hint***: Let $(h, k)$ be center of circle.

☞ The distance between the center $(h, k)$ and point on the circle (2, 2) and (4, 0) would be:
$(h-2)^2+(k-2)^2=(h-4)^2+(k-0)^2$-
$\Rightarrow h^2-4h+4+k^2+4=h^2-8h+16+k^2$ ← canceling
$\Leftrightarrow 4h-4k=8\Rightarrow h-k=2\ \ldots(*)$

☞ The slope of the line passing through $(h, k)$ and (2, 2) is perpendicular to the line $y = x$. i.e., $\dfrac{k-2}{h-2}=-1$

$\Rightarrow k-2=-h+2\Leftrightarrow -h+2\Leftrightarrow k+h=4\ \ldots(**)$

Solving (*) and (**) will be:

$$\begin{cases}h-k=2\\h+k=4\end{cases}\Rightarrow 2h=6, h=3, k=1$$

Radius: $r^2=(h-2)^2+(k-2)^2=(3-2)^2+(1-2)^2=2$

$\therefore$The equation of circle $(x-3)^2+(y-1)^2=2$

9. What is the equation of the directix for the parabola whose equation is $y^2 + 8x + 6y + 25 = 0$?

A. $y=3$ B. $x=2$ C. $x=0$ D. $x=4$

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***Hint***:
$y^2 + 6y + 8x + 25 = 0 \Leftrightarrow y^2 + 6y + 9 = -8x - 25 + 9$
$(y+3)^2 = -8(x+2)$← opens to the left.
∴Directx: $x + 2 = 2 \Rightarrow x = 0$.

10. A semi elliptical arc over a tunnel for a road through a mountain has a major axis of length 80 meters and a height of 30 meters at the center. What is the equation of the semi – elliptical arc over the tunnel, if the center is considered as the origin?

A. $\frac{x^2}{6400} + \frac{y^2}{900} = 1$ C. $\frac{x^2}{1600} + \frac{y^2}{900} = 1$

B. $\frac{x^2}{900} + \frac{y^2}{6400} = 1$ D. $\frac{x^2}{8100} + \frac{y^2}{6400} = 1$

***Hint***: Let $2a$ = length of major axis = 80 m
$2b$ = length of semi minor axis = 60 m
$\Rightarrow 2a = 80\ m \Rightarrow a = 40$m and $b = 30$m

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{x^2}{1600} + \frac{y^2}{900} = 1$ **Answer: C**

11. Let the equation $x^2 + 2x + y^2 = 8$ represents a circle. Then which one of the following lines cut the circle at exactly two points?

A. $4x + 3y + 19 = 0$ C. $2y = 5x + 43$
B. $3x + 4y + 14 = 0$ D. $2x = y - 50$

***Hint***: $x^2 + 2x + y^2 = 8 \Rightarrow x^2 + 2x + 1 + y^2 = 8 + 1$
$\Rightarrow (x+1)^2 + (y+0)^2 = 9$, center = $(-1, 0)$, $r = 3$
The distance between the center (-1, 0) and a given line is:

A. $d = \frac{|Ax_o + By_o + C|}{\sqrt{A^2 + B^2}} = \frac{|4(-1) + 3(0) + 19|}{\sqrt{4^2 + 3^2}} = \frac{15}{5} = 3$

Since: $d = r = 3$, therefore the circle and the line $4x + 3y + 19 = 0$ intersect exactly at one point.

B. $d = \frac{|3(-1) + 4(0) + 14|}{\sqrt{3^2 + 4^2}} = \frac{14}{5} < r$

Therefore the line $3x + 4y + 14 = 0$ and the circle intersect exactly at two point.

C. $d = \frac{|5(-1) + (-2)(0) + 43|}{\sqrt{5^2 + (-2)^2}} = \frac{38}{\sqrt{29}} > r$

Therefore the line do not intersect the circle.



D. $d = \frac{|2(-1) - 1(0) + 50|}{\sqrt{(2)^2 + (-1)^2}} = \frac{48}{\sqrt{5}} > r$ →do not intersect

**Answer: B**

12. If $(P \vee q) \Rightarrow (\neg r \wedge r)$ is true, then which one of the following is necessarily true?

A. $(P \vee r) \Rightarrow q$ B. $\neg q \wedge r$ C. $\neg P \Rightarrow r$ D. $\neg P \vee r$

***Hint***: $\neg r \wedge r$ is always false. Thus $P \vee q \Rightarrow (\neg r \wedge q)$ is true.
$P \vee q \equiv F$, therefore, $P \equiv F$ and $q \equiv F$.
A. $(F \vee F) \Rightarrow F \equiv T$ **Answer: A**

13. Which one of the following is a valid logical argument?

A. $P \Rightarrow q,\ q \vdash P$ C. $\neg P \wedge q,\ q \Rightarrow r \vdash r$
B. $P \Leftrightarrow q,\ P \Rightarrow q \vdash q$ D. $\neg P,\ P \vee q,\ r \Rightarrow q \vdash r$

***Hint***:
1) $\neg P \wedge q,\ q \equiv T$, ⟵ premise
$\neg P \equiv T$ and $q \equiv T$
2) $q \Rightarrow r \equiv T$ ⟵ premise
$T \Rightarrow r \equiv T$, so that $r \equiv T$
$\neg P \wedge q,\ q \Rightarrow r \vdash r$ is valid argument. **Answer: C**

14. Which one of the following is equivalent to $\neg[(\forall x)(P(x) \Rightarrow Q(x))]$?

A. $(\forall x)(\neg P(x) \Rightarrow \neg Q(x))$ C. $(\exists x)(\neg P(x) \wedge Q(x))$
B. $(\exists x)(\neg P(x) \Rightarrow \neg Q(x))$ D. $(\exists x)((P(x) \wedge \neg Q(x))$

***Hint***: Recall that: $P \Rightarrow Q \equiv \neg P \vee Q$
$\neg(P \Rightarrow Q) \equiv (\neg P \vee Q) \equiv P \wedge \neg Q$
$\neg[(\forall x)(P(x) \Rightarrow Q(x))] \equiv (\exists x)[P(x) \wedge \neg Q(x)]$ **Answer: D**

15. If the list of a measurement is 10, α, 5, α, 5, 10, 20, 15, 20, 5 with mean $\bar{x}$, then what is the value of α in terms of $\bar{x}$?

A. $10\bar{x} - 90$ B. $9\bar{x} - 90$ C. $5\bar{x} - 90$ D. $5\bar{x} - 45$

***Hint***:
$\bar{x} = \frac{\alpha + \alpha + 5 + 5 + 5 + 10 + 10 + 15 + 20 + 20}{10} = \frac{2\alpha + 90}{10}$.
$10\bar{x} = 2\alpha + 90 \Rightarrow \alpha = 5\bar{x} - 45$ **Answer: D**

16. The following is the frequency distribution of a grouped data.

| Class interval | 3 – 7 | 8 – 12 | 13 – 17 | 18 – 22 |
|---|---|---|---|---|
| Frequency (f) | 2 | 2 | 10 | 6 |

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What is the mean and standard deviation of the distribution respectively?

A. $15, 2\sqrt{5}$ B. $15, 2\sqrt{7.5}$ C. $12.5, 5\sqrt{2}$ D. $12.5, \sqrt{15}$

***Hint:***

| Class | $f$ | Class mark $(x_C)$ (Midpoint) | $Fx_C$ | $(x_C - \bar{x})^2$ | $f(x_C - \bar{x})^2$ |
|---|---|---|---|---|---|
| 3 – 7 | 2 | 5 | 10 | $(-10)^2 = 100$ | 200 |
| 8 – 12 | 2 | 10 | 20 | $(-5)^2 = 25$ | 50 |
| 13 – 17 | 10 | 15 | 150 | $0^2 = 0$ | 0 |
| 18 – 22 | 6 | 20 | 120 | $5^2 = 25$ | 150 |

- Mean $(\bar{x}) = \dfrac{\sum fx_C}{\sum f} = \dfrac{300}{20} = 15$
- Variance $\dfrac{\sum f(x_C - \bar{x})^2}{\sum f} = \dfrac{200 + 50 + 0 + 150}{20} = 20$
- Standard deviation = $\sqrt{\text{variance}} = \sqrt{20} = 2\sqrt{5}$.

**Answer: A**

17. If $Q_1$, $D_i$ and $P_i$ are respectively the $i^{th}$ quartile, decile and percentile of data arranged in an increasing order, then which one of the following is necessarily true?

A. $Q_2 = \dfrac{Q_1 + Q_3}{2}$ B C. $P_{25} > Q_1$

B. $D_3 > P_{25}$ D. $Q_2$= mean of the data

***Explanation:***

**Bear in mind that:**

The set of a given data:

i) Divided into two equal part is **Median.**
ii) Divided into four equal part is **Quartile.**
iii) Divided into ten equal part is **Decile.**
iv) Divided into hundred equal part is **percentile.**

**Recall that:**

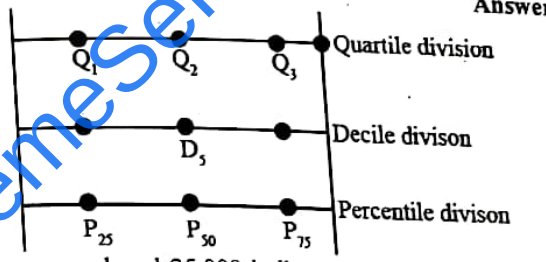
i) $Q_2 = D_5 = P_{50}$ = Median of the data
ii) $Q_1 = P_{25}$
iii) $Q_3 = P_{75}$

- $D_3 = \dfrac{3}{10}(100) < 30\%$ of the data from the lower.



- $P_{25} = \dfrac{25}{100}(100\%) < 25\%$ of the data from the lower.

**Answer: B**




18. A company produced 25,000 bulbs and randomly started 2% of the product. Among the tested bulbs, if 40 have defect to type $D_1$, 60 have defect of type $D_2$ and 25 have both types of defects. What is the probability that a bulb produced by the company has none of the defects?

A. 0.95 B. 0.80 C. 0.85 D. 0.20

***Solution:***

2% of 25,000 is 500 P (non – defects) = 1 – P (all defect)

$= 1 - (P(D_1) + P(D_2) - P(D_1 n D_2))$

$= 1 - \left[\dfrac{40}{500} + \dfrac{60}{500} - \dfrac{25}{500}\right]$

$= 1 - 0.15 = 0.85$

**Answer: C**

19. If S is a set with 10 elements and ACS, what is the probability that A has 3 or more elements?

A. $\dfrac{7}{10}$ B. $\dfrac{8}{11}$ C. $\dfrac{121}{128}$ D. $\dfrac{7}{128}$

***Solution:***

P (3 or more elements) = 1 – P(1 – element) – P (2 – element)

$= 1 - \dfrac{1}{10} - \dfrac{2}{10} = 1 - \dfrac{3}{10} = \dfrac{7}{10}$

**Answer: A**

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20. If $A=\begin{pmatrix}2&0&-1\\1&2&0\\0&0&-1\end{pmatrix}$ and $(2A+B)^T=A^TA$ then which one fo the following is equal to B.

A. $\begin{pmatrix}1&0&-2\\2&0&0\\0&0&4\end{pmatrix}$ B. $\begin{pmatrix}0&0&0\\0&0&0\\0&0&0\end{pmatrix}$ C. $\begin{pmatrix}8&0&-4\\4&8&0\\0&0&-4\end{pmatrix}$ D. $\begin{pmatrix}1&2&0\\0&0&0\\-2&0&4\end{pmatrix}$

***Solution:***

**Recall that:**

i) $(A-B)^t=A^t+B^t$

ii) $(AB)^t=B^tA^t$

$(2A+B)^T=A^TA \Leftrightarrow 2A^T+B^T=A^TA$

To find B make transpose both side

$(2A^T+B^T)^T=(A^TA)^T \Leftrightarrow 2A+B=A^TA$

$$2\begin{pmatrix}2&0&-1\\1&2&0\\0&0&-1\end{pmatrix}=(A^TA)^T\Leftrightarrow 2A+B=A^TA$$

$$2\begin{pmatrix}2&0&-1\\2&4&0\\-2&4&0\end{pmatrix}+B=\begin{pmatrix}2&1&0\\0&2&0\\-1&0&1\end{pmatrix}\begin{pmatrix}2&0&-1\\1&2&0\\0&0&-1\end{pmatrix}=\begin{pmatrix}5&2&-2\\2&4&0\\-2&0&2\end{pmatrix}$$

$$B=\begin{pmatrix}5&2&-2\\2&4&0\\-2&0&2\end{pmatrix}-\begin{pmatrix}4&0&-2\\2&4&0\\0&0&-2\end{pmatrix}=\begin{pmatrix}1&2&0\\0&0&0\\-2&0&4\end{pmatrix}$$

**Answer: D**

21. If $M=\begin{pmatrix}0&1&2\\3&-1&0\\5&2&4\end{pmatrix}$ and $A^TM=2I$, where A is $3\times 3$ matrix and I is the identity matrix of order 3, then what is det (A) ?

A. 0.2 B. $\frac{4}{17}$ C. 0.8 D. $\frac{1}{17}$

***Solution:*** **Recall that:** $|A^T|=|A|$

$A^TM=2I \Rightarrow |A^TM|=|2I|$



$\Leftrightarrow |A^T||M|=2^3|I|=8(1)$. Also $|M|=0\begin{vmatrix}-1&0\\2&4\end{vmatrix}-1\begin{vmatrix}3&0\\5&4\end{vmatrix}+2\begin{vmatrix}3&-1\\5&2\end{vmatrix}$

$=0-12+2(6+5)=10$

$\Leftrightarrow |A|(10)=8$, therefore, $|A|=\frac{8}{10}=0.8$

**Answer: C**

22. What should be the values of k so that the system of equation $\begin{cases}x-y+z=1\\-x+5y-4z=1\\2x+2y-z=k\end{cases}$ has a solution?

A. 0 B. 1 C. -4 D. 4

***Solution:***

The augmented matrix for system is $\left[\begin{array}{ccc|c}1&-1&1&1\\-1&5&-4&1\\2&2&-1&k\end{array}\right]$ reduce this matrix to echelon form. Add the first row to the second row an -2 times the first row to the third row. That is $R_2\to R_1+R_2$ and $R_3\to 2R_1+R_3$.

$$\left[\begin{array}{ccc|c}1&-1&1&1\\0&4&-3&2\\0&4&-3&-2+k\end{array}\right]\ R_3\to -R_2+R_3\ \left[\begin{array}{ccc|c}1&-1&1&1\\0&4&-3&2\\0&0&0&-4+k\end{array}\right]$$

Therefore, $-4+k=0\Rightarrow k=4$

**Answer: D**

23. Suppose $A_x=b$, where A is $3\times 3$ matrix, $b=(b_1,b_2,b_3)^T$ and $x=(x,y,z)^T$. Which one of the following is necessarily true about this system of linear equation?

A. The system has a solution only when det (A) $\neq 0$.

B. The cramer's ruel is suitable to solve the system if two rows of A are identical.

C. If det (A) $\neq 0$ and the second column of A is a multiple of b then $x=0$.

D. If $b=0$, then $x=(0,0,T)^T$ is the only solution of the system.

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***Solution:***

Let $\begin{bmatrix} a_1 & a_{12} & a_{12} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_2 \end{bmatrix}$

A. False, because if $D + 0, D_x = D_z = 0$ has infinite solution.

B. False

C. True, $D \neq 0$, and $x = \frac{D_x}{D} = 0$ because, first and second column of A are proportional, so that $D_x = 0$.

**Answer: D**

24. If $z = x + yi$ is a complex number, then $|z|^2 + \frac{1}{2}(z - \bar{z})^2 = 1$ is equivalent to which one of the following equations?

A. $x^2 - y^2 = 1$ C. $x^2 - y^2 = 2$

B. $x^2 - 3iy^2 = 1$ D. $2x^2 - y^3 = 2$

***Solution:*** $|z| = \sqrt{x^2 + y^2}$

$|z|^2 + \frac{1}{2}(z - \bar{z}) = 1$ implies that

$\left(\sqrt{x^2 + y^2}\right) + \frac{1}{2}(x + yi - (x - yi))^2 = 1$

$\Rightarrow x^2 + y^2 + \frac{1}{2}(2yi)^2 = 1$

$\Rightarrow x^2 - y^2 = 1$

***Explanation:***

$(2yi)^2 = -4y^2$

**Answer: A**

25. If $W = \frac{16i}{1+i} + (1 - 3i)^2$ and $Z = |W| + \bar{W}$ the simplest form of Z is _____.

A. $\sqrt{2} + 2i$ B. $2 + 2i$ C. $4 - 2i$ D. $2 - 2i$

***Solution:***

W is equivalent to

$W = \frac{16i(1-i)}{(1+i)(1-i)} + 1 - 6i - 9 = \frac{16i + 16}{1+1} - 8 - 6i$

$\Rightarrow W = 8i + 8 - 8 - 6i = 2i, z = |W| + \bar{W} = 2 - 2i$ **Answer: D**



26. What is the equation of a line that pass through the point $(-1, 2)$ and parallel to the vector $v\ (1, -1)$?

A. $2x - y = 1$ C. $x - 2y = 3$

B. $x + y - 1 = 0$ D. $y - 2x + 1 = 0$

***Solution:***

$(x, y) = (x_o, y_o) + t(a, b)$ ← vector equation line

$\Rightarrow (x, y) = (-1, 2) + t(1, -1)$ ← vector equation

$\Rightarrow x = -1 + t$ and $y = 2 - t$

Solving for $t$, $t = x + 1$ and $t = 2 - y$

$\Rightarrow x + 1 = 2 - y \Leftrightarrow x + y - 1 = 0$

**Answer: B**

27. Suppose $A = 3i - 4j$ an dB is a vector in the $xy$ – plane such that the angle between A and B is $\pi/3$. If $\vec{u}$ is the unit vector in the direction of B, then $\vec{A}.(\vec{A} - 2u)$ is equal to:

A. 20 B. 5 C. 15 D. 30

***Solution:*** Here recall that: $|u| = 1$.

$\vec{A}.(\vec{A} - 2u) = \vec{A}.\vec{A} - 2\vec{A}\vec{u} = |A|^2 - 2|A||u|\cos\left(\frac{\pi}{3}\right)$

$\Rightarrow \left(\sqrt{3^2 + 4^2}\right)^2 - 2\sqrt{3^2 + 4^2}(1)\left(\frac{1}{2}\right) = 25 - (5) = 20$

**Answer: A**

28. If $\ell$ is the line passing through (0, 2) and parallel to $v = i + 3j$, which one of the following is true about $(\ell)$ and the circle $(x - 2)^2 + (y - 1)^2 = 5$?

A. $\ell$ is tangent of the circle at (0, 2).

B. $\ell$ is tangent to the circle to the at some point P, where $P \neq (0, 2)$.

C. $\ell$ intersects the circle at two distinct points.

D. The distance between l and the center of the circle is greater than $\sqrt{5}$.

***Solution:***

$(x, y) = (a, b) + \overrightarrow{tv}$

$\Rightarrow (x, y) = (0, 2) + t(1, 3)$

$\Leftrightarrow x = 0 + 1$ and $y = 2 + 3t$, solving for $t$

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$y = 2 + 3x \Leftrightarrow 3x - y + 2 = 0$

center of circle = (2, 1) and radius = $\sqrt{5}$.

The perpendicular distance from the center (2, 1) to the line ($\ell$):

$3x - y + 2 = 0,\ d = \dfrac{|Ax + By + C|}{\sqrt{A^2 + B^2}} = \dfrac{|3(2) - 1(1) + 2|}{\sqrt{3 + (1)}} = \dfrac{7}{\sqrt{10}}$

We can see that: $d < r \Rightarrow \dfrac{7}{\sqrt{10}} < \sqrt{5}$

We conclude that $\ell$ intersect the circle at two point.

**Answer: C**

29. If A = (−2, 3), B = (3, 1) and C s any other point on the plane, then which one of the following is coordinate form of $\overrightarrow{AC} - \overrightarrow{BC}$.

A. (-5, 2) B. (5, -2) C. (1, 4) D. (-1, -4)

***Solution***: Recall that BC = –CB

$\overrightarrow{AC} - \overrightarrow{BC} = \overrightarrow{AC} + \overrightarrow{CB} = \overrightarrow{AB}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (3, 1) - (-2, 3) = (5, -2)$

**Answer: B**

30. Suppose $A = 2i - j + 2k$ and B is a vector in space such that $|\vec{B}| = \vec{A}.\vec{B}$. If $\vec{u}$ is the unit vector in the direction of $\vec{B}$, then $|\vec{A} + \vec{u}|^2$ is equal to:

A. 16 B. 12 C. 10 D. 14

***Solution***: We have, $|\vec{A}| = \sqrt{4 + 1 + 4} = 3$

$|\vec{B}| = |\vec{A}.\vec{B}| = |\vec{A}||\vec{B}| \cos\theta \Rightarrow \cos\theta = \dfrac{|B|}{|\vec{A}||\vec{B}|} = \dfrac{1}{|\vec{A}|}$

$|\vec{A} + \vec{u}| = \left(\sqrt{|A| + 2|\vec{A}||\vec{u}|\cos|\vec{u}|}\right)$

$= (3)^2 + 2(3)(1)\left(\dfrac{1}{3}\right) + 1^2 = 12$

Explanation:

$\cos\theta = \dfrac{1}{|A|} = \dfrac{1}{3}$

31. If a translation T takes the circle $x^2 + y^2 - 2x + 6y + 3 = 0$ into the circle whose equation is $(x + 2)^2 + (y - 4)^2 = 7$, then what is the image of the origin under T?

A. (−3, 7) B. (1, 2) C. 1, -3 D. (-2, 4)

***Solution***: Applying completing the square

$x^2 - 2x + 1 + y^2 + 6y + 9 = -3 + 1 + 9$



$\Leftrightarrow (x - 1)^2 + (y + 3)^2 = 7$

Translating the center of circle: (1, -3)→(-2, 4),

Translating vector: v = (-2 − 1, 4 − (-3)) = (-3, 7)

$(0, 0) \xrightarrow{(-3,7)} (-3, +0, 7 + 0) = (-3, 7)$ **Answer: A**

32. What is the image of the line given by (x, y) = (1, 0) + t (3, 6), under the translation that takes (1, 0) to (0, 1) followed by the reflection about the line $y = 2x$?

A. $y = 2x + 3$ B. $y = 2x - 3$ C. $y = 2x + 6$ D. $y = 2x - 5$

***Solution***: $(x, y) = (1, 0) + t(3, 6)$

$\Rightarrow x = -1 + 3t$ and $y = 0 + 6i$, solving for t

$\dfrac{x+1}{3} = \dfrac{y-0}{6} \Leftrightarrow y = 2x + 2$

$(1, 0) \rightarrow (0, 1)$, translating vector v = (-1, 1)

Taking any point on line $y = 2x + 2$, let (1, 4)

$(1, 4) \xrightarrow{-1,1} (0, 5)$

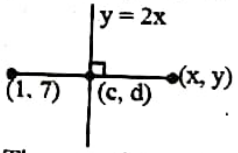
Translation is parallel movement so that the image of $y = 2x + 2$ is $y = 2x + b$.

To find b, put (0, 5) on $= 2x + b$

$\Rightarrow 5 = 2(0) + b \Leftrightarrow b = 5$

$\therefore y = 2x + 5$

To find the image of $y = 2x + 5$ under reflecting line $y = 2x$ is let $y = 2x + b$. Let take (1, 7) on $y = 2x + 5$.




The equation of line through point (1, 7) and (x, y) is

$\dfrac{y-7}{x-1} = -\dfrac{1}{2} \Leftrightarrow y = -\dfrac{1}{2}x + \dfrac{15}{2}$

To find (c, d): $\begin{cases} y = 2x \\ y = -\dfrac{1}{2}x + \dfrac{15}{2} \end{cases} = \begin{cases} d = 2C \\ d = -\dfrac{1}{2}C + \dfrac{15}{2} \end{cases}$

$2C = -\dfrac{1}{2}C + \dfrac{15}{2} \Leftrightarrow 2C + \dfrac{1}{2}C = \dfrac{15}{2}$

C = 3 and d = 6

(C, d) = (3, 6) are midpoint between (1, 7) and (x, 7) and (x, y).

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$\frac{x+1}{2}=3$ and $\frac{y+7}{2}=6$

$\Rightarrow x=5$ and $y=5$

To find b, in $y=2x+b$ is

$\Rightarrow 5=2(5)+b \Rightarrow b=-5$

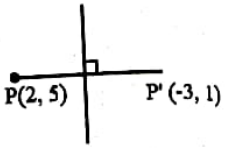
$\therefore$The image: $y=2x-5$

**Answer; D**

33. If a point (2, 5) is reflected under a line to the point (-3, 1) what si the line of reflection?

A. $2x+3y=7$ C. $8y+10x=19$

B. $x+3y=7$ D. $2x+3y+5=0$




***Solution***: Let P = (2, 5) and P' = (-3, 1)

- Slope of $PP'=\frac{1-5}{-3-2}=\frac{-4}{-5}=\frac{4}{5}$
- Slope of lien perpendicular to PP' is $\frac{5}{4}$
- Midpoint of PP' is $\left(\frac{-3+2}{2},\frac{1+5}{2}\right)=\left(-\frac{1}{2},3\right)$
- Equation of line passing through is $\frac{y-3}{x+\frac{1}{2}}=-\frac{5}{4}$

$\Rightarrow y-3=-\frac{5}{4}\left(x+\frac{1}{2}\right)\Leftrightarrow 4y-12=-5x-\frac{5}{2}$

$\Rightarrow 8y+10x=19$

**Answer: C**

34. If $\theta=2\arctan\left(\frac{1}{2}\right)$, then which one of the following is equal to $\sec\theta$?

A. $\frac{25}{3}$ B. $\frac{4}{5}$ C. $\frac{5}{3}$ D. $\frac{5}{4}$

***Solution***: $\theta=2\tan^{-1}\left(\frac{1}{2}\right)\Leftrightarrow\tan\left(\frac{\theta}{2}\right)=\frac{1}{2}$

Recall that: $\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)=1$



Dividing both side by $\cos^2\left(\frac{\theta}{2}\right)$ will get

$\Rightarrow\tan^2\left(\frac{\theta}{2}\right)+1=\sec^2\left(\frac{\theta}{2}\right)$

$\Rightarrow\frac{1}{4}+1=\sec\left(\frac{\theta}{2}\right)\Leftrightarrow\sec\left(\frac{\theta}{2}\right)=\frac{\sqrt{5}}{2}$

$\Rightarrow\cos\left(\frac{\theta}{2}\right)=\frac{2}{\sqrt{5}}$ since we have $\cos\left(\frac{\theta}{2}+\frac{\theta}{2}\right)=\cos\theta$

$\Rightarrow\cos\theta=\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)-\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\theta}{2}\right)=\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}$

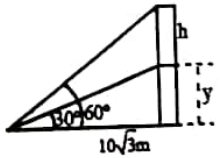
$=\cos^2\left(\frac{\theta}{2}\right)-\left(1-\cos^2\left(\frac{\theta}{2}\right)\right)=2\cos^2\left(\frac{\theta}{2}\right)-1$

$\Rightarrow\cos\theta=2\left(\frac{4}{5}\right)-1=\frac{3}{5}\Rightarrow\sec\theta=\frac{5}{3}$

**Answer: C**

35. An observer on level ground is at a distance $10\sqrt{3}$m from a building. The angles of elevation to the bottom of the windows on the second and third floors are 30° and 60°, respectively. What is the distance h between the bottoms of the windows?

A. 15m B. 20m C. $15\sqrt{3}$m D. 32m




***Solution***: $\tan 30°=\frac{y}{10\sqrt{3}}$

$\Rightarrow y=\left(10\sqrt{3}\right)\tan 30°=10m$

- $\tan 60°=\frac{h+y}{10\sqrt{3}}$

$\Rightarrow h+y=10\sqrt{3}\tan 60°=30$m

$\therefore h=30\text{m}-y=30-10\text{m}=20\text{m}$

**Answer: B**

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36. Suppose that an air plane is descending at a speed of 50 miles per hour at an angle of 30° below the horizontal line. What is the x and y components respectively, of the velocity of the plane?

A. $(50\sqrt{3}, 25)$ C. $(25, -25\sqrt{3})$

B. $(-25, \sqrt{50}, 3)$ D. $(-25, \sqrt{3}, -25)$

***Solution:***

- $\cos 30° = \frac{x}{50} \Rightarrow x = 50\cos 30° = -25\sqrt{3}$
- $\sin 30° = \frac{y}{50} \Rightarrow y = -50\cos 30° = -25$

$\therefore (x, y) = (-25\sqrt{3}, -25)$

37. Suppose P and Q are points in space such that the mid – point of PQ is on the negative z – axis and the distance between P and Q is 6. If P = (2, -1, 0), then what is the coordinate of Q?

A. (-2, 1, 4) B. (2, -1, 6) C. (2, -1, -6) D. (-2, 1, -4)

***Solution:***

Let Q = (x, y, z) and P = (2, -1, 0)

Midpoint of PQ $= \left(\frac{x+2}{2}, \frac{y-1}{2}, \frac{z}{2}\right) = (0, 0, -k)$

$\Rightarrow x = -2, y = 1, z = -2k \Rightarrow Q = (x, y, z) = (-2, 1, -2k)$

To find k, use distance between P and Q

$(-2-2)^2 + (-1-1)^2 + (-2k)^2 = 36$

$4k^2 = 36 - 20 = 16 \Rightarrow k = 2$, therefore, Q = (-2, 1, -4)

38. The sequence $\left\{\frac{(n-1)(2n+1)}{1-n^2}\right\}_{n=1}^{\infty}$ converges to:

A. $-\infty$ B. $-2$ C. 0 D. 1

***Hint:*** Here: Multiplying the numerator will be:

$$\left\{\frac{(n-1)(2n+1)}{1-n^2}\right\}_{n=1}^{\infty} = \left\{\frac{2-\frac{1}{n}-\frac{1}{n^2}}{\frac{1}{n^2}-1}\right\}_{n=1}^{\infty} \quad \ldots \text{(dividing by } n^2\text{)}$$

As, $n \to \infty$; $\frac{2-0-0}{0-2} = -2$

**Answer: B**



39. Which one of the following represents a geometric sequence?

A. $3, 1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \ldots$ C. $1, 3, 6, 10, 15, \ldots$

B. $\frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}, \frac{1}{6}, \ldots$ D. $-3, 6, -9, 12, -15, \ldots$

***Hint:*** Geometric sequence characterized by common ratio between successive term.

Use: $r = \frac{G_2}{G_1} = \frac{G_3}{G_2} = \frac{G_4}{G_3} = \ldots = \frac{G_n}{G_{n-1}}$

Thus $r = \frac{\frac{1}{3}}{1} = \frac{\frac{1}{9}}{\frac{1}{3}} = \frac{\frac{1}{27}}{\frac{1}{9}} = \ldots = \frac{1}{3}$

**Answer: A**

40. What is the value of the sum: $\sum_{n=1}^{\infty}\left(\frac{2^n + 5^n}{10^n}\right)$?

A. 0.325 B. 1 C. $\frac{5}{4}$ D. $\frac{37}{9}$

***Hint:***

The expression is the same as:

$$\sum_{n=1}^{\infty}\left(\frac{2^n}{10^n}\right) + \sum_{n=1}^{\infty}\left(\frac{5^n}{10^n}\right) = \sum_{n=1}^{\infty}\left(\frac{2}{10}\right)^n + \sum_{n=1}^{\infty}\left(\frac{5}{10}\right)^n$$

(i) $\sum_{n=1}^{\infty}\left(\frac{2}{10}\right)^n = \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \ldots, G_1 = \frac{1}{5}, r = \frac{1}{5}$

$$S_{\infty} = \frac{G_1}{1-r} = \frac{\frac{1}{5}}{1-\frac{1}{5}} = \frac{1}{4}$$

(ii) $\sum_{n=1}^{\infty}\left(\frac{5}{10}\right)^n = \frac{G_1}{1-r} = \frac{\frac{1}{10}}{1-\frac{1}{10}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$

Therefore from (i) and (ii): $\frac{1}{4} + 1 = \frac{5}{4}$

**Answer: C**

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41. The sum of the series $\sum_{n=1}^{\infty}(-1)^n 3^{-2n}$ is equal to _____

A. $-\frac{1}{8}$ B. $-0.13$ C. $-0.1$ D. $\frac{1}{8}$

***Hint***: Here $\sum_{n=1}^{\infty}(-1)^n 3^{-2n} = -\frac{1}{3^2}+\frac{1}{3^4}-\frac{1}{3^6}+\ldots \Rightarrow G_1 = -\frac{1}{9}, r = -\frac{1}{9}$

$$S_\infty = \frac{G_1}{1-r} = \frac{-\frac{1}{9}}{1-\left(-\frac{1}{9}\right)} = \frac{-1}{10} = -0.1$$

**Answer: C**

42. Let $a_n = \left(\frac{n+3}{n+1}\right)^n$, then $\lim_{n\to\infty} a_n$ is equal to:

A. 1 B. $\frac{1}{2}e$ C. $e^2$ D. $\infty$

***Hint***: Use: $\lim_{n\to\infty}\left(1+\frac{a}{kn}\right)^n = e^{\frac{a}{k}}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}\left(\frac{n+3}{n+1}\right)^n = \lim_{n\to\infty}\left(\frac{\frac{n}{n}+\frac{3}{n}}{\frac{n}{n}+\frac{1}{n}}\right)^n$$

$$\Rightarrow \lim_{n\to\infty} a_n = \frac{\lim_{n\to\infty}\left(1+\frac{3}{n}\right)^n}{\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n} = \frac{e^3}{e^1} = e^2$$

**Answer: C**

43. Given that $\lim_{x\to 3} f(x) = 5$ and $\lim_{x\to 3} g(x) = 11$. What is the value of

$$\lim_{x\to 3}\left[\frac{(f(x)-g(x))(g(x)-2f(x))}{(g(x)-(f(x))^2}\right]?$$

A. $\frac{-66}{96}$ B. $\frac{-1}{6}$ C. 0 D. does not exist

***Hint***: The expression is the same as:

$$\lim_{x\to 3}\left[\frac{(f(x)-g(x))(g(x)-2f(x))}{(g(x)-(f(x)(g(x)f(x))}\right] = \lim_{x\to 3}\frac{g(x)-2f(x)}{g(x)-f(x)}(-1)$$



$$\Rightarrow (-1)\left[\frac{\lim_{x\to 3} g(x) - 2\lim_{x\to 3} f(x)}{\lim_{x\to 3} g(x) + \lim_{x\to 3} f(x)}\right] = (-1)\left(\frac{11-2(5)}{11+5}\right) = \frac{-1}{6}$$

**Answer: B**

44. $\lim_{x\to\infty}\left(\frac{x}{x+2}\right)^{-3x}$ is equal to _____

A. $e^6$ B. $e^{-3}$ C. $e^{-3/2}$ D. $e^{-6}$

***Hint***: The expression is the same as:

$$\lim_{x\to\infty}\left(\frac{1}{1+\frac{2}{x}}\right)^{-3x} = \frac{1}{\lim_{x\to\infty}\left(1+\frac{2}{x}\right)^{-3x}} = \frac{1}{(e^2)^{-3}} = e^6$$

45. If $f(x) = x^2\sqrt{2x+12}$, what is the slope of the tangent line to the graph of $f$ at $x = 2$.

A. $-4$ B. 2 C. 18 D. 17

***Hint***: Recall that $\sqrt{x} = x^{1/2}$

$$f(x) = x^2\sqrt{2x+12} = \sqrt{x^4(2x+12)} = \sqrt{2x^5+12x^4}$$

$$f'(x) = \frac{1}{2}(2x^5+12x^4)^{-1/2}10x^4 + 48x^3$$

$$\Rightarrow f'(2) = \frac{1}{2}[2(2)^5+12(2^4)]^{-1/2}[10(2)^4+48(2^3)]$$

$$\Rightarrow f'(2) = \frac{1}{2}[64+192]^{-1/2}(544) = \frac{1}{2}\cdot\frac{544}{\sqrt{256}} = 17$$

46. Suppose, $g(2) = 1$, $g'(2) = 10$ and $f(x) = \frac{x^2}{1+xg(x)}$, then the value of $f'(2)$ is equal to _____

A. $-8$ B. $-\frac{8}{9}$ C. $\frac{4}{3}$ D. $\frac{8}{9}$

***Hint***: Use quotient rule

$$f'(x) = \frac{(x^2)'[1+xg(x)] - x^2(1+xg(x))'}{[1+xg(x)]^2}$$

$$\Rightarrow f'(x) = \frac{2x'[1+xg(x)] - x^2[1'+x'g(x)+xg'(x))]}{[1+xg(x)]^2}$$

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$$f'(x)=\frac{2x[1+xg(x)]-x^2[0+1g(x)+xg'(x)]}{[1+xg(x)]^2}$$

$$f'(2)=\frac{2(2)[1+2g(2)]-(2)^2[0+g(2)+2g'(2)]}{[1+2g(2)]^2}$$

$$f'(2)=\frac{4[1+2(1)]-(4)[0+1+2(10)]}{[1+2(1)]^2}=\frac{12-84}{9}=-8$$

47. Let $f(x)=\begin{cases}\dfrac{a\sin x}{x-|x|}, \text{if } x<0\\ e^{-x}+\cos x, \text{if } x\geq 0\end{cases}$

If f is continuous at $x=0$, then what is the value of $a$?

A. 4 B. 2 C. $\frac{1}{2}$ D. $-4$

***Hint***: Since f is cont. at x = 0

$$\lim_{x\to 0^-} f(x)=f(0)\Leftrightarrow \lim_{x\to 0^-}\frac{a\sin x}{x-|x|}=e^{-0}+\cos(0)$$

$$\lim_{x\to 0^-}\frac{a\sin x}{2x}=1+1=2\Rightarrow\frac{a}{2}=2\Rightarrow a=4$$

48. The derivative of $f(x)=\dfrac{1+\sin x}{\cos x}$ is

A. $\sec x+\tan x$ C. $\dfrac{1}{1+\tan x}$

B. $\dfrac{1+\sin x}{\cos^2 x}$ D. $\dfrac{\cos x}{\sin^2 x}$

***Hint***: $f'(x)=\dfrac{(1+\sin x)'(\cos x)-(1+\sin x)(\cos x)'}{(\cos x)^2}$

$$\Rightarrow f'(x)=\frac{(\cos x)(\cos x)-(1+\sin x)(-\sin x)}{\cos^2 x}$$

$$\Rightarrow f'(x)=\frac{\cos^2 x+\sin x+\sin^2 x+\sin^2 x)}{\cos^2 x}=\frac{1+\sin x}{\cos^2 x}$$

49. If F $(x) = f(2x + 2)$. $g(1 - x^2)$, with $f(2) = -3$, $f'(2) = 4$, $g(1) = -5$, and $g'(1) = 1$, then what is the value of F '(0)?

A. $-40$ B. $-20$ C. 0 D. 19



***Hint***:

F $'(x) = (f(2x + 2))'$. $g(1 - x^2) + f(2x + 2)(g(1 - x^2))'$

F $'(x) = f'(2x + 2)(2)g(1 - x^2) + f(2x + 2)\, g'(x)$

F $'(0) = [f'(2)](2)\, g(1 - 0) + f(2)\, g'(1)[0]$

F $'(0) = 4(2)(-5) + 0 = -40$

**Answer: A**

50. If $f(x) = e^{2x}\sin x$, then $f''(x)$ is _____.

A. $3e^{2x}\sin x - 4e^{2x}\cos x$ C. $e^{2x}(3\sin x + 4\cos x)$

B. $4e^{2x}\sin x + 2e^{2x}\cos x$ D. $e^{2x}(4\sin x - \cos x)$

***Hint***: $f'(x) = (e^{2x})'\sin x + (e^{2x})(\sin x)$

$\Rightarrow f'(x) = 2e^{2x}\sin x + e^{2x}\cos x = e^{2x}(2\sin x + \cos x)$

$\Rightarrow f''(x) = (e^{2x})'(2\sin x + \cos x) + e^{2x}(2\sin x + \cos x)'$

$\Rightarrow f''(x) = 2e^{2x}(2\sin x + \cos x) + e^{2x}(2\cos x - \sin x)$

$= e^{2x}[4\sin x + 2\cos x + 2\cos x - \sin x] = e^{2x}[3\sin x + 4\cos x]$

51. Let $y = \sin(3x^2)$, then $\dfrac{d^2y}{dx^2}$ equal to _____

A. $-6\sin(3x^2)$

B. $\cos(6x) - 6\sin(3x)^2$

C. $6\cos(3x^2) - 36x^2\sin(3x^2)$

D. $x^2\cos(3x^2) + 6\sin(3x^2)$

***Hint***:

$$\frac{dy}{dx}=\frac{d}{dx}[\sin(3x^2)]=\cos(3x^2)\frac{d}{dx}(3x^2)$$

$$\Rightarrow \frac{dy}{dx}=6x\cos(3x^2) \text{ and } \frac{d^2y}{dx^2}=\frac{d}{dx}[6x\cos(3x^2)]$$

$$=\frac{d}{dx}(6x)\cos(3x)^2+\frac{d}{dx}[\cos(3x^2)]$$

$$=6\cos(3x^2)-6x\sin(3x^2)\frac{d}{dx}(3x^2)$$

$$\therefore \frac{d^2y}{dx^2}=6\cos(3x^2)-36x^2\sin(3x^2)$$

**Answer: C**

52. Which one of the following is the set of all critical numbers of $f(x)=\frac{1}{3}x^3-|4x-1|$?

A. $\left\{\frac{1}{4},2\right\}$ B. $\left\{-2,\frac{1}{4},2\right\}$ C. $\{-2, 2\}$ D. $\left\{\frac{1}{4}\right\}$

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**Hint:** $f(x)=\frac{1}{3}x^3-|4x-1|=\begin{cases}\frac{1}{3}x^3-(4x-1) & \text{if } x\geq\frac{1}{4}\\ \frac{1}{3}x^3+4x-1 & \text{if } x<\frac{1}{4}\end{cases}$

$$f'(x)=\begin{cases}x^2-4 & \text{if } x>\frac{1}{4}\\ \nexists & \text{if } x=\frac{1}{4}\\ x^2+4 & \text{if } x<\frac{1}{4}\end{cases}$$

$f'(2)=0$ and $f'(-2)=0$ And $f'_+\left(\frac{1}{4}\right)\neq f'_-\left(\frac{1}{4}\right)$, since $f'\left(\frac{1}{4}\right)$ does not exist. Therefore, the critical numbers are -2, $\frac{1}{4}$ and 2.

**Answer: B**

53. The total cost (in Birr) of producing x iron sheets per day is $C(x) = 1{,}000 + 100x - 0.5x^2$, $0 \leq x \leq 100$. What is the marginal (rate of change of) cost at production level of 80 iron sheet?

A. 8.5 B. 20 C. 1800 D. 5800

**Hint:** $C(x) = 1{,}000 + 100x - 0.5x^2$

The rate of change of cost at x = 80 is equal to $C'(x) = 100 - x$.

At $x = 80$, $C'(80) = 100 - 80 = 20$. **Answer: B**

54. If a box with square base and open top is made from 1200 $cm^2$ material, what is the largest volume of the box in $cm^3$?

A. 4000 B. 8000 C. 15000 D. 3000

**Hint:** Let $x$ = the side of the square

$h$ = height of the box

V = (base area)(height) = $x^2h$

Surface area = (Area of base) + (Area of 4 face)

$\Rightarrow s = x^2 + 4xh = 1200\ cm^2 \Rightarrow x^2 + 4xh = 1200\ cm^2$

$h=\frac{1200-x^2}{4x}$, however

$$v=x^2h=x^2\left(\frac{1200-x^2}{4x}\right)=300x-\frac{x^3}{4}$$



$$v(x)=300x-\frac{x^3}{4}\Rightarrow v'(x)=300-\frac{3x^3}{4}$$

$v'(x) = 0$, when $\frac{3x^3}{4}=300\Rightarrow x^2=400$

$\Rightarrow x = 20$ cm, $h=\frac{1200+400}{80}=10$ cm

$\therefore V_{max}$ = (Base area)(height) = (400)(10) $cm^3$ = 4000 $cm^3$

55. A water tank is a circular cylinder with base radius 2m and height 3m. If the tank is empty and water is pumped into it at rate of $2m^3/min$, how long does it take for the tank to be full?

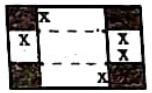
A. 1.5min B. $\frac{3}{2}\pi$ min C. 6πmin D. 12 min

**Hint:** We have: rate =2 $m^3$/min; $V_{cylinder} = \pi r^2 h = \pi(4m^2)(3m) = 12\pi m^3$; Time taken = $\frac{\text{Vol}}{\text{rate}}=\frac{12\pi m^3}{2m^3/min}=6\pi$ min

56. Suppose that equal squares cut from each of the four corners of a square card board whose sides are 72 cm long. The resulting flaps are then folded up to form a box without a top. How long should be each of the four squares that has to be cut off to maximize the volume of the box?

A. 6 cm B. 12 cm C. 15 cm D. 24 cm

**Hint:**




$V(x) = (72-2x)^2(x) = (5184 - 288x + 4x^2)(x)$

$V(x) = 5184x - 288x^2 + 4x^3 \Rightarrow v'(x)$

$= 12x^2 - 576x + 5184$

$V''(x) = 24x - 576$ and $V''(x) = 0$, when $x = 24$ **Answer: D**

57. If $F(x)$ is an ant derivative of $f(x) = 1-\frac{2}{x^2}$ and F (1) = 0 then F (2) is equal to:

A. 0 B. $\frac{1}{2}$ C. $-\frac{1}{2}$ D. 3

**Hint:** $F(x)=\int\left(1-\frac{2}{x^2}\right)dx=\int(1-2x^{-2})dx$

$$\Rightarrow F(x)=x-\frac{2x^{-2+1}}{-2+1}+C=x+\frac{2}{x}+C$$

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$F(1) = 1 + \frac{2}{1} + C = 0 \Rightarrow C = 0 - 3 = -3$

$\Rightarrow F(x) = x + \frac{2}{x} - 3$, therefore $F(2) = 2 + \frac{2}{2} - 3 = 0$ **Answer: A**

58. $\int \frac{x + \ln(x+1)}{(x+1)^2} dx$ is equal to:

A. $\ln(x+1) + \frac{x}{x+1} + C$ C. $(x+1)^2 - \frac{1}{x+1} + C$

B. $(x+1)^2 + \frac{1}{x+1} + C$ D. $\frac{x\ln(x+1)}{x+1} + C$

***Hint***: The expression is the same as:

$\int \frac{x}{(x+1)^2} dx + \int \frac{\ln(x+1)}{(x+1)^2} dx$, let $u = x + 1 \Rightarrow du = dx, x = u - 1$

$\Rightarrow \int \frac{u-1}{u^2} du + \int \frac{\ln u}{u^2} du = \int \frac{1}{u} du - \int \frac{du}{u^2} + \int u^{-2} \ln u du$

$= \ln u + \frac{1}{u} + \int u^{-2} \ln u du \ldots (*)$

Applying integration by part for $\int u^{-2} \ln u du$.

$f(u) = \ln u$ and $g'(u) = u^{-2} du$ $f'(u) = \frac{1}{u}$ and $g(u) = \frac{1}{u}$ Therefore

$\int u^{-2} \ln u du = \frac{-\ln u}{u} + \int \frac{du}{u^2} = \frac{-\ln u}{u} - \frac{1}{u} + C$

Substituting this in (*) would be:

$\ln u + \frac{1}{u} - \frac{\ln u}{u} - \frac{1}{u} + c = \ln u - \frac{\ln u}{u} + C$

$\Rightarrow \frac{u \ln u - \ln u}{u} + C = \frac{(\ln u)(u-1)}{u} + C$

$\Rightarrow \frac{\ln(x+1)(x+1-1)}{x+1} + C = \frac{x\ln(x+1)}{x+1} + C$ **Answer: D**

59. $\int_0^{\pi/2} \frac{x - \sin x}{\sec x} dx$ is equal to:

A. $\frac{\pi - 3}{2}$ B. $\frac{\pi - 1}{2}$ C. $\frac{3 - \pi}{2}$ D. $\frac{\pi + 3}{2}$



***Hint***: The expression is the same as: $\int_0^{\pi/2} x \cos x \, dx - \int_0^{\pi/2} \sin x \cos x \, dx$

**Part 1: Evaluating** $\int_0^{\pi/2} x \cos x \, dx$ **by parts.**

Let $u = x$, $dv = \cos x \, dx$; $du = 1$; $v = \sin x$ $\| \boxed{\int u dv = uv - \int v du}$

$x \sin x - \int \sin x dx = x \sin x + \cos x \Big|_0^{\pi/2}$

$\Rightarrow \frac{\pi}{2} \sin \frac{\pi}{2} + \cos \frac{\pi}{2} - (0 + \cos 0) = \frac{\pi}{2} + 0 - 1 = \frac{\pi}{2} - 1$

**Part (2): Evaluating** $\int_0^{\pi/2} \sin x \cos x \, dx$ **by subtraction.**

Let $u = \sin x \Rightarrow du = \cos x \, dx$. $\int u du = \frac{u}{2} + c = \frac{\sin x}{2} \Big|_0^{\pi/2} = \frac{1}{2}$

From part (1) and part (2): $\frac{\pi}{2} - 1 - \frac{1}{2} = \frac{\pi - 3}{2}$. **Answer: A**

60. What is the derivate of $f(x) = \int_0^{x^2 + \pi} \frac{dt}{\sin t + 1}$?

A. $\frac{\cos x}{\sin(x^2 + \pi) + 1}$ C. $\frac{2x \cos x}{\sin(x^2 + \pi) + 1}$

B. $\frac{2x}{\sin(x^2 + \pi) + 1}$ D. $\int_0^{2x} \frac{dt}{\sin t + 1}$

***Hint***: Note: $\frac{d}{dx} \int_{h(x)}^{g(x)} f(t) dt = f(g(x)) g'(x) - f(h(x)) h'(x)$

Let $f(t) = \frac{1}{\sin t + 1}$, $g(x) = x^2 + \pi$, $h(x) = 0$ and $g'(x) = 2x$.

$f(g(x)).g'(x) - f(h(x))h'(x)$

$\frac{1}{\sin g(x) + 1} . 2x - \frac{1}{\sin 0 + 1} . 0 = \frac{2x}{\sin(x^2 + \pi) + 1}$ **Answer: B**

61. What is the area of the region between the graphs of $y = -x^2 + 2$ and $y = |x|$ where $-1 \le x \le 2$?

A. $\frac{11}{6}$ B. $\frac{25}{6}$ C. $\frac{7}{3}$ D. $\frac{11}{3}$

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***Hint***: The intersection of $y = -x^2 + 2$ and $y = |x|$ is $x = -1$ and $x = 1$. The area of the shaded region would be:

$$A = \int_{-1}^{1}[-x^2+2-(-x)]dx + \int_{0}^{1}(-x^2+2-x)dx + \int_{1}^{2}(x+x^2-2)dx$$

$$= \frac{-x^3}{3} + 2x + \frac{x^2}{2}\Big|_{-1}^{0} + \frac{-x^3}{3} + 2x - \frac{-x^2}{2}\Big|_{0}^{1} + \frac{-x^2}{2} + \frac{-x^3}{2} - 2x\Big|_{1}^{2}$$

$$= (0+0+0) - \left(\frac{1}{3} - 2 + \frac{1}{2}\right) + \left(\frac{-1}{3} + 2 - \frac{1}{2} - 0\right) + \left(\frac{4}{2} + \frac{8}{3} - 4 - \frac{1}{2} - \frac{1}{3} + 2\right)$$

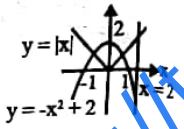
$$= \frac{5}{3} - \frac{3}{2} + 4 = \frac{25}{6}$$

**Answer: B**

62. What is the area of the region between the graph of $f(x) = -x^2 + 4x - 3$ and the $x$ – axis from $x = 0$ to $x = 3$?

A. $\frac{-2}{3}$ B. $\frac{2}{3}$ C. $\frac{4}{3}$ D. $\frac{8}{3}$

***Hint***: The area of the bounded region is




$$A = \int_{0}^{3} f(x)dx = \int_{0}^{1} f(x)dx + \int_{1}^{3} f(x)dx$$

Let $A = |A_1| + |A_2|$

i) $A_1 = \int_{0}^{1}[0-(-x^2+4x-3)dx = \int_{0}^{1} x^2 - 4x + 3)dx$

$$= \frac{x^3}{3} - \frac{4x^2}{2} + 3x\Big|_{0}^{1} = \frac{1}{3} - 2 + 3 - 0 = \frac{4}{3}$$

ii) $A_2 = \int_{1}^{3}(-x^2+4x-3)dx = \frac{-x^3}{3} + \frac{4x^2}{2} - 3x\Big|_{1}^{3} = -9 + 18 - 9 - \left(-\frac{1}{3} + 2 - 3\right) = \frac{4}{3}$ Therefore, $A = A_1 + A_2 = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$

**Answer: D**



1. Which of the following functions is a non-to-one correspondence?

A. $f: \mathbb{R} \to \mathbb{R}, f(x) = \tan x$, where $\mathbb{R}$ is the domain of $f$

B. $g: \mathbb{R} \to \mathbb{R}, g(x) = 2^x$

C. $h: [0,\infty) \to [0,\infty), h(x) = x^2$

D. $r: [0,\infty) \to [0,\infty), r(x) = x + 5$

***Explanation:***

(i) $h: [0,\infty) \to [0,\infty)$ ← Range of $h = [0, \infty)$

(ii) Range of $h(x) = x^2$ is $[0, \infty)$

Range of $h$ in first set (i) is equal to range of $h(x)$ in 2nd set. (ii) → One-to-one correspondence exists between two sets h: $[0, \infty)$ and $h(x) = x^2$

**Answer: C**

**Exercise:** For what value of K:

a) $g: \mathbb{R} \to K, g(x) = 2^x$ is one-to-one correspondence

b) $r: [0,\infty) \to K, r(x) = x + 5$ is one-to-one correspondence?

2. Different codes, each of which consisting of five characters, are to be generated in such a way that the first two characters are any of the English capital letters (A to Z) and the remaining three are any of the digits (0, 1,…,9). How many distinct codes can be generated so?

A. 468,000 B. 260 C. 676,000 D. 25!× !10

***Explanation:*** English letters A to Z are 26 and the digit are 10, then:

→ The first two characters are $26 \times 26 = 676$

→ The next three characters are $10 \times 10 \times 10 = 1000$

→ Using multiplication rule we can generate $676 \times 1000 = 676{,}00$

**Answer: C**

3. A certain meeting hall has 20 rows of seats. There are 20 seats in the first row, 22 seats in the second row, 24 seats in the third row, and so on. How many seats are there on the last ($20^{th}$) row of the hall?

A. 46 B. 58 C. 760 D. 5240

***Explanation:*** $A_1 = 20, A_2 = 22, A_3 = 24, A_{20} = ?$

$A_n = A_1 + (n-1)d \Leftrightarrow A_{20} = 20 + (20-1)(2)$

$\Rightarrow A_{20} = 20 + 38 = 58$

**Answer: B**

4. Two perpendicular lines $\ell_1$ and $\ell_2$ are intersecting at (-1, 2). If the angle of inclination of $\ell_1$ is 45°, then what is the equation of $\ell_2$?

A. $y = -x + 3$ C. $y = -x + 1$

B. $y = x + 3$ D. $y = x + 1$

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***Explanation:*** slope of $\ell_1 = \tan 45° = 1$ and

slope of $\ell_2 = \tan 135° = -1$

$\therefore$ The equation of $\ell_2 : \dfrac{y-2}{x-(-1)} = -1$

$\Rightarrow y - 2 = -(x+1)$ therefore, $\ell_2 : y = -x + 1$ **Answer: C**

5. What is the area of the region between the graph of $y = \sin x$ and $x$-axis where $0 \le x \le 2\pi$?

A. 4 B. $4\pi$ C. 2 D. $2\pi$

***Explanation:***

The intersection of $x$-axis ($y = 0$) and $y = \sin x$ on $[0, 2\pi]$

→ $\sin x = 0 \Rightarrow x = 0, \pi, 2\pi$

The area bounded by $x$-axis and $y = \sin x$ is

$$A = \int_0^{\pi} (\sin x - 0)\,dx + \int_{\pi}^{2\pi} (0 - \sin x)\,dx =$$

$$\Rightarrow A = -\cos x\Big|_0^{\pi} + \cos x\Big|_{\pi}^{2\pi} = -(-1-1) + (-1(-1)) = 4$$

**Answer: A**

6. The following is set of data representing the average mark of 13 students: 91, 89, 93, 91, 87, 94, 92, 85, 91, 90, 96, 93, 89. Then which one of the following statements is true about the data?

A. The median is 90.5.
B. The upper quartile is 92.
C. The range of the marks is 11.
D. The mean is 91.5.

***Explanation:***

→ To find the median arrange the data from least to greatest. Median of a data is the "middle" value of data

85, 87, 89, 89, 90, 91 **91** 91, 92, 93, 93, 94, 96

$\therefore$ Median = 91

→ The median of the upper part of a data is called upper quartile

$(Q_3) = 91, 92, \boxed{93, 93,}\ 94, 96$

$$\therefore Q_3 = \frac{93+93}{2} = \frac{186}{2} = 93 \leftarrow \text{upper quartile.}$$

→ Range = upper extreme – lower extreme = 96 – 85 = 11

→ $\text{Mean} = \dfrac{\text{sum of all data}}{\text{number}} = \dfrac{1181}{13} = 90.85$

**Answer: C**



**Exercise:** What is the value of lower quartile

7. If $f(x) = \ell n\left(\sqrt{x^2+1}\right)$, which of the following is equal to $f'(x)$?

A. $\dfrac{x}{\sqrt{x^2+1}}$ C. $\dfrac{2x}{\sqrt{x^2+1}}$

B. $\dfrac{x}{x^2+1}$ D. $\dfrac{2x}{x^2+1}$

***Explanation:*** Let $f(x) = \ln\sqrt{g(x)}$

Recall that $\left(\sqrt{g(x)}\right)' = \dfrac{g'(x)}{2\sqrt{g(x)}}$ and $(\ln x)' = \dfrac{1}{x}$

$$f'(x) = \frac{1}{\sqrt{x^2+1}} \cdot \left(\frac{2x}{2\sqrt{x^2+1}}\right) = \frac{x}{x^2+1}$$

→ since $\left(\sqrt{x^2+1}\right)' = \dfrac{2x}{2\sqrt{x^2+1}}$

**Answer: B**

8. If $z = \dfrac{3+i}{i-2}$ is a given complex number, then what is the conjugate, $\overline{z}$, of z?

A. $\overline{z} = \dfrac{-3+i}{2+i}$ C. $\overline{z} = -6-2i$

B. $\overline{z} = \dfrac{i-3}{2-i}$ D. $\overline{z} = -1-i$

***Explanation:*** $\overline{z} = \dfrac{3-i}{-2-i} = \dfrac{(3-i)}{-(2+i)} = \dfrac{-3+i}{2+i}$

9. What is the principal argument of $(5+5i)^{11}$?

A. $\dfrac{\pi}{2}$ B. $\dfrac{2\pi}{3}$ C. $\dfrac{\pi}{4}$ D. $\dfrac{3\pi}{4}$

***Explanation:*** Let Z = 5 + 5i,

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$$Arg(z)=\tan^{-1}\left(\frac{5}{5}\right)=\tan^{-1}(1)=\frac{\pi}{4}$$

$$\therefore Arg(z)^{11}=11\left(Arg(z)\right)=11\left(\frac{\pi}{4}\right)=\frac{11\pi}{4}$$

$$\therefore \text{Principal } Arg(z)=\frac{11\pi}{4}-2\pi=\frac{3\pi}{4}$$

**Answer: D**

10. An object is moving along the parabola $y=\sqrt{2x}$ in $xy$-plane. At what point on its path does the object becomes closest to the point (2, 0)?

A. $(3,\sqrt{6})$ C. $(1,\sqrt{2})$

B. (1, 1) D. (2, 2)

***Explanation:*** Let $d$ be the closest distance between the point (2, 0) and $(x, y)$ on the parabola $y=\sqrt{2x}$.

$$\therefore d=\sqrt{(x-2)^2+(y-0)^2}\Rightarrow d(x)=\sqrt{(x-2)^2+\left(\sqrt{2x}\right)^2}$$

$$d(x)=\sqrt{x^2-4x+4+2x}=\sqrt{x^2-2x+4}$$

Let $g(x)=x^2-2x+4\Rightarrow g'(x)=2x-2$

$$g'(x)=0\Rightarrow 2x-2=0\Rightarrow x=1$$

$\therefore$ The closest point will be when $x=1$; $(1,\sqrt{2})$ **Answer: C**

11. For arbitrary propositions $p$ and $q$, which one of the following is a valid equivalence?

A. $\neg(p\Rightarrow q)\equiv(p\Rightarrow q)$

B. $[\neg(p\Rightarrow q)\wedge p]\equiv(p\wedge\neg q)$

C. $[p\vee\neg q]\equiv(p\Rightarrow q)$

D. $[(p\vee q)q\Rightarrow p]\equiv(p\Rightarrow\neg q)$

***Explanation:*** A. $\neg(p\Rightarrow q)\equiv(q\Rightarrow p)$

$\neg[\neg p\vee q]\equiv\neg q\vee p$

$p\wedge\neg q\neq p\vee\neg p$

B. $\neg[(p\Rightarrow q)\wedge p]\equiv p\wedge\neg q$

$\neg(\neg p\vee q)\wedge p$



$(p\wedge\neg q)\wedge p$

$(p\wedge q)\wedge\neg q\equiv p\wedge\neg q$

**Answer: B**

12. Which one of the following is to $\lim_{x\to 9}\frac{x-9}{3-\sqrt{x}}$?

A. 6 B. −3 C. −6 D. ∞

***Explanation:*** $\lim_{x\to 9}\frac{x-9}{3-\sqrt{x}}=\lim_{x\to 9}\frac{x-9}{3-\sqrt{x}}\cdot\frac{3+\sqrt{x}}{3+\sqrt{x}}$

$$\lim_{x\to 9}\frac{(x-9)\left(\sqrt{x}+3\right)}{9-x}=-\lim_{x\to 9}\left(\sqrt{x}+3\right)=-6$$

**Answer: C**

13. The ozone level (in ppb – parts per billion) on a sunny day in a metropolitan area is given by the formula; $p(t)=80+12t-t^2$, where $t$ is time in hours and $t=0$ corresponds to 9 A.M. What is the rate of increase of the ozone level after 3 hours (i.e. at 12 A.M.)?

A. 6 ppb C. 107 ppb

B. 12 ppb D. 113 ppb

**Explanation:** $p(t)=80+12t-t^2$

$p'(t)=12-2t$ and $p'(3)=12-2(3)=6$

**Answer: A**

14. Suppose that $p$ represents the statement "*He missed the tournament*", $q$ represents the statement "*He got the gold medal.*" And $r$ represents the statement "*He took a trip abroad.*". Then which one of the following symbolic expression represents the statement: "If he takes a trip abroad and he does not miss the tournament, then he will get the gold medal."?

A. $(r\Rightarrow q)\vee\neg p$ C. $(r\Rightarrow q)\wedge\neg p$

B. $r\wedge(p\Rightarrow q)$ D. $\neg(r\vee p)\vee q$

**Answer: C**

15. Let $f(x)=\begin{cases}a\dfrac{\sin 2x}{x}, & \text{if } x<0\\ e^{2x}-2, & \text{if } x\geq 0\end{cases}$

If $f$ is continuous at $x=0$, then what is the value of $a$?

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A. $\frac{1}{2}$ B. 2 C. $-\frac{1}{2}$ D. $-2$

**Solution:** If $f$ is cont. at $x = 0$, then

$$\lim_{x\to 0} f(x) = f(0) \Leftrightarrow \lim_{x\to 0^-} f(x) = f(0)$$

$$\therefore \lim_{x\to 0} a\frac{\sin 2x}{x} = e^{2(0)} - 2 \Rightarrow 2a = 1 - 2 = -1$$

$$a = -\frac{1}{2}$$

**Answer: C**

16. Which one of the following is an arithmetic sequence?
   A. 3, 5, 7, 9, 11, …
   B. 3, 6, 12, 24, 48,…
   C. −3, 6, −9, 12, −15,…
   D. 1, 3, 6, 10, 15, 21,…

**Answer: A**

17. If $f(x) = 2x(x^2+1)^4$, then which of the following is an anti derivative of $f(x)$?

A. $\frac{2x}{5}(x^2+1)^5 + c$ C. $\frac{x}{5}(x^2+1)^5 + 1$

B. $\frac{2}{5}(x^2+1)^5 + c$ D. $\frac{1}{5}(x^2+1)^5 - 1$

**Solution:** $\int 2x(x^2+1)^4 dx$; Let $U = x^2+1 \Rightarrow dU$

$$\therefore \int U^4 dU = \frac{U^5}{5} + c = \frac{(x^2+1)^5}{5} + c$$

**Answer: D**

18. Which one of the following sequence is a convergent sequence?

A. $1, \frac{1}{2}, 1, \frac{1}{3}, 1, \ldots$ C. $\left\{10^{109} - \frac{1}{100}n\right\}_{n=1}^{\infty}$

B. $\{(-1)^n\}_{n=1}^{\infty}$ D. $\left\{\sin\left(\frac{1}{n}\right)\right\}_{n=1}^{\infty}$

**Solution:** A sequence $\{a_n\}$ is convergent if $\lim_{n\to\infty} f(a_n)$ exist

$$\therefore \lim_{n\to\infty} \sin\left(\frac{1}{n}\right) = 0$$

**Answer: D**



19. What is the absolute maximum value of $f(x) = 2x^2 - x^4 - 4$ on $[0, 2]$?

A. -3 B. 3 C. -4 D. 12

**Solution:** $f'(x) = 4x - 4x^3 = 4x(1 - x^2)$

$f'(x) = 0, \Rightarrow x = 0, x = 1$, but $x = -1 \notin [0,2]$

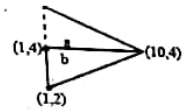
Critical numbers are 0 and 1

$f(0) = -4$, $f(1) = -3$ and $f(2) = -12$

$\therefore f(1) = -3 \leftarrow$ Absolute maximum.

**Answer: A**

20. The equation of an ellipse with center at (1, 4), vertices at (10, 4) and (1, 2) is:




A. $4(x-1)^2 + 81(y-4)^2 = 324$

B. $(x-1)^2 + 9(y-4)^2 = 4$

C. $9(x-1)^2 + 4(y-4)^2 = 1$

D. $2(x-1)^2 + 9(y-4)^2 = 4$

**Solution:** Center (1, 4)

a = 10 − 1 = 9 and b = 4 − 2 = 2, (h, k) = (1, 4)

$$\therefore \frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1 \Leftrightarrow \frac{(x-1)^2}{81} + \frac{(y-4)^2}{4} = 1$$

$$324\left(\frac{(x-1)^2}{81} + \frac{(y-4)^2}{4}\right) = 324 \Leftrightarrow 4(x-1)^2 + 81(y-4)^2 = 324$$

**Answer: A**

21. If $p(x) = 3x^2$ and $q(x) = x^2 + x$, then what is solution set of $\frac{p(x)}{3q(x)} - \frac{1}{x} = \frac{1}{q(x)}$?

A. $\{-1, 2\}$ B. $\{2\}$ C. $\{-1, -2\}$ D. $\{-3\}$

**Solution:** Multiplying both side by $q(x)$, get:

$$\Rightarrow \frac{p(x)}{3} - \frac{q(x)}{x} = 1 \Leftrightarrow \frac{3x^2}{3} - \left(\frac{x^2+x}{x}\right) = 1$$

$$\Rightarrow x^2 - (x+1) = 1 \Leftrightarrow x^2 - x - 1 - 1 = 0$$

$$\Rightarrow x^2 - x - 2 = 0 \Leftrightarrow (x-2)(x+1) = 0$$

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$\Rightarrow x=2$ and $x=-1$, but $q(-1)=0$

$\therefore$ The solution set is $\{2\}$ only.

Answer: B

22. What are the values of $u$ and $v$ that satisfy the equation: $\frac{u+3i}{4-2i}=\frac{2+vi}{20}$?

A. $u=2, v=3$ C. $u=2, v=16$

B. $u=-6, v=10$ D. $u=-4, v=6$

***Solution:*** Multiplying both side by $q(x)$

$$\frac{u+3i}{4-2i}=\frac{2+vi}{20}\Leftrightarrow\frac{(u+3i)(4+2i)}{(4-2i)(4+2i)}=\frac{2+vi}{20}$$

$$\Rightarrow\frac{4u+2ui+12i-6}{16+4}=\frac{2+vi}{20}$$

$$\Rightarrow(4u-6)+(2u+12)i=2+vi$$

$\therefore u=\frac{8}{4}=2$ and $v=2(2)+12=16$ **Answer: C**

23. What is the sum of all multiples of 3 between 20 and 200?

A. 7,227 C. 6,150

B. 6,570 D. 5,166

***Solution:***

Multiple of 3 between 20 and 200 are 21, 24, 27, 30, ..., 198

$$A_n=A_1+(n-1)d\Leftrightarrow 198=21+(n-1)(3)\Rightarrow n=\frac{180}{3}=60$$

$$S=\frac{60}{2}(21+198)=6,570$$ **Answer: B**

24. The inverse of the function defined by $g(x)=\frac{2x}{x+3}$ is equal to

A. $g^{-1}(x)=\frac{2x}{x-3}$ C. $g^{-1}(x)=-\frac{x+3}{2x}$

B. $g^{-1}(x)=-\frac{3x}{x-2}$ D. $g^{-1}(x)=-\frac{x+2}{3x}$

***Solution:*** $g(x)=\frac{2x}{x+3}\Leftrightarrow y=\frac{2x}{x+3}$ interchange $\boxed{\begin{matrix}x\to y\\ y\to x\end{matrix}}$



$\Rightarrow x=\frac{2y}{y+3}\Leftrightarrow xy+3x=2y$, solving for $y$

$\Rightarrow 3x=2y-xy$ implies $y(2-x)=3x\Rightarrow y=\frac{3x}{2-x}=\frac{-3x}{x-2}$

$\therefore g^{-1}(x)=-\frac{3x}{x-2}$ **Answer: B**

25. If $A=\begin{pmatrix}0 & x & 0\\ 1 & -1 & 1\\ 0 & y & -1\end{pmatrix}$ and $A^{-1}=\begin{pmatrix}1 & 1 & 1\\ 3 & 0 & 0\\ 2 & 0 & -1\end{pmatrix}$, then what are the values of $x$ and $y$?

A. $x=3, y=-2$ C. $x=-3, y=2$

B. $x=\frac{2}{3}, y=\frac{1}{3}$ D. $x=\frac{1}{3}, y=\frac{2}{3}$

***Solution:*** We know that $A.A^{-1}=I$

$$AA^{-1}=\begin{pmatrix}0 & x & 0\\ 1 & -1 & 1\\ 0 & y & -1\end{pmatrix}\begin{pmatrix}1 & 1 & 1\\ 3 & 0 & 0\\ 2 & 0 & -1\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$$

$$AA^{-1}=\begin{pmatrix}3x & 0 & 0\\ 0 & 1 & 0\\ 3y-2 & 0 & 1\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\Rightarrow 3x=1 \text{ and } 3y-2=0$$

$x=\frac{1}{3}$ and $y=\frac{2}{3}$ **Answer: D**

26. If $f(2)=-3$, $f'(2)=4$, $g(1)=-5$, $g'(1)=1$ and $F(x)=f(2x+2).g(1-x^2)$, then what is the value of $F'(0)$?

A. 19 B. 0 C. –20 D. –40

***Solution:*** Here at

$$F'(x)=f'(2x+2)(2)g(1-x^2)+f(2x+2)g'(1-x^2)\cdot(-2x)$$

$$\therefore F'(0)=f'(0+2)(2)g(1-0)+f(2)\cdot g'(1)(0)$$

$$=f'(2)(2)g(1)+f(2)g'(1)(0)$$

$$=(4)(2)(-5)+(-3)(1)(0)=-40+0=-40$$

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27. A ladder 6 m long rests against a vertical wall. If the bottom of the ladder slides away form the wall at a rate (speed) of $\frac{1}{2}$ m/sec, how fast is the angle between the top of the ladder and the wall changing when the angle is $\frac{\pi}{4}$ rad?

A. $\frac{\sqrt{2}}{12}$ rad / seo
C. $\frac{\sqrt{2}}{6}$ rad / sec
B. $\frac{\sqrt{2}}{2}$ rad / sec
D. $\frac{\sqrt{2}}{3}$ rad / sec

***Solution:***

$\sin\theta = \frac{x}{6}$ and $\frac{dx}{dt} = \frac{1}{2}$ m / sec

$$\frac{d}{dt}(\sin\theta) = \frac{d}{dt}\left(\frac{x}{6}\right) \Leftrightarrow \cos\theta.\frac{d\theta}{dt} = \frac{1}{6}\frac{dx}{dt}$$

$$\frac{d\theta}{dt} = \frac{1}{6}\frac{dx}{dt}\cdot\frac{1}{\cos\theta} = \frac{1}{6}\cdot\frac{1}{2}\cdot\frac{1}{\cos\theta\frac{\pi}{4}} = \frac{1}{6\sqrt{2}}$$

$$\therefore \frac{d\theta}{dt} = \frac{1}{6\sqrt{2}} = \frac{1}{6\sqrt{2}}\cdot\frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{12}\text{rad/sec}$$

**Answer: A**

28. The value (s) of $x$ where the graph of the function $y = \frac{x^2-1}{x^3}$ cross its horizontal asymptote is (are);

A. $x = -2$
B. $x = -1$ and $x = 1$
C. $x = 1$
D. $x = -\sqrt{2}$ and $x = 1 + \sqrt{2}$

***Solution:*** The horizontal asymptote is $y = 0$, so set $\frac{x^2-1}{x^3} = 0$

$\Rightarrow x^2 - 1 = 0$ at $x = 1$ and $x = -1$

$\therefore$ The graph crosses its horizontal asymptote at $x = 1$ and $x = -1$

**Answer: B**



29. If $A = \begin{pmatrix} 3 & -2 & 8 \\ 0 & 6 & 7 \\ 0 & 4 & 5 \end{pmatrix}$, then $\det(A^T A)$ is equal to

A. 12
B. 36
C. 30
D. 15

***Solution:***

$\det(A^T A) = \det(A^T)\det(A)$

det(A) by expanding first column:

$$\det(A) = 3\begin{vmatrix} 6 & 7 \\ 4 & 5 \end{vmatrix} - 0\begin{vmatrix} -2 & 8 \\ 4 & 5 \end{vmatrix} + 0\begin{vmatrix} -2 & 8 \\ 6 & 7 \end{vmatrix}$$

$$= 3(30 - 28) - 0 + 0 = 6$$

However, $\det(A) = \det(A^T)$

$\therefore \det(A^T A) = \det(A^T)\det(A) = 6 \times 6 = 36$

**Answer: B**

30. Suppose that a function $f$ has the property that $f(x+y) = f(x)f(y)$ for all values of $x$ and $y$ and that $f(0) = 2$, $f'(0) = 1$. Then which one of the following represents the formula for the derivative $f'(x)$?

A. $f'(x) = 2f'(x) + 1$
B. $f'(x) = f(x) + 2f'(x)$
C. $f'(x) = f(x) + 2$
D. $f'(x) = 2f(x) - 1$

***Solution:*** $f(x+y) = f(x).f(y)$ Let $y = 0$

$f(x+0) = f(x)f(0)$

$\Rightarrow f'(x) = f(x)f'(0) + f'(x)f(0)$

$\Rightarrow f'(x) = f(x)(1) + f'(x)(2)$

$\therefore f'(x) = f(x) + 2f'(x)$

**Answer: B**

31. A ball is thrown vertically from ground up to a height of 16m. Each time it drops h meters, it rebounds 0.80h m. Nothing that the ball travels every height of h twice. What is the total vertical distance travelled by the ball before it comes to rest?

A. 40m
B. 80m
C. 160m
D. 320m

***Solution:*** $h = 16\text{m}, r = 0.8$

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If ball is thrown vertically upward:

→ Upward series = $16 + (0.8) + 16 + (0.8)^2 16 + \ldots$

→ Downward series = $16 + (0.8) + 16 + (0.8)^2 16 + \ldots$

∴ Total vertical distance = sum of the two series

$$= 2\left(\frac{G_1}{1-r}\right) = \frac{2(16)}{1-0.8} = \frac{32}{0.2} = 160\text{m}.$$

**Answer:** C

**Exercise:** From Q #31, if a ball is dropped from the same height 16m and it rebounds 0.8m. what is the total vertical distance before comes to rest.

32. Suppose that the first 3 letters (A, B, and C) and number digits are to be used to form car plates in small town. How many different plates can be formed in total that contain 1, 2 or 3 letters and then followed by 3 digits?

A. 3,000 C. 39,000

B. 27,000 D. 100,000

***Solution:*** The car plate has the form:

| i | ii | iii | iv |
|---|---|---|---|

→ in the (i) place the digit can be $0-9 = 10$ way

→ in the (ii) place the digit can be $0-9 = 10$ way

→ in the (iii) place the digit can be $0-9 = 10$ way

→ in the (iv) place either A, B or C = 3 way

∴ Using multiplication rule, the car plates can be $3 \times 10 \times 10 \times 10 = 3{,}000$

**Answer:** A

33. The derivative of the function $F(x) = \int_{-x}^{x} \frac{1}{1+t}\,dt$ ?

A. $\frac{2}{1-x^2}$ C. $\frac{1}{1+x}$

B. $\ln|1+x|$ D. $\ln\left|\frac{1+x}{1-x}\right|$

***Explanation:*** Fundamental theorem of calculus II.

i) $F(x) = \int_a^{g(x)} f(t)$ then $F'(x) = f\left(g(x)\right)\cdot g'(x)$

$$\Rightarrow F(x) = \int_{-x}^{x} \frac{1}{1+t}dt = \int_{-x}^{0} \frac{1}{1+t}dt + \int_{0}^{x} \frac{1}{1+t}dt.$$



$$\Rightarrow F(x) = -\int_0^{-x} \frac{1}{1+t}dt + \int_0^x \frac{1}{1+t}dt$$

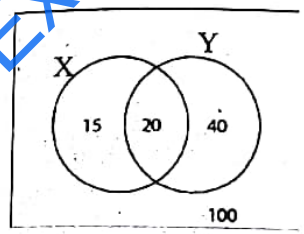
$$F'(x) = -\left(\frac{1}{1-x}(-1)\right) + \frac{1}{1+x}\cdot 1$$

$$= \frac{1}{1-x} + \frac{1}{1+x} = \frac{1+x+1-x}{1-x^2} = \frac{2}{1-x^2}$$

**Answer:** A

34. A city has two daily newspapers. X and Y. The following information was obtained from a survey of 100 residents of the city: 35 people subscribe to X, 60 people subscribe to Y and 20 subscribe to both newspapers. Then how many of the people in the survey do not subscribe to either of the newspapers?

A. 5 B. 25 C. 40 D. 55




***Explanation:***
People in the survey do not subscribe to either of the newspapers.
$= 100 \quad (15 + 20 + 40) = 25$

35. Which one of the following is necessarily true?

A. If $f'(x) = 0$ for all x in an interval $I$, then $f(x) = 0$ for all $x$ in 1.

B. If $f(x) = x^2 \sin x + 5$, then there is $c \in (0, \pi)$ such that $f'(c) = 0$.

C. If $f(x) = e^x + x^2$ is increasing on $(-\infty, \infty)$.

D. If $f'(c) = 0$, then $f$ attains its maximum or minimum value at $x = c$.

***Explanation:*** From Roll's theorem $f(0) = 5$ and $f(\pi) = 0 + 5 = 5$

Since $f(0) = f(\pi)$, then there is a number $c \in (0, \pi)$ such that $f'(c) = 0$

**Exercise:**

a) If $f'(x) = 0$ for all x in interval, I, then what can you conclude about $f(x)$?

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b) On what interval $f(x)=e^x+x^2$ is increasing and decreasing.

**Answer:**
a) $f$ is constant for all $x$.
b) increasing on $[0,\infty)$ and decreasing on $(-\infty,0]$

36. What is the focus of the parabola $y^2+4y+8x=4$?

A. $(1,-2)$ C. $(3,-2)$
B. $(-1,-2)$ D. $(-3,-2)$

***Solution:*** $y^2+4y=-8x+4$

$y^2+4y+4=-8x+4+4=-8x+8$

$(y+2)^2=-8(x-1)\rightarrow(h,k)=(1,-2)$

$\therefore 4p=8\Rightarrow p=2$, and

Focus is $(h,-p,k)=(1-2,-2)=(-1,-2)$

**Answer: B**

37. Which one of the following is equal to $\int\frac{\ell n x+x^2e^x}{x}dx$?

A. $\frac{1}{2}\ell n^2x+e^x(x^2-1)+c$

B. $\frac{1}{2}\ell n^2x+e^x(x-1)+c$

C. $\frac{1}{x^2}\ell nx+e^x(x-1)+c$

D. $\frac{-1}{x^2}\ell nx+e^x(x^2-1)+c$

***Solution:*** $\int\left(\frac{\ell n x+x^2e^x}{x}\right)dx=\int\frac{\ell n x}{x}dx+\int(xe^xdx)$

**First :** $\int\frac{\ell n x}{x}dx$ let $U=\ell nx\Rightarrow du=\frac{1}{x}dx$

$\Rightarrow\int UdU=\frac{U^2}{2}+c=\frac{(\ell nx)^2}{2}+c$

**second :** $\int xe^xdx$, Let $U=x$, $dV=e^xdx$

$dU=1dx$ $V=e^x$



$\therefore\int UdV=Uv-\int VdU=xe^x-\int e^xdx=xe^x-e^x$

Therefore, $\int\left(\frac{\ell n x^2+x^2e^x}{x}\right)dx=\frac{(\ell nx)^2}{2}+xe^x-e^x+c$

38. For what values of $a$ and $b$ is the function $f(x)=\begin{cases}1-3x^2, & \text{for } x\le 1\\ ax+b, & \text{for } x>1\end{cases}$ differentiable at $x=1$?

**Answer: B**

A. $a=6, b=0$
B. $a=-3, b=1$
C. $a=0, b=-2$
D. $a=-6, b=4$

***Explanation:***
Here, differentiable at $x=1$
$f'_+(1)=f'_-(a)$ and $f(x)=ax+b$
$\Rightarrow f'(x)=a$, for $x>1$
Also, $f(x)=1-3x^2\Rightarrow f'(x)=-6x$, for $x<1$
$\Rightarrow f'_+(1)=a=-6=f'_-(1)$
Differentiability implies continuity
$\therefore\lim_{x\to1}f(x)=f(1)$
$\Rightarrow\lim_{x\to1}(ax+b)=1-3(1)^2=-2$
$\Rightarrow a+b=-2$, but $a=-6$
$-6+b=-2\Rightarrow b=-2+6=4$

**Answer: D**

39. Consider the following system of equations: $\begin{cases}ax+by=2\\ x+3y+2z=0\\ 2x+y+z=0\end{cases}$

If the determinant of the coefficient matrix is 2, then what is the solution set of the system?

A. $\{(1,3,-5)\}$ C. $\{(-2,-6,10)\}$
B. $\left\{\left(\frac{1}{a},\frac{1}{b},0\right)\right\}$ D. $\varnothing$

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**Solution:** Here at

$$D=\begin{vmatrix}a & b & 0\\ 1 & 3 & 2\\ 2 & 1 & 1\end{vmatrix}=2 \text{ coefficient of matrix}$$

$$\Rightarrow x=\frac{D_x}{D}=\frac{\begin{vmatrix}2 & b & 0\\ 0 & 3 & 2\\ 0 & 1 & 1\end{vmatrix}}{2}=\frac{2\begin{vmatrix}3 & 2\\ 1 & 1\end{vmatrix}-b\begin{vmatrix}0 & 2\\ 0 & 1\end{vmatrix}+0}{2}=\frac{2}{2}=1$$

$$\Rightarrow y=\frac{D_y}{D}=\frac{\begin{vmatrix}a & 2 & 0\\ 1 & 0 & 2\\ 2 & 0 & 1\end{vmatrix}}{2}=\frac{a\begin{vmatrix}0 & 2\\ 0 & 1\end{vmatrix}-2\begin{vmatrix}1 & 2\\ 2 & 1\end{vmatrix}+0}{2}=-\frac{6}{2}=3$$

$$z=\frac{D_z}{D}=\frac{2(1-6)}{2}=-\frac{10}{2}=-5$$

$$\therefore S.S=\{(1,3,-5)\}$$

**Answer: A**

40. If $f$ and $g$ are continuous on $\mathbb{R}$ and $a, b \in \mathbb{R}$, which one of the following is necessarily true?

A. If $\int_a^b f(x)dx=\int_a^b g(x)dx$, then $f(x)=g(x)$ for all $x\in[a,b]$.

B. If $f'(x)=g'(x)$ for all $x\in[a,b]$, then $\int_a^b f(x)dx=\int_a^b g(x)dx$.

C. $f(x)\geq 3$ for all $x\in[-2,2]$, then $\int_{-2}^{2} f(x)dx\geq 12$.

D. $\int_a^b f(x)dx=\int_b^a f(x)dx$.

**Answer: A**



41. Which of the following functions could most likely be drawn as in the Figure 1 below?

A. $\frac{x+3}{x+2}$

B. $\frac{x^2-2x}{x^2-4}$

C. $\frac{-x^2-x+6}{x^2-4}$

D. $\frac{x^2+x-6}{x^2-4}$

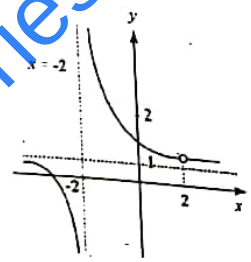


Figure 1

**Solution:** $\frac{x^2+x-6}{x^2-4}=\frac{(x+3)(x-2)}{(x-2)(x+2)}=\frac{x+3}{x+2}$

Domain $=\mathbb{R}\setminus\{-2,2\}$

- Vertical asymptote is only $x=-2$
- The graph has hole at $x=2$
- Horizontal asymptote is $y=1$,

**Answer: D**

42. A measurement is grouped into five class intervals with the following frequency distribution.

| Class Interval | 5–15 | 15–25 | 25–35 | 35–45 | 45–55 |
|---|---|---|---|---|---|
| Frequency | 22 | 40 | 68 | 50 | 204 |

What are the first quartile $Q_1$ and the 75th percentile $p_{75}$ of the measurement?

A. $Q_1=20, P_{75}=40$ C. $Q_1=20, P_{75}=39$

B. $Q_1=22, P_{75}=40$ D. $Q_1=22, P_{75}=39$

**Solution:**

| Class Interval | 5–15 | 15–25 | 25–35 | 35–45 | 45–55 |
|---|---|---|---|---|---|
| Frequency | 22 | 40 | 68 | 50 | 20 |
| Cumulative frequency | 22 | 62 | 130 | 180 | 200 |

First quartile class $=\left(\frac{1\times 200}{4}\right)$ item $=50^{th}$ item

$\rightarrow$ $50^{th}$ item lies on $2^{nd}$ class $=15-25$

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$\Rightarrow B_L = 15, \frac{n}{4} = 50, cf_b = 22, f_c = 40$, and $i = 25 - 15 = 10$

$$\therefore Q_K = B_L + \frac{\left(\frac{k \times n}{4} - cf_b\right) i}{f_c} \text{ where, } K = 1,\ n = 200$$

$$\therefore Q_1 = 15 + \left(\frac{50 - 22}{40}\right) \times 10 = 22$$

**Exercise:** What is the value of $Q_3$?

Percentile, $Pt = B_L + \left[\frac{tn}{100} - cf_b\right] i$, where $t = 75, n = 200$

$$P_{75} \text{ class} = \left(\frac{t \times n}{100}\right)^{\text{item}} = \left(\frac{75 \times 200}{100}\right)^{\text{item}} = \left(150^{\text{th}}\right)\text{item}$$

$\rightarrow$ $150^{\text{th}}$ item lies on $4^{\text{th}}$ class $= 35 - 45$

$$B_L = 35; \frac{tn}{100} = 150,\ cf_b = 130, f_c = 50,\ i = 10$$

$$\therefore P_{75} = 35 + \left[\frac{150 - 130}{50}\right] \times 10 = 39$$

**Exercise:** calculate

a) $P_{25}$, $P_{50}$ and $P_{90}$.

**Answer: D**

43. The number of shoes $s$ that a factory can produce per day is a function of the number of hours $t$ it operates:

$$s(t) = 40t \text{ for } 0 \le t \le 12.$$

The daily cost $c$, in Birr, to manufacture $s$ shoes is given by the function

$$c(s) = 0.1s^2 + 90s + 800.$$

If the factory operates for 10 hours, what is the cost it incurs in producing as much shoes it can within this time?

A. Birr 400 C. Birr 52,800

B. Birr 1,600 D. Birr 124,600

***Solution:*** $S(10) = 40(10) = 400$

$$C(400) = (0.1)(400)^2 + 90(400) + 800$$
$$= 16{,}000 + 36{,}000 + 800$$
$$= 52{,}800,$$

**Answer: C**



44. Three persons $P_1$, $P_2$ and $P_3$ are firing at a target independently, and have a probability 0.7, 0.5 and 0.4, respectively of hitting the target. What is the probability that at least one of them hits the target?

A. 0.95 B. 0.85 C. 0.91 D. 0.99

***Solution:*** Here.

$P(P_1) = 0.7 \Rightarrow P(\text{not } P_1) = 1 - 0.7 = 0.3$

$P(P_2) = 0.5 \Rightarrow P(\text{not } P_2) = 1 - 0.5 = 0.5$

$P(P_3) = 0.4 \Rightarrow P(\text{not } P_3) = 1 - 0.4 = 0.6$, then

$P$(at least one of them hit)

$= 1 - P$ (none)

$= 1 - P$ (not $P_1$) $\times P$ (not $P_2$) $\times P$ (not $P_3$)

$= 1 - P(0.3 \times 0.5 \times 0.6)$

$= 1 - 0.09 = 0.91$

**Answer: C**

45. A company manufactures $x$ computer sets per month. The monthly marginal profit (in Birr) is given by:

$$P'(x) = 165 - 0.1x, \text{ for } 0 \le x \le 400.$$

The company is currently manufacturing 10 sets of computers per month, but is planning to increase production. What is the total change in the monthly profit if the monthly production increased to 60 sets?

A. Birr 500 C. Birr 8,075

B. Birr 1,865 D. Birr 18,635.

***Explanation:*** Marginal-profit is the derivative of the profit function.

$$P(x) = \int P'(x)dx = \int (165 - 0.1x)dx = 165x - 0.1\frac{x^2}{2} + c$$

$\therefore$ Change in profit $= P(60) - P(10)$

$$= \left[165(60) - 0.1\left(\frac{60^2}{2}\right)\right] - \left[165(60) - \frac{0.1}{2}(10^2)\right]$$

$$= 9720 - 1645 = 8075$$

**Answer: C**

46. Which one of the following is equal to $\lim_{x \to \infty}\left(\frac{3x}{3x+2}\right)^{-3x}$?

A. $e^2$ B. $e^{-3}$ C. $e^{-2}$ D. $e^3$

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**Solution:** $\lim_{x\to\infty}\left(\frac{3x}{3x+2}\right)^{-3x} = \lim_{x\to\infty}\left(\frac{1}{1+\frac{2}{3x}}\right)^{-3x}$

$$= \frac{1}{\lim_{x\to\infty}\left(\frac{1}{1+\frac{2}{3x}}\right)^{-3x}} = \frac{1}{\left(e^{\frac{2}{3}}\right)^{-3}} = \frac{1}{e^{-2}} = e^2$$

**Answer: A**

47. The total cost (in birr) of producing $x$ radio sets per day is given by the expression $\frac{1}{4}x^2+35x+25$ and the price per set at which they may be sold is given by $50-\frac{1}{2}x$. What should be the daily output to obtain a maximum total profit?

A. 50 sets per day C. 10 sets per day
B. 23 sets per day D. 7 sets per day

***Explanation:*** Profit ($P$) = sale – total cost

$$= (x)(P) - \text{cost}$$

$$\Rightarrow P = x\left(50-\frac{1}{2}x\right) - \left(\frac{1}{4}x^2+35x+25\right)$$

$$= 50x - \frac{1}{2}x^2 - \frac{1}{4}x^2 - 35x - 25$$

$$P(x) = 15x - \frac{3}{4}x^2 - 25$$

For maximizing profit:

$$\frac{dP}{dx} = 0 \Rightarrow 15 - \frac{3x}{2} = 0$$

$$\therefore \frac{3}{2}x = 15 \Rightarrow x = \frac{2\times 15}{3} = 10$$

**Answer: C**



48. Let $f(x) = \begin{cases} 3-e^{2x} & \text{if } x < 0.5 \\ \frac{2^x-5}{x+1}, & \text{if } x \geq 0.5 \end{cases}$. If $c$ is a zero of $f$, that is, $f(c) = 0$, then which one of the following intervals must contain $c$?

A. $(-\infty, 0]$ B. $[0, 1]$ C. $[1, 2]$ D. $[2, 3]$

***Solution:***

For $x < 0.5$ let $x = 0$

$f(0) = 3 - e^0 = 3 - 1 = 2 \leftarrow$ Positive

For $x \geq 0.5$, let $x = 1$

$f(1) = \frac{2^1-5}{2+1} = \frac{2-5}{3} = \frac{-3}{3} = -1 \leftarrow$ negative

$\therefore$ We can find $c \in (0,1)$ such that $f(c) = 0$

**Answer: B**

49. What is the solution set of the system $\begin{cases} x+y-z=1 \\ x+2y-3z=1 \\ 2x+3y-4z=2 \end{cases}$ ?

A. $\{(0,2,1)\}$ C. $\{(2k+1,-k,k)k \in \mathbb{R}\}$
B. $\{(1-k,2k,k)k \in \mathbb{R}\}$ D. $\varnothing$

***Solution:*** Consider the system:

$$\begin{cases} x+y-z=1 & \ldots\ldots \text{A} \\ x+2y-3z=1 & \ldots\ldots \text{B} \\ 2x+3y-4z=2 & \ldots\ldots \text{C} \end{cases}$$

We choose to add equation (A) and (B) $+\begin{cases} x+y-z=1 \\ x+2y-3z=1 \end{cases}$

$$2x+3y-4z=2 \ldots (*)$$

Since equation (C) and (*) are the same, therefore, the system must be **dependent**, provided that the system has infinitely many solutions:

→ From equation A solve for $x$:

$$x+y-z=1 \Rightarrow x=1-y+z \ldots\ldots (**)$$

Now replace $x$ be $1-y+z$ in equation B.

$$\Rightarrow (1-y+z)+2y-3z=1$$

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$y-2z=0$, hence $\boxed{y=2z.}$

Replace $y=2z$ in (**)

$\Rightarrow x=1-y+z=1-(2z)+z=1-z$

$\therefore x=1-z$

Thus, for any real number $z=k\in\mathbb{R}$

$(x,y,z)=(1-z,2z,z)=\{(1-k,2k,k)/k\in\mathbb{R}\}$

Answer: B

50. Which one of the following is NOT a tautology?

A. $[p\vee(q\Rightarrow r)]\Rightarrow[\neg p](q\Rightarrow r)$

B. $p\vee(q\Rightarrow\neg p)$

C. $p\Rightarrow(p\Rightarrow q)\vee q$

D. $[p\Rightarrow(q\wedge\neg r)]\Leftrightarrow[\neg p\Rightarrow(q\neg\vee r)]$

***Explanation:*** At autology is true value for any possible combination.

Recall that $a\Rightarrow b\equiv\neg a\vee b$

A. $[p\vee(q\Rightarrow r)]\Rightarrow[\neg p](q\Rightarrow r)$

$p\vee(\neg q\vee r)\Rightarrow[p\vee(\neg q\vee r)\leftarrow$ tautology

B. $p\vee(q\Rightarrow\neg p)\equiv p\vee(\neg q\vee\neg p)$

$=(p\vee\neg p)\vee\neg q$

$=T\vee\neg q$

T $\leftarrow$ tautology

C. $p\Rightarrow(p\Rightarrow q)\vee q$

$p\Rightarrow(\neg p\vee q)\vee q$

$p\Rightarrow(\neg p\vee(q\vee q)$

$p\Rightarrow(\neg p\vee q)$

$\neg p\vee(\neg p\vee q)$

$\neg p\vee q\leftarrow$ not tautology

Answer: C



51. Which one of the following is true about a conic section represented by the equation $\frac{x^2}{k}+\frac{y^2}{k-9}=1$?

A. It is a circle whose center is at origin for some $k\in\mathbb{R}$.

B. It is an ellipse whose major axis is vertical when $k>9$.

C. It is a hyperbola whose foci are at (-3, 0) and (3$k$, 0) when $0<k<9$.

D. It is a hyperbola whose foci are at (-3$k$, 0) and (3$k$, 0) when $0<k<9$.

***Solution:*** $\frac{x^2}{k}+\frac{y^2}{k-9}=1$

A. It cannot be a circle, because $k\neq k-9$

B. When $k>9$, the expression will be horizontal ellipse because $k>k-9$

C. For $0<k<9$, the equation becomes

$\frac{x^2}{k}+\frac{y^2}{k-9}=1\leftarrow$ hyperbola

Now, $a^2=k$ and $b^2=9-k$

$c^2=a^2+b^2=k+9-k=9\Rightarrow c=\pm3$

$\therefore$ The foci are (-3, 0) and (3, 0)

**Answer: C**

52. Two ships, one with angle of depression 60° due east and the other with 30° due west are observed from a plane 1,000 meters above a sea. If the two ships are on the same line, what is the distance between the two ships?

[*you may use the values:*

$\sin 30°=\cos 60°=\frac{1}{2}$ *and* $\sin 60°=\cos 30°=32.$]

A. $\frac{4{,}000}{\sqrt{3}}$m

B. 2,000 m

C. $\sqrt{3}500$m

D. $\frac{1}{\sqrt{3}}600$m

***Solution:*** Here at,

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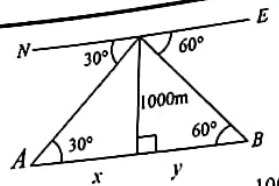


$$\tan 30° = \frac{1000}{x} \text{ and } \tan 60° = \frac{1000}{y}$$

$$\Rightarrow x = \frac{1000}{\tan 30°} = 1000\sqrt{3}\text{m and } y = \frac{1000}{\tan 60°} = \frac{1000\text{m}}{\sqrt{3}}$$

- The distance between A and B $= x + y$

$$= 1000\sqrt{3} + \frac{1000}{\sqrt{3}} = 1000\left(\sqrt{3} + \frac{1}{\sqrt{3}}\right) = \frac{4000}{\sqrt{3}}\text{m}.$$

**Answer: A**

53. What is the image of the ellipse $(x-1)^2 + 4y^2 = 1$ under the translation that takes (1, 1) to (0, 2) followed by the reflection through the $x$-axis?

A. $x^2 + 4(y-1)^2 = 1$ C. $x^2 + 4(y+1)^2 = 1$

B. $4x^2 + (y-1)^2 = 1$ D. $4x^2 + (y+1)^2 = 1$

***Solution:*** Translating vector (v) $= (0-1, 2-1) = (-1, 1)$
When you translating the center of an ellipse (1, 0)

$$(1,0) \xrightarrow{v(-1,1)} (0,1)$$

When you reflect (0, 1) on $x$-axis is (0, −1)
Since translation is parallel movement, hence the given ellipse is horizontal.

$\therefore$ the image become $x^2 + 4(y+1)^2 = 1$

54. What is the amplitude and period, respectively, of the graph of $f(x) = -6\sin x.\cos x$?

**Answer: C**

A. $3, \pi$ C. $3, \frac{\pi}{2}$

B. $6, \pi$ D. $6, 2\pi$

***Solution:*** Recall that: $\sin 2x = 2\sin x \cos x$



$$\therefore -3\sin(2x) = -6\sin x \cos x$$

$$\Rightarrow \text{Amplitude} = |-3| = 3 \text{ and Period} = \frac{2\pi}{2} = \pi$$

**Answer: A**

55. What is the work done (in Joules) when a force of 50 Newton is used to pull a crate 20 meters along a level path if the force is at an angle of 60°?
[Remember that: Work = (Force) × (distance travelled).]

A. 360 B. 500 C. 760 D. $1500\sqrt{2}$

***Solution:*** $W = F \cdot S = |F||S|\cos\theta$

$$\therefore W = (50N)(20)\cos 60° = 500J.$$

**Answer: B**

56. Suppose $\ell$ is the line through the center of the sphere $x^2 + y^2 + (z-2)^2 = 9$ and the intersects sphere at (1, 2, 4). What is the cosine of the angle between $\ell$ and positive $z$-axis?

A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{3}{5}$ D. $\frac{4}{5}$

***Solution:*** Center of sphere $c$ (0, 0, 2) point on the sphere $P$(1, 2, 4) are on $\ell$.

→ Position vector $\overline{CP} = (1-0, 2-0, 4-2) = (1,2,2)$

→ Point on $z$-axis R = (0, 0, $z$).

Angle between $\ell$ and $z$-axis.

$$\cos\theta = \frac{\overline{CP} \cdot \overline{R}}{|\overline{CP}||\overline{R}|} = \frac{(1,2,2)\cdot(0,0,z)}{\sqrt{1^2+2^2+2^2}\cdot\sqrt{0^2+0^2+z^2}}$$

$$\therefore \cos\theta = \frac{0+0+2z}{\sqrt{1+4+4}\cdot\sqrt{z^2}} = \frac{2z}{\sqrt{9}\cdot\sqrt{z^2}} = \frac{2}{3}.$$

**Answer: A**

57. Suppose that the equation $x^2 + y^2 + z^2 + 2x + Bz = 6(y+1)$, represents a sphere. Where is the point (1, -1, 4) located relative to the sphere?

A. Inside the sphere
B. On the sphere
C. At the center of the sphere
D. Outside the sphere

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**Solution:** Here at,

$x^2+y^2+z^2+2x+8z-6y=6$

$\Rightarrow (x^2+2x+1)+(y^2-6y+9)+z^2+8z+16$

$=6+1+9+16$

$\Rightarrow (x+1)^2+(y-3)^2(z+4)^2=32$; Replace the

Point $(1,-1,4):(1+1)^2+(-1-3)^2+(4+4)^2=84>32$

$\therefore$ the point (1, -1, 4) lies outside the sphere

**Answer: D**

58. If $P=(3,\alpha-1,\alpha+2)$ and $Q=(2\alpha+1,3,3\alpha)$ are points in space, what should be the value (s) of $\alpha$ so that the distance between the two points is 6?

A. $\alpha=-2$ or $\alpha=5$ C. $\alpha=-1$ or $\alpha=3$

B. $\alpha=0$ or $\alpha=5$ D. $\alpha=-3$ or $\alpha=2$

**Solution:** Here distance $PQ=d=6$

$\Rightarrow d=\sqrt{(2\alpha+1-3)^2+(3-(\alpha-1))^2+(3\alpha-\alpha-2)^2}=6$

$\Rightarrow \sqrt{(2\alpha-2)^2+(4-\alpha)^2+(2\alpha-2)^2}=6$

$\Rightarrow \sqrt{(4\alpha^2-8\alpha+4)+(16-8\alpha-\alpha^2)+(4\alpha^2-8\alpha+4)}=6$

Squaring both side:

$9\alpha^2-24\alpha+24=36 \Leftrightarrow 3\alpha^2-8\alpha+8=12$

$3\alpha^2-8\alpha-4=0$

$\therefore \alpha=\frac{8\pm\sqrt{64+48}}{6}=\frac{8\pm4\sqrt{7}}{6}=\frac{4\pm2\sqrt{7}}{3}$

59. If (−1, 2, 2) and (1, 0, -2) are endpoints of a diameter of a sphere, then which one of the following is true about the sphere?

A. (0, 1, 0) is a point on the sphere

B. The equation of the sphere is $x^2+(y-1)^2+z^2=6$

C. The equation of the sphere is $x^2+(y-2)^2+z^2=6$

D. The radius of the sphere is 6.

**Solution:** Center $=\left(\frac{1+(-1)}{2},\frac{2+0}{2},\frac{-2+2}{2}\right)=(0,1,0)$



radius $r=\sqrt{(0-1)^2+(1-0)^2(0+2)^2}=6$

Equation: $x^2+(y-1)^2+z^2=6$ **Answer: B**

60. What is the possible value of $x$ that solves the equation: $\sin^{-1}x+\cos^{-1}\left(\frac{3}{5}\right)=\pi$?

A. $\frac{\pi}{3}$ B. $\frac{3}{5}$ C. $\frac{5\pi}{2}$ D. $\frac{4}{5}$

**Solution:** $\sin^{-1}x+\cos^{-1}\left(\frac{3}{5}\right)=\pi$

$\Rightarrow \sin^{-1}x=\pi-\cos^{-1}\left(\frac{3}{5}\right)$

$\Rightarrow x=\sin\left(\pi-\cos^{-1}\left(\frac{3}{5}\right)\right)$

$\Rightarrow x=\sin\pi\cos\left(\cos^{-1}\left(\frac{3}{5}\right)\right)-\cos\pi\sin\left(\cos^{-1}\left(\frac{3}{5}\right)\right)$

$\Rightarrow x=0-(-1)\sin\left(\cos^{-1}\left(\frac{3}{5}\right)\right)$, Let $\cos^{-1}\left(\frac{3}{5}\right)=\theta$

$\Rightarrow \cos\theta=\frac{3}{5}$, therefore $\sin\theta=\frac{4}{5}$

$\therefore x=\frac{4}{5}$ **Answer: D**

61. If $\vec{A}=4\vec{i}-3\vec{j}$ and $\vec{u}$ is a unit vector such that $\left|\vec{A}+\vec{u}\right|^2=27$, then the cosine of the angle between $\vec{A}$ and $\vec{u}$ is equal to ___.

A. 0.1 B. 0.2 C. 0.3 D. 0.4

**Solution:** $\cos\theta=\frac{\vec{A}\cdot\vec{U}}{\left|\vec{A}\right|\left|\vec{U}\right|}$, Recall that

$\left|\vec{A}+\vec{U}\right|^2=\left(\vec{A}+\vec{U}\right)\cdot\left(\vec{A}+\vec{U}\right)=27=|A|^2+2A\cdot U+|U|^2=27$

$\Rightarrow \sqrt{4^2+3^2}^2+2A\cdot U+1=27$

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$$\therefore \vec{A}\cdot\vec{U} = \frac{27-26}{2} = \frac{1}{2}$$

$$\cos\theta = \frac{\vec{A}\cdot\vec{U}}{\|\vec{A}\|\|\vec{U}\|} = \frac{\frac{1}{2}}{\sqrt{25}\cdot 1} = \frac{1}{2\times 5} = \frac{1}{10} = 0.1$$

**Answer: A**

62. If each the compound propositions $P \vee Q, P \Rightarrow R$ and is $\neg R$ is True, then which one of the following is True?

A. $P$ B. $Q$ C. $Q \Rightarrow P$ D. $P \wedge \neg R$

***Solution:*** $\neg R \equiv T, R = F$

i) $P \Rightarrow R \equiv T$

$P \Rightarrow F \equiv T, P \equiv F$

ii) $P \vee Q \equiv T$ and $P \equiv F$

$F \vee Q \equiv T$, therefore, $Q \equiv T$

**Answer: B**

63. If $A = (1,-2), B = (-3,2)$ and $\vec{V}$ is a position vector such $2\vec{V} + \overrightarrow{AB} = \vec{0}$, then $\vec{V}$ is equal to _____.

A. (2, 0) B. (−1, 0) C. (−2, 2) D. (2, −2)

***Solution:*** $2\vec{V} + \overrightarrow{AB} = \vec{0}$

$2\vec{V} = -\overrightarrow{AB} \Rightarrow 2\vec{V} = \overrightarrow{BA} = (1-(-3), -2-2) = (4,-4)$

$\therefore V = (2,-2)$

**Answer: D**

64. If angle $\theta$ is an acute angle of a right triangle, what is the length of the side adjacent to $\theta$, given that the hypotenuse has 6 units length $\sec\theta = 10/3$?

A. 1.8 units C. 18 units

B. 2 units D. 20 units

***Solution:*** $\sec\theta = \frac{10}{3} \Rightarrow \cos\theta = \frac{3}{10}$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenus}} = \frac{x}{6} = \frac{3}{10}$$

$$\therefore x = 6 = \left(\frac{3}{10}\right) = \frac{18}{10} = 1.8 \text{ units.}$$

**Answer: A**



1. If the truth value of a proposition $p$ is false, then which one of the following compound proposition has a truth value true?

A. $\neg p \wedge p$ C. $\neg(\neg p \vee p)$

B. $\neg p \Rightarrow p$ D. $p \Rightarrow \neg p$

***Hint:*** $\neg p \Rightarrow p \equiv \neg(\neg p) \vee p \equiv p \vee p \equiv F \vee F \equiv F$

- $\neg(\neg p \vee p) \equiv p \wedge \neg p \equiv F \wedge T \equiv F$
- $p \Rightarrow \neg p \equiv \neg p \vee p \equiv T \vee F \equiv T$

**Answer: D**

2. What is the contra positive of "If $x \in \mathbb{N}$, then $x$ is integer and $x > 0$."?

A. If $x$ is not integer or $x < 0$, then $x \notin \mathbb{N}$.

B. If $x$ is integer or $x > 0$, then $x \in \mathbb{N}$.

C. If $x$ is not integer or $x \leq 0$, then $x \notin \mathbb{N}$.

D. If $x \notin \mathbb{N}$, then $x$ is not integer and $x \leq 0$.

***Hint:*** The contra positive of "If $x \in \mathbb{N}$, then $x$ is integer and $x > 0$."? Let $p \equiv x \in \mathbb{N}$, $q \equiv$ is an integer, $r \equiv x > 0$.

- The equivalent statement: $p \Rightarrow (q \wedge r)$
  The contra positive of $p \Rightarrow (q \wedge r)$ is $\neg(q \wedge r) \Rightarrow \neg p$
  $\equiv (\neg q \vee \neg r) \Rightarrow \neg p$, the same as
- $\neg p \equiv x \notin \mathbb{N}$, $\neg r \equiv x \leq 0$.

If $x$ is not integer or $x \leq 0$ then $x \notin \mathbb{N}$. **Answer: C**

3. Which one of the following compound propositions is a tautology?

A. $(q \vee \neg q) \Rightarrow p$ C. $p \vee (q \wedge \neg q)$

B. $p \Rightarrow (q \vee \neg q)$ D. $p \Rightarrow (q \wedge \neg q)$

***Hint:***

A. $(q \vee \neg q) \Rightarrow p \equiv T \Rightarrow p \equiv T$

B. $p \Rightarrow (q \vee \neg q) \equiv p \Rightarrow T \equiv T$....(tautology)

C. $p \vee (q \wedge \neg q) \equiv p \vee F \equiv F$

D. $p \Rightarrow (q \wedge \neg q) \equiv p \Rightarrow F$

**Answer: B**

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4. The following is a simple frequency distribution of a data with variable $X$.

| X | 3 | 5 | 6 | 7 |
|---|---|---|---|---|
| Frequency | 2 | 5 | 2 | 1 |

What are the mean $(\bar{x})$ and variance $\delta^2$ of the data?

A. $\bar{x}=5,\ \delta^2=0.7$
B. $\bar{x}=6,\ \delta^2=1.4$
C. $\bar{x}=6,\ \delta^2=0.7$
D. $\bar{x}=5,\ \delta^2=1.4$

***Hint:***

- Mean $(\bar{x})=\dfrac{3\times2+5\times5+6\times2+7\times1}{2+5+2+1}=\dfrac{50}{10}=5$
- Variance $=\dfrac{2(3-5)^2+5(5-5)^2+2(6-5)^2+1(7-5)^2}{10}$

$$=\frac{14}{10}=1.4$$

**Answer: D**

5. Which one of the following is a one-to-one correspondence function from A = [0, 1] to B = [1, 2]?

A. $f(x)=x$
B. $f(x)\ \frac{1}{3}x^3+1$
C. $f(x)=2x+1$
D. $f(x)=x^2+1$

***Hint:*** Suppose $f: A\to B$
$\Rightarrow f:[0,1]\to[1,2]$ defined by $f(x)=x^2+1$ is one to one correspondence because, $f(0)=0^2+1=1$ and $f(1)=1^2+1=2$

**Answer: D**

6. What is the solution set of $\dfrac{1-\frac{1}{x}}{1-\frac{1}{x^2}}=3x^2-\dfrac{x}{1+\frac{1}{x}}$?

A. $\left\{-1,\frac{1}{3}\right\}$ B. $\left\{\frac{1}{3}\right\}$ C. $\left\{3,\frac{-1}{3}\right\}$ D. $\left\{\frac{-1}{3}\right\}$

***Solution:*** Simplifying each side:



$$\frac{1-\frac{1}{x}}{1-\frac{1}{x^2}}=\frac{1-\frac{1}{x}}{\left(1-\frac{1}{x}\right)\left(1+\frac{1}{x}\right)}=\frac{1}{1+\frac{1}{x}}=\frac{x}{x+1}\ldots(*)$$

and

$$3x^2-\frac{x}{1+\frac{1}{x}}=3x^2-\frac{x^2}{x+1}=x^2\left(3-\frac{1}{x+1}\right)\ldots(**)$$

From (*) and (**) we have

$$\frac{x}{x+1}=x^2\left(3-\frac{1}{x+1}\right)\Rightarrow\frac{1}{x+1}=x\left(3-\frac{1}{x+1}\right)$$

$$\Rightarrow\frac{1}{x+1}=3x-\frac{x}{x+1}\Leftrightarrow 3x=\frac{1}{x+1}+\frac{x}{x+1}$$

$$\Rightarrow 3x=\frac{x+1}{x+1}=1\Rightarrow x=\frac{1}{3}$$

**Answer: B**

7. Suppose $f(x)=\dfrac{Q(x)}{x\left(x^2-1\right)}$ where $Q(x)$ is a quadratic function. Which one of the following is necessarily true about the graph of $f$?

A. $x=0$, $x=1$ and $x=-1$ are the vertical asymptotes of the graph of $f$.
B. The graph of f does not interest with its horizontal asymptote.
C. The vertical asymptote of the graph of f is only $x=-1$ if $Q(x)=x^2-x$.
D. The vertical asymptote of the graph of f is only $x=1$ if $Q(x)=2x^2$.

**Answer: C**

8. Which one of the following is equation of a circle whose center is on y-axis and radius is 3?

A. $x^2+y^2+6y=0$
B. $(x-2)^2+y^2=9$
C. $x^2+(y-2)^2=3$
D. $x^2-2x+y^2=8$

***Hint:*** $x^2+y^2+6y=0\Leftrightarrow x^2+y^2+6y+9=0+9$
$x^2+(y+3)^2=9,\Leftrightarrow x^2+(y+3)^2=3^2$

**Answer: A**

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9. The planet mercury's orbit around the sun is an ellipse with eccentricity 0.206, length of the major axis $1.16\times10^8$ km and the sun at one focus. what is the maximum distance from mercury to the sun?

A. $6.99\times10^8$ km  C. $9.66\times10^7$ km
B. $6.99\times10^7$ km  D. $9.66\times10^8$ km

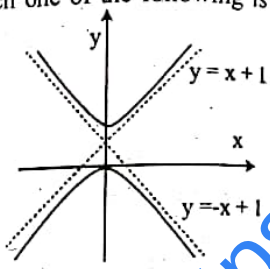
*Hint*: $2a = 1.16\times10^8$ km $\Rightarrow a = 0.58\times10^8$ km $\Rightarrow a = 5.8\times10^7$ km

$e=\frac{c}{a}\Rightarrow c = ea = (0.206)(0.58\times10^8\text{ km}) = \mathbf{1.195\times10^7\ km}$

Max-distance = a + c = $(5.8 + 1.95)\times10^7$ km = $\mathbf{6.99\times10^7\ km}$

**Answer: B**

10. The graph of a hyperbola and the lines of its asymptotes are as shown in the following figure. which one of the following is an equation of the hyperbola?




A. $y^2-2y-x^2=0$
B. $y^2-3y-x^2=0$
C. $x^2-(y-1)^2=1$
D. $(x-1)^2-y^2=1$

*Hint*: To find the center, solve

$\begin{cases} y = x+1 \\ y = -x+1 \end{cases} \Rightarrow x+1 = -x+1 \Rightarrow 2x = 0 \Rightarrow x = 0$ and $y = 0+1 = 1$

Thus, center (h, k)=(0, 1). Also $2a = 2 \Rightarrow a = 1$

Because transverse axis is parallel to y-axis, hence standard equation.

$$\frac{(y-k)^2}{a^2}-\frac{(k-h)^2}{b^2}=1\Leftrightarrow\frac{(y-1)^2}{1}-\frac{(x-0)^2}{1}=1$$

$$\Rightarrow y^2-2y+1-x^2=1\Leftrightarrow y^2-2y-x^2=0$$

$$\Rightarrow y^2-2y+1-x^2=1\Leftrightarrow y^2-2y-x^2=1-1=0$$

**Answer: A**

11. If $\{A_n\}_{n=1}^{\infty}$ is an arithmetic sequence such that its 1st term $A_1=-5$ and its 5th term $A_5=15$, then its 11th term $A_{11}$ is equal to:

A. 40  B. 50  C. 45  D. 55



*Hint*: $d=\frac{A_n-A_m}{n-m}=\frac{A_5-A_1}{5-1}$

$$\Rightarrow d=\frac{15-(-5)}{5-1}=\frac{20}{4}=5.$$

$$A_n = A_1+(n-1)d \Leftrightarrow A_{11} = -5+(11-1)(5) = 45$$

**Answer: C**

12. What is the sum of all multiples of 4 that are between 30 and 301?

A. 12,882  B. 11,288  C. 6,288  D. 6,882

*Hint*: 32, 36, 40…300. Here, $A_1 = 32$, d = 4

$$\Rightarrow A_n = A_1+(n-1)d \Leftrightarrow 300 = 32+(n-1)4$$

$$\Rightarrow n=\frac{300-32+4}{4}=\frac{272}{4}=68.$$

Hence, $S_{68}=\frac{68}{2}[32+300]=\mathbf{11{,}288}$  **Answer: B**

13. The left hand side limit, $\lim\limits_{x\to0^-}\frac{xe^x-|x|}{x}$

A. 0  B. 2  C. 1  D. Does not exist

*Hint*: $\lim\limits_{x\to0^-}\frac{xe^x-|x|}{x}=\lim\limits_{x\to0^-}\left(\frac{xe^x}{x}-\frac{|x|}{x}\right)$

$$=\lim_{x\to0^-}e^x+(1)=e^0+1=1+1=2$$

**Answer: B**

14. Which one of the following is equal to $\lim\limits_{x\to0}\frac{2x+\tan x}{x\sec x}$?

A. 2  B. 0  C. 1  D. 3

*Hint*: $\lim\limits_{x\to0}=\frac{2x+\tan x}{x\sec x}=\lim\limits_{x\to0}\frac{\frac{2x+\tan x}{x}}{\frac{x\sec x}{x}}$

$$=\lim_{x\to0}\left(\frac{2+\frac{\tan x}{x}}{\sec x}\right)=\frac{2+1}{1}=3$$

**Answer: D**

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15. A box contains 10 items of which 3 are defective. If 2 items are randomly taken out of the box, what is the probability that both items are not defective?

A. $\frac{7}{10}$ B. $\frac{4}{7}$ C. $\frac{7}{15}$ D. $\frac{49}{100}$

*Hint:* If 3 are defective then 7 are non-defective.

$$P(\text{both non-defective}) = \frac{C(7,2)}{C(10,2)} = \frac{7\times 6}{10\times 9} = \frac{7}{15}$$

**Answer: C**

16. Items produced by a certain company are subjected to two kinds of defects $D_1$ and $D_2$. Out of the total product, 5% have the defect $D_1$, 10% have the defect $D_2$, 2% have both defects. What is the probability that a randomly both defects. What is the probability that a randomly selected item has neither defect $D_1$ nor defect $D_2$?

A. 0.13 B. 0.5 C. 0.98 D. 0.87

*Hint:* $P(D_1) + P(D_2) - P(D_1 \cap D_2) + x = 100\%$

$\Rightarrow 5\% + 10\% - 2\% + x = 100\%$

$\Rightarrow x = 100\% - 13\% = 87\% = 0.87$

**Answer: D**

17. If $A = \begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix}$ and $B^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, then $(AB)^{-1}$ is equal to

A. $\begin{pmatrix} 4 & -3 \\ 4 & -5 \end{pmatrix}$ C. $\begin{pmatrix} -3 & 11 \\ 1 & -3 \end{pmatrix}$

B. $\begin{pmatrix} -2 & 5 \\ 2 & -4 \end{pmatrix}$ D. $\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$

*Hint:* If $x = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $D = ad - bc \neq 0$ then $x$ is invertible and

$$A^{-1} = \begin{pmatrix} \frac{d}{D} & -\frac{b}{D} \\ -\frac{c}{D} & \frac{a}{D} \end{pmatrix}$$



If $A = \begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix}$ and $D = 6 - 7 = -1$, then

$$A^{-1} = \begin{pmatrix} \frac{3}{-1} & \frac{-7}{-1} \\ \frac{-1}{-1} & \frac{2}{-1} \end{pmatrix} = \begin{pmatrix} -3 & 7 \\ 1 & -2 \end{pmatrix}$$

We have $(AB)^{-1} = B^{-1}A^{-1}$

$$\Rightarrow \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} -3 & 7 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} -3+1 & 7+(-2) \\ 0+2 & 0+(-4) \end{pmatrix} = \begin{pmatrix} -2 & 5 \\ 2 & -4 \end{pmatrix}$$

**Answer: B**

18. Let $A = \begin{pmatrix} 0 & \alpha & \beta \\ 2 & 2 & 1 \\ 3 & 1 & 2 \end{pmatrix}, b = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$ and $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. If det(A) = 3, then what is the solution of the system $AX = b$?

A. $\{(6, -2, -8)^T\}$ C. $\{(3, 1, 4)^T\}$

B. $\left\{\left(0, \frac{1}{a}, \frac{5}{b}\right)^T\right\}$ D. $\varnothing$

*Hint:* $AX = b \Rightarrow (A^{-1}A)X = A^{-1}B$

$X = A^{-1}b$

- Co-factor of $a_{11}$ is $A_{11} = (-1)^{1+1}\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 4 - 1 = 3$
- Co-factor of $a_{12}$ is $A_{12} = -\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} = -(4 - 3) = -1$
- Co-factor of $a_{13}$ is $A_{13} = 2 - 6 = -4$
- Co-factor of $a_{21}$ is $A_{21} = -(2\alpha - \beta) = \beta - 2\alpha$
- Co-factor of $a_{22}$ is $A_{22} = -3\beta$
- Co-factor of $a_{23}$ is $A_{23} = -3\alpha$

Cofactor of $a_{31} = \alpha - 2\beta$,
Cofactor of $a_{32}$ is $A_{32} = -2\beta$
Cofactor of $a_{33}$ is $A_{33} = 2\alpha$.

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Matrix cofactor of $A=\begin{bmatrix} 3 & -1 & -4 \\ \beta-2\alpha & -3\beta & -3\alpha \\ \alpha-2\beta & -3\alpha & 2\alpha \end{bmatrix}$

$\therefore$ adj of $A=\begin{bmatrix} 3 & \beta-2\alpha & \alpha-2\beta \\ -1 & -3\beta & -2\beta \\ -4 & -3\alpha & 2\alpha \end{bmatrix}$

$$\therefore X=A^{-1}b=\left(\frac{1}{|A|}adj.A\right)(b)=\frac{1}{3}\begin{pmatrix} 3 & \beta-2\alpha & \alpha-2\beta \\ -1 & -3\beta & -2\beta \\ -4 & -3\alpha & 2\alpha \end{pmatrix}\begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$$

$X=(6 \quad -2 \quad -8)^T$

**Answer: A**

19. In the set of complex numbers, what is the solution set of $x^2+4x+5=0$?

A. $\varnothing$
B. $\{1-2i, 1+2i\}$
C. $\{2-i, 2+i\}$
D. $\{-2-i, -2+i\}$

***Solution***: $x^2+4x+5=0$

$\Rightarrow x^2+4x+4+5-4=0 \Leftrightarrow (x+2)^2+1=0$

$\Rightarrow (x+2)^2=-1 \Rightarrow x+2=\sqrt{-1}=\pm i$

$\Rightarrow x=-2-i$ or $x=-2+I$

**Answer: D**

20. If $z=(1+i)^{10}$, then which of the following is equal to $z$?

A. $1+32i$ B. $32i$ C. $10i$ D. $1+10i$

***Hint***: Modulus of $1+i$ is $r=\sqrt{1^2+1^2}=\sqrt{2}$

$$\text{Arg}\,(1+i)=\theta=\tan^{-1}\left(\frac{1}{1}\right)=\tan^{-1}(1)=\frac{\pi}{4}$$

$$(1+i)^{10}=\left(\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)\right)^{10}$$

$$\left(\sqrt{2}^{10}\right)\left(\cos\frac{10\pi}{4}+i\sin\frac{10\pi}{4}\right)=32(0+i)=32i$$

**Answer: B**



21. A closed cylindrical can is to be made to hold $1000\text{cm}^3$ of oil. What are the dimensions (radius $r$ and height $h$) that will minimize the total surface area of the can?

A. $r=\frac{\sqrt[3]{50}}{\pi}, h=2\frac{\sqrt[3]{50}}{\pi}$
B. $r=\frac{\sqrt[3]{500}}{\pi}, h=2\frac{\sqrt[3]{500}}{\pi}$
C. $r=\sqrt[3]{\frac{50}{\pi}}, h=2\sqrt[3]{\frac{50}{\pi}}$
D. $r=\sqrt[3]{\frac{500}{\pi}}, h=2\sqrt[3]{\frac{500}{\pi}}$

***Hint***: Volume $=\pi r^2h=1000 \Rightarrow h=\frac{1000}{\pi r^2}$

Surface area: $A=2\pi r^2+2\pi rh$

$$A(r)=2\pi r^2+2\pi r\left(\frac{1000}{\pi r^2}\right)=2\pi r^2+\frac{2000}{r}$$

$$\Rightarrow A'(r)=4\pi r-\frac{2000}{r^2},\ set\ A'(r)=0$$

$$\Rightarrow r^3=\frac{500}{\pi} \Rightarrow r=\sqrt[3]{\frac{500}{\pi}}$$

**Answer: D**

**Extreme Tip 1:** The volume of a closed a cylindrical tin can is $16\pi$ cubic inches. If a minimum amount of tin is to be used to construct the can, what must be the height of the can in inches?

A. $2\sqrt[3]{2}$ B. $2\sqrt[3]{4}$ C. 4 D. 8

***Solution***: Volume $=\pi r^2h=16\pi \Rightarrow h=\frac{16}{r^2}$

Surface area: $A=2\pi r^2+2\pi rh$

$$A(r)=2\pi r^2+2\pi r\left(\frac{16}{r^2}\right)=2\pi r^2+\frac{32\pi}{r}$$

$$A'(r)=4\pi r-\frac{32\pi}{r^2} \longleftarrow \text{set } A'(r)=0$$

$$r^3=\frac{32\pi}{4\pi}=8 \Rightarrow r=\sqrt[3]{8}=2$$

$$h=\frac{16}{r^2}=\frac{16}{4}=4$$

**Answer: C**

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22. If $f(x) = 3x^2\sqrt{x^3-1}$, then which of the following is an anti derivative of $f(x)$?

A. $\frac{3x^2}{2}(x^3-1)^{3/2}+c$

B. $\frac{2}{3}(x^3-1)^{3/2}$

C. $\frac{3}{2}(x^3-1)^{3/2}$

D. $\frac{3}{2}(x^3-1)^{2/3}+c$

**Hint:** Anti derivative of $f(x)$ is $\int f(x)\,dx$

$=\int 3x^2\sqrt{x^3-1}\,dx$, let $U = x^3 - 1$, thus $du = 3x2dx$

$$\Rightarrow \int \sqrt{U}\,dU = \frac{U^{1/2+1}}{\frac{1}{2}+1} = \frac{U^{3/2}}{\frac{3}{2}} + c = \frac{2}{3}U^{3/2} + c$$

$$\Rightarrow \int 3x^2\sqrt{x^3-1}\,dx = \frac{2}{3}(x^3-1)^{3/2} + c$$

Answer: B

**Extreme Tip 2:** The value of $\int \frac{x^2}{e^{x^3}}dx$ _____

**Solution:** Let $U = x^3 \Rightarrow dU = 3x^2dx$

$$\Rightarrow x^2dx = \frac{dU}{3}$$

$$\therefore \int \frac{x^2}{e^{x^3}}dx = \int \frac{dU}{3e^U} = \left(\frac{1}{3}\right)\left(\frac{e^{-U}}{-1}\right) = \frac{-1}{3}e^{-x^3} = \frac{-1}{3e^{x^3}} + c$$

23. Which of the following is equal to $\int \frac{(\ln x)^2 + x^2\cos x}{x}dx$?

A. $\frac{1}{x^2}\ln x + x\sin x - \cos x + c$

B. $\frac{1}{3}(\ln x)^3 + x\sin x - \cos x + c$

C. $\frac{1}{3}(\ln x)^3 + x\sin x + \cos x + c$

D. $\frac{1}{x^2}\ln x + x\sin x + \cos x + c$



**Hint:** $\int \frac{(\ln x)^2 + x^2\cos x}{x}dx$

$$= \int \frac{(\ln x)^2}{x}dx + \int \frac{x^2\cos x}{x}dx$$

**First** $\int \frac{(\ln x)^2\,dx}{x}$, Let $U = \ln x \Rightarrow dU = \frac{dx}{x}$

$$\int U^2 dU = \frac{U^3}{3} + c = \frac{(\ln x)^3}{3} + c$$

**The second part** $\int \frac{x^2\cos x}{x}dx = \int x\cos x\,dx$

Using integration by part

$\int U dv = Uv - \int v dU$

Let $U = x \Rightarrow dU = dx$

$dv = \cos x dx \Rightarrow \int dv = \int \cos x dx \Rightarrow v = \sin x$

$$\therefore \int x\cos x dx = x\sin x - \int \sin x dx$$

$$= x\sin x + \cos x + c$$

$$\therefore \int \frac{(\ln x)^2 + x^2\cos x}{x}dx = \frac{(\ln x)^3}{3} + x\sin x + \cos x + c$$

**Answer: C**

24. The value of the solid generated when the region bounded between the graph of $y = \begin{cases} x^2, & 0 \le x \le 2 \\ 4, & 2 \le x \le 3 \end{cases}$ and x-axis is rotated about the x-axis is:

A. $\frac{32\pi}{5}$ B. $\frac{112\pi}{5}$ C. $\frac{112\pi}{3}$ D. $\frac{64\pi}{5}$

**Hint:** $v_1 = \int_0^2 \pi\left(x^2\right)^2 dx = \pi\left[\frac{x^5}{5}\right]_0^2 = \frac{32\pi}{5}$

$$v_2 = \int_2^3 \pi(4)^2 dx = \pi[16x]_2^3 = 48\pi - 32\pi = 16\pi$$

$$\therefore v = v_1 + v_2 = \frac{32\pi}{5} + 16\pi = \frac{112\pi}{5}$$

**Answer: B**

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**Extreme Tip 3:** The region is enclosed by the graph of $y = x^2$, the line $x = 2$, and the $x$-axis is revolved about the $y$-axis. The volume of solid generated is:

A. $8\pi$ B. $\frac{32\pi}{5}$ C. $\frac{16\pi}{3}$ D. $4\pi$

***Solution:*** Since it is revolved about y-axis.

$\Rightarrow y = x^2 \Rightarrow x = \sqrt{y}$ when $x = 2, y = 4$

$$\therefore v = \int_0^4 \pi f(y)^2\,dy = \int_0^4 \pi\left(\sqrt{y}\right)^2 dy = \pi\left[\frac{y^2}{2}\right]_0^4$$

$$= \pi\left[\frac{16}{2} - 0\right]_0^4 = 8\pi$$

**Answer: A**

25. Which one of the following is true about the horizontal asymptote(s) of the graph of $y = \frac{|x|+2}{x}$?

A. $y = 2$ is the only horizontal asymptote of the graph.
B. $y = 1$ and $y = -1$ are horizontal asymptotes of the graph.
C. $y = 2$ and $y = -2$ are the horizontal asymptotes of the graph.
D. $y = 1$ is the only horizontal asymptote of the graph.

***Hint:*** $y = \frac{|x|-2}{x} = \begin{cases} 1 + \frac{2}{x} & \text{when } x > 0 \\ -1 + \frac{2}{x} & \text{when } x < 0 \end{cases}$

- As $x \to \infty$, $y = 1$ and
- As $x \to -\infty$, $y = -1$

**Answer: A**

26. If $f(x) = 2x^5 - 3x$, then $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1}$ is equal to _____.

A. 1 B. −1 C. 7 D. $\infty$

***Hint:*** $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} = f'(1)$

and $f'(x) = 10x^4 - 3$, so $f'(1) = 10(1)^4 - 3 = 7$

**Answer: C**



27. If $f(x) = e^{3x}\cos x - \frac{x+\pi}{x^2+2}$, then $f'(0)$ is equal to _____.

A. $3 - \frac{\pi}{2}$ B. $\frac{3}{2}$ C. $\frac{7}{4}$ D. $\frac{5}{2}$

**Answer: D**

28. If $f(x) = \ln\left(\sqrt{x^2-5}\right)$, which one of the following is equal to $f'(x)$?

A. $\frac{x}{x^2-5}$ B. $\frac{-x}{\sqrt{x^2-5}}$ C. $\frac{2x}{\sqrt{x^2-5}}$ D. $\frac{-x}{x^2-5}$

***Hint:***

Use $= \left(\ln\left(g(x)\right)\right)' = \left(n \ln g(x)\right)' = \frac{n\,g'(x)}{g(x)}$

$$\therefore \left(\ln\sqrt{x^2-5}\right)' = \left(\frac{1}{2}\ln\left(x^2-5\right)\right)' = \frac{1}{2}\left(\frac{2x}{x^2-5}\right) = \frac{x}{x^2-5}$$

**Answer: A**

29. Let $f$ be twice differentiable function on $\Re$. Which one of the following is necessarily true?

A. If $f'(c) = 0$, at some $c \in \Re$, then $f$ has a relative extreme value at $x = c$.
B. If $f'(x)$ is increasing, then the graph of $y = f(x)$ is concave upward.
C. If $f'(x) = 0$ for all $x \in \Re$, then $f(x) = 0$ for all $x \in \Re$.
D. If $f(x)$ is increasing, then $f''(x) \geq 0$ for all $x \in \Re$.

***Hint:***

If the graph of $f'(x)$ is increasing, so that $f''(x) > 0$, then the graph of $y = f(x)$ is concave upward.

**Answer: B**

*For further explanation refer extreme maths page 716. Q#61*

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**Extreme Tip 3:** The region is enclosed by the graph of $y = x^2$, the line $x = 2$, and the $x$-axis is revolved about the $y$-axis. The volume of solid generated is:

A. $8\pi$ B. $\frac{32\pi}{5}$ C. $\frac{16\pi}{3}$ D. $4\pi$

***Solution:*** Since it is revolved about y-axis.

$$\Rightarrow y = x^2 \Rightarrow x = \sqrt{y} \text{ when } x = 2, y = 4$$

$$\therefore v = \int_0^4 \pi f(y)^2\, dy = \int_0^4 \pi\left(\sqrt{y}\right)^2 dy = \pi\left[\frac{y^2}{2}\right]_0^4$$

$$= \pi\left[\frac{16}{2} - 0\right]_0^4 = 8\pi$$

**Answer: A**

25. Which one of the following is true about the horizontal asymptote(s) of the graph of $y = \frac{|x|+2}{x}$?

A. $y = 2$ is the only horizontal asymptote of the graph.
B. $y = 1$ and $y = -1$ are horizontal asymptotes of the graph.
C. $y = 2$ and $y = -2$ are the horizontal asymptotes of the graph.
D. $y = 1$ is the only horizontal asymptote of the graph.

***Hint:*** $y = \frac{|x|+2}{x} = \begin{cases} 1+\frac{2}{x} & \text{when } x > 0 \\ -1+\frac{2}{x} & \text{when } x < 0 \end{cases}$

- As $x \to \infty$, $y = 1$ and
- As $x \to -\infty$, $y = -1$

**Answer: A**

26. If $f(x) = 2x^5 - 3x$, then $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1}$ is equal to _____.

A. 1 B. −1 C. 7 D. $\infty$

***Hint:*** $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} = f'(1)$

and $f'(x) = 10x^4 - 3$, so $f'(1) = 10(1)^4 - 3 = 7$

**Answer: C**



27. If $f(x) = e^{3x}\cos x - \frac{x+\pi}{x^2+2}$, then $f'(0)$ is equal to _____.

A. $3 - \frac{\pi}{2}$ B. $\frac{3}{2}$ C. $\frac{7}{4}$ D. $\frac{5}{2}$

**Answer: D**

28. If $f(x) = \ln\left(\sqrt{x^2-5}\right)$, which one of the following is equal to $f'(x)$?

A. $\frac{x}{x^2-5}$ B. $\frac{-x}{\sqrt{x^2-5}}$ C. $\frac{2x}{\sqrt{x^2-5}}$ D. $\frac{-x}{x^2-5}$

***Hint:***

Use $= \left(\ln\left(g(x)\right)^n\right)' = \left(n \ln g(x)\right)' = \frac{n g'(x)}{g(x)}$

$\therefore \left(\ln\sqrt{x^2-5}\right)' = \left(\frac{1}{2}\ln\left(x^2-5\right)\right)' = \frac{1}{2}\left(\frac{2x}{x^2-5}\right) = \frac{x}{x^2-5}$

**Answer: A**

29. Let $f$ be twice differentiable function on $\Re$. Which one of the following is necessarily true?

A. If $f'(c) = 0$, at some $c \in \Re$, then $f$ has a relative extreme value at $x = c$.
B. If $f'(x)$ is increasing, then the graph of $y = f(x)$ is concave upward.
C. If $f'(x) = 0$ for all $x \in \Re$, then $f(x) = 0$ for all $x \in \Re$.
D. If $f(x)$ is increasing, then $f^n(x) \geq 0$ for all $x \in \Re$.

***Hint:***

If the graph of $f'(x)$ is increasing, so that $f^n(x) > 0$, then the graph of $y = f(x)$ is concave upward.

**Answer: B**

*For further explanation refer extreme maths page 716. Q#61*

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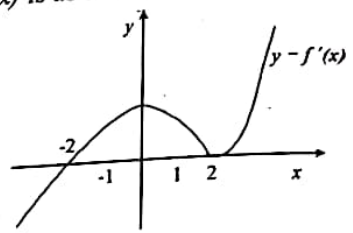
30. Suppose $f(x)$ is differentiable on $(-\infty, \infty)$ and the graph of its derivative, $y = f'(x)$ is as shown in the following figure.




Which one of the following is true about $f(x)$?

A. $f(x)$ is increasing on $(-\infty, 0) \cup (2, \infty)$.
B. $f(x)$ has a local extreme value at $x = 2$.
C. $f(x)$ has a local minimum value at $x = -2$.
D. $f(x)$ has a local maximum value at $x = 0$.

*Hint:* Using sign chart.

| The sign of $y = f'(x)$ | $-\infty$ | $---$ | $-2$ | $+++$ | $2$ | $+++$ | $\infty$ |
|---|---|---|---|---|---|---|---|
| $y = f(x)$ | | decrease | | increase | | increase | |

- $f(x)$ is decreasing on $(-\infty, -2]$
- $f(x)$ is increasing on $[-2, 2]$, $[2, \infty]$
- The sign of $f'(x)$ change from negative to positive at $x = -2$
  $\Rightarrow f(x)$ attains a local min at $x = -2$.

**Answer: C**

☞ *Refer extreme maths page 685.*

31. For any n × n square matrix A, which one of the following is true?

A. $\det(A) = -\det(A^T)$, where $A^T$ is the transpose of A.
B. If $k$ is a scalar, then $\det(kA) = k^n \det(A)$
C. If B is a matrix obtained from A by interchanging of two rows of A, then $\det(B) = \det(A)$
D. If A is invertible, then $\det(A) = \det(A^{-1})$

**Answer: B**



32. The solution of the system of linear equation of $\begin{cases} x - 3y - 2z = 6 \\ 2x - 4y - 3z = 8 \\ -3x + 6y + 8z = -5 \end{cases}$ is:

A. $x = -1, y = -3, z = -2$
B. $x = -1, y = -3, z = 2$
C. $x = 1, y = -3, z = 2$
D. $x = 1, y = 3, z = -2$

*Hint:* By direct substitution $x = -1, y = -3, z = 2$

**Answer: C**

33. If $Z = \frac{2-4i}{1+i}$ then the modulus of the conjugate of Z, $|Z|$ is:

A. $\sqrt{10}$ B. $3\sqrt{2}$ C. $2\sqrt{3}$ D. $2\sqrt{2}$

*Hint:* $\bar{z} = \frac{z+4i}{1-i}$ then $|\bar{z}| = |z| = \sqrt{10}$

$$|Z| = \left|\frac{2-4i}{1+i}\right| = \frac{|2-4i|}{|1+i|} = \frac{\sqrt{2^2+4^2}}{\sqrt{1^2+1^2}} = \sqrt{10}$$

**Answer: A**

34. The $n^{th}$ term of the sequence: 1, −4, 9, −16,...is:

A. $a_n = (-2)^n$
B. $a_n = (-1)^n n^2$
C. $a_n = (-1)^{2n} n^2$
D. $a_n = (-1)^{n-1} n^2$

**Answer: D**

35. If $f: A \to B$ and $g: B \to C$ are functions, then which one of the following is true about the composition function?

A. Domain of $(gof) \subseteq$ Domain of $f$
B. Range of $(gof) \not\subseteq$ Range of $g$
C. Domain of $(gof) \not\subseteq$ Domain of $f$
D. Range of $(gof) \subseteq$ Range of $g$

**Answer: A**

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36. If the point (3, -2) is on the graph of $y = f(x)$, which point is on the graph of $y = f^{-1}(x)$?

A. $\left(\frac{1}{3}, -2\right)$ B. $(3, -1)$ C. $(-2, 3)$ D. $\left(3, -\frac{1}{2}\right)$

Answer: C

**Extreme Tip 4:** Let $f(x) = 1 + \frac{1}{2}\ln(x-3)$ then $f^{-1}(x)$ is equal to

***Solution:*** We set $y = f(x) \Rightarrow y = 1 + \frac{1}{2}\ln(x-3)$

Interchange $x \to y$ and $y \to x$

$\Rightarrow x = 1 + \frac{1}{2}\ln(y-3)$, solve for $y$

$\Rightarrow \frac{1}{2}\ln(y-3) = x - 1 \Leftrightarrow \ln(y-3) = 2x - 2$

$y - 3 = e^{2(x-1)} \Leftrightarrow y = e^{2(x-1)} + 3$

$\therefore f^{-1}(x) = e^{2(x-1)} + 3$

37. The equation of the line that passes through (2, -1) and perpendicular to $3x + 4y = 6$ is:

A. $-4x + 3y = 5$ C. $4x + 3y = 11$

B. $4x - 3y = 5$ D. $-4x + 3y = -11$

***Hint:*** Slope of $3x + 4y = 6$ is:

$m_1 = -\frac{3}{4}$, so that $m_2 = \frac{4}{3}$.

Thus, the equation through $(2, -1)$ is

$y + 1 = \frac{4}{3}(x-2) \Leftrightarrow -4x + 3y = -11$ Answer: D

38. There are three children in a room ages three, four, and five. If a four-year-old child enters the room then which one of the following is true?

A. mean age will stay the same but the standard deviation will increase.

B. mean age will stay the same but the standard deviation will decrease.

C. mean age and standard deviation will increase

D. mean age and standard deviation will stay the same.



***Hint:*** mean age $= \frac{3+4+5}{3} = 4$

- Sd. dev. $= \frac{(3-4)^2+(4-4)^2+(5-4)^2}{3} = \frac{2}{3}$
- mean age and standard deviation after four year child enters.

$\rightarrow$ mean $= \frac{3+4+5+4}{4} = \frac{16}{4} = 4$

$\rightarrow$ S.D $= \sqrt{\frac{(3-4)^2+2(4-4)^2+(5-4)^2}{4}} = \sqrt{\frac{2}{4}} = \sqrt{\frac{1}{2}}$

**Answer: B**

39. In how many more ways can 4 people be arranged in a row than if they were arranged in a circle?

A. 1 B. 6 C. 18 D. 12

***Hint:*** $4! - 3! = 4 \times 3 \times 2 \times 1 - (3 \times 2 \times 1) = 24 - 6 = 18$

**Answer: C**

40. Two machines A and B work independently. The probability that both machines A and B work is 0.4. If the conditional probability that machine B works given that machine A works is 0.5, then the conditional probability that machine A works given that machine B works is _____ .

A. 0.8 B. 0.3 C. 0.5 D. 0.7

***Hint:*** $P(B/A) = \frac{P(A \cap B)}{P(A)} \Rightarrow 0.5 = \frac{0.4}{P(A)}$

$\therefore P(A) = \frac{0.4}{0.5} = 0.8$

Since, they are independent: $P(A/B) = P(A) = 0.8$ **Answer: A**

41. Which one of the following is ture about the fucntion $f$ defined by $f(x) = x^2 + e^{2x}$?

A. $f$ is decreasing for $x \geq 0$

B. $f$ is increasing for $x \geq 0$

C. $f$ has relative minimum at $x = 0$.

D. $f$ has relative maximum at $x = 0$.

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**Hint:** $f(x)=x^2+e^{2x} \Rightarrow f'(x)=2x+2e^{2x}$
$f'(x)>0$ for all $x \geq 0$.
$\therefore f$ is increasing on $x \geq 0$.

**Answer: B**

42. The value of $\int_0^1 (x+1)e^{x^2+2x}\,dx$ is:

A. $\frac{e^3}{2}$ B. $\frac{e^4-e}{2}$ C. $\frac{e^3-1}{2}$ D. $e^3-1$

**Hint:** Let $U=x^2+2x \Rightarrow dU=(2x+2)\,dx$
$\Rightarrow dU=2(x+1)\,dx$
$\therefore (x+1)dx=\frac{dU}{2}$

$$\int_0^1 (x+1)e^{x^2+2x}\,dx=\int e^U\frac{dU}{2}=\frac{e^U}{2}+c$$

$$\Rightarrow \int_0^1 (x+1)e^{x^2+2x}\,dx=\left.\frac{e^{x^2+2x}}{2}\right|_0^1=\frac{e^3}{2}-\frac{1}{2}=\frac{e^3-1}{2}$$

**Answer: C**

43. $\int_0^3 (x+1)^{\frac{1}{2}}\,dx =$ _____

A. $\frac{21}{2}$ B. $\frac{14}{3}$ C. 7 D. $\frac{16}{3}$

**Hint:** $\int_0^3 (x+1)^{\frac{1}{2}}\,dx=\frac{2}{3}(x+1)^{3/2}\Big|_0^3$
$=\frac{16}{3}-\frac{2}{3}=\frac{14}{3}$

**Answer: B**

44. Given $f(x)=\begin{cases} x+1, \text{for } x<0 \\ \cos \pi x, \text{for } x \geq 0 \end{cases}$, then $\int_{-1}^1 f(x)\,dx =$ _____.

A. $\frac{1}{2}+\frac{1}{\pi}$ B. $\frac{1}{2}-\frac{1}{\pi}$ C. $\frac{1}{2}$ D. $-\frac{1}{2}$



**Hint:** $\int_{-1}^1 f(x)\,dx=\int_{-1}^0 f(x)\,dx+\int_0^1 f(x)\,dx$
$=\int_{-1}^0 f(x)\,dx+\int_0^1 \cos \pi x\,dx$
$=\frac{x^2}{2}+x\Big|_{-1}^0+\frac{\sin \pi x}{\pi}\Big|_0^1=0-\left(\frac{1}{2}-1\right)+0=\frac{1}{2}$

**Answer: C**

45. The sum of $\sum_{n=0}^{\infty} 5\left(\frac{2}{3}\right)^n$ is _____.

A. 0 B. 15 C. $\frac{10}{3}$ D. 5

**Hint:** $G_1=5, r=\frac{2}{3}<1$. Convergent
$\therefore S=\frac{G_1}{1-r}=\frac{5}{1-\frac{2}{3}}=\frac{5}{\frac{1}{3}}=5\times 3=15$

**Answer: B**

46. In which interval the sequence $\left\{\frac{(-1)^n}{3n}\right\}_{n=1}^{\infty}$ is bounded?

$\left[\frac{-1}{9},\frac{1}{12}\right]$ B. $\left[\frac{-1}{3},\frac{1}{6}\right]$ C. $\left[\frac{-1}{6},\frac{1}{3}\right]$ D. $\left[\frac{-1}{12},\frac{1}{9}\right]$

**Hint:** $\left\{\frac{(-1)^n}{3n}\right\}_{n=1}^{\infty}=\left[\frac{-1}{3},\frac{1}{6},\frac{-1}{9},\frac{1}{12},\frac{-1}{15}\right]$

$\therefore \frac{-1}{3} \leq \left\{\frac{(-1)^n}{3n}\right\}_{n=1}^{\infty} \leq \frac{1}{6}$

**Answer: B**

47. Which one of the following is true about the function

$$f(x)=\begin{cases} \frac{x^2}{x}, x \neq 0 \\ 0, x=0 \end{cases}?$$

A. $f$ is continuous except at $x=0$
B. $f$ has an infinite discontinuity at $x=0$

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C. $f$ is continuous everywhere.
D. $f$ has $x=0$ as a vertical asymptote.

*Hint:* $\lim_{x\to 0} f(x)=f(0)\Rightarrow \lim_{x\to 0^+} f(x)=0=f(0)$

$\lim_{x\to a} f(x)=f(a)\Rightarrow a=a$

Answer: C

48. $\lim_{x\to\infty} x\sin\frac{1}{x}$ is equal to:

A. 0 B. 1 C. $\infty$ D. $-1$

*Hint:* Let $U=\frac{1}{x}\Rightarrow x=\frac{1}{U}$, as $x\to\infty$, $U\to 0$.

$\therefore \lim_{x\to\infty} x\sin\frac{1}{x}=\lim_{U\to 0}\frac{\sin U}{U}=1$

Answer: B

49. $\frac{d}{dx}\left(\ln e^{2x}\right)$ is equal to:

A. $\frac{1}{e^{2x}}$ B. $\frac{2}{e^{2x}}$ C. $2x$ D. 2

*Hint:* $\frac{d}{dx}\left(\ln e^{2x}\right)=\frac{d}{dx}(2x\ln e)=\frac{d}{dx}(2x)=2$

Answer: D

50. If $f(x)=2+|x-3|$ for all $x$, then the value of the derivative $f'(x)$ at $x=3$ is _____.

A. $-1$ B. does not exist C. 1 D. 2

Answer: B

51. The graph of $y=5x^4-x^5$ has a point of inflection at:

A. (3, 162) only C. (0, 0) only
B. (4, 256) only D. (0, 0) and (3, 162)

*Hint:* $f'(x)=20x^3-5x^4$

$f''(x)=60x^2-20x^3$

$\Rightarrow f''(x)=0$ when $x=0$ or $x=3$

$\therefore (3,f(3))=(3,162)$ is inflection point.

Answer: A



52. Let the angle between $\vec{V}=-2\vec{i}-\vec{j}+2\vec{k}$ and $\overrightarrow{PQ}$ be 60°, where $P$ and $Q$ are points in space. If $\vec{V}\cdot\overrightarrow{PQ}=2$, then what is that distance between $P$ and $Q$?

A. $\frac{3}{4}$ B. $\frac{4}{5}$ C. $\frac{4}{3}$ D. $\frac{5}{4}$

*Hint:* $\vec{V}\cdot\overrightarrow{PQ}=|V||PQ|\cos\theta=2$

$\Rightarrow\left|\overrightarrow{PQ}\right|=\frac{2}{|V|\cos\theta}=\frac{2}{\left(\sqrt{4+1+4}\right)\left(\frac{1}{2}\right)}=\frac{4}{3}$

**Answer: C**

53. Which one of the following is a valid assertion that can be proved by the principle of mathematical induction?

A. $2^n>10n$ for every integer $n$ such that $n\geq 6$.
B. $r^2>0$ for every real number $r$ such that $r\geq 1$.
C. $n^2>10n>2n^2$ for every natural number $n\geq 1$.
D. $2^n>8n$ for every integer $n$ such that $n\geq 3$.

**Answer: A**

54. The image of a figure with vertices A(1, 2), B(3, 6), C(−1, 2) and D(−2, −2) after reflection across the x-axis is:

A. A′(1, −2), B′(−3, −6), C′(1, −2), and D′(2, 2)
B. A′(−1, 2), B′(−3, 6), C′(1, −2), and D′(2, −2)
C. A′(1, −2), B′(3, −6), C′(−1, −2), and D′(−2, 2)
D. A′(1, −2), B′(3, 6), C′(−1, 2), and D′(−2, −2)

*Hint:* $(x,y)\xrightarrow{x-axis}(x,-y)$

**Answer: C**

55. If $\vec{V}=\overrightarrow{AB}+3\overrightarrow{BA}$, where A and B are distinct points in the coordinate plane, then which one of the following is equal to $3\vec{V}$

A. $6\overrightarrow{AB}$ B. $-6\overrightarrow{AB}$ C. $12\overrightarrow{AB}$ D. $-12\overrightarrow{AB}$

*Hint:* We know that $3\overrightarrow{BA}=-3\overrightarrow{AB}$

$\therefore \vec{V}=\overrightarrow{AB}+3\overrightarrow{BA}=\overrightarrow{AB}-3\overrightarrow{AB}=-2\overrightarrow{AB}$

$3\vec{V}=3\left(-2\overrightarrow{AB}\right)=-6\overrightarrow{AB}$

**Answer: B**

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56. If $\vec{u}$ is a unit vector in the direction of $\vec{A}$ and $|\vec{A}|=4$, then $\vec{A}\cdot\vec{u}$ is equal to _____.

A. $\frac{1}{4}$ B. 4 C. $\frac{1}{2}$ D. 2

*Hint:* $\vec{A}\cdot\vec{u} = |\vec{A}||\vec{u}|\cos 0° = (4)(1)(1) = 4$

**Answer: B**

57. If $\vec{A}$ and $\vec{B}$ are parallel vectors with opposite directions and $|\vec{B}|=|2\vec{A}|$, then $\vec{B}-\vec{A}$ is equal to.

A. $\vec{A}$ B. $-\vec{A}$ C. $-3\vec{A}$ D. $3\vec{A}$

*Hint:* $|\vec{B}|=|2\vec{A}|$ since parallel and opposite then $\vec{B}=-2\vec{A}$.

$\therefore \vec{B}-\vec{A}=-2\vec{A}-\vec{A}=-3\vec{A}$

**Answer: C**

58. If $\cot(\theta)=2$, then which of the following is equal to $\csc(\theta)$?

A. $\sqrt{5}$ B. $\frac{2}{\sqrt{5}}$ C. $\frac{1}{\sqrt{5}}$ D. $\frac{1}{2}$

*Hint:* $\csc^2\theta = \cot^2\theta + 1 = 4 + 1 = 5$

$\therefore \csc\theta = \sqrt{5}$

**Answer: A**

59. What is the amplitude and period, respectively, of the graph of $f(x)=4\sin\left(\frac{x}{3}\right)\cos\left(\frac{x}{3}\right)$?

A. $4, \frac{\pi}{3}$ B. $2, 3\pi$ C. $2, \frac{2\pi}{3}$ D. $4, 3\pi$

*Hint:* $f(x)=4\sin\left(\frac{x}{3}\right)\cos\left(\frac{x}{3}\right) = 2\sin\left(\frac{2x}{3}\right)$

- Amplitude 2, period, $P=\frac{2\pi}{\frac{2}{3}}=3\pi$

**Answer: B**



60. A boat on a sea sailed from its station toward North with constant speed of 80km/h. Another boat from the same station sailed 60°NE (North East) with constant speed of 100km/h. If the two boats started sailing at the same time, what is the straight distance between them after they have sailed for just 30 minutes?

A. $10\sqrt{42}km$ B. 90km C. $10\sqrt{41}km$ D. $10\sqrt{21}km$

*Hint:* We use $c^2 = a^2 + b^2 - 2ab\cos\theta$.

$\Rightarrow c^2 = 80^2 + 100^2 - 2(80)(100)\cos 60°$.

$= 6400 + 10{,}000 - 8000 = 8400$.

$c = \sqrt{8400} = 20\sqrt{21}km/hr$

distance between them $20\sqrt{21}\text{km} \times \frac{1}{2}\text{hr}$

$=10\sqrt{21}\text{km}$

**Answer: D**

61. What is the value $k$, for which the two vectors $\vec{u}=\begin{pmatrix}1\\k\\-3\end{pmatrix}$ and $\vec{v}=\begin{pmatrix}2k\\-5\\4\end{pmatrix}$ are perpendicular?

A. 4 B. –4 C. 3 D. –3

*Hint:* $\vec{u}\cdot\vec{u}=\begin{pmatrix}1\\k\\-3\end{pmatrix}\cdot\begin{pmatrix}2k\\-5\\4\end{pmatrix}=0$

$\Rightarrow 2k-5k-12=0 \Leftrightarrow -3k=12$

$\therefore k=-4$

**Answer: B**

62. If one of the end point of the line segment is (3, 2, -4) and the mid-point is (4, 1, -2), then the coordinate of the other end point is:

A. (5, 0, 0)
B. (2, 0, 5)
C. (5, 1, 2)
D. (3, 1, 0)

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**Hint:** $\frac{3+x}{2}=4,\ \frac{2+y}{2}=1$ and $\frac{-4+z}{2}=-2$

$x=5, y=0$ and $z=0$

63. The valid conclusion form the premises:

$P \vee Q, Q \Rightarrow R, P \Rightarrow M, \neg M$ is _____

A. $P \wedge (R \vee R)$ C. $Q \wedge (P \wedge R)$

B. $P \wedge (R \wedge R)$ D. $R \wedge (P \vee Q)$

**Hint:** $\neg M \equiv$ True so, M = false

$P \equiv F, Q \equiv T, R \equiv T$

Answer:

64. What is the value of $\arcsin\left(\frac{-\sqrt{2}}{3}\right)$

A. $\frac{\pi}{4}$ B. $\frac{\pi}{2}$ C. $-\frac{\pi}{4}$ D. $-\frac{\pi}{2}$

**Hint:** $\sin^{-1}\left(-\frac{\sqrt{2}}{2}\right)=-\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)=\frac{-\pi}{4}$

Answer: D

Answer: C

**Extreme Tip 9:** Let $f(x)=\begin{cases}\frac{\sin kx}{x}, & \text{if } x<0\\ e^{2x}-2 & \text{if } x\geq 0\end{cases}$

If $f$ is continuous at $x=0$, then what is the value of $k$.

A. 2 B. −1 C. 2 D. −2

**Solution:** Since $f$ is constant at $x=0$. Then

$\lim_{x\to 0^-} f(x)=\lim_{x\to 0^+} f(x)$

$\lim_{x\to 0^-}\frac{\sin kx}{x}=\lim_{x\to 0^+} e^{2x}-2$

$k=e^0-2=1-2=-1$

Answer:



**Extreme Tip 5:** An object is moving along the parabola $y=2\sqrt{x}$ in $xy$-plane. At what point on its path does the object becomes closest to the point (5, 0)?

A. (4, 4) B. (1, 2) C. $(3,\sqrt{6})$ D. $(3, 2\sqrt{3})$

**Solution:** Let $d$ be the distance between the point (5, 0) and a point $(x, y)$ on the graph $y=2\sqrt{x}$ is given by

$d=\sqrt{(x-5)^2+(y-0)^2}$

$\Rightarrow d(x)=\sqrt{(x-5)^2+\left(2\sqrt{x}\right)^2}=\sqrt{x^2-10x+25+4x}$

$\Rightarrow d(x)=\sqrt{x^2-6x+25}$, let $g(x)=x^2-6x+25$.

$g'(x)=2x-6 \Rightarrow g'(x)=0$ when $x=3$.

When $x=3,\ y=2\sqrt{3}$

$\therefore (3, 2\sqrt{3})$ is the closest point.

**Answer: D**

**Extreme Tip 6:** What is the area of the region between the graph of $y=-x^2+1$ and $x$-axis where $0\leq x\leq 2$?

A. $\frac{2}{3}$ B. $\frac{7}{2}$ C. 3 D. 2

**Solution:**

$\int_0^2 f\left(-x^2+1\right)dx=\left[\frac{-x^3}{3}+x\right]_0^2=\left|\frac{-8}{3}+\frac{6}{3}\right|=\frac{2}{3}$

**Answer: A**

**Extreme Tip 7:**

What is the equation of the tangent line to the graph of $f(x)=\frac{4}{x^2}-2x+4$ at $(2, f(2))$?

A. $3x+y=6$ C. $6x+y=13$

B. $3x+y=7$ D. $-6x+y=13$

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**Solution:** $f(x) = 4x^{-2} - 2x + 4$

$f'(x) = -8x^{-3} - 2$ and $f'(2) = \frac{-8}{8} - 2 = -3$

$\therefore y - f(2) = f'(2)(x-2)$ is equation of tangent line

$y - 1 = -3(x-2) \Rightarrow 3x + y = 7$

**Answer: B**

**Extreme Tip 5:**

If $f(x) = \ln(x+1)$ and $g(x) = x^3 + 7$, then what is the domain of $(fog)(x)$?

A. $(-2, \infty)$ B. $(-1, \infty)$ C. $[-2, \infty)$ D. $\varnothing$

**Solution:** Domain of $(fog)(x)$ is $x \in Dg$ and $g(x) \in D$

$= \mathbb{R}$ and $x^3 + 7 \in (-1, \infty)$

$= \mathbb{R} \cap (-2, \infty) = (-2, \infty)$

**Answer: A**



1. Let p and q stand for the statements "Nejat is intelligent" and "Almaz is hardworking", respectively. Which of the following represent the statement "Almaz is hard working if Nejat is intelligent?"

A. $\neg p \wedge q$ B. $\neg p \vee q$ C. $p \wedge q$ D. $\neg q \vee q$

**Hint:** If Nejat is intelligent, then Almaz is hardworking.

$p \Rightarrow q = \neg P \vee q$

**Answer: B**

2. Suppose the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is a reciprocal to that of the eccentricity of the ellipse $x^2 + 4y^2 = 4$. If the hyperbola passes through a focus of the ellipse, then what is the equation of the hyperbola?

A. $x^2 - 2y^2 = 2$ C. $\frac{x^2}{3} - \frac{y^2}{2} = 1$

B. $x^2 - 3y^2 = 3$ D. $\frac{x^2}{2} - \frac{y^2}{3} = 1$

**Hint:** $x^2 + 4y^2 = 4 \Rightarrow \frac{x^2}{4} + \frac{y^2}{1} = 1$, so, $m^2 = 4, n^2 = 1$

- Eccentricity of ellipse: $e = \frac{c}{m} = \frac{\sqrt{m^2 - n^2}}{m}$

$= \frac{\sqrt{4-1}}{2} = \frac{\sqrt{3}}{2}$

- Foci of ellipse: F(C, 0) and (-C, 0) $= \left(-\sqrt{3}, 0\right)$ and $\left(\sqrt{3}, 0\right)$
- Since the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is intersect the ellipse at $\left(\sqrt{3}, 0\right)$ and $\left(-\sqrt{3}, 0\right)$.

$\frac{\left(\sqrt{3}\right)^2}{a^2} - \frac{0^2}{b^2} = 1$, then $a^2 = 3 \ldots (x)$

Given that eccentricity of hyperbola

$= \frac{1}{\text{eccentricity of ellipse}} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$

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$\Rightarrow$Eccentricity of hyperbola $= \frac{\sqrt{a^2+b^2}}{a} = \frac{2}{\sqrt{3}}$, but $a^2 = 3$

$\Rightarrow \frac{\sqrt{3+b^2}}{\sqrt{3}} = \frac{2}{\sqrt{3}} \Rightarrow 3+b^2 = 4$, then $b^2 = 1$

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ will be $\frac{x^2}{3} - \frac{y^2}{1} = 1 \Leftrightarrow x^2 - 3y^2 = 3$

3. What is the radius of the largest possible circle that can be inscribed the ellipse given by $5(x-1)^2 + 3y^2 = 15$? **Answer: B**

A. $\sqrt{3}$ B. $\sqrt{5}$ C. 3 D. 5

***Hint:*** $5(x-1)^2 + 3y^2 = 15 \Rightarrow \frac{(x-1)^2}{3} + \frac{y^2}{5} = 1$

Circle is inscribed if the radius of circle is equal to the length of semi minor axis.

Now, $a^2 = 5 \Rightarrow a = \sqrt{5}$ and $b^2 = 3 \Rightarrow b = \sqrt{3}$

4. What are the values for the center (C) and radius (r) of the circle $x^2 + y^2 - 4x + 6y = 5$? **Answer: A**

A. $C = (-2, 3), r = 3\sqrt{2}$ C. $C = (2, -3), r = 2\sqrt{3}$

B. $C = (2, -3), r = 3\sqrt{2}$ D. $C = (-2, 3), r = 2\sqrt{3}$

***Hint:*** $x^2 + y^2 - 4x + 6y = 5 \Rightarrow x^2 - 4x + y^2 + 6y = 5$

$\Rightarrow x^2 - 4x + 4 + y^2 + 6y + 9 = 5 + 4 + 9$

$\Rightarrow (x-2)^2 + (y+3)^2 = 18$

C (2, -3) and $r^2 = 18$, $r = 3\sqrt{2}$ **Answer: B**

5. What is the equation of line that passes through point (a, a) in the $xy$ – plane if it is a parallel to a line that passes through points (a, b) and (b, a), where $a \neq b$? **Answer: C**

A. $y = x$ C. $y = -x + 2a$

B. $y = -x$ D. $y = 2x - a$

**Hint:** Slope: $m = \frac{a-b}{b-a} = -1$

Equation of parallel line is $y - a = -1\ (x - a)$

$\Rightarrow y = -x + 2a$



6. Which one of the following is true about the graph of $f(x) = \frac{x^3 - x}{x^3(x-1)}$? **Answer: C**

A. The vertical asymptotes of the graph are $x = 0$ and $x = 1$.

B. A horizontal asymptote of the graph is $y = 1$.

C. The graph intersects its horizontal asymptote at point (-1, 0).

D. The graph intersects the vertical line $x = 1$ at point (1, 2).

**Hint:** $f(x) = \frac{x^3 - x}{x^3(x-1)} = \frac{x(x^2-1)}{x^3(x-1)} = \frac{x(x-1)(x+1)}{x^3(x-1)} = \frac{x+1}{x^2}$

- Horizontal asymptote is $y = 0$
- $\frac{x+1}{x^2} = 0 \Rightarrow x = -1$

What is the solution set of $\frac{1}{1+\frac{1}{x}} - \frac{1}{1-\frac{1}{x}} = \frac{x+\frac{1}{x}}{x-\frac{1}{x}}$? **Answer: A**

A. {} B. {-1} C. {1} D. {-1, 1}

***Solution:*** the expression is equivalent to $\frac{x}{x+1} - \frac{x}{x-1} = \frac{x^2+1}{x^2-1}$

$\Rightarrow \frac{x^2 - x - x^2 - x}{x^2 - 1} = \frac{x^2+1}{x^2-1} \Rightarrow -2x = x^2 + 1$

$\Rightarrow x^2 + 2x + 1 = 0 \Rightarrow (x+1)(x+1) = 0$

$x = -1 \leftarrow$ not in the domain

$\therefore$ S. S = { } = $\phi$

8. Which one of the following is the inverse of $f(x) = 8x^3 + 2$? **Answer: B**

A. $f^{-1}(x) = \frac{1}{8x^3 + 2}$ C. $f^{-1}(x) = 8x^{-3} - 2$

B. $f^{-1}(x) = \frac{1}{2}\sqrt[3]{x-2}$ D. $f^{-1}(x) = \frac{1}{8}\sqrt[3]{x-2}$

***Solution:*** $f(x) = 8x^3 + 2$, set $y = f(x)$

$\Rightarrow y = 8x^3 + 2$, interchange $x \rightarrow y$ and $y \rightarrow x \Rightarrow x = 8y^3 + 2$, solve for $y$

$\Rightarrow 8y^3 = x - 2$, then $y = \frac{\sqrt[3]{x-2}}{2}$

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9. If $f(x)=\sqrt{x^3}$ and $(fog)(x)=\sqrt[4]{x}$, then what is the value of $g(8)$?
A. $\sqrt[3]{2}$ B. 2 C. $\sqrt{2}$ D. $2\sqrt{2}$

**Hint:** $(fog)(x)=f(g(x))=\sqrt[4]{x}\Rightarrow\sqrt{(g(x))^3}=\sqrt[4]{x}\Leftrightarrow g(x)=$
$\Rightarrow g(x)=x^{2/12}=x^{1/6}$, therefore, $g(8)=8^{1/6}=(2^3)^{1/6}=2^{1/2}$

**Answer: C**

10. Which of the following functions is a one to one correspondence?
A. $f:[0,\infty]\to\mathbb{R}$ defined by $f(x)=|x|$.
B. $f:\mathbb{R}\to[0,\infty)$ defined by $f(x)=x^2$.
C. $f:\mathbb{R}\to[0,\infty)$ defined by $f(x)=3^x$.
D. $f:(0,\infty)\to\mathbb{R}$ defined by $f(x)=\log_2^x$.

***Hint:***
i) If $f:A\to B$ is on to if and only if the range of $f=B$.
$\therefore B=\mathbb{R}$ and range of $f(x)=\log_2^x$ is $\mathbb{R}$
$\therefore$ Range of $f=B=\mathbb{R}$
$\therefore$ f is on to
ii) For all $x>0$, $f(x)=\log_2^x$ is one – to – one. Since f is both one – to – one and onto, then f is one – to – one correspondence.

**Answer: D**

11. If $A=(a_{ij})_{3\times3}$ is a square matrix with $A^{-1}=\begin{pmatrix}1&3&2\\1&4&3\\0&4&5\end{pmatrix}$, then what is the cofactor of $a_{23}$?
A. $-\frac{3}{4}$ B. $-\frac{2}{7}$ C. $\frac{2}{7}$ D. $-\frac{3}{7}$

***Hint:*** Inverse of $A^{-1}$ is A. i.e $(A^{-1})^{-1}=A$.

$$\text{Det } A^{-1}=|A^{-1}|=1\begin{vmatrix}1&3\\4&5\end{vmatrix}-1\begin{vmatrix}3&2\\4&5\end{vmatrix}+0=-7-7=-14$$

The cofactor of each element:

$$C_{11}=\begin{vmatrix}1&3\\4&5\end{vmatrix}=-7,\quad C_{12}=-\begin{vmatrix}1&3\\0&5\end{vmatrix}=-5,\quad C_{13}=4$$

$$C_{21}=-\begin{vmatrix}3&2\\4&5\end{vmatrix}=-7,\quad C_{22}=\begin{vmatrix}1&2\\0&5\end{vmatrix}=5,\quad C_{23}=-4$$



$$C_{31}=\begin{vmatrix}3&2\\1&3\end{vmatrix}=7,\quad C_{32}=-\begin{vmatrix}1&2\\1&3\end{vmatrix}=-1,\quad C_{33}=-2$$

$$\begin{bmatrix}-7&-5&4\\-7&5&-4\\7&-1&-2\end{bmatrix}^t=\begin{bmatrix}-7&-7&7\\-5&5&-1\\4&-4&-2\end{bmatrix}$$

$$\therefore A=\frac{\text{Adj.}A^{-1}}{|A^{-1}|}=-\frac{1}{14}\begin{bmatrix}-7&-7&7\\-5&5&-1\\4&-4&-2\end{bmatrix}=\begin{bmatrix}\frac{1}{2}&\frac{1}{2}&-\frac{1}{2}\\\frac{5}{14}&\frac{-5}{14}&\frac{1}{14}\\\frac{-2}{7}&\frac{2}{7}&\frac{1}{7}\end{bmatrix}$$

$$\therefore \text{Cof of } a_{23}=(-1)\begin{vmatrix}\frac{1}{2}&\frac{1}{2}\\\frac{-2}{7}&\frac{2}{7}\end{vmatrix}=-\left(\frac{1}{7}+\frac{1}{7}\right)=\frac{-2}{7}$$

**Answer: B**

12. A team of 10 researchers consists 4 biologists and 6 chemists. If 3 persons are chosen randomly from the team, what is the probability that at least one is a biologist?
A. $\frac{2}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{7}{10}$

***Hint:*** P (at least one is biologist) = 1 – P(all chemist)

$$=1-\frac{C(6,3).C(4,0)}{C(10,3)}=1-\frac{1}{6}=\frac{5}{6}$$

**Answer: C**

13. The probability that an electronic device produced by a company does not function properly is equal to 0.1. If devices are bought, then what is the probability that at least 1 device function properly?
A. 0.81 B. 0.09 C. 0.18 D. 0.99

***Hint:*** P(x successes in n trials) = $C(n,x).P^x.(1-P)^x$
P(at least 1 device function) = P(1) + P(2)
$= C(2,1)(0.9)^1(1-0.9)^1 + C(2,2)(0.9)^2(1-0.9)^{2-2}$

$$=2\left(\frac{9}{10}\right)\left(\frac{1}{10}\right)+(1)\left(\frac{9}{10}\right)^2(1)=\frac{18}{100}+\frac{81}{100}=0.99$$

**Answer: D**

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14. Two machines A and B produce respectively 60% and 40% of the total number of items of a factory. The percentages of defective output of these machines are 2% and 5%, respectively. If an item is selected at random, then what is the probability that the item is defective?

A. 0.032 B. 0.07 C. 0.32 D. 0.426

***Hint:***

$P(A) = 0.6$, $P(B) = 0.4$, $P(D_1|A) = 0.02$ and $P(D_2/B) = 0.05$

$P(D) = P(A).P(D_1/A) + P(B).P(D_2|B)$

$= (0.6)(0.02) + (0.4)(0.05) = 0.012 + 0.02 = 0.032$

**Answer: A**

15. In how many ways can a committee of 3 members be formed from 7 candidates?

A. 7 B. 21 C. 28 D. 35

***Hint:*** $C(7,3) = \frac{7!}{4!3!} = \frac{7\times6\times5}{3\times2} = 35$

**Answer: B**

16. The following is a frequency distribution table of a grouped data with variable X.

| $x$ | 3 – 7 | 8 – 12 | 13 – 17 | 18 – 22 |
|---|---|---|---|---|
| Frequency | 4 | 6 | 8 | 2 |

What is the mean $(\bar{x})$ and the variance $(s^2)$ of the data respectively?

A. $\bar{x} = 12, s^2 = 21$ C. $\bar{x} = 13, s^2 = 9$

B. $\bar{x} = 12, s^2 = 25$ D. $\bar{x} = 13, s^2 = 16$

***Hint:*** Class mark: $X_C = 5, 10, 15$ and 20

$$\text{mean}(\bar{x}) = \frac{\Sigma \text{ of class mark}}{\Sigma \text{ of freq}} = \frac{\Sigma fx_C}{\Sigma f}$$

$$\therefore \bar{x} = \frac{4\times5+6\times10+8\times15+2\times20}{4+6+8+2} = \frac{240}{20} = 12$$

ii) Variance $S^2 = \frac{\Sigma f(X_C - \bar{X})^2}{\Sigma f}$

$$= \frac{4(5-12)^2+6(10-12)^2+8(15-12)^2+2(20-12)^2}{20} = 21$$

**Answer: A**



17. The expenditure of 100 families is given below.

| Expenditure | 0 – 9 | 10 – 19 | 20 – 29 | 30 – 39 | 40 – 49 |
|---|---|---|---|---|---|
| No. of families | 14 | 23 | $f_1$ | 21 | $f_2$ |

If the mode of the data is 23.5, what are the values of $f_1$ and $f_2$?

A. $f_1 = 27, f_2 = 15$ C. $f_1 = 25, f_2 = 17$

B. $f_1 = 15, f_2 = 27$ D. $f_1 = 17, f_2 = 25$

***Hint:*** $m_o = B_L + i\left(\frac{d_1}{d_1+d_2}\right)$, where i = 10

$B_L = 19.5$, $d_1 = f_1 - 23$, and $d_2 = f_1 - 21$

$$\Rightarrow 23.5 = 19.5 + 10\left(\frac{f_1-23}{f_1-23+f_1-21}\right)$$

$$10\left(\frac{f_1-23}{2f_1-44}\right) = 23.5 - 19.5 = 4$$

$$\Rightarrow \frac{f_1-23}{2f_1-44} = 0.4 \Leftrightarrow f_1 - 23 = 0.8f_1 - 17.6$$

$\Rightarrow f_1 - 0.8f_1 = -17.6 + 23 = 5.4 \Leftrightarrow f_1 = 27$

To find $f_2$: $14 + 23 + 27 + 21 + f_2 = 100$

$\therefore f_2 = 100 - 85 = 15$

**Answer: A**

18. The first group of 10 children has a mean weight of 15.6kg, the second group of another 10 children has a mean weight of 16kg, and the third group of children has a mean weight of 20 kg. If the mean weight of all the children is 17 kg, what is the total number of children in all of the three groups?

A. 28 B. 29 C. 30 D. 32

***Hint:*** Combined mean $(Z) = \frac{10\times15.6+10\times16+20x}{10+10+x} = 17$

$$\Leftrightarrow \frac{156+160+20x}{20+x} = 17 \Leftrightarrow 316+20x = 340+17x$$

$\Leftrightarrow 3x = 24$, then $x = 8$

$\therefore$ Total number of children = 10 + 10 + 8 = 28

**Answer: A**

19. Which of the following is a valid argument?

A. $\neg p \Rightarrow \neg q, q \vdash \neg p$ C. $\neg p \vee q, r \Rightarrow p, r \vdash \neg q$

B. $p \Rightarrow \neg q, p, r \Rightarrow q \vdash \neg r$ D. $\neg p, p \vee q, r \Rightarrow q \vdash \neg r$

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*Hint*: P ≡ T, P ⇒ ¬q ≡T, then ¬q ≡ T, q ≡ F
r ⇒ q ≡ T, since q is false, then r is false
∴¬r is True

20. Consider the following open propositions: $P(x) = x$ is a prime number, $C(x) = x$ is a composite number, and $E(x) = x$ is an even number. Which one of the following has a truth value of True in the set of positive integers? **Answer: B**

A. $(\forall x)[P(x) \Rightarrow \neg E(x)]$
B. $\neg(\forall x)[C(x) \Rightarrow \neg P(x)]$
C. $(\exists x)[\neg P(x) \wedge \neg C(x)]$
D. $\neg(\exists x)[E(x) \wedge \neg C(x)]$

*Hint*: Let $x = 4$, $E(4) = T$, $C(4) = T$
$\neg[E(4) \wedge \neg(4)] = \neg(T \wedge F) = T$
$\therefore \neg(\exists x)[E(x) \wedge \neg C(x)]$ is true

21. Which one of the following is a convergent sequence? **Answer: D**

A. $\left\{\left(\frac{5}{3}\right)^n\right\}$ B. $\left\{\frac{2n}{n+1}\right\}$ C. $\left\{\frac{n^2}{n+1}\right\}$ D. $\left\{\frac{(-1)^n}{3}\right\}$

*Hint*: Bounded monotonic sequence are convergent.

22. Suppose a radioactive material loses one-third of its mass per year. If its current mass is 81g, then how much will its mass be just after 7 years? **Answer: B**

A. $27g$ B. $\frac{1}{27}g$ C. $\frac{128}{27}g$ D. $\frac{128}{81}g$

*Hint*: $81, 27, 9, 3, 1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \ldots$

23. Which of the following is the sum of the series $5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + - \ldots$? **Answer: B**

A. -5 B. -3 C. 3 D. 5

*Hint*: $G_1 = 5, r = \frac{-2}{3}$

$$S_n = \frac{G_1}{1-r} = \frac{5}{1-\left(\frac{-2}{3}\right)} = \frac{5}{1+\frac{2}{3}} = \frac{5}{\frac{5}{3}} = 5 \times \frac{3}{5} = 3$$

Answer: C



24. What is the sum of $\sum_{n=1}^{30}(-1)^n\left(\frac{1}{n}+\frac{1}{n+1}\right)$?

A. $-\frac{29}{30}$ B. $\frac{29}{30}$ C. $\frac{30}{31}$ D. $-\frac{30}{31}$

*Hint*: $S_n = -1 - \frac{1}{n+1}$ when n is odd

$S_n = -1 + \frac{1}{n+1}$ when is even

$S_{30} = -1 + \frac{1}{31} = \frac{-30}{31}$ **Answer: D**

25. Which of the following relations holds for the sequence: -10, -3, 4, 11, ...?

A. $a_n = a_{n-1} - 8$
B. $a_n = a_{n-1} + 7$
C. $a_n = a_{n-1} - 7$
D. $a_n = a_{n-1} + 8$

*Hint*: $a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \ldots = 7$
$\Rightarrow a_n - a_{n-1} = 7 \Leftrightarrow a_n = a_{n-1} + 1$ **Answer: B**

26. If $z = \sqrt{2}\cos\left(\frac{\pi}{12}\right) + i\sqrt{2}\sin\left(\frac{\pi}{12}\right)$, then what is the value of $z^3$?

A. $2 + 2i$
B. $\sqrt{2} + \sqrt{2}i$
C. $2\sqrt{2} + 2\sqrt{2}i$
D. $3\sqrt{2} + 3\sqrt{2}i$

*Hint*: $Z^3 = \left(\sqrt{2}\right)^3\left[\cos\left(\frac{3\pi}{12}\right) + i\sin\left(\frac{3\pi}{12}\right)\right]$

$$= 2\sqrt{2}\left[\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right] = 2 + 2i$$

**Answer: A**

27. Which one of the following is the conjugate of $|3+4i| - \frac{25i}{3+4i}$?

A. $5 + 3i$ B. $1 + 3i$ C. $3 - 5i$ D. $1 - 3i$

*Hint*: $|3+4i| = \sqrt{3^2+4^2} = \sqrt{9+16} = 5$

$$5 - \frac{25i}{3+4i} = \frac{(3+4i)(5)}{3+4i} - \frac{25i}{3+4i} = \frac{15+20i-25i}{3+4i}$$

$$= \frac{15-5i}{3+4i} \Leftrightarrow \frac{5(3-i)}{3+4i} = \frac{5(3-i)}{3+4i}\cdot\frac{3-4i}{3-4i}$$

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$$= \frac{5(9-3i-12i-4)}{9+16} = \frac{5(5-15i)}{25} = 1-3i$$

Conjugate will be $1 + 3i$

**Answer: B**

28. If $z_1 = \frac{2-i}{1+i}$ and $z_2 = \frac{1+i}{1-i}$, then what is the value of $z_1 + 2z_2$?

A. $1+i$ B. $\frac{i+1}{2}$ C. $\frac{1-i}{2}$ D. $1-i$

***Hint***: $z_1 + 2z_2 = \frac{2-i}{1+i} + 2\left(\frac{1+i}{1-i}\right) = \frac{2-i}{1+i} + \frac{2+2i}{1-i}$

$$= \frac{(2-i)(1-i)+(1+i)(2+2i)}{1+1} = \frac{1+i}{2}$$

**Answer: B**

29. When $\begin{vmatrix} a & b & c \\ a & -a & a \\ a & a & -a \end{vmatrix} = a^3$, and $a \neq 0$, what is the solution of

$$\begin{pmatrix} a & b & c \\ a & -a & a \\ a & a & -a \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}?$$

A. $x = 0, y = 2a, z = 2a$

B. $x = \frac{1}{a}, y = 2a, z = 2a$

C. $x = \frac{1}{a}, y = \frac{2}{a}, z = \frac{2}{a}$

D. $x = 0, y = \frac{2}{a}, z = \frac{2}{a}$

**Hint:** $D_x = \begin{vmatrix} 1 & b & c \\ 0 & -a & a \\ 0 & a & -a \end{vmatrix} = 0, x = \frac{D_x}{D} = 0$

$$D_y = \begin{vmatrix} a & 1 & c \\ a & 0 & a \\ a & 0 & -a \end{vmatrix} = -(a^2 - a^2) = 2a^2, y = \frac{D_y}{D} = \frac{2}{a}$$

**Answer: D**

$$D_z = \frac{2}{a}$$



30. What are the values of $\lambda$ and $\mu$ so that the system $\begin{cases} x+y+z=6 \\ x+2y+3z=10 \\ x+2y+\lambda z=\mu \end{cases}$ has infinitely many solutions?

A. $\lambda \neq 3$ and $\mu \in \mathbb{R}$ C. $\lambda = 3$ and $\mu = 10$

B. $\lambda = 3$ and $\mu \neq 10$ D. $\lambda \in \mathbb{R}$ and $\mu = 10$

**Hint:** In matrix notation, the given system of equations can be written as $AX = B$ where $A = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$.

Augmented matrix [A: B]

$$= \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right] \text{ To change } a_{21} = 0 \text{ and } a_{31} = 0$$

$R_2 - R_1, R_3 - R_1$

$$= \left[\begin{array}{ccc:c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-3 & \mu-6 \end{array}\right] \text{ To change } a_{32} = 0$$

$R_1 - R_2, R_3 - R_2$

$$= -\left[\begin{array}{ccc:c} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & x-3 & \mu-10 \end{array}\right]$$

**Case I:** If $\lambda = 3$, $\mu \neq 10$, the system has no solution

**Case II:** If $\lambda \neq 3$, $\mu$ may have any value then the system has unique solution.

**Case III:** If $\lambda = 3$, $\mu = 10$, then the system has an infinite solution.

**Answer: C**

31. Let f be a differentiable function with f(1) = -1 and f '(1) = 1. If $g(x) = [f(2x + 1) + 2]^2$, then what is the value of g'(0)?

A. 4 B. 2 C. -2 D. -4

**Hint:** $g'(x) = 2[f(2x + 1) + 2].f'(2x + 1).(2)$

$g'(0) = 2[f(1) + 2].f'(1).2 = 2(-1 + 2).2 = 4$

**Answer: A**

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32. If $f(x) = \ell n\,(x^2 + 2)$, then what is the value of $f''(1)$?

A. $\frac{3}{2}$ B. $\frac{5}{9}$ C. $\frac{2}{3}$ D. $\frac{2}{9}$

**Hint:** If $f(x) = \ell n\,(x^2 + 2)$ then

$$f'(x) = \frac{1}{x^2+2}.2x = \frac{2x}{x^2+2}$$

$$f''(x) = \frac{2(x^2+2) - 2x(2x)}{(x^2+2)^2} = \frac{-2x^2+4}{(x^2+2)^2}$$

$$\therefore f''(1) = \frac{-2(1)^2+4}{(1+2)^2} = \frac{-2+4}{9} = \frac{2}{9}$$

**Answer: D**

33. If $x^2 + xy = 10$, then what is the value of $\frac{dy}{dx}$ when $x = 2$?

A. $-\frac{7}{2}$ B. $\frac{2}{7}$ C. $\frac{3}{2}$ D. $\frac{7}{2}$

***Hint:*** $x^2 + xy = 10 \Rightarrow y = \frac{10 - x^2}{x}$

$$\frac{dy}{dx} = \frac{(-2x)(x) - (10 - x^2)}{x^2} = \frac{-x^2 - 10}{x^2}$$

$$\left.\frac{dy}{dx}\right|_{x=2} = \frac{-4-10}{4} = \frac{-14}{4} = \frac{-7}{2}$$

**Answer: A**

34. What is the equation of the tangent line to the graph of $f(x) = 3x^2 + 4x - 5$ at $(1, 2)$?

A. $10x - y - 8 = 0$ C. $-10x - y - 8 = 10$

B. $-10x + y - 8 = 0$ D. $10x + y - 8 = 0$

***Hint:*** $f'(x) = 6x + 4$ and $f'(1) = 6 + 4 = 10$

Equation of tangent line: $y - 2 = f'(1)(x - 1)$

$\Rightarrow y - 2 = 10(x - 1) \Leftrightarrow 10x - y - 8 = 0$

**Answer: A**

35. If $f(x) = \pi^2 + 1$, then what is the value of $f'(x)$?

A. $2\pi + 1$ B. $2\pi$ C. 2 D. 0

***Hint:*** Derivative of constant function is zero.

**Answer: D**

36. Suppose $f$ is continuous on [2, 6] and the only solutions of the equation $f(x) = 7$ are $x = 2$ and $x = 5$. If $f(3) = 9$, then one of the following **CANNOT** be the value of $f(4)$:

A. 5 B. 7.5 C. 8 D. 9



**Answer: A**

37. What is the value of k so that $f(x) = \begin{cases} \frac{\tan 2x}{x}, & \text{if } x > 0 \\ k - e^{2x}, & \text{if } x \le 0 \end{cases}$ is continuous at $x = 0$?

A. 0 B. 2 C. 3 D. 5

***Hint:*** $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^-} f(x)$

$$\Leftrightarrow \lim_{x \to 0^+} \frac{\tan 2x}{x} = \lim_{x \to 0^-} k - e^{2x}$$

$\Rightarrow 2 = k - 1 \Leftrightarrow k = 3$

**Answer: C**

38. If $f$ is continuous at $x = 0$ and $g(x) = \sqrt{x}\left(2f(x) + \frac{3}{\sqrt{x}}\right)$ for all $x \neq 0$, then what is the value of $\lim_{x \to 0^+} g(x)$?

A. 0 B. 2 C. 3 D. 5

***Hint:*** $\lim_{x \to 0^+} \sqrt{x}\left(2f(x) + \frac{3}{\sqrt{x}}\right) = 0$

**Answer: A**

39. What is the value of $\lim_{x \to 0} \frac{\sin x \cos 2x}{x^2 + 3x}$?

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. 1 D. 2

***Hint:*** $\lim_{x \to 0} \frac{\sin x \cos 2x}{x^2 + 3x} = \lim_{x \to 0} \frac{\sin x}{x} \cdot \frac{\cos 2x}{(x+3)}$

$$\lim_{x \to 0} \frac{\sin x}{x} . \lim_{x \to 0} \frac{\cos 2x}{x+3} = 1 . \frac{1}{3} = \frac{1}{3}$$

**Answer: A**

40. If $a \neq 0$, then what is the value of $\lim_{x \to a} \frac{x^2 - a^2}{x^4 - a^4}$?

A. $\frac{1}{a}$ B. $\frac{1}{2a^2}$ C. $\frac{1}{6a^2}$ D. 0

***Hint:*** $\lim_{x \to a} \frac{x^2 - a^2}{x^4 - a^4} = \lim_{x \to a} \frac{(x-a)(x+a)}{(x-a)(x+a)(x^2+a^2)}$

$$\lim_{x \to a} \frac{1}{x^2 + a^2} = \frac{1}{a^2 + a^2} = \frac{1}{2a^2}$$

**Answer: B**

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41. If $f(0) = -1, f(1) = 2$ and $f'(x)$ us continuous on $[0, 1]$, then which of the following is equal to $\int_0^1 f'(x)\sqrt{2+f(x)}dx$?

A. $\frac{16}{3}$ B. $\frac{14}{3}$ C. $\frac{8}{3}$ D. $\frac{4}{3}$

**Hint:** Let $u = 2 + f(x) \Rightarrow du = f'(x)dx$

When $x = 1 \Rightarrow u = 2 + f(1) = 2 + 2 = 4$

When $x = 0 \Rightarrow u = 2 + f(0) = 2 + (-1) = 1$

$$\int_1^4 \sqrt{u}du = \frac{2}{3}u^{3/2}\Big|_1^4 = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}$$

**Answer: A**

42. What is the value of $\int_0^{\ln\sqrt{3}} \frac{e^x}{e^{-x}+e^x}dx$?

A. $\frac{1}{2}\ln 2$ B. $\ln 4$ C. $\ln 2$ D. 1

**Hint:** $\frac{e^x}{e^{-x}+e^x} = \frac{e^x}{e^x}\left[\frac{e^x}{e^{-x}+e^x}\right] = \frac{e^{2x}}{1+e^{2x}}$

Let $u = 1 + e^{2x}$, $du = 2e^{2x}dx$

When $x = 0$, $u = 1 + e^0 = 1 + 1 = 2$

When $x = \ln\sqrt{3}$, $u = 1 + e^{2\ln\sqrt{3}} = 1 + (e^{\ln\sqrt{3}})^2 = 1 + 3 = 4$

Using integration by substitution.

$$\int_0^{\ln\sqrt{3}} \frac{e^{2x}}{1+e^{2x}}dx = \int_2^4 \frac{du}{2u} = \frac{1}{2}\ln u\Big|_2^4 = \frac{\ln 4 - \ln 2}{2} = \frac{\ln\left(\frac{4}{2}\right)}{2} = \frac{\ln 2}{2}$$

**Answer: A**

43. What is the value of the area of the region enclosed by the graph of $f(x) = e^x$ and $g(x) = x$ between the lines $x = -1$ and $x = 1$?

A. $\frac{e^2-1}{e}$ C. $e^2 - \frac{1}{e} + 2$

B. $e^2 - \frac{1}{e}$ D. $e - \frac{1}{e} + 2$

**Hint:** $A = \int_{-1}^1 (e^x - x)dx = e^x - \frac{x^2}{2}\Big|_{-1}^1$

$$= \left(e - \frac{1}{2}\right) - \left(e^{-1} - \frac{1}{2}\right) = e - \frac{1}{e} = \frac{e^2-1}{e}$$

**Answer: A**



44. Which of the following is equal to the volume of the solid generated when the region bounded by the graph of $y = 2\sqrt{x+1}$ and the $x$ – axis when $0 \le x \le 2$, rotates the $x$ – axis.

A. $8\pi$ B. $8\pi^2$ C. $16\pi$ D. $16\pi^2$

**Hint:** $V = \pi\int_a^b (f(x))^2 dx = \pi\int_a^b y^2 dx$

$$V = \pi\int_0^2 \left(2\sqrt{x+1}\right)^2 dx = \pi\int_0^2 4(x+1)dx$$

$$= 4\pi\left(\frac{x^2}{2} + x\right)\Big|_0^2 = 4\pi(4) = 16\pi$$

**Answer: C**

45. A particle moves along the $x$ – axis with velocity given by $v(t) = 3t^2 + 6t$ for time $t \ge 0$. If the particle is at position $x = 2$ at time $t = 0$, what is the position of the particle at $t = 1$?

A. 6 B. 9 C. 11 D. 12

**Hint:** $V = \frac{dx}{dt} \Rightarrow dx = Vdt \Leftrightarrow \int dx = \int Vdt$

$$x(t) = \int (3t^2 + 6t)dt = \frac{3t^3}{3} + \frac{6t^2}{2} + C$$

$x(0) = 0 + 0 + C = 2 \Rightarrow C = 2 \Rightarrow x(t) = t^3 + 3t^2 + 2$

$x(1) = 1 + 3 + 2 = 6$

**Answer: A**

46. Which of the following is equal to $\int x(e^x + \sin(x^2))dx$?

A. $e^x(x+1) + \frac{1}{2}\sin(x^2) + C$

B. $e^x(x-1) - \frac{1}{2}\sin(x^2) + C$

C. $e^x(x-1) - \frac{1}{2}\cos(x^2) + C$

D. $e^x(x-1) + \frac{1}{2}\cos(x^2) + C$

**Hint:** The expression is the same as $\int xe^x dx + \int x\sin(x)^2 dx$

i) $\int xe^x dx$: Let $u = x$, $dv = e^x dx$

$du = dx$ $V = e^x$

$\int udv = uv - \int vdu = xe^x - e^x + c$

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ii) $\int x\sin(x)^2\,dx$; Let $u = x^2$ then $du = 2x\,dx$

$$\int \frac{\sin u}{2}\,du = \frac{-\cos u}{2} + C = \frac{-\cos x^2}{2} + C$$

$$\therefore \int xe^x dx + \int x\sin(x)^2\,dx = xe^x - e^x - \frac{\cos x^2}{2} + C$$

47. Which one of the following is the set of critical numbers of $f(x) = \frac{4}{3}x^3 + |x|$? **Answer: C**

A. $\left\{\frac{1}{2}\right\}$ B. $\left\{0, \frac{1}{2}\right\}$ C. $\left\{0, -\frac{1}{2}\right\}$ D. $\left\{-\frac{1}{2}, 0, \frac{1}{2}\right\}$

***Hint:***

Recall that $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

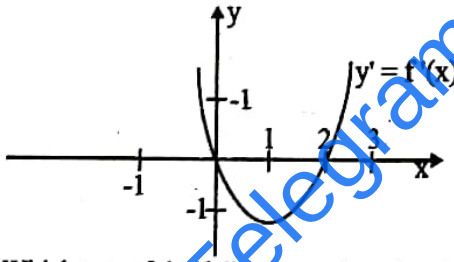
$f(x) = \frac{4}{3}x^3 + x$ for $x \geq 0$ and

$f(x) = \frac{4}{3}x^3 - x$ for $x \geq 0$

For $x < 0$: $f'(x) = 4x^2 - 1$ and $f'(x) = 0$

When $x = \frac{1}{2}$ and $-\frac{1}{2}$ **Answer: D**

48. Suppose $f: (-\infty, \infty) \to \mathbb{R}$ is differentiable and the graph of its derivative, $y' = f'(x)$, is as shown in the figure below.




Which one of the following is true about $f$?

A. $f$ is increasing on $(1, \infty)$
B. $f$ is concave upward on $(0, \infty)$
C. $f$ has no relative maximum value
D. $f$ has a relative minimum value at $x = 2$



***Hint:*** Since $f'$ changes from negative to positive at $x = 2$ then $f$ has relative minimum at $x = 2$.

- Since $f' > 0$ on the interval $(-\infty, -1)$ and $(2, \infty)$, then $f$ is increasing on $(-\infty, -1) \cup (2, \infty)$.
- $f' < 0$ on $(0, 2)$, then f is decreasing on $(0, 2)$.

**Answer: D**

49. A tin can of volume $54\pi$ cm$^3$ is to be made in the form of a right circular cylinder that has both flat top and flat bottom. What is the base radius of the tin if it is to be made of the least amount of metal?

A. 2cm B. 3cm C. 4cm D. 6cm

***Hint:*** $V = \pi r^2 h$ and $A = 2\pi rh + 2\pi r^2$

$$\text{Vol} = \pi r^2 h = 54\pi \text{cm}^3 \Rightarrow h = \frac{54}{r^2}$$

$$A = 2\pi r\left(\frac{54}{r^2}\right) + 2\pi r^2 = \frac{108\pi}{r} + 2\pi r^2$$

$$A'(r) = \frac{-108\pi}{r^2} + 4\pi r = \frac{4\pi r^3 - 108\pi}{r^2}$$

$A'(r) = 0$ when $r = 3$ **Answer: B**

50. Air is being pumped into a spherical balloon so that its volume increases at a rate of 50cm$^3$/s. How fast is the radius of the balloon increasing when the diameter is 5cm?

A. $\frac{1}{50\pi}$cm/s B. $\frac{1}{25\pi}$cm/s C. $\frac{5}{\pi}$cm/s D. $\frac{2}{\pi}$cm/s

***Hint:*** $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

$$\Leftrightarrow 50\text{cm}^3/\text{s} = 4\pi\left(\frac{25}{4}\right)\frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{50}{25\pi} = \frac{2}{\pi}$$

**Answer: D**

51. If $2 \leq f'(x) \leq 4$ for all values of $x$, then the value of $f(8) - f(2)$ is between which of the following numbers?

A. 14 and 24 C. 12 and 18
B. 12 and 24 D. 8 and 10

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**Hint:** $f'(x) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$

$\Rightarrow 2 \le f'(x) \le 4 = 2 \le \frac{f(8) - f(2)}{8 - 2} \le 4$

$\Leftrightarrow 2 \le \frac{f(8) - f(2)}{6} \le 4 \Leftrightarrow 12 < f(8) - f(2) \le 24$

**Answer: B**

52. Which of the following is true?

A. The amplitude of $f(x) = \sin 3x$ is 3.

B. The period of $f(x) = 2 \sin 4x$ is $\pi$.

C. The period of $f(x) = 3 \cos\left(\frac{1}{2}x - \frac{\pi}{3}\right)$ is $4\pi$.

D. The amplitude of $f(x) = -5 \cos(3x + 2) - 2$ is 7.

***Hint:*** $y = a \sin(bx + c) + d$, and $y = a \cos(bx + c)$

- Period is $\frac{2\pi}{b}$ and amplitude is $|a|$
- For $f(x) = 3 \cos\left(\frac{1}{2}x - \frac{\pi}{3}\right)$, the period $= \frac{2\pi}{\frac{1}{2}} = 4\pi$

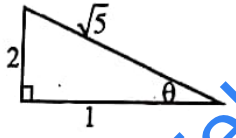
**Answer: C**

53. If $\theta = \arctan(2)$, then what is the value of $\sin(2\theta)$?

A. $\frac{2}{5}$ B. $\frac{4}{5}$ C. $4\sqrt{5}$ D. $\frac{2}{\sqrt{5}}$

***Hint:***

$\theta = \tan^{-1} 2 \Rightarrow \tan\theta = 2$




$\sin 2\theta = \sin(\theta + \theta)$

$= \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$

$\therefore \sin 2\theta = 2\left(\frac{2}{\sqrt{5}}\right)\left(\frac{1}{\sqrt{5}}\right) = \frac{4}{\sqrt{25}} = \frac{4}{5}$

**Answer: B**

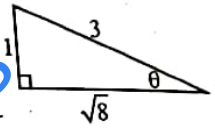
54. If $\cot\theta = \sqrt{8}$, and $\theta$ is a first quadrant angle, then what is the value of $\csc\theta$?

A. $\frac{1}{3}$ B. 3 C. $\frac{\sqrt{8}}{3}$ D. $\frac{1}{\sqrt{8}}$



***Hint:*** $\cot\theta = \sqrt{8}$

- $1^2 + \sqrt{8}^2 = r^2 \Rightarrow r = \sqrt{9} = 3$
- $\sin\theta = \frac{1}{3}$, and $\csc\theta = 3$




**Answer: B**

55. What is the translation vector $u = (h, k)$ so that the equation $x^2 + 2y^2 + 6x - 8y + 15 = 0$ is transformed to an equation of the form $x^2 + 2y^2 + d = 0$, where d is a constant?

A. $u = (-3, 2)$ B. $u = (3, -2)$ C. $u = (-2, 3)$ D. $u = (2, -3)$

***Hint:*** $x^2 + 2y^2 + 6x - 8y + 15 = 0$

$\Rightarrow x^2 + 6x + 9 + 2(y^2 - 4y + 4) = -15 + 9 + 8$

$\Rightarrow (x + 3)^2 + 2(y - 2)^2 = 2$

**Answer: A**

56. A line given by the vector equation $(x, y) = (-t, 6 + 2t)$, $t \in \mathbb{R}$, is tangent to a circle at a point (1, 4). What is the radius of the circle if its center is on the y-axis?

A. $\sqrt{5}$ B. $\frac{1}{2}\sqrt{5}$ C. $2\sqrt{5}$ D. $\sqrt{10}$

***Hint:*** $V = (-t, 2t) = (-1, 2)t$

The position vector through center C (0, y) and tangent point T(1, 4) is $\overrightarrow{CT} = (1, 4 - y)$.

Then V. $\overrightarrow{CT} = 0 \Rightarrow (-1, 2).(1, 4 - y) = 0$

$\Rightarrow -1 + 8 - 2y = 0 \Leftrightarrow 2y = 7,\ y = \frac{7}{2}$

Hence, radius between point (1, 4) and $\left(0, \frac{7}{2}\right)$ is

$r = \sqrt{(1-0)^2 + \left(4 - \frac{7}{2}\right)^2} = \sqrt{1 + \frac{1}{4}} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}$

**Answer: B**

57. For two nonzero vectors $\vec{a}$ and $\vec{b}$ if $\|\vec{a} + \vec{b}\| = \|\vec{a}\|$, then which of the following is true?

A. $2\vec{a}.\vec{b} = \vec{b}.\vec{b}$

B. $2\vec{a} + \vec{b}$ and $\vec{b}$ are perpendicular

C. $2\vec{a} + \vec{b}$ and $2\vec{a} + \vec{b}$ are parallel

D. $\vec{a}.\vec{b} - \vec{b}.\vec{b} = 0$

***Hint:*** $\|a + b\|^2 = (a + b).(a + b) = a.a + 2a.b + b.b$

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$\|a\|^2 = a.a.$
$\|a+b\|^2 = \|a\|^2 \Leftrightarrow a.a + 2a.b + b.\ b = a.\ a$
$\Rightarrow 2a.b + b.\ b = 0 \Leftrightarrow b.\ (2a+b) = 0$

58. Let $\overrightarrow{PQ}$ be a vector with initial point P = (1, 5) and terminal point Q = (4, 0). If $\vec{v} = x\vec{i} + 2\vec{j}$ is parallel to $\overrightarrow{PQ}$, then what is the value of x?

A. $-\frac{6}{5}$ B. $-\frac{2}{5}$ C. $-3$ D. $3$

**Answer: B**

***Hint***: $\overrightarrow{PQ} = (4-1, 0-5) = (3, -5)$ and $\vec{V} = (x, 2)$
$\overrightarrow{PQ}$ and $\vec{V}$ are paralle if $\overrightarrow{PQ} = \alpha\vec{V}$

Let $x = -\frac{6}{5}$, then $\alpha = \frac{-5}{2}$

$\Rightarrow -\frac{5}{2}\left(\frac{-6}{5}, 2\right) = (3, 5)$

**Answer: A**

59. Which one of the following is a valid assertion that can be proved by the principle of mathematical induction?

A. The sum of any two positive rational numbers is positive.
B. $r^2 \geq 1$, for every real number $r \geq 1$.
C. $n^2 \geq 4n$, for every integer $n \geq 4$.
D. $2^n \leq 4^n$, for every integer $n \leq 100$.

**Answer: C**

0. Consider the assertion: "The sum of positive irrational numbers is positive irrational number". Which one of the following is correct about the assertion?

A. Taking irrational numbers such as $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}$, and so on, if we add any two of them, the sum is irrational. Therefore, the assertion is true.

B. The sum of $1+\sqrt{2}$ and $1-\sqrt{2}$ is 2, which is rational. This is a counter example that disproves the assertion.

C. The sum of $\sqrt{7}$ and $\sqrt{2}$ is a counter example that shows the assertion is false.

D. The assertion can be disproved by taking the sum of $1+\sqrt{2}$ and $2-\sqrt{2}$ as counter example.

**Answer: D**



61. Suppose "If $x \in A$, then $y \in B$" is a True statement. Then, which one of the following is necessarily true?

A. $y \in B$
B. If $x \notin A$, then $y \notin B$
C. If $y \in B$, then $x \in A$.
D. If $y \notin B$, then $x \notin A$

**Answer: B**

62. Let $\vec{A}$ and $\vec{B}$ be vectors in space such that $\vec{A}.\vec{B} = -2$ and $\vec{B} = 6\vec{i} - 7\vec{j} + \sqrt{15}\vec{k}$. If θ is the angle between $\vec{A}$ and $\vec{B}$, then what is the value of $|\vec{A}|$?

A. $\frac{1}{5}\cos\theta$ B. $\frac{1}{5\cos\theta}$ C. $\frac{-1}{5}\cos\theta$ D. $\frac{-1}{5\cos\theta}$

**Hint**: $\vec{A}.\vec{B} = |A||B|\cos\theta = -2$

$|B| = \sqrt{6^2 + 7^2 + \sqrt{15}^2} = \sqrt{36+49+15} = \sqrt{100} = 10$

$\Rightarrow 10|A|\cos\theta = -2$, then $|A| = \frac{-2}{10\cos\theta} = \frac{-1}{5\cos\theta}$.

**Answer: D**

63. If $P\left(2, \sqrt{5}, 1\right)$ and Q (3, 0, 9) are points on a sphere whose center is on z – axis, then which one of the following points is outside of the sphere?

A. (−4, 3, 5) B. (2, −2, 1) C. (3, 1, 1) D. (0, 0, 0)

***Hint***: Let (0, 0, z) center of sphere. Then:

$(0-2)^2 + \left(0-\sqrt{5}\right)^2 + (z-1)^2 = (0-3)^2 + 0 + (z-9)_2$

$\Rightarrow 4 + 5 + z^2 - 2z + 1 = 9 + 0 + z^2 - 18z + 81$

$\Rightarrow 16z = 90 - 10 = 80,\ z = 5$

Equation of sphere with center (0, 0, 5) is $(x-0)^2 + (y-0)^2 + (z-5)^2 = (0-3)^2 + (0-3)^2 + (0-0)^2 + (5-9)^2$

$x^2 + y^2 + (z-5)^2 = 25$

C (3, 1, 1): $3^2 + 1^2(1-5)^2 = 9 + 1 + 16 = 26 > 25$

**Answer: C**

64. If A(x, 0, 2), B(3, 0, 2) and $(C2, \sqrt{3}, 2)$ are vertices of equilateral triangle in space, then what is the value of x?

A. 5 B. 3 C. 2 D. 1

***Hint***: Side $\overline{AB} = \overline{AC} = \overline{BC}$

Side $\overline{AB} = \sqrt{(x-3)^2 + 0^2 + 0} = \sqrt{(x-3)^2}$

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Side $\overline{AC} = \sqrt{(x-2)^2 + \left(0-\sqrt{3}\right)^2 + 0} = \sqrt{(x-2)^2 + 3}$

$\overline{AB} = \overline{AC} \Leftrightarrow (x-3)^2 = (x-2)^2 + 3$

$\Rightarrow x^2 - 6x + 9 = x^2 - 4x + 4 + 3$

$9 - 7 = -4x + 6x = 2x$

$\Rightarrow 2x = 2, x = 1$

**Answer: D**

65. A patrol boat on a sea sailed from its statin 7 km to the North; and changed its course and sailed $4\sqrt{2}$ km in the direction of 45° South – East. What is the shortest (straight) distance the boat should travel in order to return to its station?

A. 5km B. 7km C. $5\sqrt{2}$km D. $5+\sqrt{2}$km

***Hint:*** $C^2 = a^2 + c^2 - 2(a)(c)\cos\theta$

$$C^2 = 7^2 + \left(4\sqrt{2}\right)^2 - 2(7)\left(4\sqrt{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

$C^2 = 49 + 32 - 56 = 25$

$\therefore C = 5$

**Answer: A**



1. If a function $f$ is differentiable at a, then what is the value of $\lim_{x\to a} f(x)$?

A. $f(a)$ B. $f'(a)$ C. $f'(a) - f(a)$ D. 0

***Solution:*** If $f$ is differentiable at $a$, then $f$ is continuous at a, the value of $\lim_{x\to a} f(x) = f(a)$

**Answer: A**

2. Which one of the following is a convergent sequence?

A. $\left\{\frac{1+2^n}{3^n}\right\}$ C. $\left\{\frac{1-3^n}{2^n}\right\}$

B. $\left\{\frac{1}{n} + \sin(n)\right\}$ D. $\left\{\frac{(-1)^n}{2}\right\}$

***Solution:***

A. $\lim_{n\to\infty}\left(\frac{1+2^n}{3^n}\right) = 0$ C. $\lim_{n\to\infty}\left(\frac{1-3^n}{2^n}\right) = -\infty$

B. $\lim_{n\to\infty}\left(\frac{1}{n} + \sin(n)\right) = \not\exists$ D. $\lim_{n\to\infty}\frac{(-1)^n}{2} = \pm\frac{1}{2}$

$\therefore \left\{\frac{1+2^n}{3^n}\right\}$ is convergent sequence.

**Answer: A**

3. What is the value of $\sum_{n=2}^{20}\left(\frac{1}{n-1} - \frac{1}{n}\right)$?

A. $\frac{17}{20}$ B. $\frac{23}{20}$ C. $\frac{21}{20}$ D. $\frac{19}{20}$

***Solution:*** $\sum_{n=2}^{20}\left(\frac{1}{n-1} - \frac{1}{n}\right)$

$$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \ldots + \left(\frac{1}{n-1} - \frac{1}{n}\right)$$

$\therefore S_n = 1 - \frac{1}{n} \Rightarrow S_{20} = 1 - \frac{1}{20} = \frac{19}{20}$

**Answer: D**

4. Let A and B be two events. Suppose that the probability that neither event occurs in $3/8$. What is the probability that at least one of the events occurs?

A. $1/8$ B. $5/8$ C. $1/4$ D. $3/4$

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***Solution***: A and B be two events P(at least one of the events A and B occur) = 1 – P(neither events occur)

P(at least one of the events A and B occur) = $1-\frac{3}{8}=\frac{5}{8}$

**Answer: B**

5. If the truth value of $(p \wedge \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r]$ is True, then which one of the following must be True?

A. q B. p C. ¬q D. ¬r

***Solution***: The truth value of

$(p \wedge \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r] \equiv$ True (T)

$p \wedge \neg p \equiv F$ from the definition of negation and conjunction

$q \vee \neg q \equiv T$ from the definition of negation and disjunction

$(q \vee \neg q) \Rightarrow r \equiv$ False because from the given

$(p \vee \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r] \equiv T$ and from the definition of bi implication $r \equiv F$ whereas $\neg r \equiv T$

**Answer: D**

6. What is the maximum value of the function $f(x) = x^4 - 2x^2$ on $[-2, 1]$?

A. 8 B. 12 C. 24 D. 40

***Solution***: Here at, we set $f'(x) = 0$

$f(x) = x^4 - 2x^2 \Rightarrow f'(x) = 4x^3 - 4x = 0$

$\Rightarrow 4x(x^2 - 1) = 4x(x-1)(x+1) = 0$

Hence, the critical numbers are –1, 0, and 1.

We evaluate $f$ at the critical points and the end points of the interval:

$f(-2) = (-2)^4 - 2(-2)^2 = 16 - 8 = 8 \rightarrow$ is the maximum value

$f(-1) = (-1)^4 - 2(-1)^2 = 1 - 2 = -1$

$f(0) = 0 - 0 = 0$, and $f(1) = 1^4 - 2 = -1$

**Answer: A**

7. What are the greatest lower bound and least upper bound of the sequence $\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\}$, respectively?

A. –2 and 2 C. –2 and $\frac{3}{2}$

B. $-\frac{3}{2}$ and 2 D. –2 and $-\frac{3}{2}$

***Solution***:

i) For even numbers: n = 2, 4, 6, 8, …,∞, then:



$$\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\} = \left\{1+\frac{1}{n}\right\} = \left\{\frac{n+1}{n}\right\} = \left\{\frac{3}{2}, \frac{5}{4}, \frac{7}{6}, \ldots 1\right\}$$

ii) For odd numbers: n = 1, 3, 5, 7, …∞, then:

$$\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\} = -\left\{\left(1+\frac{1}{n}\right)\right\} = \left\{-2, \frac{-3}{2}, \frac{-5}{4}, \frac{-7}{6}, \ldots -1\right\}$$

$$\therefore -2 \le \left\{(-1)^n\left(1+\frac{1}{n}\right)\right\} \le \frac{3}{2}$$

**Answer: C**

8. Which one of the following is an equation of the circle whose endpoints of a diameter are (0, –2) and (2, 2)?

A. $x^2 + y^2 = 4$ C. $(x-1)^2 + y^2 = 4$

B. $x^2 + y^2 - 2y - 4 = 0$ D. $x^2 + y^2 - 2x - 4 = 0$

***Solution***:

✓ Diameter: $D = \sqrt{(2-(-2))^2 + (2-0)^2} = \sqrt{20} = 2\sqrt{5}$

✓ Radius $= \frac{D}{2} = \frac{2\sqrt{5}}{2} = \sqrt{5}$ and center $= \left(\frac{0+2}{2}, \frac{-2+2}{2}\right) = (1,0)$

✓ Equation of circle: $(x-1)^2 + (y-0)^2 = \sqrt{5}^2$

$\Rightarrow (x-1)^2 + y^2 = 5 \Leftrightarrow x^2 - 2x + 1 + y^2 = 5$

$\therefore x^2 + y^2 - 2x - 4 = 0$

**Answer: D**

9. What is the equation of the line that passes through (1, 1) and is parallel to the line $3y - x = 1$?

A. $x - 3y + 2 = 0$ C. $3y - x + 2 = 0$

B. $x + 3y = 4$ D. $3x - y = 2$

***Solution***: Any line parallel to the line $3y - x = 1$ is of the form $3y - x = k$, where $k \in \mathbb{R}$. If $3y - x = k$, passes through (1, 1).

$\Rightarrow 3(1) - 1 = k \Rightarrow k = 2$

$\therefore$ The required equation $3y - x = 2 \Leftrightarrow x - 3y + 2 = 0$

**Answer: A**

10. Which one of the following is true about signum, absolute value and greatest integer functions?

A. $\text{sgn}(x) = \pm|x|$, for all $x \in \mathbb{R}$

B. $\text{sgn}(x) \le [x]$, for all $x \le 0$

C. $|x| = x\,\text{sgn}(x)$, for all $x \in \mathbb{R}$

D. $\text{sgn}(x) \le [x]$, for all $x \ge 0$

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**Solution:** Properties of signum; absolute value and greatest integer function

A. $\text{sgn}(x)=\begin{cases}1 \text{ for } x>0\\0 \text{ for } x=0\\-1 \text{ for } x<0\end{cases}$ then $\text{sgn}(x)\neq\pm|x|$ for $x\in\mathbb{R}$

B. $\text{sgn}(x)=\begin{cases}-1 \text{ for } x<0\\0 \text{ for } x=0\end{cases}\Rightarrow \text{sgn}(x)\geq[x]$ for all $x\leq 0$

C. $|x|=\begin{cases}x \text{ for } x>0\\0 \text{ for } x=0\\-x \text{ for } x<0\end{cases}$ implies to $|x|=x\,\text{sgn}\,x$ for $x\in\mathbb{R}$

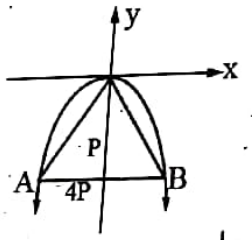
D. Let $x=\frac{1}{2}, \text{sgn}\left(\frac{1}{2}\right)\leq\left[\frac{1}{2}\right]\Rightarrow 1\leq 0$ is false

**Answer: C**

11. What is the area of the triangle (in sq. units) formed by the lines joining the vertex of the parabola $x^2=-36y$ to the end points of the latus rectum?

A. 126 B. 261 C. 216 D. 162

**Solution:**




✓ The length of the latus rectum is $AB=|4P|=36$ unit

✓ The distance between the focus and the vertex is $P=\frac{36}{4}=9$

$\therefore \text{Area}=\frac{1}{2}(\text{base})(\text{height})=\frac{1}{2}(36)(9)=162$

**Answer: D**

12. What is the partial fraction decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)}$?

A. $\frac{3}{5(x+2)}+\frac{2x+1}{5(x^2+1)}$ C. $\frac{2}{5(x+2)}+\frac{3x+1}{5(x^2+1)}$

B. $\frac{5}{3(x+2)}+\frac{2x+1}{3(x^2+1)}$ D. $\frac{2}{3(x+2)}+\frac{2x+1}{3(x^2+1)}$



**Solution:** Partial fraction decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)}$ is

$\frac{x^2+x+1}{(x+2)(x^2+1)}=\frac{A}{x+2}+\frac{Bx+C}{x^2+1}=\frac{Ax^2+A+Bx^2+Cx+2Bx+2C}{(x+2)(x^2+1)}$

$\frac{x^2+x+1}{(x+2)(x^2+1)}=\frac{Ax^2+Bx^2+(C+2B)x+A+2C}{(x+2)(x^2+1)}$

$\Rightarrow A+B=1$ .....*

$C+2B=1\Rightarrow C=1-2B$ .....**

$A+2C=1$ .....***

Substitute ** in to ***

$A+2(1-2B)=1$

$A+2-4B=1$

$-A+4B=1$ .....****

Solve simultaneously * and ****

$\begin{cases}A+B=1\\-A+4B=1\end{cases}$

$5B=2\Rightarrow B=2/5, A=3/5$ and $C=1-\frac{4}{5}=\frac{1}{5}$

$\therefore$decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)}=\frac{3}{5(x+2)}+\frac{2x+1}{5(x^2+1)}$

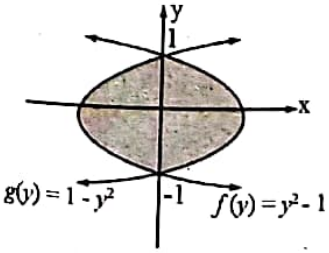
**Answer: A**

13. What is the area of the region enclosed by the graph of $y^2=x+1$ and $y^2=-x+1$?

A. $\frac{3}{8}$sq.units C. $\frac{8}{3}$sq.units

B. $\frac{4}{3}$sq.units D. $\frac{3}{4}$sq.units

**Solution:** $y^2=x+1\Rightarrow x=y^2-1$ and $y^2=-x+1\Rightarrow x=1-y^2$




Let $f(y)=y^2-1$ and $g(y)=1-y^2$

The intersection of $f(y)$ and $g(y)$ is

$f(y)=g(y)\Rightarrow y^2-1=1-y^2$

$\Rightarrow 2y^2=2\Rightarrow y=1$ or $y=-1$

$\therefore A=\int_{-1}^{1}[g(y)-f(y)]dy$

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$$\therefore A=\int_{-1}^{1}[1-y^2-(y^2-1)]dy=\int_{-1}^{1}(-2y^2+2)dy$$

$$\therefore A=\frac{-2}{3}y^3+2y\Big|_{-1}^{1}=\left(\frac{-2}{3}+2\right)-\left(\frac{2}{3}-2\right)=4-\frac{4}{3}=\frac{8}{3}$$

**Answer: C**

14. What is the value of $\int_1^2 \frac{x+4}{x(x+2)}dx$ ?

A. 2 B. ln 3 C. ln 2 D. ln 4 − ln 3

***Solution***: $\int_1^2 \frac{x+4}{x(x+2)}dx$ but partial fraction decomposition of

$$\frac{x+4}{x(x+2)}=\frac{A}{x}+\frac{B}{x+2}=\frac{Ax+2A+Bx}{x(x+2)}$$

$\Rightarrow A+B=1$ and $2A=4$, $A=2$, $B=1-2=-1$

$$\int_1^2 \frac{x+2}{x(x+2)}dx=\int_1^2 \frac{2}{x}dx-\int_1^2 \frac{1}{x+2}dx=2\ln|x|-\ln|x+2|$$

$$\ln\left|\frac{x^2}{x+2}\right|=\ln\frac{x^2}{x+2}\Big|_1^2=\ln\frac{4}{4}-\ln\left(\frac{1}{3}\right)$$

$$=\ln 1-(\ln 1-\ln 3)=\ln 1-\ln 1+\ln 3$$

$$\therefore \int_1^2 \frac{x+4}{x(x+2)}dx=\ln 3$$

**Answer: B**

15. What is the slope of the tangent line to the graph of $f(x)=3e^x+\sin x+2$ at (0, 5)?

A. 2 B. 3 C. 5 D. 4

***Solution***:

$f(x)=3e^x+\sin x+2$ at (0, 5) then the slope of the tangent line is

$f'(x)=3e^x+\cos x \Rightarrow f'(0)=3e^0+\cos 0=4$

The slope of tangent line is 4 and equation of the tangent line:

$$y-f(0)=f'(0)(x-0)$$

$$\Rightarrow y-5=4(x-0)\Rightarrow y=4x+5$$

**Answer: D**

16. The population of certain city is increasing at a rate of 3% per year. If the population was 100,000 in 2010 E.C., then what will be the population in 2020 E.C? (Given $(1.03)^9=1.30$, $(1.03)^{10}=1.34$, $(1.3)^9=10.60$, $(1.3)^{10}=13.78$)

A. 134,000 C. 1,060,000

B. 130,000 D. 1,378,000



***Solution***: Population in certain country increasing at the rate of 3%

$P_0=100{,}000$ then the population in 2020

$P=P_0(1+r)^{10}\Rightarrow P=100{,}000(1.03)^{10}=134{,}000$

**Answer: A**

17. If $z=-3+4i$ and $w=1+2i$, then what is the value of $\frac{2z}{w}+\bar{w}$?

A. $2+3i$ B. $3+5i$ C. $3+2i$ D. $3-2i$

***Solution***: $z=-3+4i$ and $w=1+2i$

$$\frac{2z}{w}+\bar{w}=\frac{2(-3+4i)}{1+2i}+1-2i=\frac{-6+8i}{1+2i}+1-2i$$

$$=\frac{-6+8i+1-2i+2i+4}{1+2i}=\frac{-1+8i}{1+2i}$$

$$\therefore \frac{2z}{w}+\bar{w}=\frac{-1+8i}{1+2i}\times\frac{1-2i}{1-2i}=\frac{-1+2i+8i+16}{5}=3+2i$$

**Answer: C**

18. A salesman sold items $x_1$, $x_2$ and $x_3$, with different rates of commissions as shown in the table below.

| Months | Sales of units | | | Total Commission (in Birr) |
|---|---|---|---|---|
| | $x_1$ | $x_2$ | $x_3$ | |
| February | 90 | 100 | 20 | 800 |
| March | 130 | 50 | 40 | 900 |
| April | 60 | 100 | 30 | 850 |

What are the rates of commission on items $x_1$, $x_2$ and $x_3$, respectively?

A. 4, 2 and 11 C. 4, 11 and 2

B. 2, 4 and 11 D. 11, 2 and 4

***Solution***: From the above table $\begin{cases}90x_1+100x_2+20x_3=800\\130x_1+50x_2+40x_3=900\\60x_1+100x_2+30x_3=850\end{cases}$

Solve by using cramer's rule $D=\begin{vmatrix}9&10&2\\13&5&4\\6&10&3\end{vmatrix}=-175$

$$D_x=\begin{vmatrix}80&10&2\\90&5&4\\85&10&3\end{vmatrix}=-350, D_y=\begin{vmatrix}9&80&2\\13&90&4\\6&85&3\end{vmatrix}=-700$$

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$$D_z = \begin{vmatrix} 9 & 10 & 80 \\ 13 & 5 & 90 \\ 6 & 10 & 85 \end{vmatrix} = -1925 \text{ then } x = \frac{D_x}{D} = \frac{-350}{-175} = 2$$

$$y = \frac{D_y}{D} = \frac{-700}{-175} = 4 \text{ and } z = \frac{-1925}{-175} = 11$$

The rates of common $x_1$, $x_2$ and $x_3$ are 2, 4 and 11 respectively.

**Answer: B**

19. If A is a square matrix of order 3 and det (A) = 5, then what is the value of det (A . adj (A))?

A. 3 B. 5 C. 125 D. 25

***Solution***: det (A. adj (A)) = det (A). det (adj. A) = $5 \times 5 = 25$

**Answer: D**

20. If $f'(x) = e^{x-1} + 3x^2 - \frac{1}{x}$ and $f(1) = 5$, what is $f(x)$?

A. $f(x) = e^{x-1} + 3x^2 + \frac{1}{x^2} + 2$

B. $f(x) = e^{x-1} - x^3 + \ln x + 5$

C. $f(x) = e^{x-1} + 3x^2 - \frac{1}{x} + 5$

D. $f(x) = e^{x-1} + x^3 - \ln x + 3$

***Solution***: If $f'(x) = e^{x-1} + 3x^2 - \frac{1}{x}$ and $f(1) = 5, f(x) =$

$$f(x) = \int f'(x)dx \Rightarrow f(x) = e^{x-1} + x^3 - \ln x + c$$

$$f(1) = e^{1-1} + 1^3 - \ln 1 + C = 5 \Rightarrow C = 3$$

$$\therefore f(x) = e^{x-1} + x^3 - \ln x + 3$$

**Answer: D**

21. Suppose the following are the premises of an argument.

*He is healthy and he is not angry.*
*He is angry or his plan fails.*
*His plan does not fail if he does not travel abroad.*

Given that the premises are true, which one of the following can be a conclusion that makes the argument valid?

A. His plan fails and he is angry.
B. His plan does not fail.
C. He travels abroad.
D. His plant fails and he does not travel abroad.

***Solution***: First translate into symbolic form:



Let P ≡ he is healthy and q ≡ he is angry
S ≡ his plan fails and r ≡ he travel abroad
Then the argument, is of the form: $p \wedge \neg q$, $q \vee s$, $\neg r \Rightarrow \neg s$

1. $p \wedge \neg q \equiv T$ if $p \equiv T$ and $\neg q \equiv T$, $q \equiv F$
2. $q \vee s \equiv T$, if $s \equiv T$, because $q \equiv F$ (from 1)
3. $\neg r \Rightarrow \neg s \equiv T$, if $\neg r \equiv F$, because $\neg s \equiv F$... (from 2)

$\therefore p \wedge \neg q, q \vee s, \neg r \Rightarrow \neg s \vdash r$ is valid

**Answer: C**

22. A man running a race-course noted that the sum of the distance from the two flag posts from him is always 10 meters and the distance between the flag posts is 8 meters. What is the equation of the path traced by the man? (Take flag posts to be on the $x$-axis with origin at their midpoint)

A. $\frac{x^2}{9} + \frac{y^2}{25} = 1$ C. $\frac{x^2}{100} + \frac{y^2}{64} = 1$

B. $\frac{x^2}{25} + \frac{y^2}{9} = 1$ D. $\frac{x^2}{64} + \frac{y^2}{100} = 1$

***Solution***: Distance between two foci is 8m

$$2C = 8m \Rightarrow C = 4m$$

$$2b = 10m \Rightarrow b = \frac{10}{2}m = 5m$$

$$a = \sqrt{b^2 - c^2} = \sqrt{25 - 16} = \sqrt{9} = 3m$$

$\therefore$ The equation of the path traced by the man $\frac{x^2}{9} + \frac{y^2}{25} = 1$

**Answer: A**

23. What is the value of $\int_1^9 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$?

A. $e^3 - e$ B. $\frac{e^3}{3} - e$ C. $e\left(e^2 - \frac{1}{3}\right)$ D. $2(e^3 - e)$

***Solution***: The value of $\int_1^9 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$ let $u = \sqrt{x}$

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow dx = 2\sqrt{x}\, du = 2udu$$

$$2\int_1^3 e^u du = 2e^u \Big|_1^3 = 2e^3 - 2e = 2(e^3 - e)$$

**Answer: D**

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24. What is the value of $\int x\sqrt{1-x^2}\,dx$?

A. $\frac{-1}{3}(1-x^2)^{3/2}+C$ C. $(1-x^2)^{3/2}+C$

B. $-2x\sqrt{1-x^2}+C$ D. $\frac{1}{2}\sqrt{1-x^2}+C$

**Solution:** $\int x\sqrt{1-x^2}\,dx = \int \sqrt{1-x^2}\,x dx$

Let $u = 1-x^2 \Rightarrow du = -2xdx \Rightarrow xdx = \frac{du}{-2}$

$= \frac{-1}{2}\int u^{1/2}du = -\frac{1}{3}u^{3/2}+c$

$\therefore \int x\sqrt{1-x^2}\,dx = \frac{-1}{3}(1-x^2)^{3/2}+c$

**Answer: A**

25. Which one of the following is **NOT** true about the function $f(x) = 3x^4 - 4x^3$?

A. (0, 0) is point of inflection of $f$.

B. 0 and 1 are critical number of $f$.

C. $f$ is concave upward on $\left(0, \frac{2}{3}\right)$ and concave downward on $(-\infty, 0)$ and on $\left(\frac{2}{3}, \infty\right)$.

D. $f$ is decreasing on $(-\infty, 1)$ and increasing on $(1, \infty)$.

**Solution:** $f(x) = 3x^4 - 4x^3$

$f'(x) = 12x^3 - 12x^2 = 12x^2(x-1) = 0$

$\Rightarrow x = 0$ and $x = 1$ are critical numbers

| | | 0 | | 1 | |
|---|---|---|---|---|---|
| $12x^2$ | − − − | 0 | + + + | | + + + |
| $x - 1$ | − − | | − − | 0 | + + + |
| $f'(x) = 12x^2(x-1)$ | − − − | 0 | − − − | 0 | + + + |

$f'(x) > 0$ on $(-\infty, 0)$ and $(1, \infty)$ is increasing

$f'(x) < 0$ on $(0, 1)$ is decreasing

To test concavity

$f''(x) = 36x^2 - 24x = 12x(3x-2) = 0 \Rightarrow x = 0$ or $x = \frac{2}{3}$



$\therefore (0, 0)$ and $\left(\frac{2}{3}, \frac{-16}{27}\right)$ are inflection points

| | | 0 | | $\frac{2}{3}$ | |
|---|---|---|---|---|---|
| $12x$ | − − − | 0 | + + + | | + + ++ |
| $3x - 2$ | − − | | − | 0 | + + + |
| $f''(x) = 12x(3x-2)$ | + + + | 0 | − − | 0 | + + ++ |

$f''(x) < 0$ on the intervals $\left(0, \frac{2}{3}\right)$ the graph is concave upward

$f''(x) > 0$ on $(-\infty, 0)$ and $\left(\frac{2}{3}, \infty\right)$ which is concave down ward

**Answer: D**

26. What is the value of $\lim_{x\to\infty}\frac{1}{x^2}\sin^2\left(\frac{x}{2}\right)$?

A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. 2 D. 4

**Solution:** $\lim_{x\to 0}\frac{1}{x^2}\sin^2\left(\frac{x}{2}\right)$

Let $u = \frac{x}{2} \Rightarrow x = 2u$ as $x \to 0, u \to 0$

$\lim_{u\to 0}\frac{\sin^2 u}{(2u)^2} = \frac{1}{4}\lim_{u\to 0}\frac{\sin^2 u}{u^2} = \frac{1}{4}$

**Answer: B**

27. If $f(x) = \frac{|x|}{x}$ and $g(x) = \frac{x+2}{x^3-4x}$ then what is the value of $\lim_{x\to -2}(f(x)+g(x))$?

A. $\frac{-9}{8}$ B. $\infty$ C. $\frac{9}{8}$ D. $\frac{-7}{8}$

**Solution:** $f(x) = \frac{|x|}{x}, g(x) = \frac{x+2}{x^3-4x}$

$\lim_{x\to -2}(f(x)+g(x)) = \lim_{x\to -2} f(x) + \lim_{x\to -2} g(x)$

$\lim_{x\to -2}(f(x)+g(x)) = \lim_{x\to -2}\frac{|x|}{x} + \lim_{x\to -2}\frac{x+2}{x(x+2)(x-2)}$

$\therefore \lim_{x\to -2}(f(x)+g(x)) = -1 + \frac{1}{8} = \frac{-7}{8}$

**Answer: D**

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28. Let $z=\left(\frac{1-i}{1+i}\right)^{18}$. Then what is the value of $z$?

A. $-1$ B. $i$ C. $-i$ D. $1-i$

***Solution:*** Let $z=\left(\frac{1+i}{1+i}\right)^{18}$

$$z=\left(\frac{1-i}{1+i}\times\frac{1-i}{1-i}\right)^{18}=\left(\frac{-2i}{2}\right)^{18}=(-i)^{18}=(-1)^{18}.((i)^2)^9=1.(-1)^9=-1$$

**Answer: A**

29. Let A be a $3\times3$ invertible matrix and B be any $3\times3$ matrix. If $|A|=a$ and, $|B|=b$, then which one of the following is **NOT** true?

A. $|A^TA|=a^2$

B. $|kA|=k^3|A|$ for any $k\in\mathbb{R}$

C. $|A^{-1}B|=ab$

D. If $b=0$, then B is not invertible

***Solution:*** A be non – singular $3\times3$ matrix and B be any matrix

If $|A|=a$ and $|B|=b$

$|A|=|A^T|=a$

A. $|A^TA|=|A^T||A|=a.a=a^2$

B. $|kA|=k^3|A|$ for any $k\in\mathbb{R}$

C. $|A^{-1}||B|=\frac{1}{a}.b=\frac{b}{a}$

D. If $b=0$ then B is not invertible

**Answer: C**

30. The time needed to type a sample of 8 business letters in an office is 7, 8, 6, 8, 9, 7, 5, 6 minutes. What are the mean $(\bar{x})$ and the standard deviation (s) of the data in minute?

A. $\bar{x}=7, s=\sqrt{1.5}$ C. $\bar{x}=8, s=\sqrt{2}$

B. $\bar{x}=7, s=\sqrt{2}$ D. $\bar{x}=8, s=\sqrt{1.5}$

***Solution:***

Sample of 8 business letters in an office is 7, 8, 6, 8, 9, 7, 5, 6 minutes

$$\text{Mean}: \bar{x}=\frac{7+8+6+8+9+7+5+6}{8}=\frac{56}{8}=7$$

Standard deviation:

$$\delta=\sqrt{\frac{(5-7)^2+2(6-7)^2+2(7-7)^2+2(8-7)^2+1(9-7)^2}{8}}$$



$$\delta=\sqrt{\frac{4+2+0+2+4}{8}}=\sqrt{\frac{12}{8}}=\sqrt{1.5}$$

$\therefore$ Mean $(\bar{x})=7$ and standard deviation $(\delta)=\sqrt{1.5}$

**Answer: A**

31. A private college has 1000 students. 60% of these students are males. 45% of these students pay their payment by credit card including 175 females. What is the probability that the student is a male or a credit card user?

A. 0.675 B. 0.225 C. 0.325 D. 0.775

***Solution:*** Let S – students; M – male students; F – Female students; S – Credit card user students

$n(S)=1000$

$n(M)=60\%$ of $1000=600$

$n(F)=40\%$ of $1000=400$

$P(M)=0.6$, $P(F)=0.4$, P (of credit card user) $=0.45$

P(male or a credit card user) $=0.6+0.45-0.275=0.775$

**Answer: D**

32. Which one of the following is true about the graph of $f(x)=\frac{2x^3+2x^2+3x}{x^2+x}$?

A. The vertical asymptote of the graph is only $x=-1$ and its oblique asymptote is $y=2x$.

B. The graph has $y$-intercept at $(0, 3)$.

C. The graph has at least one $x$-intercept.

D. The vertical asymptotes of the graph are at $x=0$ and $x=-1$ but it has not horizontal asymptote.

***Solution:*** $f(x)=\frac{2x^3+2x^2+3x}{x^2+x}=\frac{x(2x^2+2x+3)}{x(x+1)}$

$f$ has a hole at $x=0$ and vertical asymptote $x=-1$ and oblique asymptote $y=2x$

The graph has no $x$ and $y$ intercept.

**Answer: A**

33. Let $f(x)=\frac{3x+1}{x-2}$. Then what is the range of $f(x)$?

A. $\mathbb{R}\setminus\{2\}$ B. $\mathbb{R}$ C. $\mathbb{R}\setminus\{3\}$ D. $\mathbb{R}\setminus\left(\frac{1}{3}\right)$

***Solution:*** Let $y=\frac{3x+1}{x-2}$

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$yx - 2y = 3x + 1$

Solve for $x$:

$yx - 3x = 2y + 1 \Rightarrow x(y-3) = 2y + 1$

$\therefore x = \dfrac{2y+1}{y-3}$

$\therefore$ Range of $f = \mathbb{R} \setminus \{3\}$

**Answer: C**

34. What is the value of $\int \frac{1}{x}(\ln x + x^2 e^{-x})dx$ ?

A. $\frac{1}{2}x^2 \ln x - (x+1)e^{-x} + C$ C. $\frac{1}{2}x^2 \ln x + (2-x)e^{-x} + C$

B. $\frac{1}{2}\ln^2 x - (x+1)e^{-x} + C$ D. $\frac{1}{2}\ln^2 x + (2-x)e^{-x} + C$

***Solution***: $u = x$; $du/dx = 1$; $dv = e^{-x}dx$; $v = -e^{-x}$

$$\int \frac{1}{x}(\ln x + x^2 e^{-x})dx = \int \frac{\ln x}{x}dx + \int xe^{-x}dx$$

$$u = \ln x \Rightarrow du/dx = \frac{1}{x} \Rightarrow dx = xdu$$

$$\int \frac{1}{x}(\ln x + x^2 e^{-x})dx = \int \frac{ux}{x}du + \int xe^{-x}dx$$

$$= \frac{u^2}{2} + (uv - \int vdu) = \frac{1}{2}(\ln x)^2 + (xe^{-x} - \int -e^{-x}dx)$$

$$= \frac{1}{2}(\ln x)^2 - xe^{-x} - e^{-x} + c$$

$$\therefore \int \frac{1}{x}(\ln x + x^2 e^{-x})dx = \frac{1}{2}(\ln x)^2 - e^{-x}(x+1) + c$$

**Answer: B**

35. If $f(x) = \frac{1}{3}x^3 + cx^2 + ax + 5$ has a local minimum value at $x = 1$, then which one of the following is true about the possible values of a and c?

A. $a = 3, c = -2$

B. $a = -2c - 1$, any real number

C. $a = -2c - 1, c < -1$

D. $a = -2c - 1, c > -1$



***Solution***:

$f(x) = \frac{1}{3}x^3 + cx^2 + ax + 5$ has a local minimum value at $x = 1$

$f(1) = \frac{1}{3} + c + a + 5 = a + c + \frac{16}{3}$

$f'(x) = \frac{3}{3}x^2 + 2cx + a \Rightarrow f'(x) = x^2 + 2Cx + a = 0$

$f'(1) = 1 + 2c + a = 0 \Rightarrow a + 2c = -1$

$a = -1 - 2c = -2c - 1$ and $c > -1$

**Answer: D**

36. Let $\{a_n\}$ be a sequence with $a_1 = a$, $a_2 = f(a_1) = f(a)$, $a_3 = f(a_2) = f(f(a))$, ... $a_n + 1 = f(a_n)$, where $f$ is a continuous function. If $\lim_{n\to\infty} a_n = 5$, what is $f(5)$?

A. 1 B. $5^n$ C. $5^{n+1}$ D. 1

***Solution***: $\{a_n\}$ – be a sequence with

$a_1 = a$, $a_2 = f(a_1) = f(a)$

$a_3 = f(a_2) = f(f(a))$

$a_{n+1} = f(a_n)$ where $f$ is a continuous function

$\lim_{n\to\infty} a_n = 5, f(5) = 5$

**Answer: A**

37. If the sum of the first three consecutive terms of an arithmetic progression $\{A_n\}$, with $A_n > 0$ for all $n$, is 9 and the sum of their squares is 35, then what is the sum $S_n$ of the first n terms?

A. $n^2 + 1$ B. $n^2 - 1$ C. $n^2$ D. $2n^2 + 1$

***Solution***: $\{A_n\}$ be an arithmetic sequence with $A_n > 0$

$$a_1 + a_2 + a_3 = 9 \Rightarrow s_3 = \frac{3}{2}(a_1 + a_3) = 9 \Rightarrow a_1 + a_3 = 6$$

$$a_1^2 + a_2^2 + a_3^2 = 35 \Rightarrow a_1^2 + a_3^2 = 26$$

$$\Rightarrow a_1 + a_3 + a_2 = 9 \Rightarrow 6 + a_2 = 9 \Rightarrow a_2 = 3$$

$$(a_1 + a_3)^2 = a_1^2 + 2a_1a_3 + a_3^2 = 36$$

$$a_1^2 + a_3^2 = 36 - 2a_1a_3$$

$$26 + 2a_1a_3 = 36 \Rightarrow 2a_1a_3 = 10$$

$$\Rightarrow a_1a_3 = 5 \Rightarrow a_3 = \frac{5}{a_1}$$

$$a_1 + 3 + \frac{5}{a_1} = 9 \Rightarrow a + \frac{5}{a_1} = 6 \Rightarrow a_1^2 - 6a_1 + 5 = 0$$

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$a_1 = \frac{6 \pm \sqrt{36-4\times5}}{2} = \frac{6\pm4}{2} = 5$ or 1

$a_3 = \frac{5}{5} = 1$ or $a_3 = \frac{5}{1} = 5$

$\therefore (a_1, a_2, a_3) = (1, 3, 5)$ or $(a_3, a_2, a_1) = (5, 3, 1)$

If $a_1 = 1$, $d = 2$ then $a_n = a_1 + (n-1)d$

$a_n = 1 + (n-1)2 = 2n - 1$

$S_n = \frac{n}{2}(1 + 2n - 1) = n^2$

$\therefore$ The sum of the first $n$ – terms is $S_n = n^2$

**Answer: C**

38. Let A be a 3 × 3 matrix and |A| = –2. Then what is the value of |adj(A)|?

A. –8 B. –2 C. $-\frac{1}{2}$ D. 4

***Solution:***

Let A be 3 × 3 matrix and |A| = –2 then |adj A| = _____

Note: $|A| = |A^T| = |\text{adj } A|$

Since |A| = –2 then |adj A| = –2

**Answer: B**

39. Let A = {1, 2, 3, 4, 5, 6, 7}, B = {7, 8, 9} and C = {8, 9, 10}. If one of the numbers is deleted randomly from each of these sets, what is the probability that all the three deleted numbers are even or all are multiples of 3?

A. $\frac{4}{5}$ B. $\frac{2}{21}$ C. $\frac{1}{9}$ D. $\frac{8}{63}$

***Solution:***

A = {1, 2, 3, 4, 5, 6, 7}, B = {7, 8, 9}, C = {8, 9, 10}

A ∪ B ∪ C both even and multiplies of 3 = {6}

$$P(\text{even or multiple of } 3) = \frac{4}{10} + \frac{2}{10} + \frac{3}{10} - \frac{1}{10} = \frac{8}{10} = \frac{4}{5}$$

**Answer: A**

40. The age distribution of students in a certain class is given below:

| Age | 10 – 14 | 15 – 19 | 20 – 24 | 25 – 29 |
|---|---|---|---|---|
| No. of students | 2 | 10 | 6 | 7 |

What is the modal value of the distribution?

A. 17.83 B. 17.38 C. 18.37 D. 18.73



***Solution:***

| Age (Class boundaries) | 9.5–14.5 | 14.5–19.5 | 19.5–24.5 | 24.5–29.5 |
|---|---|---|---|---|
| No. of students | 2 | 10 | 6 | 7 |

Modal class 14.5 – 19.5

$d_1 = 10 - 2 = 8$ and $d_2 = 10 - 6 = 4$ and i = 5

then $M_o = B_L + \left(\frac{d_1}{d_1 + d_2}\right) i$

$M_o = 14.5 + \left(\frac{8}{3+4}\right)5 \Rightarrow M_o = 14.5 + \left(\frac{8}{3+4}\right)5$

$M_o = 14.5 + \frac{40}{12} = 17.83$

**Answer: A**

41. If $f(x) = k \ln x + e^{\sin x}$ and $f''(\pi) = -1$, then what is the value of k?

A. $2\pi^2$ B. $\pi^2$ C. $\pi$ D. $2\pi$

***Solution:*** $f(x) = k \ln x + e^{\sin x}$, $f''(\pi) = -1$ then $k =$ _____

$f'(x) = \frac{k}{x} + \cos x e^{\sin x}$ and let $u = e^{\sin x}$: $dv = \cos x\, dx$, $v = \sin x$

$f''(x) = \frac{-k}{x^2} - \sin x e^{\sin x} + \cos^2 x e^{\sin x}$

$f''(\pi) = \frac{-k}{\pi^2} - 0 + 1 = -1 \Rightarrow \frac{-k}{\pi^2} = -2 \Rightarrow k = 2\pi^2$

**Answer: A**

42. Let $f(x) = x - x^2$ and $g(x) = \frac{1}{x}$. Then what is $g\left(f\left(\frac{1}{x}\right)\right)$ equals to

A. $x - x^2$ B. $\frac{x^2}{x-1}$ C. $\frac{1}{x^2 - x}$ D. $\frac{x-1}{x^2}$

***Solution:*** $f(x) = x - x^2$ and $g(x) = \frac{1}{x}$ then

$g\left(f\left(\frac{1}{x}\right)\right) = g\left(\frac{1}{x} - \frac{1}{x^2}\right) = g\left(\frac{x-1}{x^2}\right) = \frac{1}{\frac{x-1}{x^2}} = \frac{x^2}{x-1}$

**Answer: B**

43. A cylindrical tank whose inner diameter is 2m contains $4\pi$ m$^3$ oil. If the oil is discharged from the tank at the rate of $\frac{2\pi}{3}$ m$^3$/min, then how long (in min) does it take for the tank to be empty?

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A. $\frac{4}{3}$ B. 4 C. 12 D. 6

**Solution:** $V = \pi r^2 h$, radius: $r = \frac{D}{2} = \frac{2}{2} = 1$

$$\frac{dv}{dt} = \pi r^2 \frac{dh}{dt}$$

$$\frac{2\pi}{3} = \pi(1)\frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{2}{3} \Rightarrow dh = \frac{2}{3}dt$$

$$\int dh = \frac{2}{3}dt \Rightarrow h = \frac{2}{3}t \text{ and } h = \frac{4\pi}{\pi(1)} = 4m$$

$$\therefore t = \frac{3}{2}h = \frac{3}{2}(4) = 6 \text{ minute}$$

**Answer: D**

44. What are the values of a and b so that the function $f(x) = \begin{cases} x+1, x<1 \\ ax+b, 1 \le x < 2 \\ 3x, x \ge 2 \end{cases}$ is continuous everywhere?

A. $a = 4, b = -2$ C. $a = 4, b = 2$

B. $a = -4, b = -2$ D. $a = -4, b = 2$

**Solution:** $f$ is continuous everywhere so that, limit exists

✓ $\lim_{x\to 1^+} f(x) = \lim_{x\to 1^-} f(x)$ and $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} f(x)$

$\Rightarrow \lim_{x\to 1^+}(ax+b) = \lim_{x\to 1^-}(x+1)$ and $\lim_{x\to 2^-} 3x = \lim_{x\to 2^-}(ax+b)$

$\Rightarrow a + b = 2$ and $6 = 2a + b$

Solving together: $\begin{cases} a+b=2 \\ 2a+b=6 \end{cases}$

$-a = -4$

$\therefore a = 4$ and $b = 2 - a = -2$

**Answer: A**

5. Let U = ℕ (the set of natural numbers) be the universe. Which one of the following proposition is true?

A. $(\exists x)(x + x = x)$ C. $(\forall x)(\exists y)(x \div y = y \div x)$

B. $(\forall x)(\exists y)(y < x)$ D. $(\forall x)(\exists y)(x - y = x)$

**Solution:** u = the set of natural numbers



A. $(\exists x)(x + x = x)$ is false, because there is no natural number $x$, which is $x + x = x$

B. $(\forall x)(\exists y)(y < x)$. For every natural number $x$, there is a natural number $y$ such that $y < x$ is false. For example, let $x=1$ and $y<1$.
There is no natural number less than 1.

C. $(\forall x)(\exists y)(x \div y = y \div x)$ is true
For every natural number $x$, there is a natural number $y$ such that $x \div y = y \div x$ is True. For example; if $x = 1$, $y = 1$; if $x = 3$, $y = 3$, and so on.

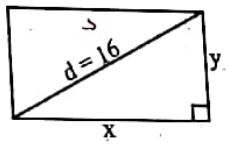
D. For every natural number $x$, there is a natural number $y$ such that $x - y = x$ is false.

**Answer: C**

46. What is the maximum possible area of a rectangle in square units with diagonal of length 16 units?

A. 48 B. 128 C. 64 D. 256

**Solution:** Area $= xy$




- $x^2 + y^2 = 16^2 = 256 \Rightarrow y = \sqrt{256 - x^2}$
- Area: $A(x) = x\sqrt{256 - x^2} = \sqrt{256x^2 - x^4}$

Let $g(x) = 256x^2 - x^4$ and $g'(x) = 512x - 4x^3$

$g'(x) = 0 \Rightarrow 512x - 4x^3 = 0 \Leftrightarrow 4x(128 - x^2) = 0$

$\therefore x = 0$ or $x = \sqrt{128} = 8\sqrt{2}$

$\therefore$ The maximum area $= \left(8\sqrt{2}\right)\left(8\sqrt{2}\right) = 128$

**Answer: B**

47. Let $f(x) = \ln\left(x\sqrt{x}\right)$. Then what is $f'(x)$ equal to?

A. $\frac{2x}{3}$ B. $\frac{\sqrt{x}}{2}$ C. $\frac{2}{x\sqrt{x}}$ D. $\frac{3}{2x}$

**Solution:** $f(x) = \ln\left(x\sqrt{x}\right) = \ln\sqrt{x^3} = \ln x^{3/2}$

$$\Rightarrow f(x) = \frac{3}{2}\ln x \text{ and } f'(x) = \frac{3}{2x}$$

**Answer: D**

48. Which one of the following is equal to $\frac{d}{dx}\log_2^{\sqrt{6x}}$?

A. $\frac{1}{2x\ln(2)}$ C. $\frac{1}{6x\ln(2)}$

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B. $\frac{3}{2x\ln(2)}$ D. $\frac{1}{6x\ln(2)}$

**Solution:** $\log_2\sqrt{6x} = \frac{\ln\sqrt{6x}}{\ln 2} = \frac{\ln(6x)^{1/2}}{\ln 2} = \frac{\ln 6x}{2\ln 2}$

$\frac{d}{dx}\log_2\sqrt{6x} = \frac{d}{dx}\left(\frac{\ln 6x}{2\ln 2}\right) = \frac{6}{(6x)2\ln 2} = \frac{6}{12x\ln 2} = \frac{1}{2x\ln 2}$

**Answer: A**

49. What is the sum of the series $\sum_{n=1}^{\infty} 3^n 4^{-n}$ ?

A. $\infty$ B. $\frac{3}{16}$ C. 3 D. 4

**Solution:** The sum of the series $\sum_{n=1}^{\infty} 3^n 4^{-n} = \frac{3}{4} + \frac{9}{16} + \ldots$

$G_1 = \frac{3}{4}, r = \frac{3}{4}$

$S_{n\to\infty} = \frac{G_1}{1-r} = \frac{3/4}{1-3/4} = \frac{\frac{3}{4}}{1/4} = 3$

**Answer: C**

50. If $z = (1+\sqrt{3}i)(1+i)$ then which one of the following is the polar representation of $z$?

A. $z = 4(\cos(105°) + i\sin(105°))$

B. $z = 2\sqrt{2}(\cos(105°) + i\sin(105°))$

C. $z = 2\sqrt{2}(\cos(15°) + i\sin(15°))$

D. $z = 4((\cos 75°) + i\sin(75°))$

**Solution:** Let $z_1 = 1+\sqrt{3}i$ and $z_2 = 1 + i$

✓ Let $\theta_1 = Arg(z_1) = Arg(1+\sqrt{3}i) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = 60°$

✓ Let $\theta_2 = Arg(z_2) = Arg(1+i) = \tan^{-1}\left(\frac{1}{1}\right) = 45°$

✓ $|z_1| = \sqrt{1^2+\sqrt{3}^2} = \sqrt{4} = 2$ and $|z_2| = \sqrt{1^2+1^2} = \sqrt{2}$

✓ Polar form of $z = z_1 . z_2 = r_1.r_2(\cos(\theta_1+\theta_2) + i(\sin(\theta_1+\theta_2))$

$\therefore z = 2\sqrt{2}[\cos(60°+45°) + i\sin(60°+45°)]$



$= 2\sqrt{2}(\cos 105° + i\sin 105°)$

**Answer: B**

51. The variance of 20 observations is 5. If each observation is multiplied by 2, then what is the variance of the resulting observations?

A. 5 B. 10 C. 20 D. 40

**Solution:** If a data multiplying by constant k:

New variance = $k^2$ (old variance)

$\therefore$ New variance = $(2)^2(5) = (4)(5) = 20$

**Answer: C**

52. What is the value of $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{2}{n}\left[\left(\frac{2k}{n}\right)^3 + 5\left(\frac{2k}{n}\right)\right]$?

A. 4 B. 14 C. 10 D. - 18

**Solution:** $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{2}{n}\left[\left(\frac{2k}{n}\right)^3 + 5\left(\frac{2k}{n}\right)\right]$

$\lim_{n\to\infty}\frac{16}{n^4}\sum_{k=1}^{n}k^3 + \frac{20}{n^2}\sum_{k=1}^{n}k$

$\lim_{n\to\infty}\frac{16}{n^4}\left(\frac{n^2(n+1)^2}{4}\right) + \lim_{n\to\infty}\frac{20(n(n+1))}{2n^2}$

$\frac{16}{4}\lim_{n\to\infty}\frac{n^4+2n^3+n^2}{n^4} + \frac{20}{2}\lim_{n\to\infty}\frac{n^2+n}{n^2} = 4+10 = 14$ **Answer: B**

53. Let A (a, 2, 5) for $a > 0$, be a point on the sphere $x^2 + y^2 + z^2 - 6z = 0$ and C be the center of the sphere. If P($k$, 2, 4) is a point in space such that $\overline{PA}$ is perpendicular to $\overline{CA}$, what is the cosine of the angle between $\overline{PA}$ and $\overline{PC}$?

A. $\frac{5}{\sqrt{35}}$ B. $\frac{7}{\sqrt{35}}$ C. $\frac{7}{\sqrt{70}}$ D. $\frac{5}{\sqrt{70}}$

**Solution:**

A (α, 2, 5) for α > 0 be point on the sphere $x^2 + y^2 + z^2 - 6z = 0$

$\Rightarrow x^2 + y^2 + z^2 - 6z + 9 = 9 \Rightarrow x^2 + y^2 + (z-3)^2 = 9$ and $r = 3$

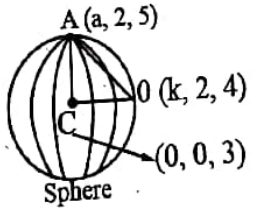
C – be center of the sphere i.e., (0, 0, 3)

P(k, 2, 4) is a point in the sphere

$\overline{PA} \perp \overline{CA}, \overline{CA} = (1,2,2)$

$r = \overline{CA} = \sqrt{\alpha^2+2^2+2^2} = 3$

$\Rightarrow \alpha^2 + 4 + 4 = 9 \Rightarrow \alpha^2 = 1 \Rightarrow \alpha = 1$

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$\overrightarrow{PA} = (1-k, 0, 1)$

$\overrightarrow{PA}.\overrightarrow{CA} = 0 \Rightarrow 1-k+0+2=0 \Rightarrow k=3$

$\cos\theta = \dfrac{PA.CP}{|PA||CP|}$

$\overrightarrow{PC} = (-3,-2,-1), |\overrightarrow{CP}| = \sqrt{3^2+2^2+1^2} = \sqrt{14}$

$\overrightarrow{PA} = (-2,0,1), |\overrightarrow{PA}| = \sqrt{4+0+1} = \sqrt{5}$

$\cos\theta = \dfrac{\overrightarrow{PA}.\overrightarrow{CP}}{|\overrightarrow{PA}||\overrightarrow{CP}|} = \dfrac{6+0+(-1)}{\sqrt{5}.\sqrt{14}} = \dfrac{5}{\sqrt{5}\times\sqrt{14}} = \dfrac{5}{\sqrt{70}}$ **Answer: D**

54. What is the period (p) and the range of $f(x) = 5\sin\left(\frac{1}{3}x+2\right)+3$?

A. p = 6π, range = [−2, 8] C. p = 6π, range = [−5, 5]

B. $p = \frac{2\pi}{3}$, range = [−5, 5] D. $p = \frac{2\pi}{3}$, range = [−2, 8]

***Solution***: $f(x) = 5\sin\left(\frac{1}{3}x+2\right)+3$

$\text{Period} = \dfrac{2\pi}{1/3} = 6\pi$

Range = $[-a+c, a+c] = [-5+3, 5+3] = [-2, 8]$

∴ The period and Range of $f$ is 6π and [−2, 8] respectively.

**Answer: A**

55. Let $\ell$ be the line given by the vector equation $(x, y) = (-2, 1) + \lambda(1, 1), \lambda \in \mathbb{R}$. Which one of the following is the equation of the image of $\ell$ after being translated by the vector u = (2, −1) followed by a rotation through 45° about the origin?

A. $y = \sqrt{2}x$ B. $y = \sqrt{2}$ C. $x = 0$ D. $y = -2\sqrt{x}$

***Solution***: Vector equation $(x, y) = (-2, 1) + \lambda(1, 1)\ \lambda \in \mathbb{R}$

$x+2 = x = y-1 \Rightarrow y = x+3$

$y = x+3 \xrightarrow{\text{translated by u}(2,-1)} y = x$ take (1, 1) be points on $y = x$ then $(1,1) \xrightarrow{\theta = 45°}$

$x' = x\cos\theta - y\sin\theta$

$x' = 1\cos 45° - 1\sin 45° = 0$

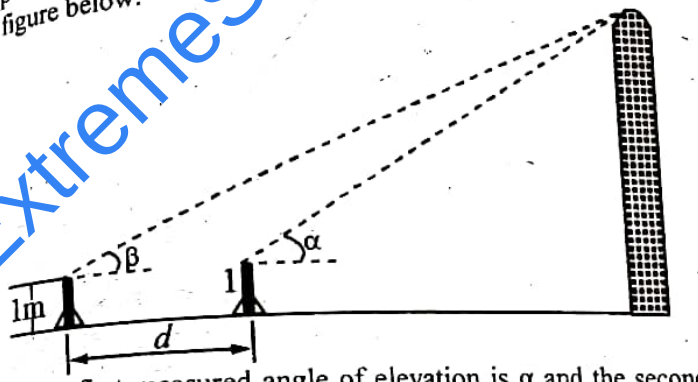
$y' = x\sin\theta + y\cos\theta$



$y' = 1\sin 45° + 1\cos 45° = \dfrac{2\sqrt{2}}{2} = \sqrt{2}$

∴ The image is $y = \sqrt{2}$. **Answer: B**

56. In order to measure the height of a tower, suppose a surveyor takes two sightings from a transit 1 meter high which are positioned d meters apart on the same ground level as in the figure below.

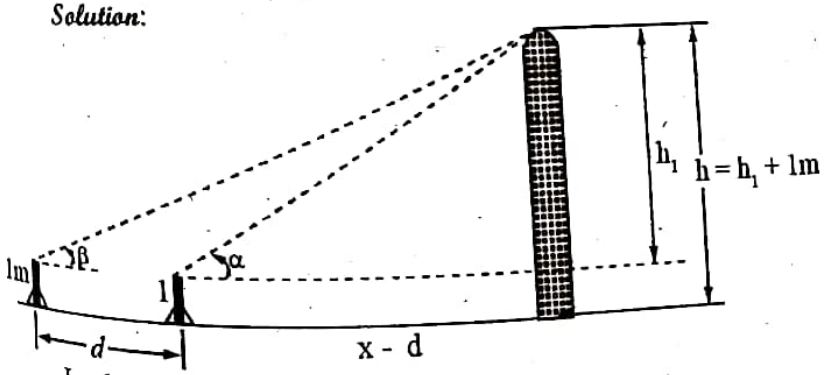



If the first measured angle of elevation is α and the second is β (see, the figure), then what is the height of the tower (in meter) in terms of α, β and d?

A. $\dfrac{\tan\alpha\tan\beta}{(\tan\alpha-\tan\beta)d}+1$ C. $\left(\dfrac{\tan\alpha-\tan\beta}{\tan\alpha\tan\beta}\right)d+1$

B. $\left(\dfrac{\tan\alpha\tan\beta}{\tan\alpha-\tan\beta}\right)d+1$ D. $\left(\dfrac{\tan\alpha\tan\beta}{\tan\alpha+\tan\beta}\right)d+1$

***Solution***:




Let $h = h_1 + 1m$

h – be the height of tower

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x – be the distance between the end surveyor sighting

$\tan\alpha = \frac{h_1}{x-d} \Rightarrow h_1 = (x-d)\tan\alpha$ ..... (1)

$\tan\beta = \frac{h_1}{x} \Rightarrow h_1 = x\tan\beta$ ..... (2)

From 1 and 2

$x\tan\beta = x\tan\alpha - d\tan\alpha$

$\Rightarrow x\tan\beta + d\tan\alpha = x\tan\alpha$

$x(\tan\alpha - \tan\beta) = d\tan\alpha \Rightarrow x = \frac{d\tan\alpha}{\tan\alpha - \tan\beta}$

$h = h_1 + 1m = \frac{d\tan\alpha}{\tan\alpha - \tan\beta}\tan\beta + 1$

$h = \left(\frac{\tan\alpha\tan\beta}{\tan\alpha - \tan\beta}\right)d + 1$ **Answer: B**

57. If the dot products of a vector $\vec{A}$ with the vectors $i - j + k$, $2i + j - 3k$ and $i + j + k$ are 4, 0 and 2, respectively, what is $\vec{A}$?

A. $\vec{A} = (2,1,1)$ C. $\vec{A} = (-2,-1,1)$

B. $\vec{A} = (-2,1,-1)$ D. $\vec{A} = (2,-1,1)$

***Solution***: Let $u = i - j + k$; $v = 2i + j - 3k$; $w = i + 5 + k$

Let $A = x_i + yj + zk$

A.u $= x - y + z = 4$ .....1

A. $v = 0 \Rightarrow 2x + y - 3z = 0$ .....2

A. $W = 2 \Rightarrow x + y + z = 2$ .....3

From 1 and 2 and 1, and 3 respectively

$3x - 2z = 4 \Rightarrow 5x = 10$

$2x + 2z = 6 \Rightarrow x = 2$

$z = 3 - x \Rightarrow z = 3 - 2 = 1$

$y = x + z - 4$

$y = 2 + 1 - 4 = -1$

∴ vector A is (x, y, z) = (2, –1, 1) **Answer: D**

58. What is the image of the circle $x^2 + y^2 - 4x - 6y + 12 = 0$ when it is reflected with respect to the line $y = -x$?

A. $(x+3)^2 + (y+2)^2 = 1$ C. $(x+2)^2 + (y+3)^2 = 1$

B. $(x-2)^2 + (y-3)^2 = 1$ D. $(x-3)^2 + (y-2)^2 = 1$

***Solution***: The image of circle $x^2 - 4x + y^2 - 6y + 12 = 0$



$x^2 - 4x + 4 + y^2 - 6y + 9 = 4 + 9 + 12$

$(x-2)^2 + (y-3)^2 = 1$

The center of circle is (2, 3)

$(2,3) \xrightarrow{\text{reflected } y=-x} (-3,-2)$

∴ Image $(x+3)^2 + (y+2)^2 = 1$ **Answer: A**

59. Let P(1, α, α) and Q = (α – 1, 1, 1) be two points in space and the distance between P and Q is 3m. Then what is the value(s) of α?

A. $\alpha = -1, \alpha = 9$ C. $\alpha = 3, \alpha = -\frac{1}{3}$

B. $\alpha = 1, \alpha = -9$ D. $\alpha = -3, \alpha = \frac{1}{3}$

***Solution***: P(1, α, α) and Q = (α – 1, 1, 1) $\overline{PQ} = 3m$

$\overline{PQ} = \sqrt{(\alpha-2)^2 + (1-\alpha)^2 + (1-\alpha)^2} = 3$

$\alpha^2 - 4\alpha + 4 + 1 - 2\alpha + \alpha^2 + 1 - 2\alpha + \alpha^2 = 9$

$3\alpha^2 - 8\alpha + 6 = 9 \Rightarrow 3\alpha^2 - 8\alpha - 3 = 0$

$3\alpha^2 - 9\alpha + \alpha - 3 = 0$

$(\alpha - 3)(3\alpha + 1) = 0$

$\alpha - 3 = 0$ or $3\alpha + 1 = 0 \Rightarrow \alpha = 3$ or $\alpha = \frac{-1}{3}$ **Answer: C**

60. If the image of the line $2x - 3y = 7$ under a translation is $2x - 3y = 0$, which one of the following is a translation vector of the translation line?

A. u = (2, –1) C. u = (–1, 2)

B. u = (–2, 1) D. u = (1, –2)

***Solution***: The image of the line $2x - 3y = 7$ under a translation is $2x - 3y = 0$ then the translation vector is equal to: $y = \frac{2}{3}x - \frac{7}{3}$

direction vector is parallel to $y = \frac{2}{3}x$

$u = (a,b) = \left(a; \frac{2}{3}a\right) = a(3,2)$ take any point on $y = \frac{2}{3}x - \frac{7}{3}$

if $x = 2, y = -1$ i.e., u = (a, b) = (2, –1) position vector. **Answer: A**

61. Let P(n) be an open proposition on the set of natural numbers (ℕ). Which one of the following is a correct application of the principle of mathematical induction?

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A. If P(1) is true for n = 1; and if both P(n) and P(n + 1) are true for certain n ∈ ℕ, then P(n) is true for all n ∈ ℕ.

B. If P(10) is true; and if P(n) is true implies that P(n + 1) is true, then P(n) is true for all n ∈ ℕ.

C. If P(1) is true; and P(n) ⇒P(n + 1) is true for any n ∈ ℕ, then P(n) is true for all n ∈ ℕ.

D. If P(10) is true; and assuming P(n) is true for any n > 10 if it follows that P(n + 1) is true, then P(n) is true for all n ≥ 10.

***Solution***: Let $P_n$ – be an open proposition on the set of natural numbers (N) then correct application of the principle of mathematical induction such that P(1) is true for n = 1 P(n) and P(n + 1) is true for some n ∈ ℕ then P(n) is true for all n ∈ ℕ.

**Answer: C**

62. If, in ΔABC, AB = 3, BC = 4 and m(∠B) = 60°, then what are the length of AC and the cosine of ∠A, respectively?

A. $\sqrt{13}$ and $\frac{1}{\sqrt{13}}$

B. $\sqrt{13}$ and $\frac{-1}{\sqrt{13}}$

C. $\sqrt{13}$ and $\frac{6}{5\sqrt{13}}$

D. $\sqrt{13}$ and $\frac{-6}{5\sqrt{13}}$

***Solution***: In triangle ABC, $\overline{AB} = 3$, BC = 4 and m(∠B) = 60° then

$\overline{AC}$ = _____ and cos (∠A) = _____

By using cosine law

$AC^2 = AB^2 + BC^2 - 2AB.BC\cos(\angle B)$

$\overline{AC} = \sqrt{3^2 + 4^2 - 2\times3\times4\cos 60°}$

$\overline{AC} = \sqrt{25-12}$

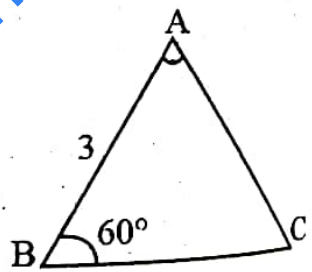
$\overline{AC} = \sqrt{13}$

$BC^2 = AB^2 + AC^2 - 2AB.AC\cos(A)$

$4^2 = 3^2 + \left(\sqrt{3}\right)^2 - 2\times3\times\sqrt{3}\cos(\angle A)$

$16 = 9 + 13 - 6\sqrt{13}\cos A$

$\cos A = \frac{1}{\sqrt{13}}$




$\therefore \overline{AC} = \sqrt{13}$ and $\cos(\hat{A}) = \frac{1}{\sqrt{13}}$

**Answer: A**



63. What is the standard equation of the line passing through the point (2, 3) and parallel to the line given by $\begin{cases} x = 1 + 2\lambda \\ y = -2 - \lambda \end{cases}, \lambda \in \mathbb{R}$?

A. $\frac{x-2}{-1} = \frac{y-3}{2}$

B. $\frac{x-1}{2} = \frac{y-3}{-2}$

C. $\frac{x-2}{2} = \frac{y-3}{-1}$

D. $\frac{x-1}{1} = \frac{y-3}{-2}$

***Solution***:

Use vector equation of the line: $r = r_o + \lambda v$.

$\lambda = \frac{x-1}{2} = \frac{y+2}{-1}$, and parallel vector v = (2, −1)

$\therefore r = r_o + \lambda v \Rightarrow r = (2, 3) + \lambda(2, -1)$

$\therefore x = 2 + 2\lambda$ and $y = 3 - x$

$\therefore \lambda = \frac{x-2}{2}$ and $\lambda = \frac{y-3}{-1}$

$\therefore$ The standard equation of the line is $\frac{x-2}{2} = \frac{y-3}{-1}$

**Answer: C**

64. Consider the following assertion:

$p + 2^n$ is an odd number for any prime p and any n ∈ ℕ.

Which one of the following is correct about a prove or disprove of the assertion?

A. The assertion can be proved by direct method; because p is odd and $2^n = 2(2^{n-1})$ is even imply that $p + 2^n$ is odd since the sum of odd and even is odd.

B. There is a counter example that disproves the assertion.

C. The assertion can be proved by the indirect method; because if n ∉ ℕ, then $2^n \notin \mathbb{N}$ and hence $p + 2^n$ is not odd.

D. The assertion can be disproved by the method of contradiction.

***Solution***:

$P + 2^n$ is odd number for any prime P and only n ∈ ℕ. The assertion can be proved by indirect method because there is a counter example that disprove the assertion p = 2 is the only even prime number $P + 2^n$ is even.

**Answer: B**

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65. If θ is a fourth quadrant angle and $\sec\theta = \sqrt{2}$ then what is $\csc\theta$ equals to?

A. $\frac{-1}{\sqrt{2}}$ B. $-\sqrt{2}$ C. $\frac{1}{\sqrt{2}}$ D. $\sqrt{2}$

***Solution:*** Let θ be fourth quadrant angle and $\sec\theta = \sqrt{2}$ then csc θ is equal to in the fourth quadrant sine, cosec, tangent and cotangent are negative.

$$\therefore \cos\theta = \frac{1}{\sec\theta} \Rightarrow \cos\theta = \frac{1}{\sqrt{2}}$$

$$\sin\theta = \pm\sqrt{1-\cos\theta} = \pm\sqrt{1-\left(\frac{1}{\sqrt{2}}\right)^2} \quad \sin\theta = \pm\sqrt{\frac{2-1}{2}} = \pm\sqrt{\frac{1}{2}}$$

$$\sin\theta = -\frac{1}{\sqrt{2}} \text{ then } \csc\theta = \frac{1}{\sin\theta} = -\sqrt{2}$$

**Answer: B**



## SECTION 2 – CHEMISTRY

1. Nitrogen – fixing bacteria convert:
A. nitrate to nitrogen gas C. nitrate to ammonia
B. nitrogen gas to nitrate D. ammonia to nitrate

***Explanation:*** In nitrogen cycle, nitrogen is taken from atmosphere by nitrogen fixing bacteria (Rhizobium bacteria) in root nodules of leguminous plants. So, the bacteria change atmospheric nitrogen to nitrate. **Answer: B**

2. What is the standard entropy change for the reaction:

$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

Given the following data; s° of $SO_2$ (g) = 248.1 J/mol. K; S° of $O_2$ (g) = 205.0 J/mol. K; S° of $SO_3$ (g) = 256.6 J/mol. K

A. -188.0 J/K C. -196.5 J/K
B. -172.0 J/K D. +60.1 J/K

***Solution:*** The entropy change, ΔS, for a chemical reaction is equal to the sum of the entropies of the products of that reaction minus the sum of the entropies of the reactants.

For the reaction,

$aA + bB \longrightarrow cC + dD$, the entropy change is

$$\Delta S = \sum S_{Products} - \sum S_{reactants}$$

$$= [(c \times S_C) + (d \times S_D)] - [(a \times S_A) + (b \times S_B)]$$

Based on the above formula;

$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

The change in entropy (ΔS) = 2 (256.6 J/mol. k) – [2 (248.1 J/mol. k) + (205 J/mol. k)]

= (513.2 J/mol. k) – (701.2 J/mol. k)

= -188 J/mol. K **Answer: A**

3. How many moles are there in 159 grams of alanine, $C_3H_7NO_2$?
A. 0.560 B. 0.992 C. 1.78 D. 3.31

***Solution:*** m = 159 grams

M = 89 g/mol

$$\text{Number of moles (n)} = \frac{\text{given mass (m)}}{\text{molar mass (M)}}$$

$$n = \frac{159 \text{ gram}}{89 \text{ gram/mol}} = 1.786 \text{ moles}$$

**Answer: C**

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4. Which one of the following is NOT true of metallic bonding?
   A. It gives rise to excellent electrical conductivity.
   B. Electrons are free to move throughout the structure.
   C. The strength of metallic bonds increases down a group.
   D. The strength of metallic bonding affects the boiling point of metals.

***Explanation***

Experimental results indicate that the strength of metallic bonding first increases with increasing number of electrons, then decreases.

**Answer: C**

5. Synthetic fibers
   A. naturally produced
   B. manufactured through the use of chemical substance
   C. produced from animal hair fiber
   D. produced from plant substances

**Answer: B**

6. An amphiprotic species is a molecule or ion that can
   A. accept protons
   B. donate more than one proton
   C. both accept and donate protons
   D. be formed into a double ion

**Answer: C**

7. Which one is true for a triportic acid, such as phosphoric acid, $H_3PO_4$?
   A. $Ka_1 > Ka_2 > Ka_3$
   B. $Ka_3 > Ka_2 > Ka_1$
   C. $Ka_1 > Ka_2 = Ka_3$
   D. $Ka_1 = Ka_2 = Ka_3$

***Explanation:*** The dissociation decreases due to the greater difficulty of removing a positively charged proton from a negatively charged ion (such as $H_2PO_4^-$) then from the neutral molecule $H_3PO_4$. And $HPO_4^{2-}$ than $H_2PO_4^-$.

In short the dissociation decreases in order $HPO_4^{2-} < H_2PO_4^{2-} < H_3PO_4$. Hence $Ka_1 > Ka_2 > Ka_3$.

**Answer: A**

8. Which group of elements are the most reactive of all the metallic elements?
   A. Alakli metals
   B. Alkali earth metals
   C. Transition metals
   D. Group 2B metals

**Answer: A**



9. All of these are characteristics of most ionic compounds in the solid phase except
   A. high melting point
   B. solubility in water
   C. high electrical conductivity
   D. insolubility in organic solvents

***Explanation:*** Ionic compounds don't conduct electricity in solid phase, because of lack of mobile ions.

**Answer: C**

10. Consider the following:

$$2NO\,(g) + Cl_2\,(g) \rightleftharpoons 2NOCl\,(g) \quad \Delta H = -78.38\ kJ$$

Which conditions of temperature and pressure will produce the highest yield of NOCl at equilibrium?
   A. high temperature – High pressure
   B. high temperature – low pressure
   C. low temperature – high pressure
   D. low temperature – low pressure

***Explanation***

- Increasing pressure will shift the equilibrium towards less number of gaseous molecules. Hence, in the above reaction the rxn will shift towards the product.
- Lowering the temperature favours the exothermic direction so for the above reaction it favours the product because the forward reaction has negative ΔH value.
- Increasing temperature favours the endothermic direction (to the reactant of the above reaction). This is because if the forward reaction is exothermic, the backward reaction is endothermic.

**Answer: C**

11. In the process known as 'roasting' a (n) _____ is chemically converted to a (an) _____
   A. carbonate/ oxide
   B. oxide/ sulfate
   C. sulfide/ oxide
   D. phosphate/ phosphide

**Answer: C**

12. Which of the following statement is NOT true according to 1st law of thermodynamics?
   A. Energy can neither be created nor destroyed although it can be changed from one form to another.
   B. The total energy of the universe is constant.

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C. The change in the internal energy of a closed system is equal to the sum of heat absorbed by the system and work done by the system.

D. The change in internal energy of a system is zero in all process.

***Explanation***

According to 1st law of thermodynamics, energy of the universe is conserved. i.e., $\Delta E_{system} + \Delta E_{surrounding} = 0$

**Answer: D**

13. How many chloride ions are in 1 mole of $CaCl_2$?

A. $3.01 \times 10^{23}\ Cl^-$ ions C. $1.81 \times 10^{24}\ Cl^-$ ions

B. $6.02 \times 10^{23}\ Cl^-$ ions D. $1.20 \times 10^{24}\ Cl^-$ ions

***Solution***

$CaCl_2 \longrightarrow Ca^{2+} + 2Cl^-$

1 mole 1 mole 2 mole (from the stoichiometry)

1 mole $Cl^- = 6.02 \times 10^{23}$ ions

2 mole $Cl^- = ?x$

Solving for the unknown

$x = 2 \times 6.02 \times 10^{23}$ ions $= 12.04 \times 10^{23}\ Cl^-$ ions $= 1.204 \times 10^{24}$

**Answer: D**

14. Which of the following types of solutions are possible?

I. solid dissolved in a liquid

II. gas dissolved in a liquid

III. gas dissolved in a gas

IV. solid dissolved in a solid

A. I and II C. I, II and IV

B. I, II, III and IV D. I

**Answer: B**

15. What is the normality of 1.0 M solution of $Na_2CO_3$?

A. 1N B. 0.5N C. 2N D. 3N

***Solution***

Normality = Molarity × charge Z (charge) = no of positive charge

$N = M \times Z$

$N = 1 \times 2 = 2N$

$Na_2CO_3 \longrightarrow 2Na^+ + CO_3$

charge = 2

**Answer: C**



16. What is the type of monomer unit in a natural protein polymer?

A. Amino acids C. α – amino acids

B. Essential amino acids D. Nucleic acids

**Answer: C**

17. Which of the following substance could be amphiprotic?

I. $NO_3^-$ II. $H_2O$ III. $NH_3$ IV. $HCO_3^-$

A. I, II and III C. II and IV

B. I and II D. II, III and IV

***Explanation***

Molecules or ions gain or lose a proton, under appropriate conditions. Such species are said to amphiprotic.

Example $H_2O$, $NH_3$, $CH_3COOH$, $OH^-$, $HSO_4^-$, $H_2PO_4^-$ . . .

**Answer: D**

18. Which one of the following does NOT form hydroxide ions when placed in water?

A. Ionic hydrides C. Nonmetal oxides

B. Ionic metal oxides D. Ionic nitrides

**Answer: C**

19. Which of the following is true for a 0.10 M solution of a weak acid, HB?

A. $[H^+] = 0.10$ M C. $[HB] > [H^+]$

B. $[B^-] = 0.10$ M D. pH = 1.0

***Explanation***

$HB + H_2O \rightleftharpoons H_3O^+(aq) + B^-(aq)$

| | | | |
|---|---|---|---|
| Initial conc. | 0.1 | - | - |
| change | -x | x | x |
| Equil.conce | 0.1 – x | x | x |

From the above table it is clear to observe that $[HB] > [H^+]$

**Answer: C**

20. What is the bond order of $O_2^+$ ?

A. 2 B. 2.5 C. 1.5 D. 3

***Explanation***

The total electrons in $O_2^+$ (Z = 8) = 15 electrons.

The molecular orbital arrangement will be

$(\delta 1s)^2\ (\delta^* 1s)^2 (\delta 2s)^2 (\delta^* 2s)^2 (\delta 2px)^2 (\pi 2py)^2 (\pi 2pz)^2 (\pi^* 2py)^1$

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Bond order $= \frac{1}{2}$(number of bonding electrons − number of anti bonding electrons) $= \frac{1}{2}(10-5) = 2.5$

Note: * represents anti bonding molecular orbitals.

21. Which of the following is characteristics of an anti bonding molecular orbital?

**Answer: B**

A. It is a molecular orbital with a high probability of finding electrons parallel to the reign between bonded atoms.
B. It has no electrons.
C. It is a molecular orbital with high probability of finding electrons away from the region between bonded atoms.
D. It is a molecular orbital with high probability of finding electrons in the region between the bonded atom.

22. Which of the following doesn't affect the rate of chemical reaction?

**Answer: C**

A. Enthalpy of the reaction
B. Concentration of reactants
C. surface area
D. Temperature

***Explanation:***

Factors that affect rate of a chemical reaction are nature of reactants, surface area, concentration, temperature and catalyst.

23. What is the molarity of a solution obtained by dissolving 0.01 moles of NaCl in 500 ml of solution?

**Answer: A**

A. 0.01 M B. 0.05M C. 0.02M D. 0.10M

***Solution:***

$$\text{Molarity (M)} = \frac{\text{number of moles of solute}}{\text{volume of solution (L)}}, V = 500\text{mL}, V = 0.5\text{L}$$

$$M = \frac{0.01\ \text{mol}}{0.5\ \text{Lit}} = 0.02M$$

**Answer: C**



24. Which of the following is NOT a valid expression for the rate of the reaction given below?

$$4NH_3 + 7O_2 \longrightarrow 4NO_2 + 6H_2O$$

A. $-\frac{1}{7}\Delta\frac{[O_2]}{\Delta t}$
B. $\frac{1}{7}\Delta\frac{[NO_2]}{\Delta t}$
C. $-\frac{1}{6}\Delta\frac{[H_2O]}{\Delta t}$
D. $-\frac{1}{4}\Delta\frac{[NH_3]}{\Delta t}$

***Explanation***

$$4NH_3 + 7O_2 \longrightarrow 4NO_2 + 6H_2O$$

$$\text{rate} = -\frac{1}{4}\frac{\Delta[NH_3]}{\Delta t} = -\frac{1}{7}\frac{\Delta[O_2]}{\Delta t} = \frac{1}{4}\frac{\Delta[NO_2]}{\Delta t} = \frac{1}{6}\frac{\Delta[H_2O]}{\Delta t}$$

**Answer: C**

25. Each of the choices below gives a reaction and the corresponding rate law. Of these choices, which one could be an elementary process or individual step in a chemical reaction?

A. $2A \longrightarrow P$ rate = k [A]
B. $A + B \longrightarrow P$ rate = k [A] [B]
C. $A + 2B \longrightarrow P$ rate = k $[A]^2$
D. $A + B + C \longrightarrow P$ rate = k [A] [C]

***Explanation***

- Molecularity is the number of reactant molecules taking part in elementary reaction.
- The rate law in an elementary step follows directly from the molecularity of that step.

**Answer: B**

26. Consider the reaction in which nitric oxide is oxidized to nitrogen dioxide.

$$2NO\ (g) + O_2\ (g) \longrightarrow 2NO_2\ (g)$$

for which of the rate law is rate = k $[NO]^2$ $[O_2]$. If the reaction takes place in a sealed vessel and the partial pressure of nitric oxide is doubled, what effect would this have on the rate of the reaction?

A. The reaction rate would increase by a factor of four.
B. The reaction rate would increase by a factor of three.
C. The reaction rate would increase by a factor of eight.
D. The reaction rate would increase by a factor of two.

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*Explanation*
If a given reactantsincreased by a factor, x, then the rate increases by a factor of $x^a$ wherea is the order the reaction with respect to the reactant. So, from the rate law equation rate = k $[NO]^2[O_2]$, the order with respect to NO is 2, and the partial pressure of NO is doubled, then the rate increases by $2^2 = 4$.

**Answer: A**

27. Which of the following will change the value of an equilibrium constant?
A. Changing the temperature.
B. Adding other substancesthat do not react with any of the species involved in the equilibrium,
C. Varying the initial concentration of reactants.
D. Varying the initial concentration of products.

**Answer: A**

28. The equilibrium constant for reaction (1) is k. What is the equilibrium constant for reaction (2)?

$$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g) \ldots (1)$$

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g) \ldots (2)$$

A. $k^2$ B. 2k C. $\frac{1}{2}$ D. $\frac{1}{k^2}$

*Explanation*
- The equilibrium expression for a reaction written in reverse is the reciprocal of that for the original reaction.
- When the balanced equation for reaction is multiplied by a factor n, the equilibrium expression for the new reaction is the original expression raised to the $n^{th}$ power. Thus

$$k_C^{new} = \left(k_C^{original}\right)^n$$

1) $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g) \ldots k$

Reversing the first equation and multiplying by 2 we get

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g) \ldots \left(\frac{1}{k}\right)^2 = \frac{1}{k^2}$$

**Answer: D**



29. Which of the following are carbohydrates?
I. nucleic acids
II. polyhydroxyl ketone
III. disaccharides
IV. aldoses
V. triglycerides
A. I and V
B. II, III and IV
C. I, III and V
D. III and IV

**Answer: A**

30. Which of the following is true concerning this acid – base reaction?

$$H\bar{S} + CH_3Cl \longrightarrow CH_3SH + Cl^-$$

A. $CH_3Cl$ is a Bronsted – Lowery acid
B. $H\bar{S}$ is a Bronsted – Lowry acid
C. $H\bar{S}$ is a Lewis acid
D. $H\bar{S}$ is a Lewis base

**Answer: D**

31. For the stepwise dissociation of aqueous, $H_3PO_4$, which of the following is NOT a conjugate acid-base pair?
A. $HPO_4^{2-}$ and $PO_4^{3-}$ C. $H_2PO_4^-$ and $HPO_4^{2-}$
B. $H_3PO_4$ and $H_2PO_4^-$ D. $H_2PO_4$ and $PO_4^{3-}$

*Explanation*

| Acid | Conjugate base |
|---|---|
| $H_3PO_4$ | $H_2PO_4^-$ |
| $H_2PO_4^-$ | $HPO_4^{-2}$ |
| $HPO_4^{-2}$ | $PO_4^{3-}$ |

**Answer: D**

32. Which solution below has the highest concentration of hydroxide ions?
A. pH = 3.21 C. pH = 7.93
B. pH = 12.59 D. pH = 9.82

*Explanation*
The higher the pH value, the lesser the pOH and the higher will its hydroxide ion concentration be.

**Answer: B**

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33. The standard reduction potential of $Zn^{2+}/Zn$ is (-0.76V) and that of $Cu^{2+}/Cu$ is +0.34V. What would be the electromotive force (emf) of the cell constructed between these two electrodes?
A. 1.10V B. 0.42V C. -1.10V D. -0.42V

***Explanation***
Zinic is a reducing agent and Cu is an oxidizing agent. Thus the reaction is as follows:

$$Zn + Cu^{2+} \rightleftharpoons Zn^{2+} + Cu$$

$$E°_{cell} = E°_{oxi} + E°_{red}$$
$$= -(-0.76V) + (0.34V) = 1.1V$$

**Answer: A**

34. At constant T and P, which one of the following statements about enthalpy (ΔH and internal energy (Δu)) changes is correct for the reaction?

$$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$$

A. $\Delta H = \Delta U$ C. $\Delta H > \Delta U$
B. $\Delta H < \Delta U$
D. ΔH and Δu are independent of physical states of the reactant.

***Explanation***
The enthalpy, H, is defined as
$H = U + PV$
$\Delta H = \Delta u + P\Delta V + V\Delta P$
For temperature constant (ideal gas)
$\Delta H = \Delta u + \Delta(nRT)$, $\Delta n = n_p - n_r$

$$\Delta n = 1 - \left(1 + \frac{1}{2}\right)$$

$$\Delta n = -\frac{1}{2} \Rightarrow \Delta H = \Delta U - \frac{1}{2}RT$$

Therefore $\Delta H < \Delta$

**Answer: A**

35. Which one of the following conditions regarding a chemical process ensures its spontaneity at all temperatures?
A. $\Delta H < 0, \Delta S < 0$ C. $\Delta H < 0, \Delta S > 0$
B. $\Delta H > 0, \Delta S < 0$ D. $\Delta H > 0, \Delta S > 0$



***Explanation***
From the Gibbs free energy formula
$\Delta G = \Delta H - T\Delta S$
- If ΔG is negative, the reaction is spontaneous.
- If ΔG is positive, the reaction is non spontaneous.
- If ΔG is zero, the reaction is at equilibrium.

And ΔG will be negative when ΔH is negative and ΔS is positive at any temperature.

**Answer: C**

36. Why silicate minerals are not commonly used as source of metals.
A. They are rare minerals.
B. They usually don't contain important metals.
C. They are difficult to reduce and concentrate.
D. They are only found at excessive depths in the ocean.

**Answer: C**

37. Which of the following metals is the best conductor of heat and electricity?
A. Copper C. Gold
B. Silver D. Tungsten

**Answer: B**

38. $HNO_3$ is used as
I. a hydrogenating agent
II. an acid
III. an oxidizing agent
IV. an ammoniating agent
A. I and II C. I, II, III and IV
B. I and II D. II, III and IV

**Answer: D**

39. Which equation represents an oxidation – reduction reaction?
A. $H_2SO_4 + 2NH_3 \longrightarrow (NH_4)_2SO_4$
B. $H_2SO_4 + Na_2CO_3 \longrightarrow Na_2SO_4 + H_2O + CO_2$
C. $2K_2CrO_4 + H_2SO_4 \longrightarrow K_2Cr_2O_7 + K_2SO_4 + H_2O$
D. $2H_2SO_4 + Cu \longrightarrow CuSO_4 + 2H_2O + SO_2$

***Explanation*** Redox reaction that occurs as a result of electron transfer. The best way to check a reaction is Redox or not is to assign the oxidation number value for each element in the

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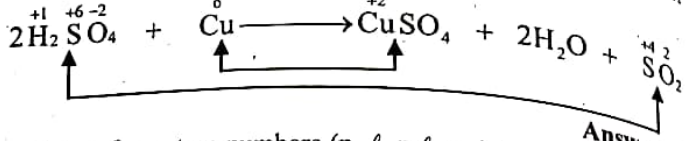
reactant and product side. Hence, if there is a change in oxidation number that reaction is Redox.



Answer: D

40. Which set of quantum numbers (n, $\ell$, m$\ell$, ms) is not permitted by the value of quantum mechanics.

A. $1, 0, 0, \frac{1}{2}$ C. $3, 3, 1, -\frac{1}{2}$

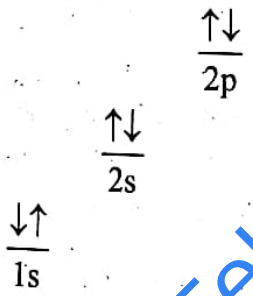
B. $2, 1, -1, -\frac{1}{2}$ D. $4, 3, 2, \frac{1}{2}$

*Explanation*

- Principal quantum number (n) takes a value of 1, 2, . . .
- Angular momentum quantum number ($\ell$) take a value from 0 to n – 1 and is always less than n.
- Magnetic quantum number (m$\ell$) takes places from $-\ell$ to $+\ell$.
- Magnetic spin quantum number (m$\ell$) takes values either $+\frac{1}{2}$ or $-\frac{1}{2}$

Answer: C

41. What can you conclude from the figure below?



A. Hund's rule has been violated.
B. The Pauli exclusion principle has been violated.
C. The Autbau principle has been violatd.
D. This is valid orbital diagram.



*Explanation*

Hund's rule states that, "For any set of orbitals that are equal in energy the configuration with the maximum number of parallel spins results in the lowest electron – electron repulsion." Besides Hnnd's rule can also be explained as:

"No pairing of electrons takes place in degenerate orbitals before orbitals are singly filled."

Pauli exclusion principle states that, "No two electrons can have the same four quantum numbers."

Aufbau principle states that, "During fulfilling of electrons in an atom oribital with lower energy will be filled first that orbitals with higher energy."

Answer: A

42. Which set contains only covalently bonded molecules?

A. $BCl_3$, $SiCl_4$, $PCl_3$ C. $I_2$, $H_2S$, NaI

B. $NH_4Br$, $N_2H_4$, HBr D. Al, $O_3$, $As_4$

Answer: A

43. How many grams of $Ca(OH)_2$ are contained in 1500 ml of 0.0250M $Ca(OH)_2$ solution?

A. 1.85 gram C. 3.17 gram

B. 2.78 gram D. 4.25 gram

*Solution*

$$\text{Molarity} = \frac{\text{mass of } Ca(OH)_2}{\text{Molar mass of } Ca(OH)_2} \times \frac{1}{\text{Volume (L)}}$$

Molar mass = 74 g/mol

Mass of $Ca(OH)_2$ = Molarity × molar mass × Volume (Liter)

$= 0.0250 \times 74 \times 1.5$

$= 2.78$ gram

Answer: B

44. The conventional equilibrium constant expression ($k_C$) for the system as described by the equation.

$2ICl(s) \rightleftharpoons I_2(s) + Cl_2(g)$

A. $[Cl_2]$ C. $[I_2][Cl_2][ICl]^2$

B. $[Cl_2]/[ICl]^2$ D. $[I_2][Cl_2]/2[ICl]$

*Explanation*

- The equilibrium is heterogeneous equilibrium because the phase of the reactant and the products are different.

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- The $k_C$ value of heterogeneous equilibrium doesn't depend on the amount of pure solids or liquids present. Thus, the $k_C$ value does not include pure solids or liquids.

$$k_C = [Cl_2]$$

45. What is the conjugate base of $HSO_4^-$?

Answer: A

A. $H^+$ B. $H_2SO_4$ C. $OH^-$ D. $SO_4^{2-}$

46. An electrochemical cell constructed for the reaction.

Answer: D

$Cu^{2+}(aq) + M(s) \longrightarrow Cu(s) + M^{2+}(aq)$, has an $E°_{cell} = 0.75V$. The standard reduction $E°_{cell}$ potential for $Cu^{2+}(aq)$, is 0.34V. What is the standard reduction potential for $M^{2+}(aq)$?

A. -1.09 V C. 0.410 V
B. -0.410V D. 1.09V

***Explanation***

$E°_{cell} = E°_{ox} + E°_{red}$
$0.75V = E°_{ox} + 0.34\ V$
$\Rightarrow E°_{red} = -E°_{ox} = -0.410V$
$E°_{ox} = +0.41\ V$

Answer: B

47. Which of the following compounds would be expected to have the highest melting point?

A. $BaF_2$ B. $BaCl_2$ C. $BaBr_2$ D. $BaI_2$

***Explanation:*** All of the compounds given are ionic. Hence, the strength of an ionic bond affects the melting point of the compounds. Ionic compounds with high lattice energies generally are harder and have higher melting point than compounds with low lattice energies.

Lattice energy increases with increasing ionic charge and decreasing ionic radii. Thus, the lattice energy decreases in the order $BaF_2 > BaCl_2 > BaBr_2 > BaI_2$

Answer: A

48. Which of the following is true about chlorofluoro carbons?

A. React directly with stratospheric ozone to destroy it.
B. Interact with UV energy and becomes free radicals which destroy ozone.
C. Become free radicals that react with oxygen to create ozone.
D. React with free radicals to remove carbondioxide.

Answer: B



49. Which metal can be found as a free element.

A. Cr B. Fe C. Mn D. pt

Answer: D

50. The enthalpy change for which reaction represents the standard enthalpy of formation for hydrogen cyanide, HCN?

A. $H(g) + C\ (graphite) + N\ (g) \longrightarrow HCN\ (g)$

B. $\frac{1}{2}H_2(g) + C(graphite) + \frac{1}{2}N_2(g) \longrightarrow HCN(g)$

C. $HCN\ (g) \longrightarrow \frac{1}{2}H_2(g) + C\ (graphite) + \frac{1}{2}N_2(g)$

D. $H_2\ (g) + 2C\ (graphite) + N_2\ (g) \longrightarrow 2HCN\ (g)$

***Explanation:*** Standard enthalpy of formation ($\Delta H°_f$) is one mole of a compound is formed from the elements in standard states. Since hydrogen and nitrogen are diatomic molecules, they should be written as $H_2$ and $N_2$.

Answer: B

51. Which one of the compounds below is most likely to be ionic?

A. $CCl_4$ B. $NO_2$ C. $ScCl_3$ D. $ClO_2$

Answer: C

52. What is the correct order when the substance $O_2$, $H_2O$, $OF_2$ and $H_2O_2$ are arranged in order of increasing oxidation number for oxygen?

A. $O_2, H_2O, OF_2, H_2O_2$ C. $H_2O_2, O_2, H_2O, OF_2$
B. $H_2O, H_2O_2, O_2, OF_2$ D. $OF_2, O_2, H_2O_2, H_2O$

***Explanation***

Molecule oxidation number of oxygen

| | |
|---|---|
| $O_2$ | 0 |
| $H_2O$ | -2 |
| $OF_2$ | +2 |
| $H_2O_2$ | -1 |

According the increasing order $H_2O < H_2O_2 < O_2 < OF_2$

Answer: B

53. A mono atomic ion that has 20 protons and a+2 charge

A. has 16 protons C. has 18 neutrons
B. has the symbol $Ar^{2+}$ D. is isoelectronic with Ar

***Explanation:*** a+2 charge indicates that 2 electrons are lost. So the number of electrons become 18, which makes a+2 isoelectronic with Ar.

Answer: D

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54. What is the molality of a solution that contains 51.2 gram of naphthalene, $C_{10}H_8$, in 500 ml of carbon tetrachloride? (Given the density of CCl = 1.60 g/ml)

A. 0.25 m B. 0.5 m C. 0.75 m D. 0.84 m

*Solution*

**Given:**

$m_{C_{10}H_8} = 51.2$ gram

Molar mass = 128 g/mol

V = 500 ml

density (s) = 1.60 g/ ml

$$\text{moles of } m_{C_{10}H_8} = \frac{51.2 \text{ gram}}{128 \text{ g/mol}} = 0.4 \text{ mole}$$

$$\text{density (s)} = \frac{m_{CCl_4}}{V_{CCl_4}}$$

$m_{CCl_4} = 1.60$ g/mol × 500 ml

= 800 gram = 0.8 kg

$$\therefore \text{molality (m)} = \frac{\text{no of moles}}{\text{mass of solvent (kg)}} = \frac{0.4}{0.8} = 0.5 \text{ m}$$

**Answer: B**

55. A homogeneous liquid reaction mixture is often heated to increase the rate of reaction. This is best explained by the fact that raising the temperature.

A. increases the heat of reaction
B. decreases the energy of activation
C. increases the vapor pressure of the liquid
D. increases the average kinetic energy of the reactants

**Answer: D**

56. Which of the following titrations will have an equivalence point pH > 7.00?

A. weak acid with strong base
B. strong acid with strong base
C. weak acid with weak base
D. strong acid with a weak base



*Explanation*

| Titration type | pH at equivalence point |
|---|---|
| Strong acid – strong base | pH = 7 |
| Weak acid – strong base | pH > 7 |
| Strong acid – weak base | pH < 7 |
| Weak acid – weak base | Difficult to determine |

**Answer: A**

57. A 0.10M solution of weak acid, Hx, is 0.059% ionized. What is the dissociation constant for the acid?

A. $3.8 \times 10^{-9}$ C. $6.5 \times 10^{-7}$
B. $3.5 \times 10^{-8}$ D. $4.2 \times 10^{-6}$

*Solution*

Percentage dissociation (%) = 0.059%

$$\text{percentage dissociation (x\%)} = \frac{\text{amount of dissociated ion (x)}}{\text{initial concentration (C)}} \times 100$$

$$x\% = \frac{x}{C} \times 100$$

$$x = \frac{\% \times C}{100} = \frac{0.059 \times 0.1}{100} = 5.9 \times 10^{-5} \text{ M} = [H_3O^+] = [A^-]$$

$$HA + H_2O \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

| | HA | $H_2O$ | $H_3O^+$ | $A^-$ |
|---|---|---|---|---|
| Initial conc. | 0.1 | - | - | |
| change | -x | | +x | +x |
| Equilibrium | 0.1 – x | | x | x |

$$Ka = \frac{[H_3O^+][A^-]}{[HA]} = \frac{x \cdot x}{0.1 - x} \cong \frac{x^2}{0.1} = \frac{(5.9 \times 10^{-5})^2}{0.1}$$

$= 3.5 \times 10^{-8}$

**Answer: B**

58. Consider the standard voltaic (or galvanic) cell: Fe, $Fe^{2+}$ versus Au, $Au^{3+}$. Which answer identifies the cathode and gives the $E°_{cell}$ for the cell?

**Given:**

$E°(Au^{3+}/Au) = 1.50$ V, $E°(Fe^{2+}/Fe) = -0.44$ V

A. Fe, -0.44V C. Au, 1.94 V
B. Au, 1.06V D. Fe, 1.94 V

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***Explanation***
The half cell with more positive electrode potential is always placed to the cathode. Therefore, gold will be reduced and iron is oxidized.

$$E^\circ_{cell} = E^\circ_{ox} + E^\circ_{red}$$
$$= 0.44\ V + (1.50V) = 1.94\ V$$

59. What is the enthalpy for the reaction **Answer: C**

$$C_2H_6\,(g) + \frac{7}{2}O_2\ (g) \longrightarrow 2CO_2\,(g) + 3H_2O\,(\ell)?$$

Given the following reaction

1. $2C(s) + 3H_2\,(g) \longrightarrow C_2H(g)$ $\Delta H^\circ_f = -84.7\ kJ.mol^{-1}$
2. $C(s) + O_2\,(g) \longrightarrow CO_2\,(g)$ $\Delta H^\circ f = -393.5\ kJ.mol^{-1}$
3. $H_2(g) + \frac{1}{2}O_2\,(g) \longrightarrow H_2O\,(l)$ $\Delta H^\circ f = -285.8\ kJ.mol^{-1}$

A. -764 kJ B. -1560 kJ C. -1664 kJ D. -312 kJ

***Explanation***

- Reversing the first equation we get,

A. $C_2H_6\,(g) \longrightarrow 2C\,(s) + 3H_2\,(g)$ $\Delta H^\circ_f = +84.7\ kJ.mol^{-1}$

- Multiplying the second equation by two, we get

B. $2C(s) + 2O_2\,(g) \longrightarrow 2CO_2\,(g)$
$\Delta H^\circ f = -393.5\ kJ.mol^{-1} \times 2 = -787\ kJ/mol$

- Multiplying the third equation by three, we get

C. $3H_2(g) + \frac{3}{2}O_2\,(g) \longrightarrow 3H_2O\,(l)$

$\Delta H^\circ f = -285.8\ kJ.mol^{-1} \times 3 = -857.4\ kJ/mol$

Adding equations A, B, C we get

$$C_2H_6\,(g) + \frac{7}{2}O_2\,(g) \longrightarrow 2CO_2\,(g) + 3H_2O\,(\ell)$$

$\Delta H^\circ_{rxn} = 84.7 + (-787) + (-857.4)$

$\Delta H^\circ_{rxn} = -1559.7\ kJ = -1560\ kJ$

**Answer: B**

60. How many atoms are present in 64 gram of $SO_2$?

A. $6.02 \times 10^{23}$ C. $3 \times 6.02 \times 10^{23}$
B. $2 \times 6.02 \times 10^{23}$ D. $4 \times 6.02 \times 10^{23}$



***Explanation***
1 mole of $SO_2$ = Avogadro's Number = Mass of $SO_2$ (in gram)
1 mole of $SO_2 = 6.02 \times 10^{23}$ molecules = 64 gram
1 molecule of $SO_2$ is made up of 3 atoms then $6.02 \times 10^{23}$ molecules of $SO_2$ are made up of $6.02 \times 10^{23} \times 3 = 12.06 \times 10^{23}$ atoms

**Answer: C**

61. What is the correct decreasing order of the atomic radius in the elements below?

A. $Mg < Al < Cl$ C. $Mg > Al > Cl$
B. $Al < Mg < Cl$ D. $Cl > Al > Mg$

***Explanation:*** Atomic radius decrease across a period.

**Answer: C**

62. The outermost electron (s) of which the following experiences, is the greatest effective nuclear charge?

A. K B. Ca C. Br D. P

**Answer: C**

63. Which of the following is not possible shape of a molecules?

A. Tetrahedral $GeCl_4$
B. A bent HCN
C. A trigonalbipyramidal $PCl_5$
D. A T – shaped $BrF_3$

**Answer: B**

64. Four elements, A, B, C, D have atomic numbers Z – 1, Z, Z + 1 and Z + 2, respectively. If Z is 9, then the bond between which pair of elements will be ionic?

A. A and C C. B and D
B. C and D D. B and C

***Explanation:*** The atomic numbers are 8, 9, 10 and 11 respectively. And the configuration of the elements will be

| | K | L | M |
|---|---|---|---|
| $8^A$ | 2 | 6 | - |
| $9^B$ | 2 | 7 | - |
| $10^C$ | 2 | 8 | - |
| $11^D$ | 2 | 8 | 1 |

From the configuration C is necessarily a noble gas i.e., Ar, which is not expected to form a compound at normal condition. Besides element B is in group VIIA and an element D is a group IA element. Therefore element B and D forms an ionic compound.

**Answer: C**

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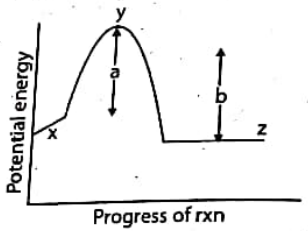
65. Which of the following statements is INCORRECT for the reaction represented by the figure below?




A. y represents the transition state of the reaction
B. the enthalpy of the reaction is given by b – a
C. the forward reaction is endothermic
D. b is the activation energy of the formed reaction

***Explanation***

'x' represents the reactant, and 'z' represents the product and y is the transition state

- b isheat content of product
- a is the activation energy
- The reaction is exothermic since heat of product (z) is lower in energy than the reactant (x).

No answer (B, C and D are incorrect)

66. Given the initial rate below the reaction?

$aA + bB \longrightarrow$ products

What is the correct rate law?

| Experiment | Concentration of A (M) | Concentration of B (M) | Initial Rate M/s |
|---|---|---|---|
| 1 | 0.100 | 0.100 | $4.00 \times 10^{-5}$ |
| 2 | 0.200 | 0.100 | $1.6 \times 10^{-4}$ |
| 3 | 0.100 | 0.200 | $8.0 \times 10+-5$ |

A. Rate = k [A] [B]
B. Rate = k $[A]^2$ [B]
C. Rate = l $[A]^2$ $[B]^2$
D. Rate = k [A] $[B]^2$

***Explanation:***

For the reaction $aA + bB \longrightarrow$ product
The rate law expression will be
rate = k $[A]^x$ $[B]^y$ where x and y are order of the rxn
To determine the order with respect to A, take experiments 1 and 2 (where the concentration of B are similar)



$rate_1 = k [A]^x [B]^y = k (0.2)^x (0.1)^y = 4.0 \times 10^{-5}$
$rate_2 = k [A]^x [B]^y = k (0.2)^x (0.1)^y = 1.6 \times 10^{-4}$
Taking the ratio of $rate_1$ and $rate_2$

$$\frac{rate_1}{rate_2} = \frac{k(0.1)^x (0.1)^y}{k(0.2)^x (0.1)^y} = \frac{4.0\times10^{-5}}{1.6\times10^{-4}}$$

$$\left(\frac{1}{2}\right)^x = \frac{1}{4}$$

$$\left(\frac{1}{2}\right)^x = \left(\frac{1}{2}\right)^2$$

x = 2 (order with respect to A)
To determine the order with respect to B, take experiments 1 and 3 (where the concentratin of A are similar).
$rate_1 = k (0.1)^x (0.1)^y = 4.0 \times 10^{-5}$
$rate_3 = k (0.1)^x (0.2)^y = 8.0 \times 10^{-5}$

$$\left(\frac{1}{2}\right)^y = \frac{1}{2}$$

y = 1 (order with respect to B)
Therefore, the rate law expression will be

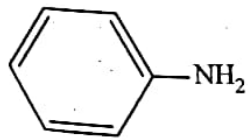
rate = k $[A]^2$ [B]

**Answer: B**

67. Which of the following compounds is the strongest base?

A. $NH_2Cl$
B. $NH_3$
C.

D.




***Solution***: Electron with drawing groups such as halogens and phenyl decrease basicity.

**Answer: B**

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68. For preparing a buffer solution of pH 6.0 by mixing sodium acetate with acetic acid. What would be the ratio of the concentratin of salt and acid (ka = $10^{-5}$)?

A. 1: 10 B. 10: 1 C. 100: 1 D. 1: 100

***Explanation***

Using Handresson's Hassel Balch formula

$$pH = p^{Ka} + \log\left(\frac{[CH_3COONa]}{[CH_3COOH]}\right)$$

$$6 = -\log(10^{-5}) + \log\left(\frac{[CH_3COONa]}{[CH_3COOH]}\right)$$

$$6 = 5 + \log\left(\frac{[CH_3COONa]}{[CH_3COOH]}\right)$$

$$1 = \log\left(\frac{[CH_3COONa]}{[CH_3COOH]}\right)$$

$$\frac{[CH_3COONa]}{[CH_3COOH]} = 10^1 = \frac{10}{1}$$

Therefore, the ratio of the salt to the acid is 10: 1.

**Answer: B**

69. Given the following datea at 25°C.

1. $2NOCl\ (g) \longrightarrow 2NO\ (g) + Cl_2\ (g)$ $\Delta H^\circ = +75kJ$
2. $2NO\ (g) + O_2\ (g) \longrightarrow 2NO_2(g)$ $\Delta H^\circ = -113.05\ kJ$
3. $2NO_2\ (g) \longrightarrow N_2O_4\ (g)$ $\Delta H^\circ = -58.03\ kJ$

CalcualteΔH° at 25°C for the reaction

$N_2O_4\ (g) + Cl_2\ (g) \longrightarrow 2NOCl\ (g) + O_2\ (g)$

A. +96.08 kJ C. 186.8 kJ

B. +299 kJ D. 85.52 kJ

***Explanation***

- Reversing the first equation, results the following

  1. $2NO\ (g) + Cl_2\ (g) \longrightarrow 2NOCl\ (g)$, $\Delta H^\circ = -75kJ$

- Reversing the second equation, results the following

  2. $2NO_2\ (g) \longrightarrow 2NO\ (g) + O_2\ (g)$, $\Delta H^\circ = 113.0\ kJ$

And reversing also the third equation, gets the following



3. $N_2O_4\ (g) \longrightarrow 2NO_2\ (g)$, $\Delta H^\circ = 58.03\ kJ$

Adding equations 1, 2 and 3 results

$N_2O_4\ (g) + Cl_2\ (g) \longrightarrow 2NOCl\ (g) + O_2(g)$

$\Delta H^\circ_{rxn} = -75\ kJ + 113.0\ kJ + 58.03\ kJ$

$= +96.08\ kJ$

**Answer: A**

70. How many moles of $KBrO_3$ are required to prepare 0.0700 moles of $Br_2$ according to the equation.

$KBrO_3 + KBr + HNO_3 \longrightarrow KNO_3 + Br_2 + H_2O$

a. 0.0233 B. 0.0732 C. 0.210 D. 0.220

***Explanation***

First balance the given equation

$KBrO_3 + 5KBr + 6HNO_3 \longrightarrow 6KNO_3 + 3Br_2 + 3H_2O$

So using the balanced equation

$$\frac{x}{1\ mole} = \frac{0.07\ moles\ of\ Br_2}{3\ mole}$$

$$x = \frac{1 \times 0.07\ moles}{3} = 0.0233\ moles$$

**Answer: A**

71. A possible mechanism for a reactin $2A + B \longrightarrow C + D$ is

| | | |
|---|---|---|
| a. | $A + A \longrightarrow A_2$ | fast, equilibrium |
| b. | $A_2 + A \longrightarrow A_3$ | slow |
| c. | $A_3 + B \longrightarrow A + C + D$ | fast |

According to the mechanism, the rate law will be

A. Rate = k [A] C. Rate = k $[A]^3$

B. Rate = k $[A]^2$ D. Rate = k $[A]^2$

***Explanation***

Rate determining step: the elementary reaction in a mechanism that is so much slower than the rest that it governs the rate of the over all reaction.

So, the rate determining step is the second step which is the slowest step

$A_2 + A \longrightarrow A_3$ (slow)

Rate = k $[A_2]$ [A]

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Let us check wheather the mechanism works or not to give the over allbalnced equation.

$A + A \rightleftharpoons A_2$

$A_2 + A \longrightarrow A_3$

$A_3 + B \longrightarrow A + C + D$

$2A + B \longrightarrow C + D$

Here $A_2$ and A are intermidate.

The rate law for the slowest step is Rate = k $[A_2]$ [A], contains an intermidate species which is $A_2$. Therefore $A_2$ should be substituted $A + A \underset{k-1}{\overset{k_1}{\rightleftharpoons}} A_2$

Rate of forward reaction = $k_1 [A]^2$

Rate of the backward reaction = $k_1 [A_2]$

At equilibrium, the rate of forward reaction is equal to the rate of backward reaction.

$$k_1 [A]^2 = k_1 [A_2]$$

Solving fir $A_2$

$$[A_2] = \frac{k_1[A]^2}{k-1}$$

Now substituting the expression for $[A]_2$ into the rate law expression for the slow step.

$$\text{Rate } k [A_2] [A_2][A] = k\left(\frac{k_1[A]^2}{k-1}\right)[A] = \frac{k.k_1}{k-1}[A]^2[A]$$

$$= k [A]^2 [A] = k [A]^3$$

Note: $k = \frac{k.k_1}{k-1}$

**Answer: C**

72. What is the $[H_3O^+]$ in a 0.050 M solution of $Ba(OH)_2$?

a. $1.0 \times 10^{-13}$ M  C. $1.0 \times 10^{-5}$ M

b. $5.0 \times 10^{-10}$ M  D. $5.0 \times 10^{-2}$ M

***Explanation***

$Ba(OH)_2 + H_2O \longrightarrow Ba^{2+}(aq) + 2OH^-(aq)$

0.05M  0.05  $2 \times 0.05$ M = 0.1

$k_w = [H_3O^+][OH^-]$



At 25°C $k_w = 1 \times 10^{-14}$

$$[H_3O^+] = \frac{k_w}{[OH^-]} = \frac{1\times10^{-14}}{0.1} = 1\times10^{-13}\text{M}$$

**Answer: A**

73. Which species has the largest F – A – F bond angle where A is the central atom?

a. $BF_3$  B. $CF_4$  C. $NF_4$  D. $OF_2$

***Explanation***

| Moelcu le | Type | Bond Angle | Shape |
|---|---|---|---|
| $BF_3$ | $AB_3$ | 120° | Trigonal planar |
| $CF_4$ | $AB_4$ | 109.5° | Tetrahedral |
| $NF_3$ | $AB_3E$ | 107° | Trigonal pyramidal |
| $OF_2$ | $AB_2E_2$ | 104.5° | Angular/V – shape |

Note: A – central atom  B – surrounding atom  E – loneparis

**Answer: A**

74. When the atoms Li, Be, B, Na are arranged inorder of increasing atomic radius. Which is correct order?

a. Li, Be, B, Na  C. Na, Li, Be, B

b. Li, Na, B, Be  D. B, Be, Li, Na

***Explanation:*** Except Na, all the other elements are found at period two. Across a period atomic radius decreases. Hence insize Li > Be > B and Na belongs IA element found next Li. Hence, Na > Li because size increases down a group.

Therefore in size Na > Li > Be > B.  **Answer: D**

75. One of the steps in the manufacture of nitiric acid is the oxidation of ammonia shown in this equation

$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$

If gaseous water appears at a rate of 0.025 mol. $min^{-1}$. At what rate does ammonia dis appear?

A. 0.0040 mol. $min^{-1}$  C. 0.038 mol. $min^{-1}$

B. 0.017 mol.$min^{-1}$  D. 0.150 mol. $min^{-1}$

***Solution:*** $4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$

The rate law of expression will be

$$\frac{-\Delta[NH_3]}{4\Delta t} = \frac{-[O_2]}{5\Delta t} = \frac{\Delta[NO]}{4\Delta t} = \frac{\Delta[H_2O]}{6\Delta t}$$

rate of decomposition of ammonia = rate of formation of water

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$$\frac{1}{4}\frac{\Delta[NH_3]}{\Delta t} = \frac{1\Delta[H_2O]}{6\Delta t}$$

$$\frac{1}{4}\left(\frac{\Delta[NH_3]}{\Delta t}\right) = \frac{1}{6}(0.025\ \text{mol/min})$$

$$\frac{\Delta[NH_3]}{\Delta t} = \frac{4(0.025\ \text{mol/min})}{6} = 0.017\ \text{mol.min}^{-1}$$

Answer: B

76. What is the half life of the irreversible first order reaction A $\longrightarrow$ B, if 75% of A is converted into B in 60 minutes?

a. 30 minutes C. 60 minutes
b. 45 minutes D. 80 minutes

***Explanation:*** If the reaction is 75% complete, 75% of the reactant has been consumed. This leaves 25% in the original form.

For the 1st order reaction $\log\frac{[A]_o}{[A]_t} = \frac{kt}{2.303}$

$$\log\left(\frac{[A]_o}{\frac{1}{2}[A]_o}\right) = \frac{k.60}{2.303}$$

$k = 0.023\ \text{min}^{-1}$

$$t\frac{1}{2}\text{ for } 1^{st}\text{ order} = \frac{0.693}{k} = \frac{0.693}{0.023\,\text{min}^{-1}} = 30\ \text{min}$$

Answer: A

77. At a certain temperature for the reaction:

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

the equilibrium constrations are $[NO] = 0.52$ M; $[O_2] = 0.24$ M $[NO_2] = 0.18$ M. What is the value of $k_C$ at this temperature?

A. 0.0693 B. 0.50 C. 1.4 D. 2.0

***Solution:*** $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

$$k_c = \frac{[NO_2]^2}{[NO]^2[O_2]} = \frac{(0.18)^2}{(0.52)^2(0.24)} = \frac{0.0324}{0.065} = 0.5$$

Answer: B



78. A 25.00 ml sample of 0.1050 M $H_2SO_4$ is titrated with a NaOH solution of unknown concentration. The phenolphthalein end point was reached when 17.23 ml of the NaOH solution had been added. What is the cncentration of the NaOH?

a. 0.07617 M C. 0.1524 M
b. 0.1447 M D. 0.3047 M

***Solution***

At the end point, number of equivalents of the acid and the base are equal. That is

Normality of the acid × Volume of the acid = Normality of the base × volume of the base

$N_{Acid} \cdot V_{Acid} = N_{Base} \cdot V_{Base}$

Normality of $H_2SO_4$ = charge × Molarity

$= 2 \times (0.1050\ \text{M}) = 0.21\ \text{N}$

$0.21\ \text{N} \cdot 25\ \text{ml} = N_{Base}(17.23\ \text{ml})$

$N_{Base} = 0.3047\ \text{N}$

$$\text{Molarity of the base} = \frac{0.3047}{\text{charge}} = \frac{0.3047}{1} = 0.3047\text{M}$$

Answer: D

79. During electrolysis of an aquesous solution of $AgNO_3$. What would happen to the mass of siliver metal deposited if the current is doubled and the electrolysis time is decreased to $\frac{1}{2}$ of its intial value?

a. It would stay the same.
b. It would increase twice the initial value.
c. It would decrease to $\frac{1}{4}$ of its initial value.
d. It would decrease to $\frac{1}{2}$ of its value.

***Explanation***

Let $m_1$ – be the mass of Ag – metal produced by the current ($I_1$)
$m_2$ – be the mass of Ag – metal produced by the current ($I_2$)
given that

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The current is increased by two. That is $I_2 = 2I$, the time is reduced by $\frac{1}{2}$. That is $t_2 = \frac{1}{2}t$.

Applying Faraday's 1st law, for $I_1$ and $I_2$

$$\frac{m_1}{M} = \frac{I_1.t_1}{nF} \ldots\ldots\ldots (1) \qquad m - \text{mass}$$

$$\frac{m_2}{M} = \frac{2I_1.t\frac{1}{2}}{nF} \ldots\ldots (2) \qquad M - \text{molar mass}$$

Taking the ratio of (1) and (2), we get

$$\frac{m_1}{M}.\frac{M}{m_2} = \frac{I_1.t_1}{nF}.\frac{nF}{2I_1.t\,1/2}$$

$$\frac{m_1}{m_2} = 1; \qquad m_1 = m_2$$

**Answer: A**

80. What is the balanced equation for the cell notation of the following gavalnci cell?

$Pb(s) / Pb^{2+}(aq) // Cl_2(g) / Cl^-(aq) / C(s)$

a. $Pb(s) + 2Cl^-(aq) \longrightarrow Pb^{2+}(aq) + Cl_2(g)$

b. $Pb(s) + Cl_2(g) \longrightarrow Pb^{2+}(aq) + 2Cl^-(aq)$

c. $Pb^{2+}(aq) + 2Cl^-(aq) \longrightarrow Pb(s) + Cl_2(g)$

d. $Pb^{2+}(aq) + 2Cl^-(aq) \longrightarrow PbCl_2(s)$

***Explanation***

- The left side of the double arrow (II) represents the anode reaction and the right side indicate the cathode reaction.
- Carbon (graphite) is an electrode.
- The double arrow (II) indicates the salt bridge.

**Answer: B**



1. Which of the following is NOT a state function?
   A. Work C. Internal Energy
   B. Enthalpy D. Pressure

***Explanation***

A thermodynamic function that depends on the initial and final state of a system, regardless of how the conditions are achieved is called a state function.

Examples: Energy, volume, pressure, temperature, internal energy, Enthalpy, Entropy etc.

Functions that depend on the path are path functions. Examples, Heat, Work

**Answer: A**

2. Which of the following metals has the highest positive oxidation state?
   A. Ti C. Mn
   B. Cr D. Cu

***Explanation***

| Element | Oxidation State |
|---|---|
| Ti | +2, +3, +4 |
| Cr | +2, +3, +4, +5, +6 |
| Mn | +2, +3, +4, +5, +6, +7 |
| Cu | +1, +2, +3 |

**Answer: C**

3. Which of the following is NOT a natural polymer?
   A. Proteins C. Cellulose
   B. Starch D. Teflon

**Answer: D**

4. The coefficients required to correctly balance the reaction below are

$$?Al(NO_3)_3 + Na_2S \longrightarrow Al_2S_3 + NaNO_3$$

   A. 2, 1, 3, 2 C. 1, 1, 1, 1
   B. 2, 3, 1, 6 D. 4, 6, 3, 2

**Answer: B**

5. One mole of $(NH_4)_2HPO_4$ contains _____ moles of hydrogen atom.
   A. one C. six
   B. five D. nine

***Solution:***

1 mol $(NH_4)_2HPO_4 = 2 \times 4 + 1 = 9$ moles H

**Answer: D**

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6. A 500 grams tooth paste sample has 0.2 g of flouride concentration. What is the concentration of fluoride in terms of ppm?

A. 200 B. 250 C. 400 D. 1000

***Solution***

Given mass of solute = 0.2 gram

Mass of sample (soln) = 500 gram

Required ppm (parts per million)

$$ppm = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6 = \frac{0.2}{500} \times 10^6 = 400 \text{ ppm}$$

**Answer: C**

7. Which solution below has the highest concentration of hydroxide ion?

A. pH = 3.21 C. pH = 9.82

B. pH = 7.93 D. pH = 12.59

***Explanation***

The pH of the solution is inversely proportional to $[H^+]$ and directly proportional to $[OH^-]$.

**Answer: D**

8. Which substance is reduced in the following reaction?

$$Cr_2O_7^{2-} + 6S_2O_3^{2-} + 14H^+ \longrightarrow 2Cr^{3+} + 3S_4O_6^{2-} + 7H_2O$$

A. $Cr_2O_7^{2-}$ C. $H^+$

B. $6S_2O_3^{2-}$ D. $Cr^{3+}$

***Explanation***

$\overset{+6}{Cr_2}O_7^{2-} \longrightarrow 2Cr^{3+}$, the oxidation state of Cr is reduced from +6 to +3.

**Answer: A**

9. What percent is gold in 18 karat gold?

A. 18 B. 84 C. 25 D. 75

***Solution***

Pure gold is 24 karat

$$\%18 \text{ karat} = \frac{18}{24} \times 100 = 75\%$$

**Answer: D**



10. Which of the following is a colligative property of a solvent in a solution?

A. Lowering of vapor pressure

B. Osmotic pressure

C. Boiling point

D. Change in entropy

***Explanation***

The lowering of vapor pressure of a solvent when non – volatile solute is added is colligative property of a solvent.

**Answer: A**

11. Which one of the following is the criterion for a spontaneous change using thermodynamic parameter entropy(s)?

A. $\Delta S_{system} < 0$ C. $\Delta S_{universe} \leq 0$

B. $\Delta S_{system} \leq \Delta S_{surrounding}$ D. $\Delta S_{universe} > 0$

***Explanation***

2nd law of thermodynamic states that, "For a spontaneous process the total entropy of a system and its surrounding process always increases."

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surrounding} > 0$$

**Answer: D**

12. Which element corresponds to the electron configuration $[Ne]3s^2 3p^1$?

A. Mg B. Al C. Si D. Ga

**Answer: B**

13. Which bond between the following groups of elements will be most polar?

A. S – F C. S – Br

B. S – Cl D. S – I

***Explanation***

- Electro negativity is the power of an atom to attract shared pair of electrons towards itself.
- Electro negativity decreases down a group. Hence the one with high electro negative element forms the most polar bond.

**Answer: A**

14. What is the unit for the rate of a chemical reaction if the concentration is expressed in moles per liter and the time in second?

A. $L\ mol^1 S^{-1}$ C. $S^{-1}$

B. $mol\ L^{-1} S^{-1}$ D. $mol\ L^{-1} S^{-2}$

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***Solution:***

$$\text{rate} = \frac{\Delta C}{\Delta t} = \frac{\text{mol/L}}{\text{s}} = \text{molL}^{-1}\text{s}^{-1}$$

**Answer: B**

15. What is the most important type of solute-solvent interaction in a solution of $CaCl_2$ in water?

A. Dipole – dipole
B. Dispersion force
C. Hydrogen bonding
D. Ion – dipole

***Explanation***

- Dipole – dipole interaction exists between polar solute and polar solvents.
- Dispersion (London force) exists between non polar solutes and non polar solvents.
- Ion – dipole exists between ionic compounds and polar solvents.
- Hydrogen bonding exists between molecules containing hydrogen together with most electronegative elements (F, O, N). Example $H_2O$, HF, $NH_3$

**Answer: D**

16. A bleaching solution contains 5.2 mass percent sodium hypochlorite, NaOCl. What is the mass of NaOCl in a bottle containing 2500 gram of bleaching solution?

A. 90
B. 130
C. 480
D. 520

***Solution***

**Given:**

Percentage by mass (%m/m) = 5.2%

Mass of solution = 2500 gram

**Required:**

Mass of NaOCl =?

$$\%\text{m/m} = \frac{\text{mass of NaOCl}}{\text{mass of soln}} \times 100$$

$$\text{mass of NaCl} = \frac{\%\text{ m/m} \times \text{mass of solution}}{100}$$

$$= \frac{5.2 \times 2500}{100} = 130 \text{ gram}$$

**Answer: B**



17. What is the correct order of increasing acid strength in the following species: $HBrO_3$, $HBrO_2$, $HBrO_3$ and $HClO_3$?

A. $HBrO < HBrO_2 < HBrO_3 < HClO_3$
B. $HBrO < HBrO_2 < HClO_3 < HBrO_3$
C. $HClO_3 < HBrO_3 < HBrO_2 < HBrO$
D. $HBrO_3 < HBrO_2 < HBrO < HClO_3$

***Explanation:*** In strength $HBrO_3 > HBrO_2 > HBrO$ and $HClO_3 > HBrO_3$ because Cl is more electronegative than Br

**Answer: A**

18. A 50.0 mL sample of 0.50 M acetic acid ($CH_3COOH$), Ka = $1.8 \times 10^{-5}$ is titrated with 0.15M NaOH solution. What is the pH of the solution when no NaOH is added?

A. 6
B. 2.25
C. 2.58
D. 3.17

***Solution:*** Because there is no added base, the pH of the solution will be simply calculated as the following.

$$CH_3COOH \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

| | | | |
|---|---|---|---|
| Initial conc | 0.5 | - | - |
| change | -x | +x | +x |
| Equil. conc | 0.5 – x | x | x |

$$Ka = \frac{x^2}{0.5 - x}$$

$$\frac{x^2}{0.5} = 1.8 \times 10^{-5}$$

$$x = \sqrt{(0.5)(1.8 \times 10^{-5})} = 0.003 = [H_3O^+]$$

$$\text{pH} = -\log [H_3O^+] = -\log^{0.003} = 2.58$$

**Answer: C**

19. The enthalpy change for the conversion of white phosphorous to red phosphorus is -17.7 $kJ.mol^{-1}$. Which of the following statement is true?

A. One mole of red phosphorus will give off more heat when born in air than one mole of white phosphorus.
B. One mole of white phosphorus will give off more heat when burned in air than one mole of red phosphorus.
C. The heat generated by both types is the same.
D. The enthalpy change for the conversion of red phosphorus to white will be -35.4 kJ/mol.

**Answer: B**

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20. Which of the following is NOT a spontaneous process?
A. The rusting of iron.
B. The dissolution of salt in water.
C. A clear solution becomes in uniform color after a few drops of dye are added.
D. Cleaning a room.

***Explanation:*** A spontaneous process is a process that occurs without an external influence.

**Answer: D**

21. What are the expected products of the electrolysis of an aqueous solution of $CuCl_2$?
A. Cu and $H_2$ at the cathode and $Cl_2$ and $O_2$ at the anode.
B. Cu at the cathode and $Cl_2$ at the anode.
C. Cu and $H_2$ at the cathode and only $Cl_2$ at the anode.
D. Cu at the cathode and $Cl_2$ and $O_2$ at the anode.

***Solution:***

Anode rxn: $2Cl^- \longrightarrow Cl_2 + 2e^-$

Cathode rxn: $Cu^{2+} + 2e^- \longrightarrow Cu$

Cell rxn: $Cu^{2+} + 2Cl^- \longrightarrow Cu + Cl_2$

**Answer: B**

22. Which is the most abundant element in the earth's crust?
A. Aluminum
B. Silicon
C. Oxygen
D. Iron

***Explanation***

The twelve most abundant elements in earth's crust are oxygen, silicon, aluminum, iron, calcium, magnesium, sodium, potassium, titanium, hydrogen, phosphorous and manganese.

**Answer: C**

23. The reaction of a dialcohol with diprotic carboxylic acid is an example of
A. a condensation polymerization
B. a copolymer
C. an addition polymerization
D. an thermoses polymer

***Explanation***

- Condensation polymerization is a combination of monomers with the elimination of simple molecules like water. Monomers are usually bifunctional.



- Addition polymerization: involves unsaturated compounds containing double or triple bonds.
- Copolymer: are prepared by polymerizing more than one kind of monomer unit.
- Thermo set polymers are polymers that can be moulded into different shapes by heating and become hard when cooled. However, once solidified and set they cannot softened or remoulded by heating. Example Bekelite.

**Note:** Polyester (polyethylene terephthalate or Dacron) is a consation polymer made from 1, 4 – Benzene dicarboxylic acid (terephthalic acid) and 1, 2 – ethanediol (ethylene glycol).

**Answer: A**

24. Which of the following is NOT a carbohydrate?
A. Glucose
B. Cellulose
C. Glycine
D. Starch

**Answer: C**

25. The exothermic step in the dissolution of a salt in water is
A. Separation of the salt into ions in gaseous state.
B. Separation of the water molecules.
C. Hydration (solvation) of the ions from the salt.
D. Separation of the salt into ions in gaseous state and separation of the water.

**Answer: C**

26. Which of the following is NOT correct?
A. Addition of a strongly solvated solute decreases the solubility of a gas is liquid.
B. For solid dissolved in liquid that is exothermic, increase in temperature increase the solubility.
C. The solubility of a substance is its concentration in the saturated solution.
D. For solution to occur solvent – solute attractions must overcome solute – solute and solvent – solvent attractions.

***Explanation:*** For exothermic mixing, increasing temperature, decreases solubility.

**Answer: B**

27. How many milligrams are in 0.2 decigrams?
A. 20 milligrams
B. 2000 milligrams
C. 0.002 milligrams
D. 0.00002 milligrams

***Solution*** 1 mg = $10^{-2}$ decigram

?x = 0.2 decigram

x = 20 mg

**Answer: A**

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28. How many grams of $CaCO_3$ will be formed by passing excess $CO_2$ gas through 100 ml of 0.25 M $Ca(OH)_2$ solution?

A. 10 grams C. 2.5 grams
B. 2 grams D. 5 grams

***Solution***

$$CO_2 + Ca(OH)_2 \longrightarrow CaCO_3 + H_2O$$

Number fo moles of $Ca(OH)_2$ = Molarity × volume (Lit)
= 0.25 × 0.1
= 0.025 moles

$$\underset{0.025}{CO_2} + \underset{x}{Ca(OH)_2} \longrightarrow CaCO_3 + H_2O$$

$$\frac{0.025 \text{ mole}}{1 \text{ mole}} = \frac{x}{1 \text{ mole}}$$

x = 0.025 mole of $CaCO_3$
Mass of $CaCO_3$ = number of mole × molar mass
= 0.025 × 100 = 2.5 grams

**Answer: C**

29. Which one of the following electromagnetic radiation has the longest wave length?

A. Gamma ray C. radio waves
B. Micro wave D. x-rays

***Explanation:***
In wave length cosmio ray < gamma ray < x – ray < UV < visible < infrared < microwave < radio wave

**Answer: C**

30. What are the possible values of azimuthal quantum number for an electron in the 4f subshell?

A. 0, 1, 2, 3 C. 3
B. 1, 2, 3 D. 4

***Explanation***

The azimuthal quantum number ($\ell$) takes a value from 0 to n – 1.

In 4f n = 4, hence the values of $\ell$ for s – sub shell, p – sub shell, d – sub shell and f – sub shell are 0,1, 2 and 3 respectively.

**Answer: C**



31. Which of the following is the correct electronic configuration of Ni?

A. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
B. $1s^2 2s^2 2p^6 3s^2 3s^2 3p^6 3d^9 4s^1$
C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^6 5s^2$
D. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^2$

**Answer: A**

32. What type of hybridization is used by carbon in $C_2H_2$ (acetylene)?

A. sp B. $sp^3$ C. $sp^2$ D. $sp^2d$

**Answer: A**

33. Which of the following rate relationships is valid for the reaction?

$$2NO_2(g) \longrightarrow 2NO(g) + O_2(g)$$

A. $\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[NO]}{\Delta t}$ C. $\frac{-1}{2}\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$

B. $\frac{-\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$ D. $\frac{-\Delta[O_2]}{\Delta t} = \frac{\Delta[NO]}{\Delta t}$

**Answer: C**

34. Which of the following statement is NOT correct regarding equilibrium constant? The value of the equilibrium constant does not depend upon the _____

A. initial concentration of the reactant
B. initial concentration of the products
C. nature of the reactants
D. presence of a catalyst

**Answer: D**

35. Consider the reaction,

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

in a closed container at equilibrium at a constant temperature. What will be the effect of adding more $PCl_5$ on the equilibrium concentration of $Cl_2$ (g)? It

A. decreases
B. cannot be predicted without the value of $k_P$
C. remains unaffected
D. increases

***Explanation:*** Increasing the concentration of the reactant shifts the equilibrium to the right.

**Answer: D**

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36. What is the mole fraction of a solute in one molal aqueous solution?
A. 0.009 B. 0.018 C. 0.027 D. 0.036

***Solution:***
Molality and mole fraction of solute ($X_B$) are related as follows:
Where
$X_B$ = mole fraction of solute
m = molality
$M_A$ = molecular mass of the solvent

$$\frac{1}{X_B} = 1 + \frac{100}{m \times M_A}$$

$$\frac{1}{X_B} = 1 + \frac{100}{1 \text{ molal} \times 18 \text{ g/mol}}, \quad \frac{1}{X_B} = 56.5556$$

$X_B = 0.018$

**Answer: B**

37. Which of the following aqueous solution will have the lowest freezing point? 0.1 molal solution of _____
A. Urea C. sodium chloride
B. acetic acid D. calcium chloride

***Explanation***

| Compound | Van't Hoff factor (i) |
|---|---|
| Urea | 1 |
| Acetic acid | 2 |
| Sodium chloride | 2 |
| Calcium chloride | 3 |

Theoretically,
Note: i = number of cations + number of anions
$T_f = T°f - ikfm$
$T°f$ for $H_2O$ is 0°C

A. $T_f = 0°C - (1)\, k_f (0.1)$
$= 0°C - 0.1\, k_f$
$= -0.1\, k_f$

B. $T_f = 0°C - (2)\, k_f (0.1)$
$= 0°C - 0.2\, k_f$
$= 0.2\, k_f$

C. $T_f = 0°C - (2) k_f (0.1)$
$= -0.2 k_f$

D. $T_f = 0°C - (3)\, k_f (0.1)$
$= 0°C - 0.3\, k_f$
$= -0.3\, k_f$

**Answer: D**



38. Which one of the following pair CANNOT be mixed together to form a buffer solution?
A. $NH_3$ and $NH_4Cl$ C. RbOH and HBr
B. $NaC_2H_3O_2$ and $HC_2H_3O_2$ D. $H_3PO_4$ and $KH_2PO_4$

***Explanation***
- Buffer solution is made from
  - A weak acid with its conjugate base.
  - A weak base with its conjugate acid.
- A buffer solution will not be formed between RbOH and HBr.

**Answer: C**

39. Which of the following aqueous solutions will be basic?
A. NaCl C. $Na_2CO_3$
B. $Na_2SO_4$ D. $FeCl_3$

***Explanation***
- NaCl is made from NaOH and HCl (strong acid and strong base).
- $Na_2SO_4$ is made from NaOH and $H_2SO_4$ (strong base and strong acid).
- $Na_2CO_3$ is made from NaOH and $H_2CO_3$ (strong base and weak acid).
- $FeCl_3$ is made from $Fe(OH)_3$ and HCl (weak base and strong acid).

A basic salt is formed between strong base and weak acid.

**Answer: C**

40. Consider the reaction:

$CH_4(g) + 4Cl_2(g) \longrightarrow CCl_4(g) + 4HCl(g), \Delta H = -434$ kJ

What energy change occurs when 1.2 moles of methane reacts?
A. $5.2 \times 10^5$ J are released
B. $5.2 \times 10^5$ are absorbed
C. $3.4 \times 10^5$ J are released
D. $3.6 \times 10^5$ J are absorbed

***Solution***
From the reaction
1 mole of $CH_4$ releases $-434$ kJ
1.2 mole of $CH_4$ releases ?
Solving for the unknown
1.2 mole of $CH_4$ release 520.8 kJ
$5.208 \times 10^5$ J (released)

**Answer: A**

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41. When electricity is passed through aqueous solution of aluminum chloride, 13.5 gram of Al are deposited. What must be the number of Faraday's of electricity passed?

A. 2.0 B. 1.5 C. 1.0 D. 0.5

***Explanation***

Given mass of Al = 13.5 gram

Charge of Al in $AlCl_3 = 3$ $\quad Al^{3+} + 3e^- \longrightarrow Al$

Molar mass of Al = 27 g/mol

$$\text{number moles of Al} = \frac{\text{mass of Al}}{\text{molar mass}} = \frac{13.5}{27} = 0.5 \text{ moles}$$

$Al^{3+} + 3e^- \longrightarrow Al$

1 mole of $Al^{3+}$ needs 3 moles of electrons. Hence 3 Faraday's of electricity are required to produce 1 mole of Al

1 mole = 3 Faraday

0.5 mole of Al =?

x = 1.5 Faraday

**Answer: B**

42. Which of the following is the anodic reaction in the discharge of lead – acid storage battery?

A. $PbSO_4(s) + 2H_2O(l) \longrightarrow PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^-$

B. $Pb(s) + SO_4^{-2}(aq) \longrightarrow PbSO_4(s) + 2e^-$

C. $PbSO_4(s) + 2e^- \longrightarrow Pb(s) + SO_4^{2-}(aq)$

D. $PbO_2(s) + SO_4^{2-}(aq) + 4H^+ + 2e^- \longrightarrow PbSO_4(s) + 2H_2O(\ell)$

***Explanation***

The anode and cathode reaction for lead – acid storage battery are

Anode reaction $\quad Pb(s) + SO_4^{2-}(aq) \rightleftharpoons PbSO_4(s) + 2e^-$

Cathode rxn

$PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^- \rightleftharpoons PbSO_4(s) + 2H_2O(\ell)$

**Answer: A**

43. With which one of the following is gold complex to recover it from its ore?

A. $NO_3^-$ B. $CO_3^{2-}$ C. $SO_4^{2-}$ D. $CN^-$

**Answer: D**



44. For the reaction, $3O_2(g) \longrightarrow 2O_3(g)$; $\Delta H = 284$ kJ. What are the signs of $\Delta G$ and $\Delta S$ respectively?

A. positive and negative
B. positive and positive
C. negative and negative
D. negative and positive

***Explanation***

$3O_2(g) \longrightarrow 2O_3(g)$

The entropy is negative because the number of moles of reactant is greater than number of moles of product.

From the formula $\Delta G = \Delta H - T\Delta S$

$\Delta H$ = positive

$\Delta S$ = negative

$\therefore \Delta G$ is always positive.

**Answer: A**

45. Which of the following is NOT true about anode of a voltaic cell?

A. Reduction occurs at it.
B. It is assigned as the negative pole.
C. Electrons flow from it.
D. Cations in solution move away from the surface of the electrode as the redox reaction occurs.

**Answer: A**

46. What is the orbital hybridization of the oxygen atom in water?

A. sp B. $sp^2$ C. $sp^3$ D. $sp^3d$

**Answer: C**

47. Within a specified period, an increase in atomic number's usually accompanied by:

A. an increase in atomic radius and an increase in electronegativity
B. a decrease in atomic radius and an increase in electronegativity
C. an increase in atomic radius and a decrease in electronegativity
D. a decrease in atomic radius and a decrease in electronegativity

**Answer: B**

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48. Which of the following sets of quantum numbers is allowed in the hydrogen atom?

A. $n = 2, \ell = 1, m\ell = -1$ C. $n = 8, \ell = 1, m\ell = -6$

B. $n = 1, \ell = 0, m\ell = 0$ D. $n = 2, \ell = 0, m\ell = 2$

***Explanation***

H (atomic no 1) = $1s^1$

$n = 1, \ell = 0$ (s-sub shell), $m\ell = 0$

**Answer: B**

49. $NF_3$, $BF_3$ and $ClF_3$ all have molecular formulae of the type $XF_3$. Which molecule(s) has (have) trigonal pyramidal molecular geometry?

A. $NF_3$ and $BF_3$ C. $NF_3$ and $ClF_3$

B. $BF_3$ and $ClF_3$ D. $NF_3$

***Explanation***

| Molecule | Molecular geometry |
|---|---|
| $NF_3$ | Trigonal pyramidal |
| $BF_3$ | Trigonal planar |
| $ClF_3$ | T-shaped |

**Answer: D**

50. Which one of the following molecules cannot be formed?

A. $NCl_3$ B. $OF_6$ C. $ONF_3$ D. $PCl_3$

**Answer: B**

51. Of $Br_2$, Ne, HCl and $N_2$ which is likely to have the largest molecular dispersion forces?

A. $Br_2$ B. Ne C. HCl D. $N_2$

***Explanation***

Molecules with higher molecular mass have large dispersion force.

**Answer: A**

52. Thiosolfate ion is oxidized by iodine according to the following reaction:

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$



If a certain experiment, 0.006 mol of $S_2O_3^{2-}$ is consumed in 1.0 L of solution each second, at what rates are $S_4O_6^{2-}$ and $I^-$ produced in this solution?

A. 0.012 mol $L^{-1}$ $s^{-1}$ ($S_4O_6^{2-}$) and 0.006 mol $L^{-1}$ $s^{-1}$ ($I^-$)

B. 0.006 mol $L^{-1}$ $s^{-1}$ ($S_4O_6^{2-}$) and 0.006 mol $L^{-1}$ $s^{-1}$ ($I^-$)

C. 0.003 mol $L^{-1}$ $s^{-1}$ ($S_4O_6^{2-}$) and 0.006 mol $L^{-1}$ $s^{-1}$ ($I^-$)

D. 0.012 mol $L^{-1}$ $s^{-1}$ ($S_4O_6^{2-}$) and 0.012 mol $L^{-1}$ $s^{-1}$ ($I^-$)

***Explanation***

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I(aq)$$

$$\text{rate} = \frac{\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{-\Delta[I_2]}{\Delta t} = \frac{\Delta[S_4O_6^{2-}]}{\Delta t} = \frac{\Delta[I^-]}{2\Delta t}$$

$$\Rightarrow \frac{-\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{\Delta[S_4O_6^{2-}]}{\Delta t}$$

$$\frac{-1}{2}(0.006) = \Delta[S_4O_6^{2-}]$$

$$[S_4O_6^{2-}] = \frac{0.006}{2} = 0.003 \text{ mol/L.s}$$

and

$$\frac{-\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{\Delta[I^-]}{2\Delta t}$$

$\Delta[I^-] = 0.006$ mol/L. s

**Answer: C**

53. A particular reaction has found to depend on the concentration of hydrogen ion, $[H^+]$. The initial rate change as a function of $[H^+]$ as follows.

| $[H^+]$ (M) | 0.005 | 0.01 | 0.02 |
|---|---|---|---|
| Initial rate (M/s) | $3.4 \times 10^{-7}$ | $1.7 \times 10^{-7}$ | $0.85 \times 10^{-7}$ |

What is the order of the reaction in $[H^+]$?

A. -1 B. 1 C. 2 D. 0

***Explanation***

$$\text{rate} = K[H^+]^x$$

Take any of the two rates with their corresponding concentrations of $[H^+]$.

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$rate_1 = k(0.005)^x = 3.4 \times 10^{-7}$

$rate_2 = k(0.01)^x = 1.7 \times 10^{-7}$

Taking the ratio of the two rates

$$\frac{rate_1}{rate_2} = \frac{k(0.005)^x}{k(0.01)^x} = \frac{3.4\times10^{-7}}{1.7\times10^{-7}}$$

$$\left(\frac{1}{2}\right)^x = 2$$

$(2^{-1})^x = 2$

$2^{-x} = 2$

$-x = 1$

$x = -1$

**Answer: A**

54. The conversion of ozone into $O_2$ proceeds in two elementary steps as follows.

$$O_3(g) \longrightarrow O_2(g) + O(g)$$

$$O_3(g) + O(g) \longrightarrow 2O_2(g)$$

What is the equation for the overall reaction?

A. $2O_3(g) \longrightarrow 3O_2(g)$

B. $2O_3(g) + O(g) \longrightarrow 3O_2(g) + O(g)$

C. $O_3(g) + O(g) \longrightarrow O_2(g) + 2O(g)$

D. $2O_3(g) + O(g) \longrightarrow 3O_2(g)$

***Explanation***

O is an intermediate so it should not appear in the overall reaction.

$$O_3(g) \longrightarrow O_2(g) + O(g)$$

$$O_3(g) + O(g) \longrightarrow 2O_2(g)$$

$$2O_3(g) \longrightarrow 3O_2(g)$$

**Answer: A**

55. Which of the following factors does not affect the rate of a chemical reaction that is of zero order?

A. concentration of reactants

B. surface area of reactants

C. temperature

D. presence of catalyst

**Answer: A**



56. How would you prepare 0.90 L of $1.2 \times 10^{-2}$ M NaCl aqueous solution?

A. weigh 0.012 g of NaCl, dissolve in a small amount of water and dilute to 0.90L.

B. weigh 0.63 gram of NaCl, dissolve in a small amount of water and dilute to 0.90L.

C. weigh 1.2 gram of NaCl, dissolve in a small amount of water and dilute to 0.90L.

D. weigh 1.4 gram of NaCl, dissolve in a small amount of water and dilute to 0.90L.

***Explanation***

$$\text{Molarity} = \frac{\text{mass}}{\text{molar mass}} \times \frac{1}{V(L)} \Rightarrow \text{mass} = \text{molarity} \times \text{molar} \times V$$

$= 1.2 \times 10^{-2} \times 58.5 \times 0.9 = 0.631$ gram

**Answer: B**

57. Which of the following solutes will produce the largest increase in boiling point upon addition to 1 kg of water?

A. 1 mole of $Co(NO_3)_2$

B. 2 mole of KCl

C. 3 mol of ethylene glycol ($C_2H_6O_2$)

D. 1 mol of $CaCl_2$

***Explanation***

$T_b = T_b^\circ + ik_b\, m$

where $T_b$: B. p of the solution

$T_b^\circ$: B. p of pure solvent

i : Vant Hoff factor

$k_b$: boiling point elevation constant

m: molality

A. 1 mol of $CO(NO_3)_2$

$i = 2$

$$T_b = T_b^\circ + ik_b\, m = 100°C + 2k_b\left(\frac{1\text{ mol}}{1\text{ kg}}\right) = 100°C + 2k_b$$

B. 2 mole of KCl

$i = 2$

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$$T_b = T_b^o + 1\,kbm \Rightarrow 100°C + 2k_b\left(\frac{2\text{ mol}}{1\text{ kg}}\right) = 100°C + 4k_b$$

C. 3 mole of $C_2H_6O_2$
i = 1 (blse it is non electrolyte)

$$T_b = T_b^o + i\,k_b m$$

$$T_b = 100°C + 1\,k_b\left(\frac{3}{1}\right) = 100°C + 3k_b$$

D. 1 mole of $CaCl_2$
i = 3

$$T_b = T_b^o + ik_b m \Rightarrow 100°C + 3k_b\left(\frac{1}{1}\right) = 100°C + 3k_b$$ Answer: B

58. What is the percentage of HF molecules ionized in a 0.10 M HF solution ($Ka = 6.8 \times 10^{-4}$)?
A. 2 B. 5 C. 8 D. 10

***Solution***

$$HF + H_2O \rightleftharpoons H_3O^+(aq) + F^-(aq)$$

| | | | |
|---|---|---|---|
| Initial conc. | 0.1 | – | – |
| Change | -x | x | +x |
| Equili. Conc | 0.1 – x | X | x |

$$Ka = \frac{x^2}{0.1-x} = 6.8\times10^{-4}$$

Using approximation method $\frac{x^2}{0.1-x} = 6.8\times10^{-4}$

$$\frac{x^2}{0.1} = 6.8\times10^{-4}$$

$$x^2 = 6.8 \times 10^{-5}$$

$$x = \sqrt{6.8\times10^{-5}} = 8.24\times10^{-3}$$

$$\text{Percentage dissociation (x\%)} = \frac{\text{amount of dissiciation (x)}}{\text{initial concentration (c)}}\times100$$

$$= \frac{8.24\times10^{-3}}{0.1}\times100 = 8.24\%$$ Answer: C

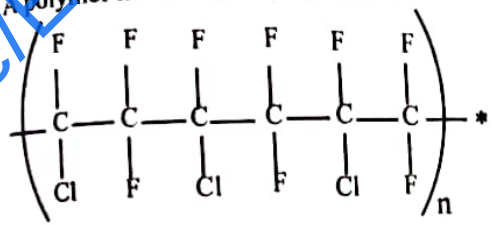


59. Which of the following reaction is a redox reaction?
A. $3Se(s) + 4HNO_3(aq) \longrightarrow 3SeO_2(s) + 4NO(g) + H_2O(\ell)$
B. $CaCl_2(aq) + 2AgNO_3(aq) \longrightarrow Ca(NO_3)_2(aq) + 2AgCl(s)$
C. $BaCO_3(s) \longrightarrow BaO(s) + CO_2(g)$
D. $3Mg(OH)_2 + 2NH_3(g) \longrightarrow Mg_3N_2(s) + 6H_2O(\ell)$

***Explanation***
The way to check whether a given reaction is redox or not is by assigning oxidation number value for reactants and products. As a result, if there is a change in oxidation number the reaction is redox otherwise it is non redox. All acid base and precipitation reactions are non redox.

**Answer: A**

60. A polymer has the following structure:



What is the monomer of this polymer?
A. $CFCl = CF_2$ C. $CF_2Cl_2$
B. $CF_3Cl$ D. $CF_4$

**Answer: A**

61. How many significant figures are there in 0.0500?
A. one B. three C. two D. Four

**Answer: B**

62. If 16 grams of $O_2$ react with excess $C_2H_6$, how many grams of $CO_2$ will be formed? The unbalanced equation is

$$C_2H_6(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$$

A. 22 grams C. 13 grams
B. 9 grams D. 12.6 grams

***Solution***

$$C_2H_6(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$$

When it is balanced

$$2C_2H_6(g) + 7O_2(g) \longrightarrow 4CO_2(g) + 6H_2O(g)$$

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Using mass – mass relationship

$$\frac{\text{mass of } O_2}{7\times \text{ molar mass}} = \frac{x}{4 \times \text{molar mass of } CO_2}$$

$$\frac{16}{7\times 32} = \frac{x}{4\times 44}$$

$$\frac{16}{224} = \frac{x}{176}$$

x = 12.57 grams

**Answer: D**

63. How does the titration of a strong acid with a strong base differ from the titration of a weak acid with a strong base with respect to the quantity of base required to reach the equivalence point?
   A. The quantity of base required to reach the equivalence point is the same in both titrations.
   B. The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is higher than for the titration of a weak acid with a strong base.
   C. The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is lower than the titration of weak acid with a strong base.
   D. The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is slightly higher than for the titration of a weak acid with a strong base.

**Answer: A**

64. Hydrogen atoms are excited to the stationary state designated by the principal quantum number, n = 4. What will be the number of lines (frequencies) emitted by the hydrogen atoms?
   A. 1 B. 4 C. 6 D. 16

**Answer: B**

65. According to the molecular orbital theory, which one of the following is paramagnetic
   A. $F_2$ B. $H_2$ C. $NO^+$ D. NO

***Explanation***

Species with odd number of electrons are paramagnetic:

$F_2 = 2 \times 9 = 18e^-$

$H_2 = 2 \times 1 = 2e^-$



$NO^+ = 7 + 8 - 1 = 14e^-$

$NO = 7 + 8 = 15e^-$

**Answer: D**

66. For the reaction

$$2NO\,(g) + O_2\,(g) \longrightarrow 2NO_2\,(g)$$

the following data were obtained

| | | | |
|---|---|---|---|
| $[NO]_0$ (M): | 0.10 | 0.10 | 0.40 |
| $[O_2]_0$ (M): | 0.10 | 0.20 | 0.20 |
| Rate (M/s): | $2.5 \times 10^{-4}$ | $5.0 \times 10^{-4}$ | $8.0 \times 10^{-3}$ |

What is the rate law for the reaction?

A. rate = $k[NO][O_2]$ C. rate = $k[NO]^2[O_2]$

B. rate = $k[NO][O_2]^2$ D. rate = $k[NO]^2[O_2]^2$

***Solution:*** $2NO\,(g) + O_2\,(g) \longrightarrow 2NO_2\,(g)$

rate = $k[NO]^x[O_2]^y$

To determine y order with respect to $O_2$, take trial 1 and 2 where the concentration of $O_2$ are the same

$rate_2 = k\,(0.1)^x\,(0.1)^y = 2.5 \times 10^{-4}$

$rate_3 = k\,(0.1)^x\,(0.2)^y = 5.0 \times 10^{-4}$

$$\frac{rate_1}{rate_2} = \frac{k\,(0.1)^x\,(0.1)^y}{k(0.1)^x\,(0.2)^y} = \frac{2.5\times10^{-4}}{5.0\times10^{-4}}$$

$$\left(\frac{1}{2}\right)^y = \left(\frac{1}{2}\right)^1$$

y = 1

∴ order with respect to $O_2$ is 1 ($1^{st}$ order)

To determine x (order with respect to NO), take trials 2 and 3 where the concentrations of $O_2$ are the same.

$rate_1 = k\,(0.1)\,(0.2)^y = 5.0 \times 10^{-4}$

$rate_2 = k\,(0.4)\,(0.2)^y = 8.0 \times 10^{-3}$

Taking the ratio of $rate_3$ to $rate_2$

$$\frac{rate_3}{rate_2} = \frac{k(0.4)^x(0.2)^y}{k(0.1)^x(0.2)^y} = \frac{8.0\times10^{-3}}{5.0\times10^{-4}} = 4^x = 16$$

$4^x = 4^2$

x = 2

∴ order with respect to NO is 2 (second order)

Hence the complete rate expression would be

rate = $k[NO]^2[O_2]$

**Answer: C**

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67. The value of $k_C$ for the reaction:

$$COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$$

is $2.5 \times 10^{-10}$ at 300k. Starting with pure $COCl_2$ (g) in a closed vessel, if the concentration of CO (g) is $1 \times 10^{-5}$ M at equilibrium. What would be the equilibrium concentration of $COCl_2$ (g)?

A. 2.5M B. $2.5 \times 10^{-5}$M C. $2.5 \times 10^{-5}$ M D. 0.4 M

***Explanation***

$$COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$$

$$k_C = \frac{[CO][Cl_2]}{[COCl_2]}$$

At equilibrium $[CO] = [Cl_2] = 1 \times 10^{-5}$ M

$$2.5 \times 10^{-5} = \frac{(1\times10^{-5})(1\times10^{-5})}{[COCl_2]}$$

$[COCl_2] = 0.4$ M **Answer: D**

68. An aqueous solution freezes at -0.186°C ($k_f$ = 1.86°C/m; $k_b$ = 0.512°C/m). What is the elevation in boiling point (in °C)?

A. 0.186 B. 0.512 C. 0.5.12/1.86 D. 0.0512

***Solution***

$\Delta T_f = T_f^\circ - T_f$

$= 0°C - (-0.186°C)$

$= 0.186°C$

Where

$T_f^\circ$ = freezing point of pure solvent

$T_f$ = freezing point of the solution

$T_b$ = boiling point of the solution

$\Delta T_f = kfm$

m = molality

$$m = \frac{\Delta T_f}{k_f} = \frac{0.186}{1.86}$$

$$\Delta T_b = k_b m = (0.512)\left(\frac{0.186}{1.86}\right) = 0.0512°C$$ **Answer: D**

69. The $P^{ka}$ of acetyl salicylic acid (aspirin) is 3.5. The pH of gastric juice in human stomach is about 2 to 3 and the pH in the small intestine is about 8. Which of the following statement is correct? Aspirin is

A. unionized in the small intestine and in the stomach

B. completely ionized in the small intestine and in the stomach



C. ionized in the stomach and almost unionized in the small intestine

D. ionized in the small intestine and almost unionized in the stomach

***Explanation:*** The stomach medium is acidic and that of the small intestine is basic, so, when aspirin is added to a gastric juice with acidic pH, it lowers the dissociation of the acid. And the acid ionization increase when it inserted in to a small intestine.

**Answer: D**

70. A 10 ml sample of tartaric acid (a diprotic acid) is titrated with 20 mL of 1.0 M NaOH. What is the molarity of the acid?

A. 2.0 B. 1.0 C. 4.0 D. 0.5

***Solution***

Number of equivalent of the acid = number of equivalent of the base

Normality of NaOH = Z × M = 1 × 1 = 1

$N_1V_1 = N_2V_2$

$$N_1 = \frac{N_2V_2}{V_1} = \frac{1\times20}{10} = 2N$$

$$\text{Molarity} = \frac{N}{\text{charge (Z)}} = \frac{2N}{2} = 1M$$

i.e., Z = no of replaceable hydrogen **Answer: B**

71. When the equation for the following reaction in basic solution is balanced, what is the sum of the coefficients?

$$MnO_2 + HO_2^- \longrightarrow MnO_4^-$$

A. 11 B. 31 C. 14 D. 9

***Explanation:*** $MnO_2 + HO_2^- \longrightarrow MnO_4^-$

1. Identify the oxidation and reduction half reaction.

oxidation half reaction reduction half reaction

$$\overset{+4}{Mn}O_2 \longrightarrow \overset{+7}{Mn}O_4^- + 3e^- \qquad HO_2^- + 2e^- \longrightarrow OH^-$$

2. Balance atom other than oxygen and hydrogen.

$$\overset{+4}{Mn}O_2 \longrightarrow MnO_4^- + 3e^- \text{ (already balanced)}$$

$$HO_2^- + 2e^- \longrightarrow OH^- \text{ (already balanced)}$$

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3. Balance oxygen atoms by adding water to side where oxygen is deficient.

$2H_2O + MnO_2 \longrightarrow MnO_4^- + 3e^-$ $HO_2^- + 2e^- \longrightarrow OH^-$

4. Balance hydrogen atoms by adding $H^+$ to side where hydrogen is deficient.

$2H_2O + MnO_2 \longrightarrow MnO_4^- + 4H^+ + 3e^-$

$2H^+ + HO_2^- + 2e^- \longrightarrow OH^- + H_2O$

5. Balance number of electrons lost and gained.
   - Multiply the oxidation half reaction by two.
   - Multiply the reduction half reaction by three.

$4H_2O + 2MnO_2 \longrightarrow 2MnO_4^- + 8H^+ + 6e^-$

$6H^+ + 3HO_2^- + 6e^- \longrightarrow 3OH^- + 3H_2O$

6. Write the oxidation and reduction half reactions together.

$4H_2O + 2MnO_2 + 6H^+ + 3HO_2 + 6e^- \longrightarrow$

$2MnO_4^- + 8H^+ + 3OH^- + 3H_2O + 6e^-$

$\Rightarrow$ $H_2O + 2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- + 2H^+ + 3OH^-$

add 2 moles of $OH^-$ to the reactant and product side $2OH^-$

$+ H_2O + 2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- +$

$2H^+ + 2OH^- + 3OH^-$

$\Rightarrow$ $2OH^- + H_2O + 2MnO_2 + 3HO_2^- \longrightarrow$

$2MnO_4^- + 2H_2O + 3OH^-$ canceling the $H_2O$ molecule

$OH^- + 2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- + H_2O + 3OH^-$

$\Rightarrow$ $2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- + H_2O + OH^-$

(in basic medium)

Number of atoms are conserved!

$\therefore$ The sum of the coefficients will be 9

**Answer: D**



72. The standard emf for the all reaction, $Zn + Cu^{2+} \longrightarrow Cu + Zn^{2+}$, is 1.1V. What would the emf for the all reaction be when 0.1 M $Cu^{2+}$ and 0.1 M $Zn^{2+}$ solutions are used at 298 K?

A. 1.10V B. 0.10V C. -1.10V D. -0.110V

***Solution***

Anode rxn: $Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

Cathode rxn: $Cu^{2+} + 2e^- \longrightarrow Cu(s)$

$E°cell = 1.1$ V

Using Nernest equation

$$E_{cell} = E°_{cell} - \frac{0.0592}{2}\log\left(\frac{[Zn^{2+}]}{[Cu^{2+}]}\right)$$

$$= 1.1 - \frac{0.0592}{2}\log\left(\frac{0.1}{0.1}\right) = 1.1 - \frac{0.0592}{2}\log 1 = 1.1$$

**Answer: A**

73. Which one of the following is a polyamide molecule?

A. Terylene C. Nylon

B. Rayon D. Polystyrene

**Answer: C**

74. Which metal atom can form ionic bonds by losing electrons from both the outermost and next to outermost principal energy levels?

A. Fe B. Mg C. Pb D. Ca

**Answer: A**

75. The energy required to remove an electron from an iron atom amounts to $7.21 \times 10^{-19}$ J. What is the maximum wave length of light that can perform this?

A. 250 nm C. 289 nm

B. 276 nm D. 300 nm

***Solution***

$$\Delta E = \frac{hC}{\lambda}$$ Where E = energy

$$\lambda = \frac{hC}{\Delta E}$$ h = plank's content

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$$= \frac{(6.626\times10^{-34})(3\times10^{8}\ m/s)}{7.21\times10^{-19}}$$ C = speed of light

$= 2.757 \times 10^{-7}$ m $= 276$ nm

76. What is the molecular geometry of $O_3$? **Answer: B**

A. Trigonal planar C. Trigonal pyramidal
B. Bent D. Tetrahedral

***Explanation***

$O_3$ is $AB_2E$ type so its molecular geometry is Bent/Angular / V-shape and its electron pair geometry is trigonal planar.

77. At 295 K, a 3.0L flask contains of mol HI, 4 mol $H_2$, and 0.6 mol $I_2$ in equilibrium. What is the value of k at 295k? **Answer: B**

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

A. 10.2 B. 20.4 C. 30.4 D. 61.2

***Solution***

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

$$[H_2] = \frac{\text{no of mole}}{\text{volume (L)}} = \frac{4\ mol}{3} = 1.3\ M$$

$$[I_2] = \frac{0.6}{3} = 0.2M$$

$$[HI] = \frac{1}{3} = 2.3M$$

$$k_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2.3)^2}{(1.3)(0.2)} = \frac{5.4}{0.26} = 20.769$$ **Answer: B**

78. The equilibrium constant is 0.09 at 295 K for the reaction

$$H_2O(g) + Cl_2O(g) \rightleftharpoons 2HOCl(g)$$

For which of the following sets of conditions is the system of equilibrium?

A. A 1.0L flask contains 1.0 mol of HOCl, 0.10 mol of $Cl_2O$ and 0.1 mol of $H_2O$.

B. A 2.0 L flask contains 0.084 mol of HOCl, 0.080 mol of $Cl_2O$, and 0.98 mol of $H_2O$.



C. A 3.0 L flask contains 0.25 mol of HOCl, 0.000 mol of $Cl_2O$, and 0.56 mol of $H_2O$.

D. A 4.0 L flask contains 0.25mol HOCl, 0.001 mol of $Cl_2O$ and 0.56 mol of $H_2O$.

***Explanation:*** For choice

$$[HOCl] = \frac{0.084\ mol}{2\ Lit} = 0.04M$$

$$[H_2O] = \frac{0.98\ mol}{2} = 0.49M$$

$$[Cl_2O] = \frac{0.080\ mol}{2\ Lit} = 0.040\,M$$

$$H_2O(g) + Cl_2O(g) \rightleftharpoons 2HOCl(g)$$

$$k_C = \frac{[HOCl]^2}{[H_2O][Cl_2O]} = \frac{(0.042)^2}{(0.040)(0.40)} = 0.09$$

$\therefore$ The $K'_C = K_C$ **Answer: B**

79. 1000 grams of a 55% (mass percentage) nitric acid solution is to be diluted to 20% (mass percentage) nitric acid solution. How many liters of water should be added to the starting solution?

A. 0.45 B. 1.55 C. 2.30 D. 2.75

***Explanation***

1 gram = 1 ml

1000 gram = 1000 ml = 1 L

Applying dilution formula

$$\%\ m/m \cdot V_1 = \%\ m/m \cdot V_2$$

$$\frac{55}{100}.1L = \frac{20}{100}.V_2$$

$$\frac{0.55}{0.2} = .V_2 = 2.75\ L$$

$V_{H_2O}$ added $= V_2 - V_1$

$= 2.75L - 1L$

$= 1.75L$

**Answer: D**

80. When pure methanol is mixed with water, the solution becomes warm. Is this an ideal solution?

A. yes, for an ideal solution $\Delta H_{soln} = 0$
B. No, for ideal solution $\Delta H_{soln} = 0$
C. yes, for ideal solution $\Delta H_{soln} > 0$
D. No, for ideal solution $\Delta H_{soln} < 0$

**Answer: B**

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1. Which of the following is the right order of the steps of a scientific method?
   A. Performing experiments – formulating hypothesis – marking observation
   B. Formulating hypothesis – making observation – performing experiment
   C. Making observation – formulating hypothesis – performing experiments
   D. Making observation – performing experiments – formulating hypothesis

***Explanation:***
Scientific method is a careful, logical and systematic approach towards answering scientific questions. The steps are stated under choice C.

**Answer: C**

2. The compound CuCl emits blue light having a wave length of 450 nm. When heated about 120°C. What is the increment in energy (quantum) that is emitted at 450 am?
   A. $2.25 \times 10^{-19}$ J
   B. $4.41 \times 10^{-19}$ J
   C. $8.20 \times 10^{-19}$ J
   D. $16.20 \times 10^{-19}$ J

***Solution:***

$$\Delta E = \frac{hC}{\lambda}$$ where h = plank's constant ($6.626 \times 10^{-34}$ J.s)

C = speed of light ($2.9979 \times 10^{8}$ m/s$^{-1}$)

$\lambda$ = wave length

$$\Delta E = \frac{(6.626\times10^{-34}\,\text{J.s})(2.9979\times10^{8}\,\text{m/s})}{450\times10^{-9}\,\text{m}} = 4.41\times10^{-19}\,\text{J}$$

**Answer: B**

3. What is the difference between chlorine -35 and chlorine 37?
   A. Chlorine – 37 has two more protons than chlorine – 35
   B. Chlorine – 37 has two more neutrons than chlorine – 35
   C. Chlorine – 37 has two more electrons than chlorine – 35
   D. Chlorine – 37 has one more proton and one more neutron than chlorine – 35

***Explanation***
Isotopes are atoms of the same elements that have the same atomic number but different atomic mass. In a neutral atom atomic number, number of electron = number of proton



$A = Z + n$ where A = mass number
Z = atomic number
n = neutron number

For chlorine 35, A = 35, Z = 17, n = A – Z = 35 – 17 = 18
For chlorine 37, A = 37, Z = 17, n = A – Z = 37 – 17 = 20

**Answer: B**

4. Which one of the following electromagnetic radiation has the shortest wavelength?
   A. x-ray
   B. uv-ray
   C. gamma ray
   D. microwave

***Explanation***
- The following order of electromagnetic radiation is based on increasing order of energy y is cosmic ray > gamma ray > x-ray > uv-ray > visible > infrared > micro wave > TV > Radio (in energy).
- The energy of the wave is:
  (1) Directly proportional to the frequency
  (2) Inversely proportional to the wave length

**Answer: C**

5. Which group of elements is characterized with $ns^2np^2$ outer-electron configuration?
   A. Group 2A
   B. Group 4A
   C. Group 4B
   D. Group 3B

***Explanation***
$ns^2np^2$ indicates the two sub shells exists at the same energy level because they have the same n-value. Hence to find the group number one should add the number of electrons s and p sub shells (2 + 2 = 4 electrons). Therefore, $ns^2np^2$ belongs to group IVA = 4A.

**Answer: B**

6. Which quantum number/s determine the energy of an electron in hydrogen atom?
   A. n
   B. n and $\ell$
   C. n, $\ell$ and m
   D. n, $\ell$, m and s

***Explanation***
Hydrogen has 1 electron, so its configuration is $1s^1$. Hence n = 1, $\ell$ = 0 (s- sub shell). Therefore, n value determines the energy of an electron in hydrogen atom.

**Answer: A**

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7. For elements in the left-most column of the periodic table, properties that have increasing values as the atomic number increases include which of the following?
I. Ionization II. Atomic radius III. Atomic mass
A. III only C. I and II only
B. I, II and III D. II and III only

***Explanation:***
Atomic size, atomic mass and metallic property increase down a group.

**Answer: D**

8. For an electron that has quantum numbers n = 4 and ml = 0, which of the following is true?
A. It must have the quantum number n = 0.
B. It must have the quantum number $\ell = 0$.
C. It must have the quantum number ms = $+\frac{1}{2}$.
D. It may have the quantum number $\ell = 0, 1, 2, 3$.

| n | $\ell$ (0 to n – 1) | m$\ell$ (– $\ell$ to+ $\ell$) | Ms |
|---|---|---|---|
| 4 | 0 | 0 | +1/2 |
| | 1 | -1, 0, +1 | +1/2 |
| | 2 | -2,-1,0,1,+2 | +1/2 |
| | 3 | -3,-2,-1,0,1,2,3 | +1/2 |
| | . | | +1/2 |

If n = 4 and m$\ell$ = 0 implies the $\ell$ – value should be 0, 1, 2 or 3.

**Answer: D**

9. Which of the following electron transition requires the smallest energy to be absorbed by the hydrogen atom?
A. from n = 1 to n = 2 C. from n = 3 to n = 4
B. from n = 2 to n = 3 D. from n = 4 to n = 5

***Explanation***
Using the following formula:

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{n_i 2} - \frac{1}{n_f 2}\right)$$ where,

$n_i$=initial quantum number
$n_f$ = final quantum number

- For electron transition from n = 1 to n = 2.





$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = 2.18\times10^{-18}\left(1 - \frac{1}{4}\right) = 1.64\times10^{-18}$$

- For an electron transition from n = 2 to n = 3.

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 2.18\times10^{-18}\left(\frac{1}{4} - \frac{1}{9}\right) =$$

$3.03 \times 10^{-19}$ J

- For an electron transition from n = 3 to n = 4.

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = 2.18\times10^{-18}\left(\frac{1}{9} - \frac{1}{16}\right)$$

$= 1.06 \times 10^{-19}$

- For an electron transition from n = 4 to n = 5.

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{4^2} - \frac{1}{5^2}\right) = 2.18\times10^{-18}\left(\frac{1}{10} - \frac{1}{25}\right)$$

$= 4.91 \times 10^{-20}$

**Answer: D**

10. Which of the following below best explains why atoms react chemically with each other?
A. When atoms react, they gain protons and are more stable.
B. When atoms react they lose all their electrons and become more stable.
C. When atoms react, they lose, gain or share electrons and are then less stable.
D. When atoms react they lose gain or share electrons to attain for outer energy level and are then more stable.

**Answer: D**

11. What did Rutherford's alpha particle experiment show?
A. Electrons have a negative charge.
B. A proton is a hydrogen atom without electron.
C. Electrons circle the nucleus of an atom in orbit.
D. Most of the mass and all of the positive charge of an atom is found in a tiny nucleus.

**Answer: D**

12. For which of the following elements is Hund's rule used in writing the electron configuration?
A. C B. B C. Be D. Li

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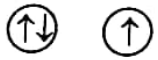
***Explanation:***
Hund's rule states that, "For any set of orbitals that are equal in energy (degenerate orbitals), the configuration with maximum number of parallel signs results in the lowest electron-electron repulsion."

$_6C = 1s^2 2s^2 2p^2$

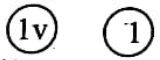
1s 2s 2px 2py 2pz

$_6B = 1s^2 2p^2 2p^1$

$_4Be = 1s^2 \quad 2s^2$



$_3Li = 1s^2\ 2s^1$



Since only carbon contains degenerate orbitals, it need Hund's rule to write its electronic configuration. **Answer: A**

13. What is the total number of valence shell electrons in $BrO_3^-$?

A. 20 B. 26 C. 32 D. 36

***Explanation***

- Valence shell is the outermost shell of an atom.
- Valence shell electrons are electrons that are found in the outer most shell.

For main group elements (Representative elements), the valence electrons determines the group to which that element belongs to i.e.,

| Valence electrons = Group number |
|---|

So, Br is group VIIA element hence its valence electrons are seven. Likewise, oxygen is group VIA element hence its valence electrons are six. Besides the negative ion on the species $(BrO_3^-)$ indicates one electron is added. Therefore, the total number of valence electrons will be 7 electrons + 6 electrons + 1 electrons = 26 electrons. **Answer: B**



14. Which term describes the units that make up compounds with covalent bonds?

A. Ions C. Salts
B. Acids D. Molecules

**Answer: D**

15. There is a strong covalent bond between the N atoms in nitrogen gas, $N_2$. Why, then, does nitrogen have such a low boiling point of –196°C?

A. The bond between the N-atom is triple.
B. N is very electronegative, only next ot F and O.
C. The strong bond an intra molecular one, determines the boiling point of the substance.
D. Boiling point is determined by intermolecular force, which is this case as the molecule is non-polar.

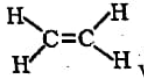
***Explanation:***
The type of inter-molecular force that holds nitrogen molecule together is London force (weakest force). **Answer: D**

16. Which molecule listed below has two sigma (σ) and two pi (π) bonds?

A. $N_2$ B. $C_2H_4$ C. $N_2F_2$ D. HCN

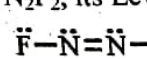
***Explanation***
To identify the type and number of bonds, the Lewis structure of each compound should be known.

- For $N_2$, it is Lewis structure is $N \equiv N$. The existence of triple bond shows 2π bonds and one sigma bond.
- For $C_2H_4$, it Lewis structure is



which implies are 5σ bonds and 2π bond

- For $N_2F_2$, its Lewis structure is



which implies there are 1π and 3σ bonds.

- For HCN

$H - C \equiv N$ which implies there are 2σ bonds and 2π bonds

Note that:

| Bond type | Number of sigma and π-bonds |
|---|---|
| Single bond (–) | has only one sigma bond |
| Double bond (=) | has one sigma and one pi bond |
| Triple bond (≡) | has one sigma and two pi bonds |

**Answer: D**

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17. Which of the following explains why, at room temperature, $I_2$ is a solid, $Br_2$ is a liquid and $Cl_2$ is a gas?
A. Ionic bonding C. Hydrogen bonding
B. Hybridization D. London dispersion forces

**Answer: D**

***Explanation:***
As size increases, the strength of London force also increases. That is why, $I_2$ is solid. $Br_2$ is liquid as $Cl_2$ is gas.

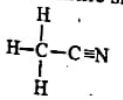
18. Which molecule has a Lewis structure that does not obey the octet rule?
A. NO B. $CS_2$ C. $PF_3$ D. HCN

**Answer: D**

***Explanation***
Molecules with an odd number of valence electrons does not obey octet rule.

| Molecule | Total no of valence electrons |
|---|---|
| NO | 5 + 6 = 11 electrons |
| $CS_2$ | 4 +2 (6) = 16 electrons |
| $PF_3$ | 5 + 3(7) = 26 electrons |
| HCN | 1 + 4 + 5 = 10 electrons |

19. What is the hybridization of carbon atom attached to nitrogen in acetonitile show below?



A. sp B. $sp^2$ C. $sp^3$ D. $sp^4$

**Answer: A**

***Explanation***
The number of sigma bonds around carbon which is attached to nitrogen atom are two sigma bonds. Hence the type of hybridization is sp.

20. What is the half life, $t_{\frac{1}{2}}$ for a zero order reaction $A \longrightarrow B$, (k is the rate constant)?
A. ln 2k B. [A]0/2k C. ln k$[A]_o$ D. ln 2$[A]_o$k

**Answer: A**

***Explanation***
Consider A reaction $A \longrightarrow B$



| Order | Rate reaction | Half life $\left(t_{\frac{1}{2}}\right)$ |
|---|---|---|
| 0 | $[A]_t = -kt + [A]_o$ | $[A]o/2\,k$ |
| 1 | $\ln [A]_t = -k + \ln [A]_o$ | $\frac{0.693}{k}$ |
| 2 | $\frac{1}{[A]_t} = kt + \frac{1}{[A]_o}$ | $\frac{1}{k[A]_o}$ |

$[A]_o$ = initial concentration, $[A]_t$ = concentration of reactant A at a certain time.

21. Consider the following reaction:

**Answer: B**

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

If, in an experiment, 0.05 mol $S_2O_3^{2-}$ is consumed in 1.0L of solution each second, at what rates are $S_4O_6^{-2}$ and $I^-$ produced in this solution?

A. $S_4O_6^{2-} = 0.025, I^- = 0.025$ C. $S_4O_6^{2-} = 0.025; I^- = 0.05$
B. $S_4O_6^{2-} = 0.05; I^- = 0.05$ D. $S_4O_6^{2-} = 0.05; I^- = 0.025$

***Explanation***
For the general hypothetical reaction
$aA + bB \longrightarrow cC + dD$ where a, b, c, d are coefficients.
The rate expression in terms of the concentration of each reactant and each product is related as:

$$\text{Rate of reaction} = \frac{\Delta[A]}{a\Delta t} = \frac{-\Delta[A]}{b\Delta t} = \frac{\Delta[C]}{c\Delta t} = \frac{\Delta[D]}{d\Delta t}$$

For the given reaction

$$2S_2O_2^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

$$\text{Rate} = \frac{-\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{-\Delta[I_2]}{\Delta t} = \frac{\Delta[S_4O_6^{2-}]}{\Delta t} = \frac{\Delta[I^-]}{2\Delta t}$$

rate of $S_2O_3^{2-} = 0.05\ mol.L^{-1}.s^{-1}$

$$-\frac{1}{2}(0.05) = \frac{\Delta[S_4O_6^{2-}]}{\Delta t}$$

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$\therefore$ rate of $S_4O_6^{2-} = \frac{\Delta[S_4O_6^{2-}]}{\Delta t} = 0.025$

rate of $I^- = \frac{\Delta[I^-]}{2\Delta t} = \frac{2}{2}(0.05) = 0.05$ **Answer: C**

22. The reaction between NO and $I_2$ is second order in NO and first order in $I_2$. What change occurs in the rate of the reaction if the concentration of NO is doubled and $I_2$ left unchanged?

A. double C. eight times
B. quadruple D. three times

***Explanation***

If the concentration of a given reactants is increased by a factor x, the rate increases by a factor $x^a$ where "a" is the order of the reaction with respect to that reactant rate = $k[NO]^2[I]$, so if the concentration NO is doubled, the rate becomes quadrupled.

**Answer: B**

23. A reaction is 50% complete in 2 hours and 75% complete in 4 hours. What is the ratio of this reaction?

A. 0 B. 1 C. 2 D. 3

***Explanation***

At t = 2hr, 50% of the reaction is complete implies $50\%[A]_o$. At t = 4 hr, 75% of the reaction is complete shows if 75% of the starting material is reacted, the amount left is (100% − 75%) which is 25% i.e., $\frac{25}{100}[A]_o$. Taking the two values ratio

$$\frac{\frac{5}{100}[A]_o}{\frac{25}{100}[A]_o} = \frac{50}{100} \times \frac{100}{25} = 2$$

**Answer: C**

24. Consider the following reaction:

$$HOCl + H_2O \rightleftharpoons H_3O^+ + OCl^-$$

What will happen if NaOCl is added to the above reaction at equilibrium?

A. The concentration of both HOCl and $H_3O^+$ would increase.
B. The concentrations of both HOCl and $H_3O^+$ would decrease.



C. The concentration of HOCl would increase and the concentration of $H_3O^+$ would decrease.
D. The concentration of HOCl would decrease and the concentration of $H_3O^+$ would increase.

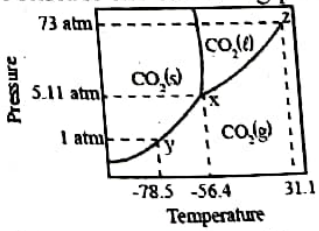
***Explanation***

Lechatelier's principle states that, "If an external stree is applied to a system at equilibrium the system will adjust itself in such away that it reduces the stress to reach a new state of equilibrium." The work stress means a change in temperature, pressure and concentration.

- Increasing concentration of the reactant shifts the equilibrium to be the right (product).
- Pressure affects only equilibrium system that involve gases.
- Increasing pressure affects the equilibrium to shift towards the least number of moles.
- A decrease in pressure affects the equilibrium to shift to greater no of gaseous molecules.
- Increasing pressure favors the endothermic reaction.
- Decreasing pressure favors the exothermic reaction.

$HOCl + H_2O \rightleftharpoons H_3O^+ + OCl^-$, hence adding NaOCl in this reaction increases the concentration of $OCl^-$ because $NaOCl + H_2O \rightleftharpoons Na^+ + OCl^-$. Therefore, according to Lechatlier's principle if one of the product concentration increases, the reaction shifts to the reactant. **Answer: C**

25. Consider the following phase diagram for $CO_2$.




What happens in a $CO_2$ sample intialling at 1 atom and -70°C, the temperature increases from -70°C to −10°C at a constant pressure of 60 atm?

A. $CO_2(g) \longrightarrow CO_2(s)$ C. $CO_2(g) \longrightarrow CO_2(g)$
B. $CO_2(s) \longrightarrow CO_2(\ell)$ D. $CO_2(g) \longrightarrow CO_2(\ell)$

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***Explanation***

- A phase diagram is defined as a graphical representation the condition when two or more phases (such as vapor-liquid, liquid-solid) of a system can exist in equilibrium.
- A substance exists in solid state at low temperature and high pressure.
- A substance exists in vapor phase at high temperature and low pressures.
- The remaining area between two regions represents liquid phase.

Hence keeping the pressure constant (p = 60 atom) and increasing temperature from –70°C to –10°C changes ($CO_2$ (s) to $CO_2$ ($\ell$)).

**Answer: B**

26. The dye bromothymol blue (HBb) is a weak acid whose ionization can be represented as follows:

$$HBb\ (aq) \rightleftharpoons H^+\ (aq) + Bb^-\ (aq)$$

Which way will the equilibrium shift when NaOH is added?

A. to the left
B. to the right
C. initially to the left and after a while to the right
D. initially to the right and after a while to the left

***Explanation***

The NaOH will dissociate as follows:

$$NaOH \longrightarrow Na^+\ (aq) + OH^-\ (aq)$$

Hence, the $OH^-$ obtained by the dissociation of NaOH reacts with $H^+$ obtained by HBb as a result its concentration decreases. As a result, decreasing one of the product concentration favors the reaction to move towards the product.

**Answer: B**

27. Consider the following equilibrium:

$$CaCO_3\ (s) \rightleftharpoons CaO\ (s) + CO_2\ (g)$$

Which of the following mixtures, each placed in a closed container and allowed to stand is not capable of reaching the equilibrium given above?

A. pure $CaCO_3$
B. $CaCO_3$ and CaO



C. some CaO and a pressure of $CO_2$ greater than the value of $K_P$
D. some $CaCO_3$ and a pressure of $CO_2$ than the value of $k_P$

28. If the following reaction is at equilibrium, which one of the following changes will shift the equilibrium to the left?

$$N_2 + 3H_2 \longleftrightarrow 2NH_3 + Heat$$

A. increasing pressure
B. decreasing temperature
C. adding more $N_2$ and $H_2$
D. increasing volume of the reaction container

***Explanation***

Factors affecting chemical equilibrium are change in concentration, temperature and pressure.

| Equilibrium factor | Position of an equilibrium |
|---|---|
| • increasing reactant concentration | • shifts the equilibrium to the left (product) |
| • decreasing reactant concentration | • shifts the equilibrium to the right (reactants) |
| • increasing pressure | • shifts the equilibrium to the less no of gas molecules |
| • decreasing pressure | • shifts the equilibrium to the higher no of gas molecules |
| • increasing temperature | • shifts the rxn towards endothermic side |
| • decreasing temperature | • shifts the equilibrium towards the exothermic side |
| • increasing product concentration | • shifts the equilibrium to the reactant |
| • decreasing product concentration | • shifts the equilibrium to the product |
| • increasing volume | • shifts the rxn to higher no of gas molecules |
| • decreasing volume | • shifts the rxn to the less no of gas molecules |

**Answer: D**

29. What is the name of a base-promoted ester hydrolysis reaction?

A. Acylation
B. Esterification
C. condensation
D. saponification

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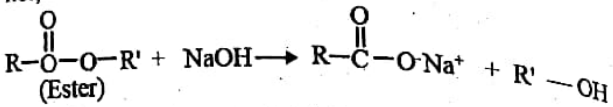
***Explanation***

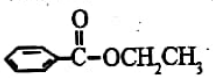
- Condensation is a reaction in which two molecule combine to form a larger one and a small molecule is eliminated.
- Esterification is a kind of condensation reaction in which an ester is formed from carboxylic acid and alcohol. i.e., carboxylic acid of alcohol → ester + water
- Saponificatioin is a base catalyzed ester hydrolysis. i.e.,




- Acylation is a kind of substitution in which aromatic rings are substituted with acyl group (RCO–).

30. What is the following compound?

**Answer: D**




A. Benzoate ester
B. Ethyl benzoate
C. Phenyl butyrate
D. Ethyl benzyl ketone

***Explanation:***

The esters are named by converting acid to oate and alcohol portion to alkyl group. The name for the given ester will be Ethylbenzoate.

**Answer: B**

31. Which of the following is a chemical formula that represents an amino acid?

A. $CH_4$
B. $CH_3NH_3$
C. $CH_3COOH$
D. $NH_2CH_2COOH$

***Explanation***

Amino acids are basic units of proteins.

- The general formula of amino acids is:

- Amino acids have both carboxylic acid (–COOH) part and basic part ($–NH_2$).

**Answer: D**

32. Which of the following statements is true about esters?

A. Esters can form intermolecular hydrogen bonds.
B. Ester molecules can form intermolecular hydrogen bonds.



C. Ester molecules cannot form intermolecular hydrogen bonds.
D. Esters have higher boiling points than alcohols of comparable molecular weight.

**Answer: C**

33. The organic compound $CH_3C(O)CH_3$ is:

A. aldehyde
B. ester
C. carbonyl
D. ketone

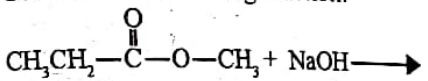
***Explanation:***

The general formula for ketone is:

$CH_3C(O)CH_3 = CH_3—C—CH_3$ - propanone

**Answer: D**

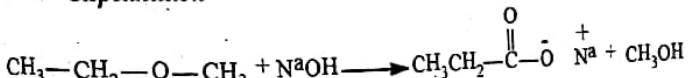
34. Consider the following reaction:




What are the products of this reaction?

A. Sodium acetate and ethanol.
B. Sodium acetate and methanol.
C. Sodium propionate and methanol.
D. Methyl prop ionic acid and methanol.

***Explanation***




**Answer: C**

35. Which catalyst is used in the hydrogenation of vegetable oils?

A. Iron
B. Nickel
C. Platnium
D. Molybednum

***Explanation***

- Hydrogenation is addition of hydrogen.
- The catalytic hydrogenation of oils (liquids) gives fats (solids). The process of converting oils or soft fats to hard fats is called hardening. The catalyst used is nickel.

**Answer: B**

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36. Which of the following is **NOT** a solution?
A. Milk C. Whisky
B. Brass D. Coca Cola drink

***Explanation*** Milk: Heterogeneous mixture
Brass: Homogeneous mixture and it is an alloy.
Whisky: Homogeneous mixture
Coca Cola: Homogeneous mixture
A solution is a homogeneous mixture of solutes and solvents.

**Answer: A**

37. How many ml of water is required to dilute 50 ml of 3.5 M $H_2SO_4$ to 2.00 M $H_2SO_4$?
A. 37.5 B. 45 C. 75 D. 87.5

***Solution:*** Applying dilution formula.
Number of moles before solution = Number of moles after dilution $(\text{Molarity} \times \text{Volume})_{\text{before dilution}} = (\text{Molarity} \times \text{volume})_{\text{after dilution}}$

$$M_1V_1 = M_2V_2$$

Given: $M_1 = 3.5\ M$ $V_1 = 50\ ml$
$M_2 = 2.00\ M$ $V_2 = ?$
$M_1V_1 = M_2V_2$

$V_{H_2O}$ added $= V_2 - V_1 = 87.50 - 50 = 37.5\ mL$

$$V_2 = \frac{M_1V_1}{M_2} = \frac{3.5M \times 50ml}{2M} = 87.5ml$$

**Answer: A**

38. How much water has to be evaporated from 250 ml of 1 M $Ca(OH)_2$ to make it 3M?
A. 100 ml B. 150 ml C. 167 ml D. 200 ml

***Solution:*** Applying dilution formula
$M_1V_1 = M_2V_2$
Given:

$V_1 = 250\ ml$ $V_2 = \frac{M_1V_1}{M_2} = \frac{250}{3} \times 1 = 83.3mL$

$M_1 = 1\ M$
$M_2 = 3\ M$
$V_2 = ?$

The amount of $H_2O$ to be evaporated is: 250 ml – 83.3mL = 166.7 ml.

**Answer: C**



39. The solubility of sodium selenate, $Na_2SiO_4$, is 84 g/ 100 g of water at 35°C. If a solution is obtained by dissolving 92 gram of $Na_2SiO_4$ in 200 gram of water at 35°C, what do you call this solution?
A. Dilute C. Unsaturated
B. Saturated D. Supersaturated

***Explanation***
Dilute: the amount of solvent is higher than the amount of solute.
Saturated solution: a solution in which dissolved and undissolved solute are in dynamic equilibrium.

undissolved solid $\underset{\text{crystallization}}{\overset{\text{dissolution}}{\rightleftharpoons}}$ dissolved ions

Rate of crystallization = Rate of dissolution
Supersaturated solution: A solution containing more solute than a saturated solution.
Solubility is the quantity of a substance dissolved in a given quantity (volume, mass) of the solvent at a given temperature to form a saturated solution. Having all these things, at 35°C 84 grams of $Na_2SiO_4$. Hence at the same temperature $2 \times 84 = 168$ gram of $Na_2SiO_4$ should be dissolved in 200 gram of water to form saturated solution. Taking amount of $Na_2SiO_4$ less than 168 gram in 200 gram of $H_2O$ makes the solution unsaturated.

**Answer: C**

40. Which of the following is most likely deviate from ideal gas behavior?
A. He B. Ar C. $Cl_2$ D. $CCl_2F_2$

***Explanation***
- The hypothetical gas that obeys the ideal gas law exactly at all temperature and pressures is called ideal gas.
- All gases are found to obey the law of an ideal gas more and more closely as the reduced to very low values.
- Real gases don't behave exactly as predicted by the ideal gas law. This deviation arises from intermolecular force. And this force is important at high temperature.
- Relatively speaking $CCl_2F_2$ will have a dipole-dipole force.

**Answer: D**

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41. Which of the following compounds would give the lowest freezing point depression when 100 g of each are dissolved in 1 kg of water ($k_f$ for water = 1.86°C/m)? Assume complete dissociation.

A. NaCl
B. $NH_4NO_3$
C. $(NH_4)_2SO_4$
D. glucose, $C_6H_{12}O_6$

***Explanation***

| Compound | Theoretical Vant Hoff factor |
|---|---|
| NaCl | 2 |
| $NH_4NO_3$ | 2 |
| $(NH_4)_2SO_4$ | 3 |
| Glucose, $C_6H_{12}O_6$ | 1 |

Note: The theoretical Van't Hoff's factor is calculated by summing up he cations and anions.

$\boxed{T_f = T°_f - ik\,fm}$ $\Delta T_f$ – freezing point depression

$T°_f$ – freezing point of pure solvent

m – molality

$k_f$ – freezing point depression constant

i – vant hoff's factor

$$\text{molality} = \frac{\text{no of moles}}{\text{mass of solvent in kg}}$$

For NaCl:

Number of moles of NaCl

$$= \frac{\text{mass}}{\text{molar mass}} = \frac{100 \text{ gram}}{58.5 \text{ gram/mol}} = 1.7 \text{ mol}$$

$$\text{molality} = \frac{\text{number of moles}}{\text{mass of solvent (kg)}} = \frac{1.7 \text{ mol}}{1 \text{ kg}} = 1.7 \text{ mol}$$

$T_f = T°_f - ik_fm = 0°C - (2)(1.86)(1.7) = -6.324°C$

For $NH_4NO_3$:

$$\text{Number of moles} = \frac{\text{mass}}{\text{Molar mass}} = \frac{100 \text{ gram}}{80 \text{ g/mol}} = 1.25 \text{ mol}$$

$$\text{molality} = \frac{\text{number of moles}}{\text{mass of the solvent in kg}} \frac{1.25}{1} = 1.25 \text{ molal}$$

$T_f = T°_f - ik_fm = 0°C - (2)(1.84)(0.76) = -4.6°C$



For $(NH_4)_2SO_4$:

$$\text{Number of moles} = \frac{\text{mass}}{\text{molar mass}} = \frac{100 \text{ gram}}{132 \text{ gram/mol}} = 0.76 \text{ moles}$$

$$\text{molality} = \frac{\text{no of moles}}{\text{mass of the solvent (kg)}^2} = \frac{0.76}{1} = 0.76 \text{ molal}$$

$\Delta T_f = T°_f - ikfm = 0°C - 3(1.86)(0.76) = -4.22°C$

For glucose, $C_6H_{12}O_6$:

$$\text{Number of moles} = \frac{100}{180} = 0.56$$

$$\text{Molality} = \frac{\text{number of moles}}{\text{mass of solvent (kg)}} = \frac{0.56}{1} = 0.56$$

$T_f = T°_f - ik_fm = 0\,°C - (1)(1.86)(0.56) = -1.03°C$

**Answer: A**

42. Consider the following balanced equation.

$3Ba(NO_3)_2(aq) + Fe_2(SO_4)_3(aq) \longrightarrow 3BaSO_4(s) + 2Fe(NO_3)_3(aq)$

the net ionic equation to describe this balance equation is?

A. $3Ba^+(aq) + SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

B. $3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

C. $6NO_3^-(aq) + 2Fe^{2+}(aq) \longrightarrow 2Fe(NO_3)_3(aq)$

D. $3Ba^{2+}(aq) + 2NO_3^-(aq) + 2Fe^{3+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s) + 2Fe^{3+}(aq) + 6NO_3^-(aq)$

***Explanation***

- By looking at the state of the reactant and product tells whether the compound is written in the form of ion or not.
- If the compound exists in aqueous (aq) form, write that compound in the form of ion otherwise leave it as it is (i.e., in its molecular form). For the reaction given below:

$3Ba(NO_3)_2(aq) + Fe_2(SO_4)_3(aq) \longrightarrow 3BaSO_4(s) + 2Fe(NO_3)_3(aq)$ the ionic equation becomes

$3Ba^{2+}(aq) + 6NO_3^-(aq) + 2Fe^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s) + 2Fe^{3+}(aq) + 6NO_3^-(aq)$.

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And to find the net ionic equation, the spectator ions should be cancelled. The net ionic equation is:

$3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

Note: The spectator ions are ions that exists at the left and right side of the reaction. For the above reaction the spectator ions are $NO_3^-$, and $Fe^{3+}$.

**Answer: B**

43. Which one of the following organic molecules has the highest water solubility?

A. $HOCH_2CH_2CH_2OH$
B. $CH_3CH_2CH_2CH_2OH$
C. $HOCH_2CH_2CH_2CH_2OH$
D. $CH_3CH_2CH_2CH_2OH$

***Explanation***

- All the choices given are alcohols. The solubility of alcohols can be affected by:

A. Number of OH (hydroxyl) groups.
- Solubility of alcohol increases with increasing number of hydroxyl groups.

B. Carbon number
- Solubility of alcohols decreases with increasing carbon number.

**Answer: A**

44. A solution is labeled 0.500 M HCl. What is $[H^+]$?

A. 0.5 B. 1.5 C. 1.6 D. 1.69

***Explanation***

Strong acids dissociates almost completely. Hence

$HCl(aq) \longrightarrow H^+(aq) + Cl^-(aq)$

0.5 M 0.5M 0.5 M

For strong mono protic acid, [Acid] = $[H^+]$ **Answer: A**

45. How is a buffer solution prepared?

**A.** By mixing a weak acid and a strong acid.
**B.** By mixing a weak acid and its conjugate base.
**C.** By mixing a strong acid and its conjugate base.
**D.** By mixing a strong base and its conjugate acid.

***Explanation***

- Buffer solution is a homogenous mixture that resists the pH change up on small addition of acids or bases.
- Buffer solution contains an acid and a base parts together that should not consume each other but react with externally added acid or base.



- Buffer solutions are made by conjugate acid-base pair. There are two kinds of buffer solutions.

A. Acidic buffer: A buffer solution made form a weak acid and its conjugate base. Example $CH_3COOH/CH_3COO^-$.

B. Basic buffer: A buffer solution made from a weak base and its conjugate acid. Example: $NH_3$/ $NH_4^+$

**Answer: B**

46. Which of the following compounds would be the most basic?

A. 0.1 M acetic acid
B. 0.1 M sodium acetate
C. 0.1 M hydrochloric acid
D. 0.1 M ammonium chloride

***Explanation***

- 0.1 M $CH_3COOH$ is an acid with pH < 7
- 0.1 M $CH_3COONa$ is a salt obtained from a weak acetic acid ($CH_3COOH$) and strong base is always basic (pH>7)
- 0.1 M HCl is an acid with pH < 7
- 0.1 M $NH_4Cl$ is a salt drived from weak base ammonia ($NH_3$) and strong acid HCl. Hence, a salt obtained from weak base and strong acid is acidic solution i.e., pH < 7.

**Answer: B**

47. Which one of the following statement is **NOT** true about acids?

A. an acid is a proton donor
B. an acid is an electron pair donor
C. an acid is an electron pair acceptor
D. an acid is a substance that is ionized in water to produce $H^+$ (aq).

***Explanation:*** According to:

1. Arrhenius Definition "An acid is a substance that is ionized in water to give $H^+$ (aq)"
2. Bronsted – Lowry definition "An acid is a proton donor"
3. Lewis "An acid is an electron pair acceptor"

**Answer: B**

48. Consider the following acids;

I. $CH_3COOH$, $Ka = 1.8 \times 10^{-5}$
II. $HCO_2H$, $Ka = 1.8 \times 10^{-4}$
III. HOBr, $Ka = 2.4 \times 10^{-9}$
IV. $C_6H_5OH$, $Ka = 1.0 \times 10^{-10}$

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